

Advances in Science, Technology & Innovation
IEREK Interdisciplinary Series for Sustainable Development



Jahan Hassan · Sara Khalifa ·
Prasant Misra *Editors*

Machine Learning for Drone-Enabled IoT Networks

Opportunities, Developments, and Trends



Advances in Science, Technology & Innovation

IEREK Interdisciplinary Series for Sustainable Development

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Jahan Hassan · Sara Khalifa · Prasant Misra
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Opportunities, Developments, and Trends

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Preface

The convergence of Unmanned Aerial Vehicles (UAVs) and the Internet of Things (IoT) is revolutionizing industries and unveiling a world of new possibilities. This integration not only enhances operational efficiency, but also fosters innovation across diverse sectors, from agriculture to disaster management. At the heart of this transformation is the power of machine learning, which serves as a crucial link between UAVs and IoT, enhancing their applications with intelligence. It enables sophisticated data analysis, real-time decision-making, and autonomous operations, paving the way for smarter, more responsive systems. As the editors of *Machine Learning for Drone-Enabled IoT Networks: Opportunities, Developments and Trends*, we are thrilled to present a comprehensive exploration of this dynamic field, highlighting the latest advancements, challenges, and future directions that will shape the landscape of UAV and IoT integration.

The first chapter “[Unmanned Aerial Vehicles Enabled IoT Platform for Effective Disaster Management](#)” explores the transformative role of Unmanned Aerial Vehicles (UAVs) integrated with Internet of Things (IoT) platforms in disaster management. It emphasizes their capacity for real-time data collection and enhanced situational awareness during emergencies, illustrating how these technologies can detect anomalies and support informed decision-making. The chapter introduces advancements in UAV–IoT networks and outlines various approaches to system architecture and data collection, featuring algorithms such as YOLO and LSTM. Additionally, it addresses critical challenges, including security, privacy, and network coverage while providing guidelines for optimizing these technologies in disaster response. Looking ahead, the first chapter “[Unmanned Aerial Vehicles Enabled IoT Platform for Effective Disaster Management](#)” envisions potential advancements in AI, deep learning, and blockchain technology, highlighting their role in improving the efficiency and effectiveness of disaster management. This comprehensive overview aims to illuminate the current and future impact of UAV-enabled IoT platforms in emergency scenarios.

The second chapter “[Exploring Machine and Deep Learning for Crop Detection and Yield Prediction in Drone-Enabled IoT Networks](#)” explores the transformative advancements in agriculture driven by the integration of drones and Internet of Things (IoT) platforms. It highlights how these technologies have revolutionized traditional practices, turning labor-intensive tasks like crop counting and production estimation into efficient, data-driven processes. By leveraging machine learning (ML) and deep learning (DL) models, the chapter demonstrates effective strategies for analyzing agricultural data tailored to specific factors such as crop types and image resolutions. Through a focused study on managing apricot crops, it showcases the capabilities of YOLOv10, which achieved an impressive F1 Score of 88%, and a linear regression model for yield prediction with a low Relative Error of 2.18%. This exploration underscores the remarkable potential of ML and DL technologies to revolutionize precision agriculture and enhance productivity.

In the third chapter “[Drone Swarm Coordination Using Machine Learning in IoT Networks](#)”, authors explore the transformative potential of drone swarm coordination using machine learning in IoT networks. As Unmanned Aerial Vehicles (UAVs) revolutionize the aviation industry, the concept of UAV swarms enables multiple drones to operate autonomously, executing tasks without human intervention. This chapter underscores the critical role of communication in managing these swarms, highlighting the need for robust systems

to tackle challenges such as coordination, control, and security. By leveraging distributed learning algorithms, drone swarms can enhance capabilities like target tracking, remote surveillance, and efficient resource allocation. The chapter also reviews the application of machine learning techniques for swarm communication, addressing both the opportunities and challenges that arise in optimizing drone performance within IoT ecosystems, ultimately paving the way for advanced applications in various fields.

The fourth chapter “[Security and Privacy Issues in Drone-Enabled IoT Networks](#)” examines the pressing security and privacy issues inherent in drone-enabled IoT networks. While the integration of drones with the Internet of Things offers remarkable potential across various industries, it also raises significant concerns regarding data interception, unauthorized access, and physical security. This chapter explores these vulnerabilities in detail, addressing the implications of data breaches and denial-of-service attacks. It highlights the pivotal role of machine learning in identifying and mitigating malicious activities, emphasizing the importance of implementing robust security measures to enhance the integrity and privacy of drone operations. By discussing advanced security protocols and innovative technologies, the chapter aims to provide insights into creating safer, more efficient drone-enabled IoT networks that protect sensitive data and ensure reliable operations in an increasingly interconnected world.

Building on the critical security and privacy issues discussed in the fourth chapter “[Security and Privacy Issues in Drone-Enabled IoT Networks](#)”, the fifth chapter “[Enhancing Capabilities and Security Features of Drone Networks Through Machine Learning: A Comprehensive Overview](#)” offers an in-depth exploration of how machine learning can significantly enhance the capabilities and security of drone networks. It examines the integration of IoT and collaborative learning techniques, empowering drones to operate autonomously and adapt seamlessly to dynamic environments. By leveraging large datasets, drones can achieve improved situational awareness, predictive maintenance, and optimized mission planning through advanced methods like adaptive reinforcement learning. The chapter also discusses the crucial role of federated learning and swarm intelligence in enhancing operational efficiency and effectiveness across various applications, including traffic management and disaster response. Additionally, it addresses pressing concerns such as privacy issues and regulatory compliance, setting the stage for future innovations that will further advance the security and functionality of drone networks in the rapidly evolving IoT landscape.

The sixth chapter “[Unlocking the Potential of Machine Learning for Enhanced Security in Drone-Enabled IoT Networks](#)” explores the pressing security vulnerabilities inherent in drone-enabled IoT networks. As drones become integral to various applications, their susceptibility to hacking, data theft, and other malicious activities poses significant risks to public safety and privacy. This chapter highlights the critical need for robust security measures and examines how sophisticated machine learning techniques can effectively address these challenges. By analyzing real-time data, machine learning algorithms can detect anomalies, predict potential threats, and autonomously adapt security protocols to evolving risks. Additionally, the chapter discusses the integration of advanced encryption methods and other security frameworks designed to protect drone operations from interference. Through this exploration, readers will understand the importance of implementing effective security solutions to ensure the safe and responsible integration of drones into the broader IoT ecosystem.

The seventh chapter “[Machine Learning Models for Drone Security: Cognitive Versus Cyber Intelligence for Safety Operations](#)” explores the critical intersection of machine learning and cybersecurity within drone networks, emphasizing the need for robust data governance to ensure the integrity and authenticity of communications among network participants. As airspace violations by drones become more prevalent, it is essential to rigorously evaluate the foundational security architectures governing their operations. This chapter delves into how artificial intelligence models can enhance trust and reliability in drone systems, addressing key security challenges such as ownership authentication, delivery verification, and operational authorization. By integrating zero trust architecture, blockchain technology, and explainable AI, the chapter highlights innovative strategies for ensuring security and accountability in drone operations. Readers will gain valuable insights into the evolving

landscape of drone security and the role of advanced technologies in fostering safe and effective operations.

The eighth chapter “[Cost–Benefit Analysis of Sensing and Data Collection with Drones for IoT Applications](#)” examines the transformative role of drones in the Internet of Things (IoT) landscape, focusing on their ability to enhance sensing and data collection capabilities. As industries increasingly adopt drone technology, understanding the economic implications of these advancements becomes crucial. This chapter presents a thorough cost–benefit analysis of using drones equipped with various sensors, highlighting how they improve operational efficiency and data accuracy in IoT applications. By evaluating initial acquisition costs, maintenance expenses, and compliance requirements alongside the substantial benefits—such as faster data collection and access to challenging environments—this chapter sheds light on the financial viability of drone integration. Furthermore, it addresses the challenges associated with this analysis, including data uncertainty and ethical considerations, while showcasing how technological advancements are driving down costs and enhancing benefits. Through this exploration, the eighth chapter “[Cost–Benefit Analysis of Sensing and Data Collection with Drones for IoT Applications](#)” underscores the strategic importance of drones in modern IoT networks, revealing their potential to optimize operations across multiple sectors.

The ninth chapter “[Economic Analysis of AI Integration in Internet of Drones \(IoD\)](#)” examines the economic ramifications of integrating artificial intelligence (AI) within the Internet of Drones (IoD). As the drone industry experiences rapid growth, projected to reach a market size of \$42.8 billion by 2025, understanding the financial implications of AI adoption becomes paramount. This chapter evaluates cost and benefit structures, emphasizing how AI enhances operational efficiency, reduces labor costs, and increases reliability in drone operations. It also addresses challenges such as initial development costs, regulatory compliance, and potential workforce displacement due to automation. By exploring various economic models, including pay-per-use services and public-private partnerships, this chapter aims to provide insights into the long-term impacts of AI integration in the IoD, setting the stage for future research in this dynamic field.

The tenth chapter “[Emerging Trends and Future Directions in ML for Drone-Assisted IoT Networks](#)” focuses on emerging trends and future directions for machine learning (ML) within drone-assisted IoT networks. As these technologies evolve, the chapter highlights key areas of progress such as navigation, real-time data processing, and autonomous operations. By addressing current challenges like security protocols and communication optimization, it outlines how ML can enhance operational efficiency and adaptability in drone ecosystems. Furthermore, the chapter discusses opportunities for future advancements, including cybersecurity, ethical considerations, and the application of meta-learning in UAV operations. This forward-looking perspective not only emphasizes the transformative potential of ML in shaping the next generation of intelligent IoT infrastructures but also encourages continued research and innovation in this rapidly advancing field.

In conclusion, this book offers a thorough exploration of the intersection between machine learning, drones, and IoT networks, showcasing their potential to transform various industries. Each chapter deepens our understanding of the advancements, challenges, and opportunities in this dynamic landscape. By integrating cutting-edge technologies and addressing critical issues like security, economic feasibility, and operational efficiency, we aim to inspire future research and practical applications that maximize the benefits of drone-enabled IoT networks.

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Unmanned Aerial Vehicles Enabled IoT Platform for Effective Disaster Management

R. Anjana Devi , V. Harshini Amutha  and Priya Thiagarajan 

Abstract

This chapter explores the revolutionary role of Unmanned Aerial Vehicles (UAVs) integrated with Internet of Things (IoT) platforms in disaster management, highlighting their potential to deliver real-time data collection and improve awareness during emergencies. It begins with an introduction to UAVs and IoT technologies, explaining their advancement and help in disaster scenarios with UAV-IoT networks being tested, emphasizing how these technologies detect anomalies and decision-making. A few adopted approaches for system architecture and data collection are given, explaining the use of algorithms such as YOLO and LSTM. This chapter also highlights the important obstacles while implementing UAV-enabled IoT devices, security and privacy, and network coverage. Guidelines for enhancing the use of these technologies in disaster management are given. Looking into the future, this chapter explains the possible improvements in AI, deep learning, and blockchain technology, which helps to improve UAV-IoT networks. It visualizes advanced applications like predictive analytics, autonomous UAV operations, and collaboration internationally, which enhance the efficiency and effectiveness of disaster response. This broad overview aims to realize and give insights into the current and future potential of UAV-enabled IoT platforms, highlighting their critical role in disaster management.

Keywords

Integration of IoT with UAV · YOLO · Real-time data · ML algorithms

1 Introduction

Over the past decade, the integration of Unmanned Aerial vehicles (UAVs) and the Internet of Things (IoT) has paved its way in various sectors, including military, agriculture, logistics, and more. These technologies are widely being used in disaster management strategies due to their effectiveness over manual inspections. The integration of UAVs and IoT provides real-time data collection and decision-making during emergencies (Pettit et al. 2022).

1.1 IoT

In recent years, the internet has made an impact in the journey towards connectivity in the era of the IoT. The concept of IoT was first introduced by Peter T. Lewis in 1985. He defined it as the fusion of individuals, processes, and technology with interconnected devices and sensors (Contributors 2023). The 2005 report of the International Telecommunications Union (ITU) suggested that “Internet of Things will connect the world’s objects in both a sensory and intelligent manner” (International Telecommunication Union (ITU) 2005). IoT is a network of interconnected devices that communicate and exchange data over the internet. It has transformed the twenty-first century by enhancing the decision-making process during emergencies and disasters through real-time data collection and exchange (Ananna and Saifuzzaman 2023). These data form the basis for analysis, decision-making, and subsequent actions, which can be physical, like adjusting a smart thermostat

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automatically, or virtual, like sending notifications to the mobile (Sethi and Sarangi 2017). The emergence of IoT has transformed the way people interact with technology by providing convenience and personalization. The substantial number of devices and data generated pose significant challenges, with a projected 41.6 billion IoT devices producing 79.4 zettabytes of data by 2025 (Internet of Things and Data Placement | Edge to Core and the Internet of Things | Dell Technologies Info Hub, n.d.).

The IoT architecture consists of objects embedded with sensors, actuators, processors, and software. It has various communication layers and IoT protocols. Based on necessities, the IoT systems follow either the three-layer architecture, four-layer architecture, or five-layer architecture. The three-layer architecture, also known as the generalized architecture, is divided into 3 layers: the application layer, the network layer, and the perception layer (Sethi and Sarangi 2017). The representation of the generic architecture, which is the most commonly used architecture, is represented in Fig. 1 (Generic architecture of IoT n.d.).

The perception layer, also known as the sensor layer, is used to collect data from the devices via sensors and transmit the data to the upper layer, which is known as the network layer (Atzori et al. 2012). The network layer, also known as the transmission layer, is responsible for the two-way transmission of data to and from the applications through interfaces and gateways (Leo et al. 2014). The application layer or the business layer is responsible for the aggregation of data from the network layer and is responsible for upholding the data's authenticity, integrity, and confidentiality (Wu et al.

2010). In the future, the applications of IoT will be further expanded by machine-to-machine (M2M) communications, Big data, and the cloud (Nayak 2019).

The use of advanced deep learning algorithms like Convolutional Neural Networks (CNN), Region-Based Convolutional Neural Networks (R-CNN), Bidirectional Encoder Representations from Transformers (BERT), You Only Look Once (YOLO), Long Short-Term Memory (LSTM), Support Vector Machine (SVM), and many other algorithms further maximizes the benefits of integrating IoT and UAV technologies in disaster management (Chirumamilla et al. 2023, Rate et al. 2023).

1.2 UAVs

UAVs, commonly known as drones or Unmanned Aircraft Systems (UAS), were initially used in the military, particularly for surveillance in the air. UAVs are generally referred to as a flying machine that are operated remotely or autonomously without a human pilot on board (Yong et al. 2022). Back in the 1800s, the UAS began with a balloon and kite. During the World War periods, there were progressive developments in the design of UAVs. Starting with the usage of unmanned aircraft in World War I, this progress led to the development of various unmanned aerial systems as weapons during World War II (Fahlstrom and Gleason 2012). Over recent decades, UAVs have been considered as a transformative technology in many sectors due to their rapidly evolving capabilities and applications beyond the military. It plays a significant role in various areas such as agriculture, remote sensing, rescue operations, and many more applications such as remote sensing, surveillance, precision agriculture, and rescue operations. The UAVs can gather photographs faster and more accurately than satellite imagery and manned systems (UAVs for Multidisciplinary Applications 2022).

Based on features such as size, altitude, landing method, weight, endurance, and purpose, the UAVs can be classified into different types. The performance of drones depends on the type of embedded electronics used and the material used to make them. Nowadays, UAVs are equipped with a large number of sensors and communication devices (Sönmez 2022). In general, UAVs can be categorized into two categories: Multi-rotor UAVs and Fixed-Wing UAVs. Fixed-wing UAVs are known for their efficient long-range surveillance, whereas multi-rotor drones are known for their aerial photography (Priyashree et al. 2023). A study on the comparison of two types of UAVs was made by P. K. Garg in 2022, and the results are tabulated in Table 1. Even though Fixed-wing provided satisfactory results, the study concluded that Multi-rotor data represented the environmental features more accurately (Garg 2022).

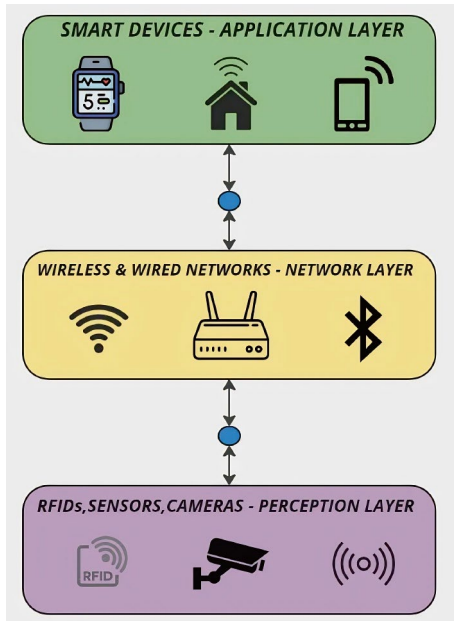


Fig. 1 Generic architecture of IoT

Table 1 Suitability for different attributes for both types of UAVs (Garg 2022)

ATTRIBUTES	FIXED-WING (SKY-WALKER)	MULTI-ROTOR (RAVEN)
PAYLOAD CAPACITY		✓
FLIGHT TIME	✓	
STABILISATION		✓
GEOREFERENCING	✓	
COST	✓	
MAINTAINANCE	✓	
VEGETATION REPRESENTATION		✓
EROSION GULLY REPRESENTATION		✓
WETLAND SLOPE MAPPING	✓	✓
CONTOUR MAPPING	✓	✓

1.3 Current Trends Due to UAVs and IoT in Disaster Management

In recent years, the usage of tools and technology for disaster management has rapidly increased. As mentioned in Fig. 2, disaster management can be divided into 3 phases, such as preparedness, response, and recovery (Sspaulding 2024). The integration of UAVs and IoT has played a significant role in mitigating the risk during all 3 phases.

Even before a crisis strikes, these technologies aid in the prevention and preparedness phase by the implementation of early warning systems and conducting infrastructure

inspection. The data collected from UAVs and sensors assists in risk assessment, enabling more effective planning to reduce the damage caused by disasters. For the early warning system, UAVs are equipped with sensors to monitor and collect data on environmental conditions such as temperature, humidity, rainfall, water levels, and seismic activities (Ejaz et al. 2019). Connected devices analyze in real-time and look for anomalies that could probably be early signs of potential disasters. If an anomaly is detected priorly, it allows the authorities to take proactive measures. Similarly, UAVs can be used to conduct aerial surveillance to identify vulnerable areas and assess risks. UAV's capability to gather high-resolution images, which helps the authorities to identify the spots that require mitigation efforts and maps the geographical features (IoT and to predict and prevent natural disasters -TagoIO (n.d.), TELECOMMUNICATION ENGINEERING CENTRE 2024).

During the disaster, UAVs and IoT technologies can be used for real-time data communication, environment monitoring, traffic management, and search and rescue operations. UAVs equipped with thermal imaging assist in locating survivors efficiently during emergencies. However, the resolution of the thermal image will be low, and information about the color and texture could not be found. The described system (Yeom 2024) identifies the targets from the video clippings taken by the UAVs during a crisis. This can cover large areas quickly, and it has the capability to identify individuals trapped under debris or in remote locations. Yeom (2024) IoT devices equipped with satellite communication are used in disaster-hit areas, even when terrestrial communication networks are down. Satellite communications provide a reliable alternative to keep the isolated people connected to the outside world (CANLA n.d.). During the disaster, UAVs and IoT devices provide real-time data, which plays a substantial role in making situational awareness to the outside world. UAVs provide aerial imagery for comprehensive damage assessment in hard-to-reach areas.

During the aftermath of the disaster, UAVs are used for rapid damage assessments, monitoring the recovery

**Fig. 2** Phases of disaster management

process, enhancing search and rescue (SAR) operations, and allocations of resources (TELECOMMUNICATION ENGINEERING CENTRE 2024). Hence, by the usage of UAVs with IoT-enabled surveillance systems, damage assessment and restoration efforts can be carried out effectively by prioritizing SAR operations and reconstruction of infrastructure. Despite the effective use of UAVs and IoT for disaster management, there exist a few challenges, such as interoperability issues, data security and privacy concerns, infrastructure limitations, and high costs. These issues are further elaborated in Sect. 4 (Admin 2024).

1.4 Legal Framework for UAVs and IoT Deployment in Disaster Management

The legal framework ensures that the usage of UAVs and IoT has to be done effectively and responsibly by abiding to the international laws and regulations. The legal framework addresses issues such as airspace sovereignty, privacy, and human rights. When deploying UAVs and IoT in disaster management, we have to consider the national and international laws while maximizing the potential of these technologies (The use of drones for commercial purposes and privacy rights of others—legal developments n.d.). The usage of UAVs has many implications with respect to the international laws. This section gives an overview of the international laws regarding the usage of UAVs (Implications of the use of drones in international law—Peace in Progress magazine 2022).

1. **Airspace Sovereignty:** Before operating in any airspace, the UAVs must have the consent from the respective state government. This ensures that operations involving UAVs are coordinated smoothly without any issues.
2. **Privacy and Data Protection:** Privacy is considered to be one of the biggest threats of UAVs due to its surveillance capabilities and data collection. IoT devices are used to store the collected data. Hence, this raises privacy concerns. Legal policies ensure that an individual's privacy and data are secured properly while transmitting and storing.
3. **International Humanitarian Law:** Aligning to the law, UAVs used for disaster management should be unarmed and should prevent harm to the civilians.
4. **Human Rights:** The data collected from the UAVs and IoT deployments during disasters should respect the individual's rights. The legal framework holds the operators accountable for any sort of violation.

These laws are broadly applicable to all nations, but however, individual countries may have additional concerns.

The legal framework defines the clear guidelines on the conditions and limitations of the usage of UAVs and other technologies (Implications of the use of drones in international law—Peace in Progress magazine 2022).

2 Machine Learning Algorithms for UAV-Enabled IOT Networks

2.1 ML in Disaster Management (Predictive Analysis)

Generally, for applications such as disaster management, we use cellular-connected drones. Cellular-connected drones are UAVs that are connected with communication modules and many other components, such as cameras. These use cellular technology to establish a connection to the internet or central control station (Bednasz 2023). However, there exist a few communication and security challenges with the usage of cellular-connected drones. To acknowledge these challenges, we use Machine Learning (ML) based solutions to improve the design of the network. ML methods are considered effective when compared to other existing approaches because of its in-built ability to predict future network states. This prediction analysis would allow the UAVs to adjust dynamically according to the network and is effective for large-sized networks (Shahbazi 2022). The main types of ML are Supervised Learning, Unsupervised Learning, Semi-supervised Learning and Reinforcement Learning (RL). These methods are used to classify and form clusters of the collected data with which we can identify patterns and make predictions (Shafik 2022, Linardos et al. 2022).

ML is used to analyze large amounts of data from various sources, such as social media, sensors, and satellite images that are preprocessed, and the respective ML methods are used to identify patterns, and make predictions. ML is also used in risk assessment, post-disaster response systems, disaster detection, and early warning systems (Linardos et al. 2022; Machine Learning for Disaster Risk Management n.d.). It also faces challenges such as the need for large datasets, which need to be preprocessed, trained, and validated to ensure accuracy. Data preprocessing is a crucial step that includes data cleaning, removing data inconsistencies, converting data to a suitable format for analysis, handling null and categorical values, and feature extraction to enhance the quality of the data and the working of the model (Shwetha 2022). Practically, while implementing ML, we must ensure that the ML model used is transparent and interpretable. Few popularly known Disaster Management Systems where ML was widely used are to predict flood risks, detect landslides, and

assess damage from natural disasters (Linardos et al. 2022; Machine Learning for Disaster Risk Management n.d.).

2.2 Anomaly Detection in Sensor Data

Anomaly Detection is the process of identifying patterns in data that deviate from the expected behavior. Anomaly Detection is also referred as detection of faults or outliers (AZoSensors 2023). ML algorithms such as Random Forest, Support Vector Machine (SVM), and Neural Networks can be used for Anomaly detection. Similarly, DL algorithms such as Autoencoders, Convolutional Neural networks (CNN), and Recurrent Neural networks (RNN) can also be used. Statistical methods such as hypothesis testing and statistical process control can also be used to detect anomalies (Models and for detecting sensor faults and outliers in the IoT—a survey 2019; Cucurull et al. 2010a).

In disaster management, anomaly detection models identify the unusual patterns or events that indicate a disaster. For example, by analyzing data from seismic sensors, weather satellites, and historic data, we can identify disasters such as earthquakes, tsunamis, or hurricanes. It is also used to identify unusual patterns in emergency response data, like changes in population distribution or sudden increase in emergency calls. Identifying such patterns helps in effective resource allocation and provides a proper response. It is also used to monitor infrastructures if there are any signs of damage or failure to prevent accidents and minimize damage (Cucurull et al. 2010a; Esmaeili et al. 2023). Anomaly detection requires high-quality data to function effectively, as poor-quality data may lead to false positives and negatives. Large volumes of data make it challenging to detect anomalies as the data may contain noises. Thus, acquiring data for Anomaly detection is a bit challenging. Anomaly detection algorithms can be difficult to interpret and to understand the reasons behind the detected anomalies (Cucurull et al. 2010a). Practically, Anomaly detection identifies extreme events like wildfires, cyclones, and floods using satellite data (Anomaly Detection in Satellite Videos using Diffusion Models 2023).

2.3 Object Detection & Semantic Segmentation

Object detection and Semantic segmentation are popular computer vision techniques. These ML techniques are used to analyze the image captured by UAVs, which helps to identify objects and environmental features during disasters.

Object detection involves localizing and classifying specific objects within the image or video. The main

objective is to identify the object of interest and locate it, then draw the bounding boxes around the object of interest. Object detection in computer vision can be achieved with many approaches, such as Viola-Jones, Scale-invariant feature transform (SIFT), or by using a histogram of oriented gradients. It can also be implemented using modern algorithms such as Haar cascade classifiers, YOLO, faster R-CNN, Residual Networks (ResNet), and Mobile Ad Hoc Networks (MANet) (Cucurull et al. 2010b; Chaitanya and Gauri 2023). Advancements such as CenterNet++ have shown superior performance over traditional algorithms such as YOLO (Computer Vision Models Application in the Current System on Object Detection Tasks 2023). Object Detection is mostly used for damage assessment during disasters. Rizwan Sadiq et al. has shown the effectiveness of using object detection models in disaster scenarios to identify relevant objects in social media or satellite images (Rizwan et al. 2023). We can also apply object-based image analysis to the Earth observation data to map the landslide features. Object-based image analysis does risk assessment for landslides, which is very helpful in areas highly susceptible to landslides (Application of Object-Based Image Analysis for Detecting and Differentiating between Shallow Landslides and Debris Flows 2023).

Semantic segmentation is the process of dividing an image or video file into semantically meaningful regions and assigning labels to each and every pixel present. It provides detailed information about object boundaries and regions (Yawar 2023). Semantic Segmentation focuses on categorizing the pixels and provides detailed information on each category, which allows better understanding of the visual content (Ejaz et al. 2019). Semantic segmentation can be challenging as it needs extensive annotated data, which is costly and consumes a lot of time to acquire. Large-scale datasets have issues such as class imbalance and data diversity, which affects the performance of semantic segmentation models. These challenges are addressed by new benchmark datasets like ImageNet-S to develop new methods like Generative Adversarial Networks (GAN). Recent advancements in semantic segmentation networks such as Generative Networks (G-Net), U-shaped Networks (U-Net), ResNet, Segmentation Networks (SegNet), Labeled Co-segmentation Networks (LCSegnet), and Fast Lightweight Segmentation Network (FLSNet) has improved the accuracy of semantic segmentation (Mander 2023).

Object detection and Semantic segmentation are techniques in computer vision that serve different purposes. Semantic segmentation provides information about object boundaries, while object detection identifies specific objects and their locations. Semantic segmentation focuses on dividing an image into meaningful regions, where else object detection localizes and classifies specific objects (Cucurull et al. 2010b).

2.4 NLP for Text Analysis

Natural Language Processing (NLP) tools are used to analyze and interpret human language by identifying patterns, sentiments, and the linguistic structure of the sentence. Social media text analysis would be very helpful in getting information on currently happening disasters. It provides information on the current state of the event, the location, and the time (Gründer-Fahrer et al. 2018). NLP models use ML algorithms such as Bidirectional Encoder Representations From Transformers (BERT), Generative pretrained Transformer 2 (GPT-2), Text-to-Text Transfer Transformer (T5), and Robustly Optimized BERT Pre-Training Approach (RoBERTa) to understand the human language. Large volumes of data can be handled using the development of more complex NLP models with the advancements in AI and ML (GeeksforGeeks 2024).

2.5 Challenges in Using ML for Disaster Management:

As discussed previously, ML provides significant advantages in disaster management. However, several challenges need to be addressed, particularly with respect to various factors such as data quality, accuracy, reliability in prediction, and the need for computational resources to process massive datasets, especially image datasets. Accurate predictions and accuracy of the algorithm are completely based on the quality and completeness of the data used. Acquiring high-quality data is particularly challenging in remote or rural places, where they have poor data access. ML algorithms are not foolproof, and there are possibilities for false positives and inaccurate predictions. In addition to these general challenges, there are limitations while using ML algorithms for disaster management. For instance, it is difficult to predict the path of a hurricane as it requires explorations of various factors such as wind, temperature, speed, and pressure. If the given criteria is too complex or if it has lots of missing and null values, the algorithm might find it challenging to forecast properly (Jain et al. 2023). Using predictive modeling without planning and oversight can introduce biases. This can lead to false responses due to the usage of skewed data for training the model, thereby reinforcing current social imbalances. Hence, predictive algorithms might perpetuate biases or discrimination against certain communities (Motunrayo 2024).

The integration of anomaly detection models in disaster management is recognized as an effective system to mitigate risk. However, aside from data quality, a significant challenge faced in the usage of these is the system's compactness. This can hinder timely responses, and the

complexity of hyperparameter tuning in the model can hinder its application in disaster scenarios (Oh et al. 2023).

The usage of computer vision algorithms for disaster management has rapidly increased in the past few years, as discussed in Sect. 2.3. However, several challenges still hinder the effective implementation of disaster response strategies. Object detection models such as YOLOv8 have shown improvements in precisions up to 85%, but these algorithms still struggle in real-time applications. In disaster scenarios, there is a need for rapid data processing. Even though the need is critical, the volume of data generated can overwhelm the system. Weather conditions can impact the performance of UAVs and object detection systems, which limits their effectiveness in disaster scenarios. Advancements in object detection offer promising solutions for disaster management, and it can be enhanced by addressing the challenges. The collection of an image dataset with high-quality images in disaster-prone zones and the system design involved are difficult (Yi et al. 2023; Salluri et al. 2020). Generally, applying complex deep learning (DL) algorithms requires significant computational resources. Disaster scenarios in remote or isolated areas may not have sufficient resources to support complex ML algorithms. Despite these existing challenges, there is a growing consensus among experts in the field to significantly improve disaster response and recovery efforts using ML (Velev and Zlateva 2023).

3 Technological Integration and Data Collection in Disaster Management

3.1 Integration of UAVs and IOT in Disaster Management

Over the last few decades, many countries have suffered considerable casualties and economic losses due to natural disasters. During and after disasters, many infrastructures, such as communication stations, can be destroyed for days, as it may take a few days to weeks to re-establish a functioning communication backbone for telecommunications and internet services to restart. Due to which people residing in the disaster areas may not have sufficient availability of sufficient resources. Therefore, it is essential to integrate emerging technologies to restore emergency communication networks (Ejaz et al. 2019, Maiti and Awasthi 2022, Alsamhi et al. 2021, Abdel-Basset et al. 2023). To address the challenge faced during the disaster periods, IoT big data analytics and UAV technologies have emerged as key elements in the solution (Yang et al. 2023). The integration of IoT and UAVs can be effectively used during disaster management scenarios as they can quickly access dangerous

regions and provide real-time data in a cost-effective manner. It can also be used to establish communication networks and for Search and Rescue (SAR) operations (Maiti and Awasthi 2022).

We can use UAVs for real-time data collection and task offloading. IoT devices can be used to overcome the limitations in communication coverage, which would lead to effective transmission rates and improves the overall system performance (Chirumamilla et al. 2023, Rate et al. 2023). This integration can provide effective and reliable communication support during the post-disaster period. Hence, this can address the immediate needs of both the affected population and logistical challenges. Biao Yang et al. has described a system model for emergency communication and flexible coverage capabilities of UAVs to restore communication networks in disaster-affected areas. The system

integrates IoT big data analytics with UAVs for efficient and reliable connectivity. Figure 3 shows the visual representation of the described model by Biao Yang and el (Yang et al. 2023).

The system architecture for integrating UAVs with IoT devices involves several key components. This provides connection between the aerial and ground users in challenging environments such as disaster scenarios (Ejaz et al. 2021). The usage of Named Data Networking (NDN) architecture enhances the performances of the ground IoT network by providing a more efficient communication system (Kumar et al. 2019). The integration of edge intelligence technology allows UAVs to process data locally, reducing the processing in cloud. This integration improves real-time decision-making (ResearchGate. (n.d.) 2024). The components of the system can be divided into UAV hardware,

Fig. 3 The system model for UAV-assisted emergency communication



IoT devices, and software. UAV hardware components consist of Drones, Flight Control Sensors, Communication Antennas, and Power Systems (Sensors (n.d.), Moheddine et al. 2019, Zhou et al. 2022). UAVs are flying devices that collect data from the sensors and transmit it to the IoT gateways. Flight sensors monitor the internal state of UAV, such as temperature, pressure, and acceleration (Atzori et al. 2018). Communication Antennas are used to communicate with the IoT gateway and the other devices. To power up the UAVs flight and operations, we use components such as batteries and motors (Moheddine et al. 2019). For the integration, we use IoT devices such as sensors, IoT gateways, LPWAN gateways, and boards such as Arduino and Raspberry Pi (Zhou et al. 2022). IoT sensors such as cameras, Global Positioning System (GPS), and thermal imaging sensors are used to collect data from the disaster site. To provide long-range, low-power connectivity for IoT devices, we use technologies such as Long Range (LoRa) for Low-Power Wide-Area Network (LPWAN) Gateways (Ejaz et al. 2021, Kumar et al. 2019).

3.1.1 Challenges in Integrating UAVs and IoT for Disaster Management

As discussed in Sect. 3.1, the integration of UAVs and IoT can be used during disaster scenarios for effective management. However, there exist a few challenges that hinder its effectiveness. One of the most faced challenges is the limited battery life of UAVs. The UAVs have to cover large areas during SAR operations. Hence, frequent charging or rotation of UAVs would delay the response time and reduce the overall efficiency. UAVs are very sensitive to extreme weather conditions such as winds and rain. This could lead to inefficient data collection and area coverage. The efficiency of data transmission between the UAVs and IoT devices plays a very important role in real-time decision-making during disaster scenarios. Natural disasters often damage the existing communication networks. The disruption of the communication environment can lead to potential data loss or delays in data transfer. We have established coordination among government, rescue teams, and technology providers. This lack of coordination can lead to delays and inefficiencies in deploying UAVs and IoT devices during disasters. Additionally, concerns on data privacy will arise due to the surveillance made with UAVs. To improve the effectiveness of this integration, especially during disaster scenarios, we have to improve the battery life, communication network, and gain public trust by ensuring data privacy. (Yang et al. 2023, View of Smart Farming Platform using IoT and UAVs n.d.).

3.2 Real-Time Data Collection Using UAVs

Collecting Real-time data is very essential during disaster periods. Real-time data provides crucial information such as human presence, structural damage, and environmental conditions. UAVs are attached with various sensors, such as GPS, cameras, and a gyrocompass, which are used to collect real-time data and monitor the internal state of the UAVs (Bhaskar and Anitha 2021; Nikhil 2020). The data processing units process and analyze the data collected by the UAVs and IoT using ML algorithms. The edge intelligence technology allows the processing of the data locally (Nikhil 2020). The real-time data collection and processing in disaster management reduces risk to human life (Bhaskar and Anitha 2021).

Shen et al. (2022) proposed a model for data collection using UAVs, Machine-type communication devices (MTCDs), and ML algorithms such as Greedy Learning Clustering (GLC) Algorithm, and the Artificial Energy Map (AEM). The UAVs act as data collectors from MTCDs. The model does trajectory and hover optimization dynamically to ensure that the best route is taken and the hovering time of the UAVs. The effective implementation of Trajectory and hovering optimization saves energy and reduces the time taken to collect the data. MTCDs play an important role in the process of real-life data collection. MTCDs are small, low-power units that collect and send the data to the UAVs. During disaster periods, they can monitor the surrounding environment conditions. In the proposed model by Lingfeng Shen et al. These devices are grouped into clusters to make the data collection process more energy efficient. As the UAVs pass over or hover near the MTCDs, the data is transmitted in real time and processed during the flight. The use of ML algorithms such as GLC and AEM helps in making real-time decisions about the optimal paths and hovering points. The collected data is either processed in the central processing unit or it goes to the cloud server for immediate analysis (Shen et al. 2022).

Shianto et al. (2019) proposed a model to collect real-time data using multiple UAVs. The model considers each UAV as an individual agent that is sent to collect data from the sensor nodes. The sensor nodes harvest energy from the environment, and data collection is only possible when the sensor has sufficient energy. The UAVs must avoid collisions, so an efficient trajectory planning is made dynamically. To limit the observations made by each UAV, the data collection problem is formulated as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP). To allow each of the UAVs to make real-time

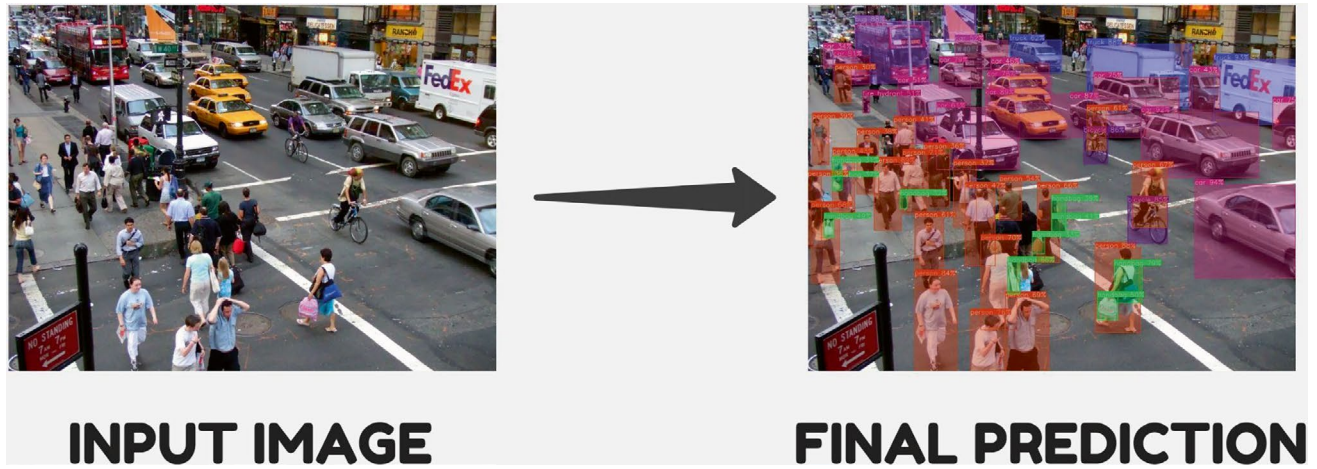


Fig. 4 Illustration of You Only Look Once (YOLO)

decisions based on their observations from the collected data, a multi-agent deep reinforcement learning algorithm with Centralized Training and Decentralized Execution (CTDE) was used. Similar to the previous model (Shen et al. 2022), this model also sends the collected data to either be processed in a central processing unit or it goes to the cloud server for immediate analysis. (Wang et al. 2023) shows the visual representation of the described model. Real-time data collection and analysis helps in damage assessment during disasters, SAR operations, to establish communication in the disaster regions, and to create early warning systems (Kumar et al. 2019, Bhaskar and Anitha 2021).

3.3 Implementation of YOLO in Disaster Scenarios

The ‘You Only Look Once’ (YOLO) is an object recognition method based on Convolutional Neural Network (CNN). Here the algorithm is implemented using the python programming language to detect objects in disaster scenarios. It is evident that the accuracy result of detecting objects of disaster victims is 89% (Sarosa et al. 2021) (Fig. 4). Shianto et al. (2019) shows a visual representation of the working of YOLO model. There are several models in the YOLO family, from YOLOv1 to YOLOv9 and PP-YOLO, including PP-YOLOv2 and PP-YOLOE. Each model in the YOLO family has its own strengths, making them suitable for different applications. Table 2 shows the comparison between the most used YOLO models (YOLOv5, YOLOv6, YOLOv7, YOLOv8, YOLOv9, YOLOv10) based on its performance on the COCO dataset. The mean average precision (mAP) score is a key metric used to evaluate the performance of the object detection model (Mehra 2023;

Table 2 Comparison of YOLO Models (Sapkota 2024)

YOLO model	Accuracy (mAP) (%)	Speed (FPS)
YOLOv5	47.4	60
YOLOv6	50.1	55
YOLOv7	51.5	>60
YOLOv8	52.8	>60
YOLOv9	54.4	<100
YOLOv10	54.4	543.8

Term 2023; YOLO Models and EfficientDet Mean Average Precision (mAP) and (n.d.)) (Fig. 4).

From the real-time data collected from the disaster site, we can preprocess the video or the image captured for the model and detect the objects from the input. At the time of a Disaster, during the evacuation process, the usage of object detections in danger-prone areas would help the SAR team (Sarosa et al. 2021).

An initially loaded image/video file is a preprocessed file that has undergone resizing and normalizing of pixel values from 0 to 1. Pre-trained YOLOv7 is initialized. The model is used on the preprocessed image to detect the objects. The class name for the detected object is assigned properly. The resultant image with bounded boxes is displayed and saved in the system (Khoa (n.d.). 2024). A system is built using the YOLOv8 model to detect the damaged and undamaged buildings in a post-disaster site. Figure 5 shows the working of the adopted model. The data collected from the UAVs with high-resolution cameras and GPS, as discussed in Sect. 3.2. The UAVs capture aerial images of the disaster sites and save these images in Georeferenced Tagged Image File Format (GeoTIFF). Each GeoTIFF file contains the image data along with the georeferencing information. The collected GeoTIFF images are transmitted from UAVs to

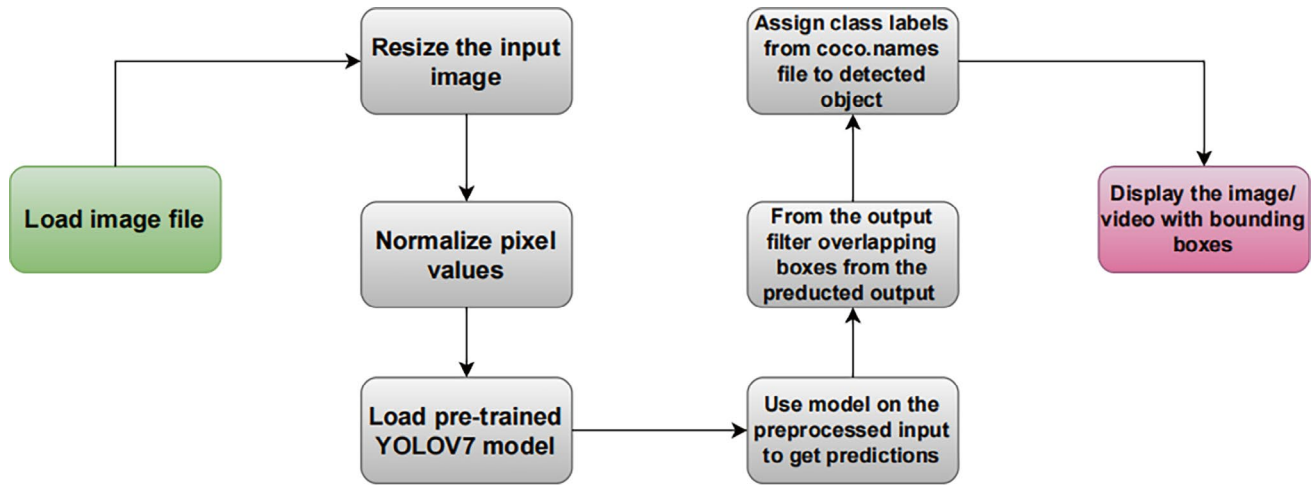


Fig. 5 Block diagram of object detection using YOLOV7

an edge computing device via WiFi. The GeoTIFF images are preprocessed. The preprocessing procedure consists of resizing the images, normalizing the images, dividing the images into smaller tiles, and converting the image to jpg (Khoa (n.d.). 2024). The code processes the captured GeoTIFF image by splitting into smaller tiles, as it is efficient in large-scale disaster management scenarios where UAVs collect vast amounts of data. Then the smaller tiles are converted to JPG format to run the model.

The pretrained YOLOv8 model is initialized for building detection, and the damage criteria are defined to differentiate between damaged and undamaged buildings. The model is trained for 60 epochs for accurate detection.

When the model is trained, the detected objects from the training dataset are analyzed and determines if they meet the damage criteria. The results from all tiles of the image are combined to form the entire image. From the final image, the bounding box location and confidence score are extracted, and the details are stored in a text file. The image of the disaster site with the detection of damaged and undamaged buildings is displayed (Khoa (n.d.). 2024) (Fig. 6).

Thermal image target tracking is an advanced technique used in disaster management. The YOLOv5 object detection model is used to identify and track individuals (targets) in challenging environments like areas affected by disasters. The YOLOv5 model is highly known for its high speed and highly accurate results during real-time processing. Initially, the YOLOv5 model is pre-trained on a visible image dataset. The YOLOv5x model performs well among its variants, and it is transferred to detect thermal objects in infrared videos. These videos were captured by UAVs equipped with thermal imaging cameras at the disaster site

To enhance the tracking and detection process, the YOLOv5 model is fine-tuned with thermal images from disaster scenarios. For fine-tuning, a dataset with 197 thermal images with a total of 548 manually annotated bounding boxes was used for training. The annotations were done using a graphical image annotation tool named “LabelImg” and each instance was named under the same class: “Walking” (Yeom 2024) (Fig. 7).

This application is effective in SAR operations, mainly in regions such as mountains and forests during disaster periods. The UAVs equipped with thermal imaging cameras can capture IR videos even during nights, low-light conditions, or regions covered with smoke. The final output from the YOLOv5 model with bounding boxes helps in identification and tracking of targets.

This integration of UAVs with models such as YOLOv5 offers a reliable solution for SAR operations during disasters.

3.4 Usage of Text Analysis for Disaster Alerts

A framework was developed to analyze social media data and identify the disaster-related posts. The frameworks used both LSTM and BERT models to identify the posts using the Twitter dataset (Asif et al. 2021). Analyzing the content of tweets can help us to identify the post that indicates danger or needs help. Figure 8 shows the working flow of the model identifying disaster-related texts in social media.

To implement Twitter data in real-time, we can use the Twitter API to stream real-time tweets. The system consists of 4 important steps, which are discussed briefly in the block diagram (Fig. 8).

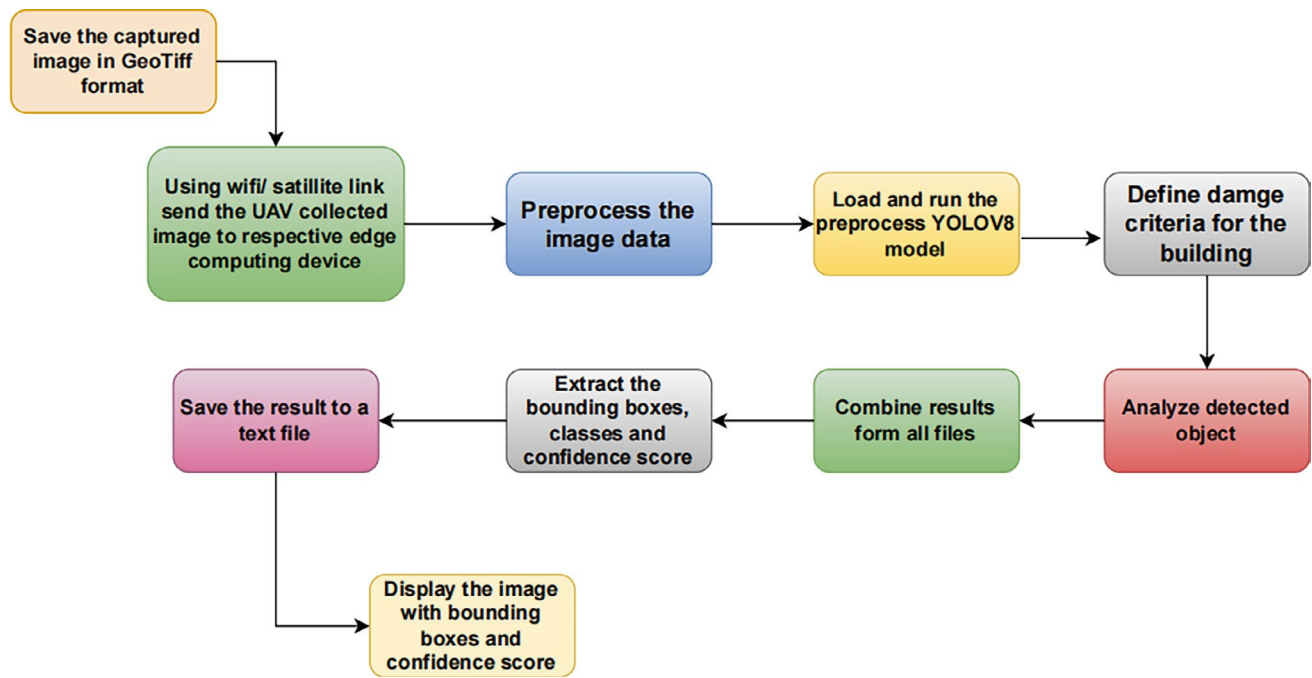


Fig. 6 Block diagram for classifying damaged versus undamaged buildings

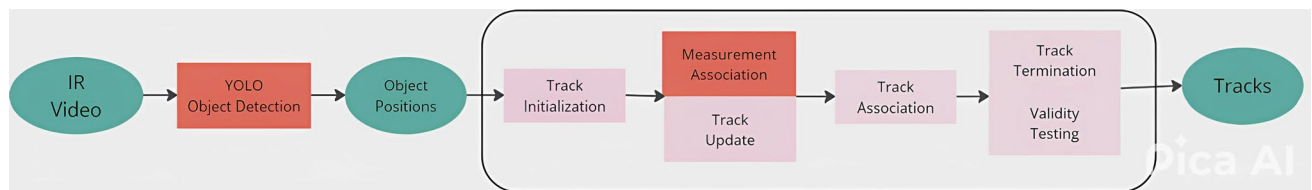


Fig. 7 Block diagram of thermal image target tracking

1. Importing libraries

We have to import certain libraries like scikit-learn, Pandas, Numpy, Matplotlib, and other needed libraries.

2. Load and preprocessing the data

We have to load the CSV file into the model. To preprocess the data, we have to split the dataset into training and testing data. Then we have to clean the data by removing all the null values and changing the format of the data according to the model. The preprocessing process consists of tokenizing the text data, converting the text to indices, and creating a pytorch data loader. We tokenize the text for BERT using BertTokenizer, if we are using LSTM or CNN, we have to convert text to indices using Word2Vec and Glove. We have to create a pytorch data loader to handle the large dataset batch by batch.

3. Model training

We have to initialize the model and then set up an optimizer and scheduler. We have to set up an optimizer and scheduler. The optimizer improves the model's performance, and the scheduler dynamically adjusts

the learning rate during the training process. The data was sent for 15 iterations in batches, which are known as epochs. This helps in improving the accuracy of the model.

4. Model Testing

Model testing is the final step of the system, which consists of testing the model with the test dataset, displaying the final result, and saving the predictions to a csv file. The accuracy of the model is displayed by evaluating the results.

4 Obstacles in Implementing UAV-Enabled IOT Devices for Disaster Management

Implementing UAV-enabled IoT devices in disaster management includes several obstacles. These challenges can remarkably affect the working and reliability of disaster response operations. This section explains technological,

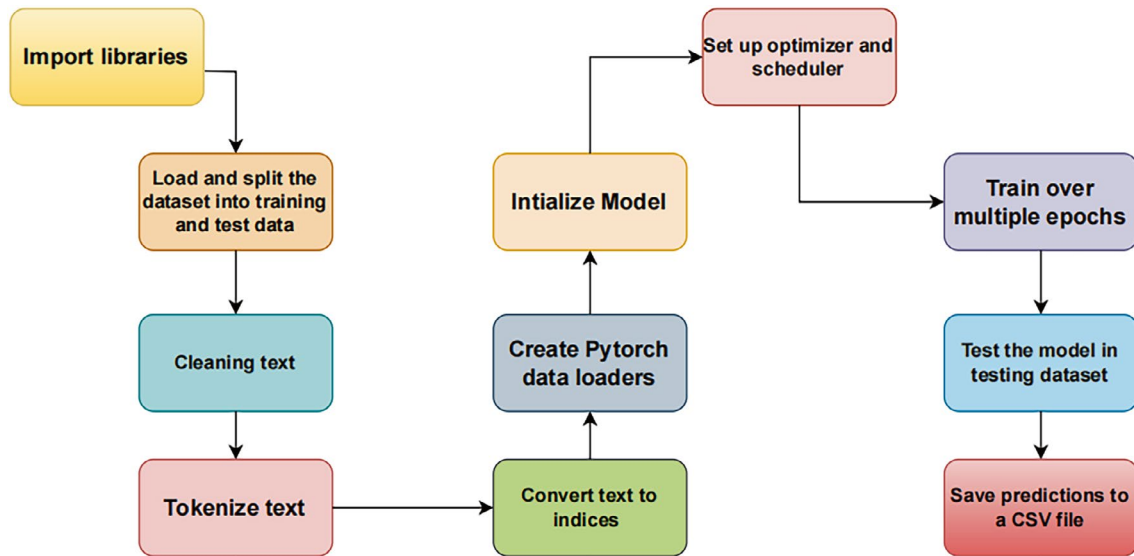


Fig. 8 Block diagram to identify the disaster-related posts

security provocation, and connectivity, as well as the guidelines for skilled usage, and gives recommendations to overcome these problems.

4.1 Technology Limitations

While talking about hardware, the biggest challenge would be limited battery life for UAVs (View of Smart Farming Platform Using IoT and UAVs n.d.-b.), which limits the length of time they can perform without recharging. The next challenge would be that the weight it carries is restricted (Floreano and Wood 2015), which means it limits the number of sensors that can hold on the UAV. The impact of environmental conditions (Erdelj et al. 2017) will affect the durability of the UAV.

Proceeding to software, the major challenge could be maintaining the effectiveness of the algorithm and monitoring the processing time and speed of decision-making (Jordan and Mitchell 2015). Using complex ML algorithms and data analytics will slow down the processing power (He et al. 2016).

While integrating IoT devices and UAVs, ensure that UAS systems from different producers can communicate with each other and give significant results (Alam and Patel 2016).

4.1.1 Previous Efforts and Recommendations

Many researchers have worked on it and developed lightweight sensors and optimized the energy consumption in UAVs in order to address hardware limitations. More

efficient AI algorithms and structured UAS platforms are recommended to overcome the software limitation.

4.2 Network Coverage Challenges

Disasters damage communication infrastructure, which basically damages the connectivity for UAVs and IoT devices (Sarosa et al. 2021). Normally network coverage is very low in certain areas, which brings major challenges. To overcome this mesh, networks and satellite communication are used.

Mesh networking improves the connectivity by allowing UAVs to communicate and enlarges the data communication range (Mehra 2023). Satellite communication is used as a backup when the ground based communication fails or is down (Term 2023).

4.2.1 Previous Efforts and Recommendations

There are some ongoing research focusing on improving the hardness and integration of these systems into UAV-IoT networks that enhance the mesh networking and satellite communication.

4.3 Guidelines for Efficient and Effective Usage

Planning, developing, and testing the UAV-IoT before the disasters can help in efficiency and response time (YOLO Models and EfficientDet Mean Average Precision

(mAP) and (n.d.)). Developing standard operating procedures (SOPs) and following them will help in improving efficiency (Evaluation and of YOLOv10, YOLOv9 and YOLOv8 on Detecting and Counting Fruitlet in Complex Orchard Environments 2024), reduce miscommunication, and increase reliability. You can use certain strategies for optimization, such as real-time data analysis and collaboration.

Applying real-time data analysis tools can quickly collect data gathered by UAVs and IoT to enhance the speed of decision-making (Asif et al. 2021). Collaborating with government agencies, private companies, and NGOs can improve the working of UAV IoT deployments (Floreano and Wood 2015). Sticking to the data management policies will help in data security, privacy, and integrity (Erdelj et al. 2017).

4.3.1 Previous Efforts and Recommendations

Constant development and refinement of collaborative frameworks and real-time data analytics practices are recommended to improve response efficiency.

4.4 Privacy and Security Concerns

Strong cyber security measures should be taken in order to prevent UAV and IoT devices from hacking, which can lead to unauthorized access and data breaches (Jordan and Mitchell 2015). And ensure security while data transfer between UAVs, IoTs, and central command centers is important (He et al. 2016).

Ensuring the data collected by UAVs is stored securely because that may contain personally identifiable information (Alam and Patel 2016). Establishing clear policies and guidelines to manage the privacy of the individuals (Albrecht and Wright 2020).

4.5 Expert Recommendation

To enhance payload capacity and battery life, invest in solid-state batteries or high-density lithium-sulfur batteries, which outstandly extend UAV flight durations in disaster scenarios. To reduce UAV weight, use carbon fiber-reinforced polymers, which allow a greater number and variety of sensors to be carried at once, including high-resolution cameras, etc. (Meng et al. 2020).

To optimize algorithms, use edge computing to create lightweight algorithms that reduce latency and processing demands. Integrating ML models that process large amounts of sensor data, such as environment readings and images captured by drones, to quickly spot the hazards like

weaknesses in collapsed buildings and spreading speed of the wildfire (Ren et al. 2021).

To improve the connectivity, give first preference to multi-band mesh networks that help consistent UAV communication in the disaster-affected areas. Incorporating low-Earth orbit (LEO) communication through satellite helps to maintain continuous UAV usage even in heavily damaged areas (Yi and Liao 2022).

To improve security measures, adapting to highly advanced encryption standards such as AES-256 for data transmission between command centers and UAV-IOT devices (Shi and Li 2021).

5 Into the Future

Advancement in technology leads to advancement in UAV-enabled IoT networks in disaster management. In this section, we will learn about the future emerging trends and innovations that optimize the present work in disaster management. The effectiveness and capabilities of UAV-enabled IoT devices can be achieved by integrating advanced AI, blockchain technology, and deep learning.

5.1 Innovation and Future Development:

In the field of artificial intelligence, building efficient machine learning algorithms to improve the UAV and IOT devices performance and generating more predictive models to forecast disasters more accurately and alert them. Using AI could help to avoid the obstacles that we are facing right now and manage the changing conditions without human interference. AI will analyze the data spontaneously and inform the disaster management team (Mohan and Ramasamy 2021).

In the field of deep learning, developing an UAV with high-resolution cameras to capture the image and record the videos in more details that helps the deep learning models to increase its efficiency. Deep learning models could be trained in the way that it understands the context, which is captured either as an image or video, and provide for efficient results. Training the NLP models in such a way that it supports multiple languages for analyzing social media posts and emergency calls to gather comprehensive information. Using sentiment analysis, we can prioritize areas where help is needed urgently (Nawaz and Rehman 2021).

In the field of Blockchain technology, data collected by UAVs and IoT devices could be converted to immutable records to maintain the confidential records. Blockchain proved secure, transparent data sharing between various stakeholders, NGOs, private organizations, and government

agencies. Based on certain predefined conditions, smart contracts on blockchain could respond automatically (Viriyasitavat et al. 2020).

5.2 Potential Future Applications

Blending UAV-enabled IoT with advanced technologies helps to develop a lot of potential application development in disaster management. Let's see some key future applications.

5.2.1 Swarm Technology for Interdependent UAV Operations

The use of swarm technology in UAVs can reshape disaster response by enabling the interdependent working of multiple drones (Yan et al. 2020). Swarm UAVs can enfold large geographical areas, giving comprehensive real-time data analysis and collection. It helps to reduce the need for constant human oversight. It improves the accuracy of disaster analysis.

5.2.2 Integrating UAV-IoT with IoMT

The integration of UAV-IoT with the Internet of Medical Things (IoMT) gives hopeful advancements in disaster management, more essential, it provides medical support during emergencies (Javaid et al. 2021). UAVs integrated with IoMT which helps the areas that are difficult to access by delivering first aid kits and medications. It also helps to carry the health assessment tools and transmit patient data to medical teams in real time.

5.2.3 Edge Computing to Enhance Decision Making in Real-Time

Integrating edge computing will improve the efficiency of the UAV-enabled IoT. The data collected by the UAV-IoT will be sent to the local network rather than being sent to centralized cloud servers. This helps to improve the decision-making speed during critical situations. This technology is most useful in the scenarios where immediate responses are needed, such as in assessing structural damage and detecting hazardous materials (Shi et al. 2020).

5.2.4 Global Collaboration

Collaborating worldwide will allow countries to share technology, resources, and expertise in disaster management (Ahmed and Ali 2021). International collaboration can lead to the improvement of data sharing and coordination during cross-border disasters. It helps to foster the innovation in UAV and IoT technologies.

6 Conclusion

In this chapter, we have explored and gained much knowledge about Unmanned Aerial vehicles (UAVs) enabled with Internet of Things (IoT) platforms to improve disaster management. During emergencies, these technologies will be so helpful in real-time data collection and spreading awareness. The fundamental concepts of UAVs and IoT, which explains the introduction, development, and causes. The integration of ML algorithms with these technologies was explained. It teaches us how predictive analytics and anomaly detection can help improve decision-making during tight corners.

Our proposed solution for system architecture and collection of data highlights the use and advancement in gathering and data analyzing from affected areas using algorithms like YOLO and LSTM to increase the reliability. Implementation cannot be done without facing any challenges. The limitations are such as technological challenges, network coverage problems in disaster areas, and security and privacy concerns. At last, this chapter explains how to improve and future advancement in the fields like AI, deep learning, and blockchain technology, which leads to further development in UAV-IoT device capabilities. Collaborating worldwide will allow countries to share technology, resources, and expertise in disaster management. Continued research and development in these fields will help to build innovative technologies that will save lives.

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Exploring Machine and Deep Learning for Crop Detection and Yield Prediction in Drone-Enabled IoT Networks

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Abstract

In recent years, agriculture has witnessed significant advancements driven by modern technology, particularly in enhancing crop productivity and optimizing orchard management. Traditionally, tasks such as crop counting and production estimation were labor-intensive and time-consuming, relying heavily on manual methods. However, the rise of precision agriculture has dramatically transformed these processes. Integrating drones with the Internet of Things (IoT) has become essential for acquiring crucial data. These advanced technologies facilitate rapid and accurate large-scale data collection, providing insights that were previously unattainable. By employing machine learning (ML) and deep learning (DL) models to analyze this data, we can effectively implement these models alongside customized training strategies tailored to factors such as agricultural object types, image resolutions, and specific crop characteristics. In this study, we explored the application of ML and DL techniques within drone-enabled IoT networks for managing apricot crops. When benchmarked against

other leading methods, YOLOv10 achieved an F1 Score of 88%. Additionally, linear regression was applied for yield prediction, resulting in a Relative Error of 2.18% across the selected trees. These results underscore the remarkable potential of ML and DL technologies in revolutionizing precision agriculture. This chapter provides a thorough exploration of these transformative applications.

Keywords

Precision agriculture · Machine learning · Deep learning · Crop detection · Yield prediction · Drone-enabled IoT

1 Introduction

Agriculture, a cornerstone of human civilization, has continuously evolved driven by technological advancements. From the early days of manual labor and rudimentary tools to the current era of digital transformation, the agricultural sector has adapted to meet the changing needs of society. The integration of modern technologies has not only enhanced productivity but also introduced new methodologies for managing farms and orchards (Tian et al. 2020). Traditionally, agricultural tasks such as crop counting and yield prediction were heavily reliant on manual methods. These processes were not only labor-intensive but also time demanding and subject to human error. With the growing global population and the growing need for food, the need for more efficient and accurate agricultural practices has never been more critical. This is where precision agriculture takes center stage. Utilizing advanced technology to improve agricultural practices (Hayajneh et al. 2024). Advanced sensor-equipped drones have become essential assets in agriculture, offering rapid and efficient data collection capabilities in the field. When integrated with IoT networks, these drones enable continuous real-time monitoring

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of crop health, soil conditions, and other vital parameters, enhancing agricultural operations (Zhang and Li 2020; Du et al. 2019). This integration facilitates the seamless flow of data, enabling more informed decision-making processes. Machine learning, a key component within the realm of artificial intelligence, holds significant importance, plays a pivotal role in analyzing the data collected by drone-enabled IoT systems. By applying sophisticated algorithms (Bhadra et al. 2024; Sarker 2021), machine learning can uncover patterns and insights that are not immediately apparent. This capability is particularly beneficial for crop detection (Ferreira et al. 2020) and yield prediction (Chlingaryan et al. 2018; Donohue et al. 2018; Kaul et al. 2005; Stiller et al. 2024), where accurate forecasting is essential for planning and resource allocation. By using the combination of machine learning and drone-enabled IoT (Aldeen and Abdulhadi 2021; Garg et al. 2024), we can achieve new levels of insight and forecasting capabilities related to growing seasons and crop productivity. This exploration aims to demonstrate how these advanced technologies have the potential to transform agriculture, making it more efficient, sustainable, and productive. The chapter is organized as follows: Sect. 2 provides foundational context on agricultural management, setting up an understanding of the broader framework within which contemporary agricultural practices are situated. Section 3 focuses on the enhancement of precision agriculture through IoT and Big Data, detailing how the merger of Internet of Things (IoT) technologies and big data analytics is reshaping precision agriculture and advancing towards Industry 4.0. Section 4 examines the different types of drones used in agriculture, exploring various drone technologies and their specific applications in agricultural environments. Section 5 covers the practical applications of drones in agriculture, including their roles in crop monitoring, field condition assessment, and resource optimization. Section 6 delves into the use of machine learning and deep learning techniques in agriculture, providing an overview of how these sophisticated methods enhance agricultural practices, such as crop prediction, pest detection, and yield estimation. Section 7 addresses real-time data collection using IoT-equipped drones and the application of machine learning, discussing how real-time data collected by drones, when analyzed with machine learning algorithms, can enhance decision-making and operational efficiency in agriculture. Section 8 focuses on Data Sources and Evaluation Techniques for deep learning applications in agriculture, covering the various data sources used and the methods employed to assess and refine model performance. Finally, Sect. 9 concludes the chapter and discusses potential future directions.

2 Background Information for Agriculture Management

Agricultural management, the cornerstone of sustainable food production, has undergone significant evolution throughout human history (Wesemael et al. 2010). Rooted in ancient civilizations, early farming practices relied on manual labor and rudimentary tools to cultivate crops and raise livestock. These traditional methods, though foundational, were limited in precision and efficiency. Farmers had to rely on their instincts and observations to manage their fields, often facing challenges such as unpredictable weather patterns and resource inefficiencies. The advent of modern technologies, including machinery, fertilizers, and pesticides, marked a pivotal transformation in agriculture (Wang 2009). While these advancements enabled larger-scale farming operations and increased productivity, they also raised concerns about environmental degradation and food safety. Recently, there's been a greater awareness of the necessity for agricultural practices that are both sustainable and environmentally friendly (Pretty 2008). This shift has prompted the integration of innovative technologies, including precision farming (Raj et al. 2021), remote sensing (Weiss et al. 2020), and data analytics (Nyoman Kutha Krisnawijaya et al. 2022). In agricultural management, the adoption of these technologies empowers farmers to make data-driven decisions, enhancing resource efficiency and reducing environmental impact, which in turn fosters more effective and sustainable farming practices (Bibri and Krogstie 2020). Thus, understanding the historical context and evolution of agriculture management is essential for appreciating the challenges and opportunities facing modern agricultural practices. The primary goal of precision agriculture is to improve efficiency, reduce costs, minimize environmental impact, and increase crop yield and quality by making decisions based on precise, site-specific data (Dholu and Ghodinde 2018). Yield monitors are vital for site-specific management strategies. Yield maps, generated from these monitors, aid in analysis and provide reliable answers to field experiments. They enable the assessment of input and agricultural practices' impact on yields (Fulton et al. 2018). Yield maps offer valuable insights into the effects of managed inputs such as fertilizer, lime, seed, pesticides, as well as cultural practices like tillage, irrigation, and drainage. A foundational approach to large-scale crop yield prediction was established by merging crop modeling principles with machine learning methods, prioritizing precision, adaptability, and scalability (Paudel et al. 2021). The devised workflow generated attributes from crop simulations, weather patterns, remote sensing observations, and

soil information. It was structured to accommodate various crops and geographical locations with minimal adjustments. Preliminary case studies forecasting yield for diverse crops across the Netherlands, Germany, and France demonstrated that our baseline achieved similar accuracy to conventional techniques, with room for enhancement via additional data sources and advanced algorithms. Additionally, yield monitoring based on image segmentation using neural networks and deep learning was explored. A U-net architecture neural network was trained, and an algorithm was created for subsequent analysis of processed images (Gapon et al. 2023). Another research endeavor assessed the effectiveness of a machine learning system tailored for analyzing soybean yield within fields. The system utilized Sentinel-2 multispectral images alongside soil parameters for prediction purposes. For crop yield prediction, researchers utilized three cloud-free Sentinel-2 multispectral images annually, strategically selected from specific vegetation periods. These images were compiled into a time-series dataset essential for the analysis (Pejak et al. 2022). Another study aimed to develop and evaluate a machine vision-based yield monitor. This monitor was designed to categorize shallots by size and continuously count them in real-time during harvesting. The objective was to enhance the evaluation of specialty crop quality and quantity, providing producers with real-time harvest information, ultimately leading to higher returns and improved crop management practices (Boatswain Jacques et al. 2021).

3 Enhancing Precision Agriculture with IoT and Big Data: Moving Towards Industry 4.0

3.1 Internet of Things (IoT)

The Internet of Things (IoT) refers to a worldwide network of distinct, addressable entities that are linked through standardized communication protocols, similar to the internet itself (Binti Mohamad Noor and Hassan 2019). Within this framework, the internet functions as both a storage and communication platform, connecting virtual models of physical objects that continuously refresh and consolidate information from multiple sources. This setup allows for the remote management of processes over the internet (Chopra et al. 2019).

3.2 Role of Agricultural IoT in Industry 4.0

Within the context of Industry 4.0, physical and virtual objects are interconnected through smart networks established throughout the value chain. This interconnection is particularly relevant in the agricultural sector, where IoT plays a crucial role. Agricultural IoT (AIoT), aligned with Agriculture 4.0 guidelines, enables the acquisition, processing, management, and dissemination of farm-related data. This data integration allows for seamless operation within Farm Management Systems (FMS), making AIoT an essential technology for smart farms (Gyamfi et al. 2024). These systems are designed to handle diverse data streams in real-time, process complex and sometimes contradictory information, and dynamically adjust network behaviors to optimize agricultural processes. AIoT facilitates the creation of active nodes within agricultural value chains, enhancing the overall efficiency and effectiveness of farm operations.

3.3 Enhancing Agriculture with Big Data Technologies

The advancement of IoT technology, which interconnects various objects and devices throughout the cyber-physical management cycle, generates large volumes of real-time data on agricultural processes and the entire supply chain. This data is essential for comprehending the operations and transactions in agriculture. Sensor devices and robots generate non-traditional data, such as images and videos, while social media contributes human-originated data. The combination of this diverse data set gives a clear picture and allows for better decision-making that was not possible before. The key to harnessing the value of this data lies in Big Data and analytics technology. Big Data is reshaping the scope and organization of agriculture by offering predictive insights into agricultural operations, enabling real-time decision-making, and redefining processes. It addresses critical issues such as improving efficiency, enhancing food safety and security, addressing climate change, and promoting sustainability. Through the application of Big Data, agriculture is becoming more informed, precise, and responsive to the challenges of modern farming (Rodríguez et al. 2019).

4 Various Types of Drones Employed in Agriculture

The adoption of drones in agriculture is rapidly increasing, driven by their precision and efficiency in collecting data (Rejeb et al. 2022). Advances in drone software and hardware have greatly enhanced their performance. Different types of drones are tailored for specific agricultural tasks, playing a vital role in data collection, monitoring, and decision-making within precision agriculture (Sharma and Hema 2021; Hassanalian and Abdelkefi 2017). The drones are discussed in the below subsections.

4.1 Fixed-Wing Drones

Fixed-wing drones, akin to small airplanes, excel in efficiently covering vast areas. Outfitted with high-resolution cameras and sensors, they possess the capacity to capture aerial images across extensive agricultural landscapes. These drones are frequently utilized for tasks such as crop mapping, terrain modeling, and monitoring crop health over expansive areas (Elijah et al. 2021).

4.2 Multi-Rotor Drones

Multi-rotor drones, such as quadcopters and hexacopters, feature multiple rotors and are highly maneuverable. They excel in capturing detailed, high-resolution imagery of smaller agricultural plots. Multi-rotor drones are versatile and can be equipped with various sensors, including RGB cameras, multispectral cameras, and thermal cameras, to assess crop health, detect pests, and monitor irrigation systems (Yang et al. 2017).

4.3 Hybrid Drones

Hybrid drones merge the advantages of fixed-wing and multi-rotor drones, offering vertical take-off and landing capabilities. This versatility enables them to hover for detailed inspections and perform long-distance flights to cover extensive areas. Consequently, farmers can efficiently capture detailed images of specific spots and conduct comprehensive surveys of their fields (Todeschini et al. 2019).

4.4 Spraying Drones

Spraying drones are specially designed for precision pesticide and fertilizer application. Equipped with spraying

systems, these drones can precisely target and deliver agrochemicals to crops, minimizing chemical drift and reducing environmental impact. Spraying drones offer significant advantages in terms of efficiency, safety, and accuracy compared to traditional spraying methods (Shahrooz et al. 2020).

4.5 Monitoring and Surveillance Drones

Monitoring and surveillance drones are equipped with advanced sensors, including thermal imaging cameras and LiDAR (Light Detection and Ranging) systems. These drones are used for monitoring crop health, identifying regions exhibiting stress or disease, and conducting aerial surveys to assess field conditions. They are particularly valuable for early detection of pest infestations, water stress, and other potential threats to crop yield (Gohari et al. 2022). Table 1 provides a comparative analysis of various types of drones utilized in agriculture, outlining their features, advantages, and limitations. In Table 2 presents a summarized overview of the primary features, applications, advantages, limitations, and typical use cases associated with various camera, imaging, and IoT technologies used in agricultural drones.

5 Applications of Drones in Agriculture

Drones have a variety of applications in agriculture, leveraging their versatility, agility, and data-capturing capabilities to revolutionize various aspects of crop management (Dutta and Goswami 2020). Here are some specific applications of drones in agriculture:

5.1 Crop Monitoring and Management

Drones equipped with RGB, multispectral, or thermal cameras are used for crop monitoring and management. They capture high-resolution aerial imagery of fields, allowing farmers to:

- Assess crop health
- Detect signs of stress or disease
- Monitor growth patterns
- Identify areas requiring intervention, such as irrigation or fertilization

5.2 Field Mapping and Surveying

Drones are employed for mapping and surveying agricultural fields, providing detailed insights into topography, soil

Table 1 Comparative analysis of different types of drones used in agriculture

Monitoring and surveillance drones	Spraying drones	Hybrid drones	Multi-rotor drones	Fixed-wing drones	Feature/type of drone
Equipped with advanced sensors	Specialized with spraying systems	Combination of fixed-wing and multi-rotor	Multiple rotors (e.g., quadcopters)	Resemble small airplanes	Design
Short to moderate (15–30 min)	Moderate (20–30 min)	Moderate (30–60 min)	Short to moderate (15–30 min)	Long (up to several hours)	Flight Time
Variable, depends on sensor range	Targeted spraying	Variable, flexible coverage	Smaller, detailed areas	Large areas	Coverage Area
High, can hover and move precisely	High, can hover and move precisely	Moderate to high, VTOL capability	High, can hover and move precisely	Limited, requires runway for landing	Maneuverability
Low to moderate, limited by rotor power	Moderate, specialized for spraying	Moderate, balanced design	Low to moderate, limited by rotor power	High, can carry multiple sensors	Payload Capacity
Thermal cameras, LiDAR, multispectral	Spraying systems, some with cameras	Various, depending on configuration	RGB cameras, multispectral, thermal	High-resolution cameras, multispectral	Primary Sensors
Crop health monitoring, field condition assessment	Precision pesticide/fertilizer application	Versatile, crop health, large area coverage	Crop health assessment, pest detection	Crop mapping, terrain modeling, crop health	Applications
Advanced monitoring, early threat detection	Precision spraying, reduced chemical use	Flexible operations, dual capabilities	High precision, detailed imagery	Long flight time, extensive coverage	Advantages
Shorter flight time, smaller payload capacity	Limited to spraying tasks, moderate payload	Complex design, moderate flight time	Limited flight time, smaller coverage	Requires large area for takeoff/landing	Limitations

characteristics, and land features. Using LiDAR-equipped drones, farmers can create precise 3D maps to:

- Identify drainage issues
- Determine soil erosion patterns
- Optimize field layout for improved water management and crop productivity

5.3 Precision Irrigation

Drones equipped with thermal cameras and moisture sensors enable precision irrigation management. By conducting aerial surveys and analyzing thermal imagery, drones can:

- Identify areas of water stress
- Optimize irrigation schedules
- Adjust water application rates

This helps conserve water resources, reduce water usage, and improve crop yield and quality.

5.4 Pest and Disease Detection

Drones equipped with multispectral or hyperspectral cameras assist in early-stage pest identification, diseases, and weed infestations in crops. By analyzing spectral signatures and vegetation indices, drones can:

- Identify anomalies indicative of pest damage, disease outbreaks, or weed encroachment
- Enable timely actions, such as targeted pesticide application or manual intervention, to mitigate crop losses and minimize chemical usage

5.5 Yield Monitoring and Prediction

Drones are used for yield monitoring and prediction by capturing aerial imagery throughout the growing season. By analyzing crop canopy density, biomass accumulation, and vegetation indices derived from multispectral imagery, drones can:

- Estimate crop yield potential
- Predict harvest outcomes

This information helps farmers optimize harvesting logistics, plan storage and transportation, and make informed marketing decisions.

5.6 Crop Spraying and Application

Drones equipped with spraying systems are employed for the precise application of agrochemicals, such as pesticides, fertilizers, and herbicides. These drones can:

Table 2 Comparative study of sensor, imaging, and IoT technologies in agricultural drones

Typical use cases	Limitations	Advantages	Applications	Data Provided	Imaging bands	Technology
Crop scouting, field mapping, general observation	Limited to visual spectrum, less detailed than other sensors	Essential for basic aerial photography, easy to interpret	General observation, crop scouting, growth monitoring	Standard color images, visual data	Red, Green, Blue	RGB Cameras
Early disease detection, nutrient deficiency assessment, water stress detection	Limited spectral bands compared to hyperspectral cameras	Provides insights into crop health, stress, and biomass accumulation	Assessing crop health, detecting disease, stress levels	Vegetation indices (e.g., NDVI), crop health data	Visible, Near-Infrared (NIR)	Multispectral Cameras
Irrigation management, identifying pest infestations, thermal stress monitoring	Less effective in cloudy conditions, requires calibration for accurate temperature	Useful for monitoring crop water stress, optimizing irrigation	Identifying heat stress, irrigation inefficiencies, pest infestations	Temperature variations	Thermal Radiation	Thermal Infrared (IR) Cameras
Land leveling, optimizing irrigation, field drainage planning	Expensive, requires specialized equipment	Enables precise terrain analysis, optimizes field drainage and irrigation	Terrain modeling, elevation mapping, topographical analysis	3D terrain maps, elevation data	Laser Pulses	LiDAR
Fertilization strategy optimization, soil fertility assessment, contamination detection	High cost, complex data processing and interpretation	Fine-grained analysis of plant and soil properties, identifies specific plant species	Analyzing plant biochemical properties, soil composition, nutrient levels	Detailed crop and soil characteristics	Hundreds of Narrow Spectral Bands	Hyperspectral Cameras
Monitoring soil moisture, weather patterns, crop health, enabling precision farming techniques	Dependent on robust network infrastructure, potential data security issues	Provides real-time data, enables automated control systems, enhances decision-making	Continuous monitoring of soil moisture, weather conditions, crop health	Real-time environmental and operational data	N/A	IoT Technologies

- Navigate fields autonomously
- Deliver targeted spray treatments to specific areas with high accuracy and efficiency

This reduces chemical drift, minimizes environmental impact, and improves worker safety compared to traditional ground-based spraying methods.

Overall, drones play a pivotal role in modern agriculture by providing farmers with actionable insights, optimizing resource management, and enhancing productivity, sustainability, and profitability in crop production.

6 Machine Learning and Deep Learning Techniques in Agriculture

In this section, we explore the utilization of machine learning (ML) and deep learning (DL) approaches within the agricultural domain. These advanced methodologies play a crucial role in processing and analyzing the extensive datasets gathered by drones, offering valuable insights for crop management and yield prediction (Singh et al. 2021). Specifically, machine learning, especially deep learning, involves training algorithms to discern patterns and generate predictions or decisions based on input data. In the context of agricultural data collection, it's crucial to extract pertinent features for effective analysis (Konig et al. 2020). These features, including geometric features like size, shape, and crown diameter, alongside texture features such as homogeneity, contrast, entropy, and variance, which help assess the regularity and complexity of the crown surface, as well as color and shape specific crown features, are essential for accurately identifying and analyzing data through our developed neural network model. During preprocessing to facilitate the training of tree crown recognition models, we focus on several types of features. Geometric features provide information about the overall dimension and shape of the crown, while texture features help assess the regularity and complexity of the crown surface. Color features are crucial for determining the health and composition of the vegetation (Kasaei et al. 2021), and shape-specific crown features provide additional details about the structure and density of the vegetation. By combining these features (Gehler and Nowozin 2009), we create a rich dataset essential for training precise and robust machine learning and deep learning models for tree crown identification and analysis. Key algorithms used in agricultural data analysis encompass a variety of techniques tailored to handle the unique challenges of agricultural data (Kamilaris et al. 2017).

Supervised learning algorithms learn from labeled examples, allowing them to identify patterns and predict results. In agriculture, this method finds utility in tasks like crop

classification and disease detection, where historical data can guide predictive models. In a related study (Malik et al. 2016) researchers applied k-means segmentation to recognize fruit, achieving accurate image segmentation for improved yield estimation. Furthermore, in this study (Xie and Niculescu 2022), Researchers presented a method for mapping winter crops such as winter wheat and barley using Sentinel-2 data. They employed a range of classification techniques, including hierarchical and conventional direct extraction methods. Additionally, they monitored the seasonal growth of these crops by analyzing Sentinel-1 SAR data through the Google Earth Engine platform. The findings demonstrated that hierarchical classification offered superior accuracy and was especially effective for agricultural land mapping and winter crop identification.

Unsupervised learning algorithms aim to identify patterns within unlabeled data. In agriculture, techniques such as clustering are employed to tasks like delineating crop health zones within fields, allowing farmers to target interventions more precisely. This study introduces an innovative methodology leveraging high-resolution satellite data for mapping and monitoring agricultural fields, providing crucial insights into their distribution, size, and temporal evolution.

The approach, which combines a convolutional neural network with clustering techniques, was successfully applied to an agricultural region in Saudi Arabia. The results demonstrated high accuracy in identifying agricultural fields and the ability to characterize their evolution over a three-decade period. However, despite these successes, certain limitations related to data availability and environmental conditions must be considered (Li et al. 2022). The researchers investigated the application of Variable Rate Application (VRA) technology in precision agriculture, utilizing Google Earth Engine (GEE) to access satellite imagery data. They explored various machine learning techniques, including Random Forest (RF) and Support Vector Machine (SVM), as well as unsupervised k-means clustering. Their objective was to generate precise Management Zones (MZs) to optimize agricultural profitability, resource efficiency, and reduce environmental impact through more accurate VRA (Gallardo-Romero et al. 2023). Furthermore, the researchers explored drone utilization for weed detection in agricultural fields, suggesting an automated approach employing Convolutional Neural Networks with unsupervised data collection from drone-captured images. The findings demonstrated comparable results to traditional supervised methods, with slight variances in accuracy observed across various field types (Bah et al. 2018).

Convolutional Neural Networks (CNNs) are highly effective in analyzing images because they can automatically learn hierarchical representations of data. In

agricultural settings, CNNs are applied to tasks such as crop detection and disease identification using high-resolution images obtained from drones. In (Puliti and Astrup 2022), researchers introduced an object detection approach to automatically recognize and categorize trees based on snow damage. They employed a YOLO CNN architecture and manually annotated UAV images from 89 study areas collected over the year. This annotation process encompassed over 55,000 individual trees, which were classified as healthy, damaged, or dead.

The objective of another study (Pandey and Jain 2022) was to create a crop classification method using drone-captured images, utilizing deep learning for multi-crop classification. The researchers introduced a dense convolutional neural network (CD-CNN) structure incorporating an activation function termed SL-ReLU. This architecture seamlessly integrates data fusion, feature extraction, and classification, achieving an impressive accuracy on the study data. Notably, this approach surpasses conventional machine learning methods and standard CNN architectures.

In a subsequent study (Wu et al. 2024), drones and advanced object detection models were utilized to efficiently identify tree diseases and pests. By integrating drone remote sensing data with the YOLO-v3 algorithm, the research attained high accuracy and rapid processing speeds. This offers significant potential for devising strategies to protect and prevent forest diseases and pests.

Additionally, in another study (Hnida et al. 2024), high-resolution images captured by a DJI Phantom 4 RTK drone were processed and segmented using four neural network models: YOLOv8, Mask R-CNN, FCN, and U-Net. The segmentation results were analyzed by dividing the images into quadrants and using Euclidean distance to determine the spatial distribution of olive buds. Flower density was estimated through normal distribution analysis. The models' performances were evaluated using precision, recall, F1 score, and Dice coefficient, with YOLOv8 showing the best results. The entire method was validated using fivefold cross validation.

Recurrent Neural Networks (RNNs) are highly effective in processing sequential data, making them a popular choice in agricultural applications. They are frequently employed to analyze time-series data, aiding in the understanding of crop growth patterns over time. Through the examination of temporal dependencies, RNNs offer valuable insights into seasonal trends and growth cycles, empowering farmers to make well-informed decisions. The study (Kumar et al. 2022), introduces a method for forecasting land subsidence in the Jharia Coalfield, India, employing a monitoring approach known as Modified PSInSAR. By utilizing data from 14 sites and incorporating 67 historical subsidence values, the researchers develop and assess predictive

models, including Vanilla and Stacked LSTM neural networks. The findings reveal concerning rates of subsidence, emphasizing the significance of the research for geological risk mitigation. In another study (Çetiner and Kara 2022), researchers use meteorological and wheat production data to estimate wheat productivity. They employ GRU and LSTM methods for this purpose. While LSTM offers better performance, its training time is longer than that of GRU. In this study (Kussul et al. 2020), researchers introduce a novel approach for accurate crop mapping using Sentinel-1 and Sentinel-2 satellite imagery, employing a recurrent neural network. Unlike conventional methods reliant on cloud-free images, this technique utilizes all satellite data, allowing the neural network to extract features independently. Experimental findings indicate superior performance over traditional machine learning methods and the Deep U-net architecture in terms of overall accuracy. Applications of machine learning in crop detection and yield prediction are numerous. ML models trained on drone-captured data can accurately classify crops, detect pests and diseases, and estimate yield potential. These models enable farmers to make informed decisions regarding crop management practices, resource allocation, and harvest planning. In summary, machine learning and deep learning techniques play a pivotal role in leveraging drone collected data to optimize agricultural practices. By analyzing this data, these algorithms empower farmers with actionable insights for enhancing crop productivity, sustainability, and profitability.

The polarimetric system for image analysis is another technique widely used in scientific research to analyze images and extract detailed information (Hoekman and Vissers 2003). In the agricultural field, remote sensing analysis is also prevalent, particularly with the use of the NDVI technique as a vegetation index (Huang et al. 2021). Additionally, regression analysis is employed for predictive modeling (Sharma et al. 2021).

Among all these techniques, we have chosen to focus on deep learning and its application in agriculture. This decision is based on several factors, some of which were discussed in the previous section. Deep learning, a subfield of artificial intelligence, involves neural networks organized in multiple layers, which makes it capable of learning at various levels, from basic to advanced. Consequently, data undergo multiple transformations from the input layer to the output layer, resulting in a greater capacity for learning.

Having been successfully applied in various domains, deep learning has recently entered the agricultural sector (Kamilaris and Prenafeta-Boldú 2018). As a recent, modern, and promising technique, deep learning in agriculture is gaining popularity. The progress and applications of deep learning in other fields underscore its immense potential in revolutionizing agricultural practices.

7 Real-Time Data Collection Using IoT-Equipped Drones and Machine Learning Applications

In the realm of modern agriculture, drones outfitted with advanced technology are revolutionizing farming practices. These drones, equipped with an array of sensors and high-quality cameras, conduct systematic flights over agricultural fields along predetermined routes. Throughout these flights, they meticulously gather crucial data on various environmental factors and capture detailed images of the crops below. Rather than storing this collected data locally, it is transmitted in real-time to a central base station through IoT networks, ensuring immediate access to valuable information. At the base station, the data undergoes rigorous processing and analysis, leveraging sophisticated machine learning algorithms. The primary objective here is multifaceted: to identify potential issues such as plant diseases or weed infestations and to estimate the expected yield of the crops. Subsequently, the results of this analysis are presented to end-users typically farmers via an intuitive interface. This interface, equipped with user-friendly dashboards and comprehensive reports, empowers farmers with actionable insights to optimize their crop management strategies.

Furthermore, this entire process operates within a feedback loop, where insights gained from data analysis are continuously utilized to enhance the accuracy and efficiency of the machine learning models. This iterative refinement ensures that farming practices evolve dynamically, leading to more effective monitoring and better estimation of crop yield. In Fig. 1, the process of utilizing drones equipped with sensors and cameras to enhance agricultural practices is illustrated. These drones follow specific flight paths over agricultural fields, collecting environmental data such as temperature, humidity, and soil moisture, as well as detailed images of the crops. The collected data is transmitted in real-time to a central base station via IoT networks. Upon arrival, the data undergoes preprocessing for cleaning and organization to prepare it for analysis. Following this, machine learning algorithms are utilized to analyze the data, extracting valuable insights. These insights encompass the detection and diagnosis of issues impacting the crops, as well as the estimation of potential crop yield. The results of the analysis are presented to farmers through a user-friendly interface, providing actionable recommendations to optimize crop management practices. This feedback loop enables farmers to make informed decisions in real-time, leading to more efficient and sustainable agricultural practices.

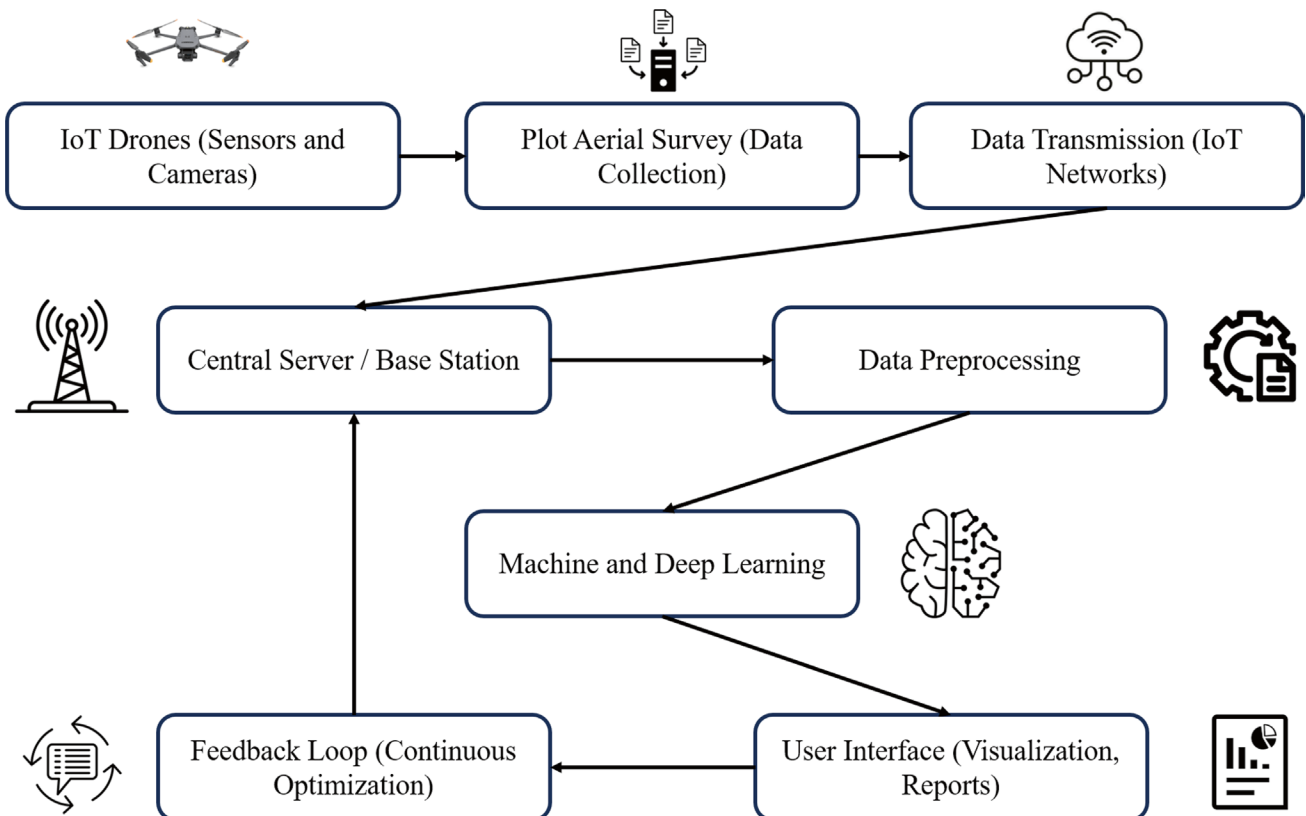


Fig. 1 Drones-enabled IoT network architecture with machine learning

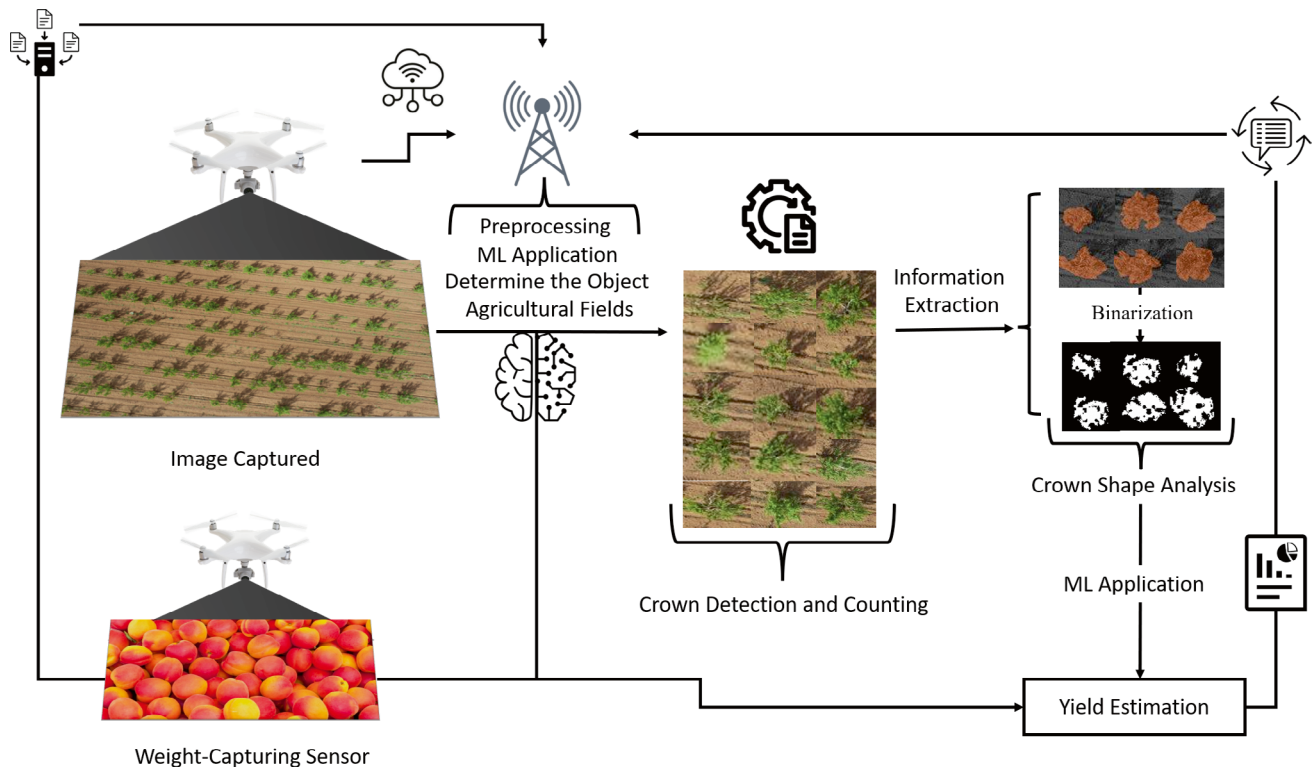


Fig. 2 A proposed scenario: integrating drones and IoT with ML applications for crop detection and yield prediction

In Fig. 2 proposed scenario, drones equipped with sensors and cameras are deployed to enhance farming practices. These drones follow predefined paths over fields, collecting environmental data and detailed crop images. The gathered data is transmitted in real-time to a central base station via IoT networks, where it undergoes preprocessing before being analyzed using machine learning algorithms. The workflow begins with data collection and preprocessing, where environmental data and crop images are acquired and refined. Machine learning applications are then employed for crop detection and counting, followed by object segmentation to extract key crop characteristics. Simultaneously, yield estimation is performed by analyzing the weight of each detected crop. Furthermore, the ML algorithms are utilized to determine the surface area of agricultural fields, leveraging this information to estimate crop production. The results are presented through a user-friendly interface, offering actionable insights to optimize crop management strategies. This iterative process continuously refines the machine learning models, leading to more accurate and efficient farming monitoring and management practices.

8 Data Sources and Evaluation Techniques in Deep Learning for Agricultural Applications

Datasets are essential for training deep learning models and vary across studies. Researchers either create custom datasets tailored to specific needs or use publicly available datasets, which often include annotated images. Typically, data is divided into training and testing/validation sets, with some studies using cross validation to evaluate models. Initially, high learning loss, influenced by network architecture and dataset size, decreases with training. Public image datasets for precision agriculture are categorized by their applications, such as crown tree detection, weed control, and fruit detection, as detailed in Table 3.

9 Comparative Study of Open-Source Object Detection Models

This section provides a comparison of different object detection models. After reviewing the design of each network. Table 4 a simple description of these models and their

Table 3 Data sources and metadata

Dataset name	Application	Size
NeonTreeCrowns (Weinstein et al. 2020)	Individual tree crown recognition	100,000
25kTrees (Li et al. 2023)	Segmentation of individual tree crowns	25,000
BAMFORESTS (Troles et al. 2024)	Detection of individual tree crowns	27,160
OAM-TCD (Veitch-Michaelis et al. 2024)	Individual tree crown delineation (TCD)	5072
Fruits 360 (Moltean 2024)	Fruit and vegetable recognition	94,110
DeepFruit (Latif et al. 2022)	Fruit recognition	21,122
Fruit and Vegetable (Suryawanshi et al. 2022)	Fruit and vegetable recognition	3825
CropDeep (Zheng et al. 2019)	Species classification and detection	31,147
Plant-Disease (Arun Pandian et al. 2019)	Plant disease classification	61,486
Indian Potato Weeds (Goyal et al. 2023)	Potato and weeds classification	240

applications. They are used in various fields, such as surveillance, where speed and accuracy are very important.

Preparing and configuring the dataset is one of the most critical phases in the object detection process. The dataset plays a crucial role in scientific research and is one of the most important factors for enhancing deep learning (Davila Delgado and Oyedele 2021; Sanaat et al. 2022). Unfortunately, data is still challenging and costly to generate and annotate. A series of operations, summarized in Fig. 3.

10 Crop Detection Metrics

Detection evaluation methods are generally measured using metrics that are commonly utilized in computer vision. These metrics assess the performance of identifying individual trees or outlining tree crowns from different angles. By integrating various metrics, a thorough analysis of the model's outcomes and potential errors is obtained. These metrics also offer valuable insights into a model's effectiveness, aiding in its proper application and interpretation within forest management and planning. Table 5 provides detailed descriptions of these metrics.

These metrics are adjusted to fit the specific goals of each study. For example, (Padilla et al. 2021) this study calculated Average Precision (AP) by averaging precision across different parts of the images. In other studies, different IoU thresholds were used instead of the usual 50%. For instance, (Yan et al. 2019) this research and this one (He et al. 2017) used higher IoU thresholds to more accurately evaluate how well objects were localized. They calculated AP and mAP over a range of 50% to 95%, following the standards set by the COCO dataset.

11 Yield Prediction Metrics

Yield estimation methods are assessed through a range of metrics that are vital for gauging the accuracy and reliability of predictions. These metrics evaluate the model's performance from various perspectives, offering insights into prediction errors and their potential effects on agricultural decision-making. By employing multiple metrics, a well-rounded view of the model's performance is obtained, allowing for the identification of strengths and weaknesses in the estimation process. This approach ensures the model is effectively applied and interpreted within the context of crop management and yield forecasting. Detailed descriptions of these metrics are provided in Table 6 quantitative metrics can be adjusted to meet the specific goals of yield estimation studies. For example, RMSE, MAE, and R^2 can be modified based on the particular context and objectives of the analysis, allowing for a more accurate assessment of model performance. Customizing these metrics provides deeper insights into prediction accuracy and improves the model's capacity to perform consistently across diverse datasets and conditions.

12 Experimental Results

The data for this study were collected on the outskirts of Meknes, Morocco, at the GPS coordinates 33°53'03.7"N 5°25'11.4"W. The images of the study area, covering an area of 0.6 hectares, were captured using a DJI Phantom 4 RTK drone, equipped with a 20 MP (1-inch) CMOS sensor, known for its high precision. The drone flew over the area at a constant altitude of 680 m, in a top-down view, ensuring uniform coverage. The sensor, with a focal length

Table 4 Comparison of different object detection models

Model	Description	Application
YOLOX (Ge et al. 2021)	An advanced object detection model by Megvii Research, released in July 2021, using an anchor-free approach for enhanced performance. YOLOX achieves state-of-the-art speed and accuracy, with 50.0% AP on COCO at 68.9 FPS	Real-time object detection in autonomous driving, video surveillance, and other practical applications
DETR (Carion et al. 2020)	An end-to-end object detection model using Transformer architecture, developed by Facebook Research, with a baseline AP score of 42.0% on COCO, known for handling complex scenes effectively	Complex object detection tasks requiring high accuracy, including scene understanding and autonomous systems
Detectron2 (Wu et al. 2019)	A PyTorch-based system by FAIR, supporting keypoint detection, object detection, and semantic segmentation, with advanced versions of models like Mask RCNN, achieving an AP up to 64.31 on COCO test-dev	Rapid research implementation, advanced object detection, keypoint detection, and semantic segmentation
YOLOv10 (Wang et al. 2024)	A real-time, state-of-the-art object detection model offering 46% less latency, NMS-Free training, and several variants (N, S, M, B, L, X) for different performance needs	Real-time object detection, autonomous systems, security applications, and various use cases requiring high efficiency
Efficient Det (Tan et al. 2019)	A compact, real-time object detection model based on EfficientNet, available in versions B0 to D7, with D7 achieving high accuracy (55.1)	Mobile and edge device deployment, applications requiring high accuracy with low computational cost, real-time object detection in resource-constrained environments
Faster R-CNN (Yang et al. 2017)	A two-step deep learning object detector developed by Ross Girshick, known for high accuracy in complex scenes, with AP between 34.9 and 36.8 on COCO test-dev	High-precision applications such as medical imaging, surveillance, and research requiring detailed object detection

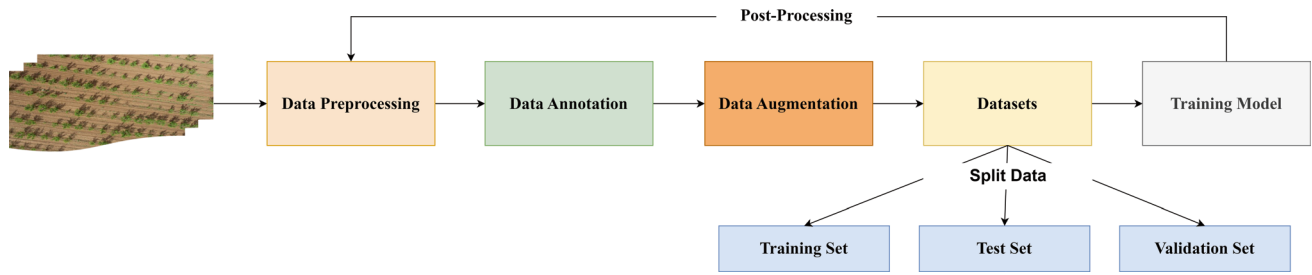


Fig. 3 Data preparation and training workflow for object detection models

Table 5 Accuracy Metrics for Crop Detection: Descriptions and Formulas from Analyzed Studies

Metric	Description	Equation
Precision	Proportion of true positives relative to all positive predictions	$\text{Precision} = \frac{TP}{TP+FP} \quad (1)$
Recall	Proportion of true positives identified among all actual cases	$\text{Recall} = \frac{TP}{TP+FN} \quad (2)$
F1-Score	Harmonic mean of precision and recall	$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$
Intersection over Union (IoU)	Measures the overlap between the prediction and the ground truth	$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}} \quad (4)$
Average Precision (AP)	Evaluates the quality of a model's predictions for a specific class by integrating the area under the precision recall curve	$\text{AP} = \int_0^1 p(r) dr \quad (5)$
Mean Average Precision (mAP)	Calculated across all categories, is used to evaluate the overall performance of object detection models	$\text{mAP} = \frac{1}{N} \sum_{i=1}^N \text{AP}_i \quad (6)$

Table 6 Precision metrics in yield prediction: overview and descriptions

Metric	Description	Equation
Root mean square error (RMSE)	Mean of the squared differences between predictions and observations	$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (7)$
Mean absolute error (MAE)	Mean of the absolute differences between predictions and observations	$\text{MAE} = \frac{1}{n} \sum_{i=1}^n \hat{y}_i - y_i \quad (8)$
Mean square error (MSE)	Mean of the squared differences between predictions and observations	$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (9)$
Coefficient of determination (R^2)	Proportion of the variance in the data explained by the model	$R^2 = 1 - \frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (10)$
Relative error (%)	Measure of absolute error relative to the true value, expressed as a percentage	$\text{Relative Error} = 100 - \left \frac{\hat{y}_i - y_i}{y_i} \right \times 100 \quad (11)$

of 9 mm and an aperture of f/5, provided images with a resolution of 5280×3956 pixels, offering great clarity and optimal depth of field. These images were enriched with precise metadata, such as GPS coordinates, flight altitude, and shooting conditions, ensuring the quality and reliability of the data for future analyses. In total, 70 images of the fields were captured. The RGB images of the apricot tree crowns, in PNG format and very high resolution, were then cropped into multi-level matrices, producing a total of 7952 images. These images were annotated using the LabelImg Annotations application, which allowed

key objects to be labeled by drawing bounding boxes around them. After annotation, the images were resized to 640×640 pixels before being used as input to a CNN model. To prevent overfitting and evaluate the model's performance, the images were divided into three sets: 70% for training, 20% for validation, and 10% for testing. The final accuracy of the model was assessed using the test set after training. During our experiments, we implemented and trained four models on our dataset. The configuration of each model varies. Each object detection model comes with its own setup shell, explicitly prepared by the creators

Table 7 Comparison of Object Detection Models

Training Duration (h)	Error Rate on Our Dataset (%)	Results on Our Dataset (F1 Score) (%)	Number of Training Iterations	Results on COCO (AP) (%)	Type	Model
12	15	85	500	50	One stage	YOLOX
24	18	80	1,000,000	42	Transformer-based	DETR
18	16	83	750	64.31	Two stage	EfficientDet
10	12	88	400	55	One stage	YOLOv10
14	14	86	500	55.1	One stage	EfficientD
30	17	81	1,200,000	36.8	Two stage	Faster R-CNN

(basic pipeline file). A pre-trained checkpoint specifies the pre-trained weights file, which is saved after the model was trained on a large dataset. We also specify certain training parameters, such as the number of iterations, training steps, and the input image size. To evaluate the performance of the neural networks and deep learning models after training them on our dataset, in our analysis, we employed several metrics to evaluate the performance of the object detection models. Accuracy was measured to assess the model's ability to correctly classify images as positive. Additionally, error rates were calculated to provide insights into the model's misclassification tendencies.

The Table 7 presents a summary of the primary performance metrics used in the training and evaluation of various open-source object detection models employed in our experiment, detailing the number of training iterations required, F1 Score, error rates, and execution times. These metrics were evaluated on both the COCO dataset and our custom dataset. A key observation is the significant variance in the number of training iterations necessary for each model. Initially, all models were trained with an equal number of iterations. However, the analysis revealed that some models required additional iterations to reach their optimal performance levels. Consequently, the iteration counts were adjusted during the training process to enhance accuracy while carefully monitoring for signs of overfitting. As the iterations increased, a notable trend emerged: both accuracy improved, and error rates decreased. This iterative refinement underscores the importance of tailoring the training process to the specific characteristics of each model to avoid overfitting and to ensure robust performance. Moreover, the dataset's size and quality were critical factors influencing model performance. Larger, high-quality datasets generally led to more reliable outcomes across all models. To further optimize performance and minimize training times, the use of a GPU rather than a CPU is strongly recommended. Ultimately, the table highlights those certain models, particularly YOLOv10 and EfficientDet, demonstrated superior adaptability. These models not only produced more accurate results but also benefited more

significantly from the optimized iteration adjustments, making them particularly effective in our experiments.

13 Linear Regression Model for Apricot Tree Weight Estimation

Once the apricot tree crown is identified, we outline its shape using edge detection methods, such as the Canny edge detector from the OpenCV library. To accurately determine the size of the apricot crown, we apply the Ground Sampling Distance (GSD), which indicates the distance between the centers of two adjacent ground pixels. GSD is vital for establishing the spatial resolution of the captured images.

After identifying the crown shape in the image, we determine the number of pixels representing the crown's width and height. The actual size of the crown is then calculated by multiplying the pixel count in each dimension by the corresponding Ground Sampling Distance (GSD) value, as follows:

$$\text{RealSize} = \text{Imagecaptured} \times \text{GSD} \quad (12)$$

where Imagecaptured indicates the pixel measurement of the crown's width or height as recorded by the imaging sensor, giving its size in pixel units. The Ground Sampling Distance (GSD), on the other hand, measures the real-world distance between two neighboring pixel centers in the image. GSD is crucial for determining the spatial resolution of the image and is calculated using the following formula:

$$\text{GSD}(x, y) = \frac{\text{sensor}(x, y) \times \text{altitude} \times 100}{\text{focal_length} \times \text{img}(x, y)} \quad (13)$$

where sensor (x, y) refers to the physical dimensions of the camera sensor along the x and y axes, typically measured in millimeters, determining the portion of the scene captured by each pixel. Altitude indicates the height at which the drone or camera is positioned above the ground, measured

Table 8 Production in kilograms and associated shape for each tree

N	Production (kg)	Associated shape
1	70	1500
2	65	1450
3	72	1600
4	68	1550
5	74	1620
6	66	1480
7	69	1530
8	71	1580

in meters, affecting the image scale and detail level. Focal length is the distance between the camera lens and the sensor, measured in millimeters, influencing the field of view; a longer focal length provides a narrower field of view, capturing less area with more detail. Image dimensions ($\text{img}(x, y)$) refer to the resolution of the captured image in pixels, representing how many pixels are used to depict the scene.

We collaborated with an expert who possesses extensive experience in the field and is also native to the studied region. A local expert was specifically appointed to conduct data collection in the area. The data collection process involved various techniques, including manual counting during field surveys and visual assessments of apricot tree growth. Leveraging their extensive skills and practical experience, the experts calculated and estimated the potential yield of each apricot tree. The collected data is presented in Table 8.

Following this, we apply linear regression to estimate the mass of each tree. We also employed a classic linear regression model to estimate the weight of apricot trees, based on the segmentation of apricot crown shapes:

$$\text{Estimated Weight} = \alpha + \beta X \quad (14)$$

where EstimatedWeight represents the predicted weight of the apricot crown, α is the intercept of the linear model, and β is the coefficient associated with the feature. The feature, denoted as X , refers to the shape of the apricot crown.

In this model, the shape of the apricot crown (X) is used as the primary feature to predict the weight. The linear regression equation relates this feature to the estimated weight by applying the coefficient (β) and adding the intercept (α) to provide the final predicted weight. This method allows us to make informed estimates based on measurable characteristics of the apricot crown.

In Fig. 4, the comparison between actual production and estimated weight for each tree is depicted. The model generally follows the trend of actual production, but discrepancies are observed. For instance, Tree 2 shows significant deviations where the estimated weight is higher than the actual production, indicating a tendency of the model to overestimate in certain cases. This observation is further supported by the residuals plot in Fig. 5, where the variability in the model's accuracy is highlighted. Positive and negative residuals indicate instances of overestimation and underestimation, respectively. The fluctuations in residuals suggest inconsistencies in the model's predictions across different trees, pointing to areas where improvements can be made. Furthermore, the scatter plot in Fig. 6 compares actual production with estimated weight, where a dotted line represents perfect estimation. The close proximity of most points to this line suggests that the model is generally accurate; however, the presence of significant outliers indicates that there is still room for enhancing the precision of the model.

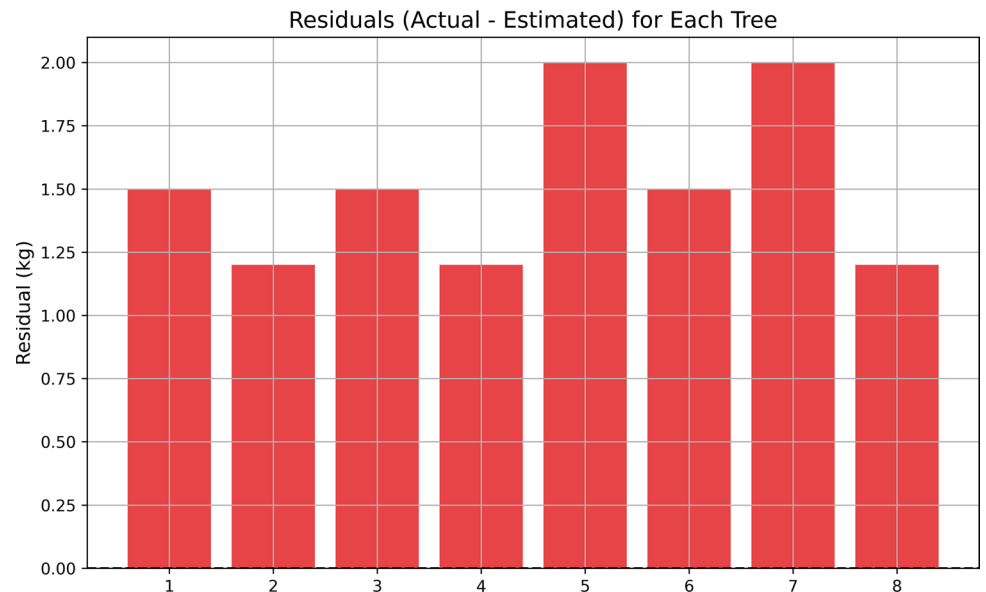
Fig. 4 Comparison of actual production versus estimated weight for each tree

Fig. 5 Residuals (Actual–Estimated) for Each Tree

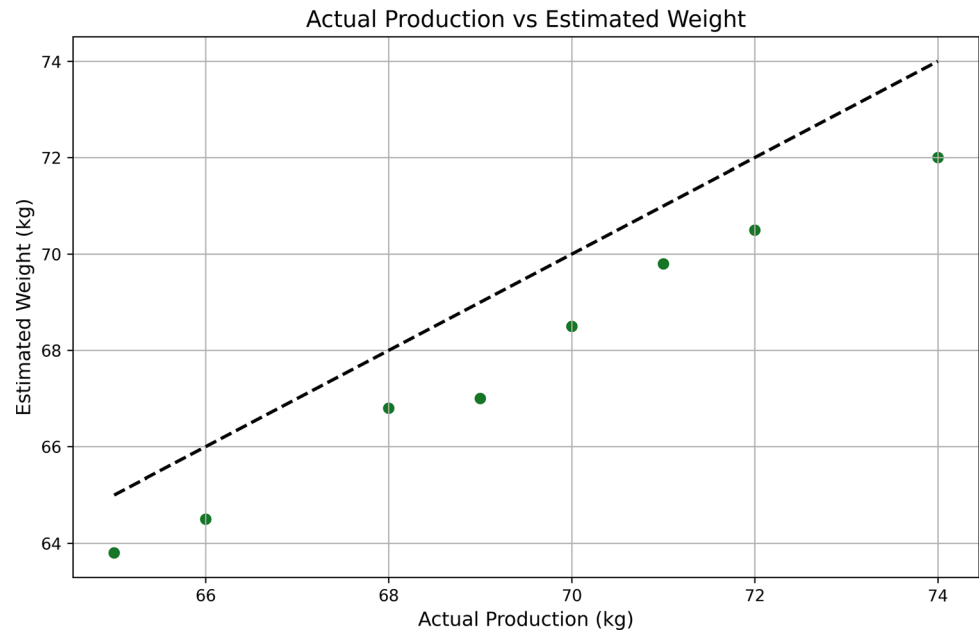
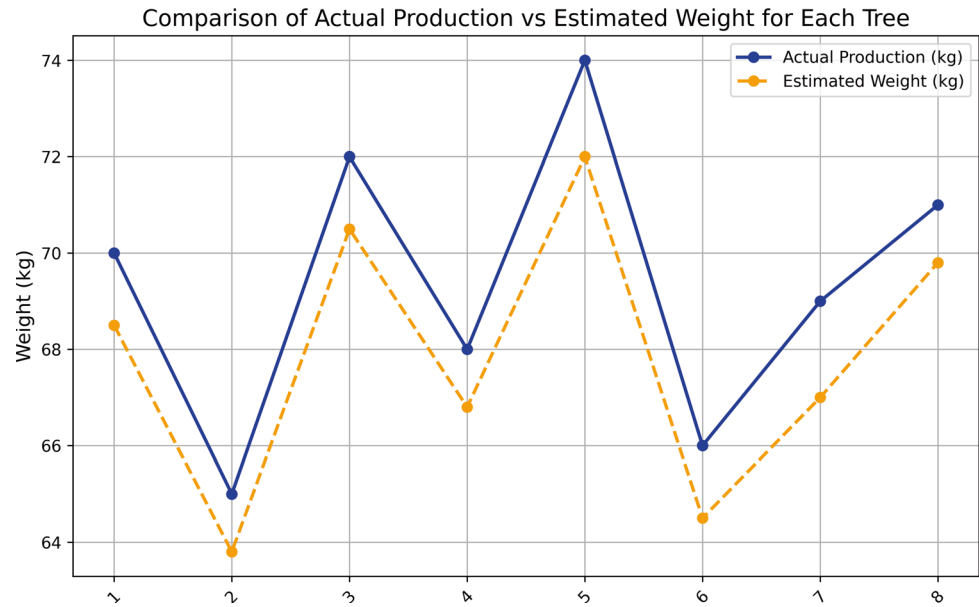


Fig. 6 Actual production versus estimated weight



The performance metrics presented in Table 9 provide a quantitative assessment of the model's accuracy. The Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Relative Error (%) are calculated for each tree. On average, the MAE and RMSE values suggest that the model's predictions are fairly accurate, with a Relative Error of approximately 2.18%. However, certain trees, such as Tree 5 and Tree 7, exhibit higher errors, indicating that the model's predictions are less reliable for these cases. This aligns with the observations from the figures, reinforcing the need for further refinement of the model to reduce these discrepancies.

Table 9 Error Metrics for Each Tree in Estimating Weight Production

N	MAE (kg)	MSE (kg ²)	RMSE (kg)	Relative error (%)
1	1.5	2.25	1.5	2.14
2	1.2	1.44	1.2	1.85
3	1.5	2.25	1.5	2.08
4	1.2	1.44	1.2	1.76
5	2.0	4.00	2.0	2.70
6	1.5	2.25	1.5	2.27
7	2.0	4.00	2.0	2.90
8	1.2	1.44	1.2	1.69
Average	1.51	2.38	1.51	2.18

14 Conclusion and Future Directions

In conclusion, this chapter highlights the transformative impact of modern technologies in agriculture, particularly in boosting efficiency, sustainability, and productivity. The integration of drones equipped with advanced sensors into IoT networks has markedly enhanced the speed and accuracy of real-time data collection. This technological convergence facilitates continuous and detailed monitoring of crops and soil conditions, thereby enabling more informed and timely decision-making.

The use of Machine Learning (ML) and Deep Learning (DL) techniques to agricultural data has uncovered complex patterns, leading to more precise forecasts and optimized resource management. Our case study on apricot crop management, which utilized drone-enabled IoT networks and advanced object detection models like YOLOv10, achieved notable precision, with an F1 Score of 88% and an error rate of 12%. Additionally, the use of linear regression for yield prediction yielded a Mean Absolute Error (MAE) of 1.51 kg and a Relative Error of 2.18%, demonstrating the effectiveness of these models in accurately predicting crop productivity.

Looking forward, several areas warrant further exploration and development. First, the design of more sophisticated, agriculture-specific ML algorithms is crucial to enhancing forecast accuracy and improving disease detection in crops. The integration of emerging technologies such as blockchain for product traceability and advanced AI for deeper data analysis offers significant potential to further revolutionize the sector.

Additionally, enhancing IoT networks, especially in terms of energy efficiency and rural coverage, will be vital for enabling more reliable and continuous data collection. Large-scale case studies across diverse agricultural and climatic environments are also needed to provide a comprehensive assessment of these technologies' impacts.

For successful adoption of these advanced technologies, it will be necessary to provide tailored training programs for farmers. Furthermore, political and regulatory support will be essential to foster an environment conducive to widespread innovation adoption while ensuring data security and privacy.

By focusing on these research and development directions, we can continue to advance agricultural innovation, build resilience, and meet the growing food demands of the global population.

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Drone Swarm Coordination Using Machine Learning in IoT Networks

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Abstract

Unmanned Aerial Vehicles, or UAVs, also known as drones, have drastically disrupted the aviation industry. The more the disruption, the more probable it is that legislation and technology will continue to advance. One such technology that could amplify this disruption is the UAV swarm. A UAV swarm is capable of assigning tasks to numerous UAVs while requiring no human interaction. Communication is vital for the management and coordination of a UAV swarm. In general, extra sensory or supplementary equipment is also required to deal with any issue with the swarm, like high-end essentially trustworthy communication to meet these challenges, including coordination and collaboration, control, security, mission planning algorithms, and more. Drone swarms benefit from the inclusion of Distributed Learning (DL) algorithms to endow them with target tracking, remote surveillance, power management, user association, perception, trajectory planning, wireless resource allocation, and communication services. This chapter presents a review of UAV swarms in IoT networks, the application of machine learning methods for swarm communication in UAVs and challenges in communication of swarm drones alongside the optimization techniques. Finally, we review their applications in UAV cluster job scheduling, federated reinforcement learning (RL) empowered aerial access networks (AANs) with low-earth orbit (LEO) satellites and UAVs, and energy-efficient wireless

communications with distributed reconfigurable intelligent surfaces.

Keywords

Distributed learning · IoT networks · Machine learning for distributed learning UAVs · UAV security · UAV swarm

1 Introduction

Drones or Unmanned Aerial Vehicles (UAVs) first emerged for military use. Their main uses were for Close Air Support (CAS), Border Surveillance, and Search and Destroy operations. Apart from their military uses, Unmanned Aerial Vehicles (UAVs) are crucial in commercial and public service domains. These include disaster relief, traffic monitoring and management, wind assessment, agriculture remote sensing, civil safety and security, and forest fire management (Meng et al. 2022). The past several years has seen a significant growth in UAV research and development due to their vast range of applications. The UAVs can be flown manually, through remote control operations, or automatically, through the use of processors installed on the drones. Figure 1 shows how swarm drones are used in different applications.

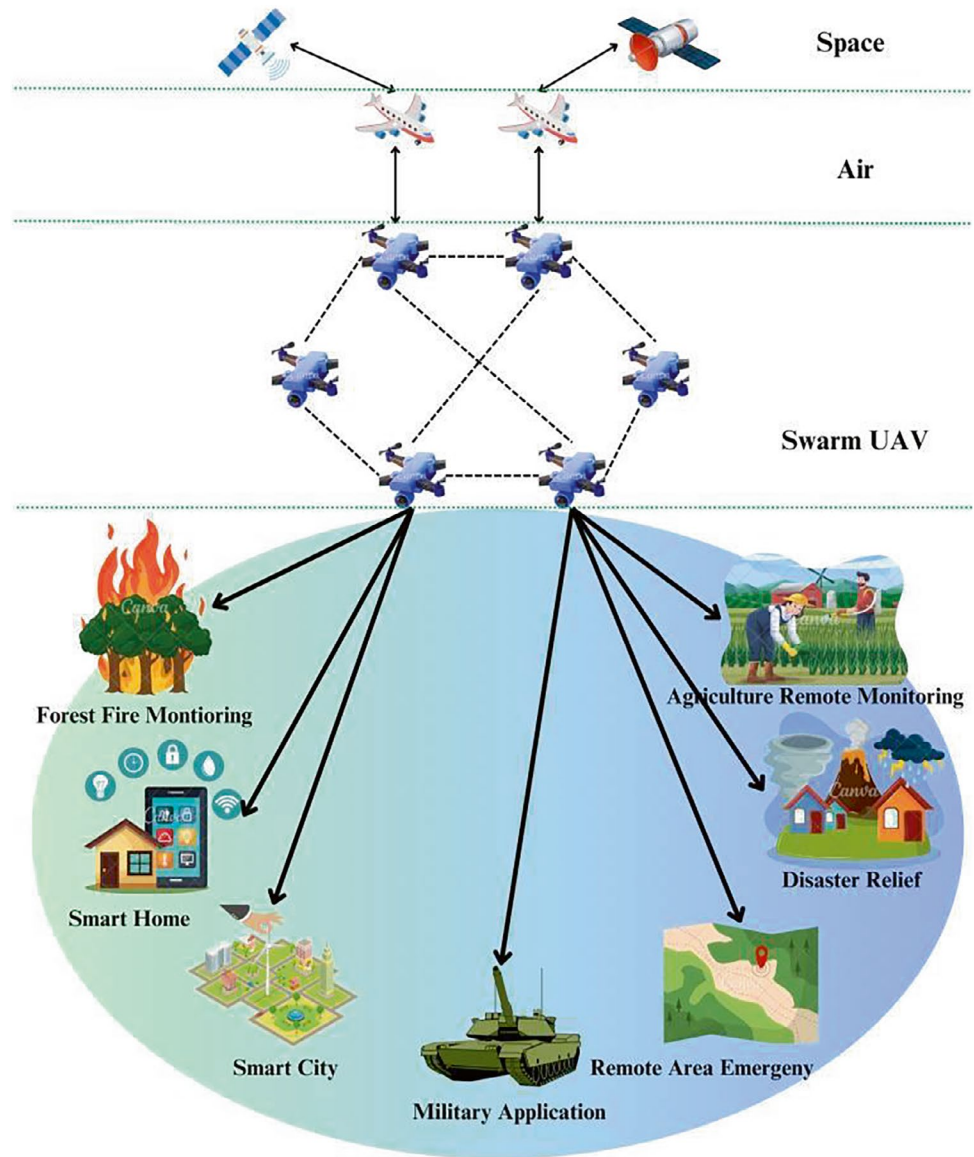
The next generation of wireless communication networks will include Unmanned Aerial Vehicles (UAVs) as a fundamental component. When compared to conventional ground-based solutions, their implementation in a variety of communication-based applications is anticipated to improve coverage and spectrum efficiency (Hashesh et al. 2022). But the increased flexibility that the network will offer will also bring with it new difficulties. In this situation, it is anticipated that the Machine-Learning (ML) framework would offer answers to the several issues that have already been found when using UAVs for communication.

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Fig. 1 Applications of swarm drones



UAVs are starting to swarm or form clusters more frequently. The swarm or formation produced by multiple UAVs offers clear advantages over a single UAV system. Similar high-performance, reliable communication is needed to address the challenges of a UAV swarm, such as coordination and cooperation methods, control strategies, security protocols, mission planning algorithms, and so on. A swarm is equipped with two types of communication: UAV-to-UAV (U-T-U) and UAV-to-Infrastructure (U-T-I) (Azari et al. 2020). Communication architecture: The communication architecture has a key impact on the intelligence control and autonomous cooperation of UAV swarms. In the early stages of swarm formation, a central station that can talk to all the UAV swarm is available. Singling site strategy

is encouraged for larger swarm which is decentralized despite the fact that it is more complex (i.e. ordered) than the more-dispersed variety, this method is less reliant on central stations for the swarm (Xu et al. 2021). This chapter gives a brief introduction about the compared four different architectures: Single-group swarm ad-hoc network architecture, multi-group swarm ad-hoc network architecture, multi-layer swarm ad-hoc network architecture and centralized architecture, and comparisons of pros and cons.

We frequently concentrate on the needs of high coverage and preserving connectivity when debating the benefits and drawbacks of UAV swarm communication topologies. When acquiring intelligence and analyzing circumstances, high coverage is crucial. The only way to guarantee the

swarm's real-time communication is to maintain connectivity. Nonetheless, UAV swarms frequently operate in uncharted territory, and dangers and impediments can arise at anytime and anywhere. As a result, it is imperative that participants leave or return. As a result, connecting without interruption is impossible. This mostly relies on the attenuation of signals caused by loss of transmission. In order to maintain a continuous connection, the UAV swarm ad hoc network configuration must, on the one hand, stay within the minimum signal-to-noise ratio (SNR) or receive signal strength indicators (RSSI) range, respectively, and not exceed the sensitivity of the receiver.

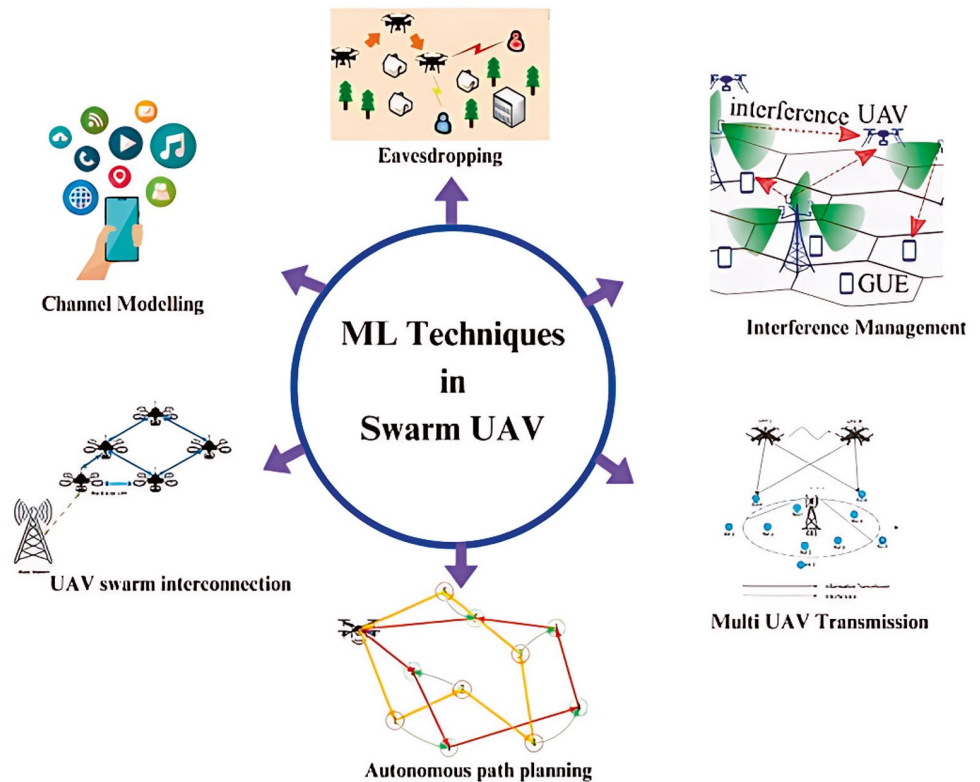
Drone swarms, with their inherent advantages including flexibility, mobility, cost-effectiveness, and the ability to deliver a range of services both independently and collaboratively, represent a promising improvement in next-generation wireless communication networks. Drone network performance is changing as a result of machine learning. There is no denying the increasing impact of AI and machine learning in this field (Asaamoning et al. 2021). Figure 2 illustrates the use of machine learning algorithms for IoT network coordination and drone swarm communication. Distributed Learning (DL) algorithms are integrated into drone swarms to provide them with intelligent capabilities for target tracking, remote surveillance, power management, user assignment, perception, trajectory planning, wireless resource allocation, and communication services

(Asaamoning et al. 2021). Four state of the art DL methods including Federated Learning, Multi-agent Reinforcement Learning, Distributed Inference and Split learning will be described in detail with analysis of their advantages and disadvantages.

It also investigates how to best allocate resources and perform joint training for federated learning in UAV swarms, demonstrating the advancement of cooperative learning methods in these networks. In path planning and optimisation tasks, deep learning methods are utilized to train neural networks to learn and adapt to dynamic settings. This allows the networks to optimize UAV trajectories based on environmental feedback and real-time data. DL models optimize spectrum allocation, power control techniques, and communication parameters to maximize network performance and dependability when it comes to wireless resource allocation. Additionally, DL improves sensing capabilities by allowing drones to identify abnormalities, patterns, and objects from sensor data, which helps them make better decisions and improves situational awareness. To further improve overall speed and intelligent autonomy in drone swarm operations, DL approaches are also used in caching optimization, computing resource allocation, and collaborative learning tactics.

The problems related to optimization in UAV swarm operations presents a challenging technical perspective. These are path optimization algorithms, which are a

Fig. 2 Machine learning techniques in swarm drones IoT network



necessary building block for navigation and trajectory planning. This way, UAVs can fly over difficult terrain and follow pre-plotted courses with the lowest energy usage and smallest possible time in the air. Wireless resource allocation algorithms are indispensable both to govern the scarce spectrum resources the UAV swarm can use for communication, and to achieve maximum network throughput, minimum interference, and optimal data transfer rate.

The improvement of drone perception abilities relies heavily upon sensing optimization techniques, by which one can collect and process data from a large number of sensors at minimum latency, and with maximal accuracy. This will utilize efficient data processing methods tailored for drone swarm use-cases, adaptive sampling techniques and innovative data fusion algorithms. Caching optimization methods operate by using smart data caching techniques, where the drones cache essential (or frequently reading) data locally to reduce the data retrieval overhead in terms of bandwidth consumption and data reply time. It is primarily designed to make use of the drone swarm's usage trends and the temporal popularity of the data, in order to optimally store and can store as much data as possible.

Engine resource allocation algorithms were also required to regulate the speed and computational responsibilities of each drone or UAV in the swarm. By optimizing task scheduling, workload distribution, and resource allocation, these algorithms also facilitate the execution of data processing, image recognition, and decision-making algorithms on-board UAVs. The optimization problems encompass not only ground-based network cases but also non-ground network cases e.g. satellite-UAV communications. Within such environments the SATCOM and UAVs are managed by the optimization of communication protocols, data routing techniques and network changes. This ensures a stable and efficient data communication under different operational conditions (Chen et al. 2020a).

This chapter provides an overview of the ways in which machine learning methods are applied to enhance communication in UAV swarm networks. The underlying technology for designing and implementing routing protocols, common UAV swarm communication routing, is covered in detail in this chapter. Router protocols are then categorized according to three factors: position/geographic, swarm intelligence, and topology. In addition, it offers an in-depth overview of every category of routing protocols, including their pros and cons. This chapter explores diverse applications of Unmanned Aerial Vehicle (UAV), the challenges they present, and open problems related to their wireless communication infrastructure. Swarm intelligence is a key component in creating resilient UAV networks. The chapter highlights how crucial test-beds and real-world experimentation are to assessing swarm intelligence methods in these networks.

With an eye on the future, the discussion delves into the benefits and problems that non-terrestrial networks bring in the 6G era, highlighting the potential applications of UAV network technology. Lastly, the use of reinforcement learning methods is discussed. We address its application in energy-efficient wireless communication with distributed reconfigurable intelligent surfaces, federated reinforcement learning-based aerial access networks with low-earth orbit satellites and UAVs, and UAV cluster task scheduling. This demonstrates how reinforcement learning may be used to improve wireless communication in UAV networks.

This chapter offers a high-level overview of the main research topics and offers insightful information about the current situation, challenges, and potential applications of swarm intelligence and drone networks. It also highlights how important intelligent and cooperative methods are becoming to maximize drone network performance.

2 Swarm Drones in IoT Networks

Swarm drones are revolutionizing the Internet of Things (IoT) network landscape by taking inspiration from the collective behavior seen in biological swarms, such as bees or birds. Swarm drones provide a novel approach to drone operations by using the combined strength of numerous drones to achieve unprecedented scalability, resilience, flexibility, and efficiency. With each drone functioning as a simple actor inside the swarm, these drones operate as a networked control system (Asaamoning et al. 2021). Compared to single large drones, the incorporation of swarm drones into IoT networks results in considerable gains, including better system stability and scalability (Abdallah et al. 2019). The significance of routing and Internet of Drones technologies is highlighted by the fact that self-organization and clustering are essential components in the development of drone-based cognitive IoT systems (Aftab et al. 2019). Swarm drone security is a crucial factor to take into account, especially when sensitive data is handled by the drones on important missions (Cabuk et al. 2021). Through a rating-based federated learning approach, it is possible to achieve efficient compute offloading management in a swarm of drones while accommodating the heterogeneous drones' different capabilities (Rahbari et al. 2021).

2.1 Improving Coverage and Scalability

Every drone in the swarm can function as an IoT network node, enabling data gathering and communication across a large area. Drones can communicate over longer distances than a single drone could by forming a mesh network, in

which case the drones relay data to other drones in the area. This makes drones ideal for applications like agricultural management and environmental monitoring that need to cover large areas (Meng et al. 2022). Optimizing coverage path planning, point-of-interest identification, and overall network efficiency depend on the cooperative behavior of several drones or a swarm of drones, a phenomenon known as Swarm Intelligence (Bezas et al. 2022). Furthermore, using a cluster-based swarm intelligence technique facilitates the development of drone networks that are self-synchronized, which guarantees critical timing synchronization for efficient swarm interaction (Alsolami et al. 2021).

Furthermore, deploying a swarm of collaborative and intelligent drones saves energy consumption and costs while increasing throughput and service quality, making it a popular alternative for IoT applications (Rahbari et al. 2021). Drone swarms have the ability to improve wireless communication services and adapt to changing communication settings, which highlights their potential to alleviate connectivity issues in remote or disaster-affected areas (Shvetsov et al. 2023).

2.2 Improved Robustness and Reliability

Because of their redundancy, swarm drones provide a dependable solution for IoT networks, especially in mission-critical applications. The technology continues to work even in the event of a drone malfunction, guaranteeing uninterrupted operation and enhancing overall dependability. In order to sustain network connectivity and functionality, these drones are able to self-correct their structure. To ensure continuous data flow and network stability, nearby drones, for example, can realign themselves to fill in any gaps left by malfunctioning drones (Chung et al. 2018). Traditional static IoT networks frequently require user intervention to restart performance; this self-healing feature is a huge improvement.

2.3 Adaptive and Autonomous Operations

By utilizing swarm intelligence methods like particle swarm optimization (PSO) and ant colony optimization, swarm drones in Internet of Things networks exhibit adaptability and autonomy. Based on basic guidelines and local data, these algorithms allow drones to work together. Due to their versatility, swarm drones can react quickly and dynamically to changing network requirements and environmental conditions. For example, they can modify their positions in real-time based on sensor data to ensure accurate data collection and timely responses (Chung et al. 2018). Functioning as a swarm of drones requires excellent

coordination and communication in order to accomplish group goals. The effectiveness of drone swarms depends on algorithms that enable cooperation, communication, task assignment, trajectory planning, and flight. Drones that can autonomously navigate challenging settings and adjust to changing conditions rely heavily on machine learning algorithms (Nwankwo Constance Obiuto 2024). Edge computing is a vital component for intelligent applications in Internet of Things networks since it can also provide low latency, quick response times, mobility assistance, context-aware services, and privacy preservation (Bourechak et al. 2023).

2.4 Enhanced Data Collection and Processing

By leveraging various sensors, each drone within the swarm can gather a wide range of data types, including temperature, humidity, air quality, and visual imagery. This collective data acquisition provides a comprehensive and multi-dimensional perspective of the environment being monitored (Chung et al. 2018). One of the key advantages of utilizing swarm drones is their ability to conduct distributed data processing through edge computing. Instead of relying on a centralized server for data analysis, swarm drones can perform local data processing on the drones themselves. This approach reduces latency, conserves bandwidth, and facilitates real-time data analysis and decision-making (Hou et al. 2020). Moreover, swarm drones can be instrumental in scenarios where immediate data collection and processing are crucial, such as in disaster response, wildlife conservation, and environmental monitoring. By leveraging AI-driven capabilities, these drones can provide real-time data for swift and informed decision-making (Bourechak et al. 2023).

2.5 Energy Efficiency and Resource Management

Swarm drones, particularly in IoT networks, play a crucial role in optimizing energy consumption, especially for battery-powered devices like drones. Collaborative strategies among swarm drones can significantly enhance energy efficiency. By employing collaborative approaches, drones can alternate between high-energy tasks, such as long-distance communication or heavy-lift operations, allowing some drones to conserve energy while others handle demanding activities. Moreover, swarm algorithms are instrumental in optimizing flight paths to minimize energy expenditure. By planning routes that avoid obstacles and capitalize on favorable wind conditions, swarm drones can extend their operational endurance, thereby enhancing overall energy

efficiency. Additionally, drones can coordinate to establish energy-efficient communication patterns. For instance, they can form temporary relay chains to transmit data over long distances without depleting individual drones' batteries, showcasing the potential for swarm drones to operate efficiently and effectively in IoT networks (Li et al. 2018).

2.6 Security and Privacy Enhancements

Security and privacy are crucial aspects in IoT networks, and swarm drones offer unique advantages in addressing these concerns. The decentralized nature of swarm drones enhances network security by making it challenging for malicious actors to compromise the entire network. In the event of a drone being hacked or captured, the remaining drones can reconfigure the network to isolate the threat (Cooley et al. 2018).

Swarm drones can utilize distributed encryption and authentication protocols to ensure secure data transmission. Each drone can act as an independent verifier, ensuring data integrity and authenticity across the network (Abdel-Malek et al. 2022). Additionally, privacy concerns can be addressed through the implementation of differential privacy techniques. By distributing data collection and anonymization tasks among multiple drones, sensitive information can be protected, safeguarding individuals' privacy while enabling comprehensive data analysis (Perera et al. 2022).

Obstacle-induced connectivity problems in multiple isolated sensor networks (MISN) are addressed by the incorporation of UAV swarms in IoT networks in the research done by Pei et al. (Pei et al. 2022). Under situations where ground-based nodes are unsuccessful, UAVs serve as mobile relay nodes to improve network accessibility. Three UAV swarm types are suggested by the authors: Bionic Echelon Mode (BEM) for swarms of smart nodes, Closed Loop Mode (CLM) for big MISNs, and Discrete Distributed Mode (DDM) for small to medium-sized MISNs.

3 The Rise of Drone Swarms: Revolutionizing Aerial Operations

The swarm or formation produced by multiple UAVs offers clear advantages over a single UAV system. A single, significant UAV was all that was needed to carry out one or more missions in the early days of the UAV system. Even with highly sophisticated and specced-out UAVs, a single system is insufficient when compared to a multi-UAV system. Multi-UAV systems offer several benefits, including redundancy-based survivability and dependability. When a

single UAV in a multi-UAV system fails, the network reorganizes to keep communication going via other nodes. A single UAV system would not be able to accomplish this. However, the protocols put in place must address problems like shifting topology, mobility, and power limitations if the many UAVs are to really benefit from their cooperative efforts. A single UAV system would need to maintain connections with the control station or stations, base stations, servers, and access functions in order to meet its communication needs. This severely limits the bandwidth and battery power that are available. Only one or two UAVs may connect to the servers and control in a multi-UAV system, feeding the other UAVs. In this manner, the majority of UAVs only need to maintain the mesh structure in order to provide simple access features for data, video, and communications. In comparison to its larger counterparts, multi-UAV systems also prove to be less expensive to purchase, maintain, and run (Gupta et al. 2015). The performance comparisons between single and multi-UAV systems are summarized in Table 1. Below is a comprehensive overview of each of the several criteria:

- **Survivability:** If a single-UAV is shot down while on a mission, it is deemed a failure in a single-UAV system. A single out-of-control UAV, however, poses no threat to multi-UAV systems since the other UAVs will keep flying.
- **Scalability:** The coverage of single-UAV systems is only partially increased by using large UAVs. On the other hand, multi-UAV systems can readily expand their operational range.
- **Mission completion speed:** Studies indicate that employing multiple UAV systems might expedite mission completion times. This is particularly true for search jobs, where the ability of multi-UAV systems to handle tasks in parallel reduces processing time.
- **Autonomy:** In a conventional single-UAV system, all aircraft systems are directly and instantaneously controlled by the pilot operating from the ground. The onboard automation of the majority of multi-UAV systems guarantees controlled flight in compliance with flight plans and other instructions obtained from infrastructures.
- **Cost:** Studies reveal that using multiple UAV systems can result in missions being finished at a reduced cost.
- **Communication requirements:** Single-UAV systems must always be in contact with infrastructure or ground pilots. A multi-UAV system, on the other hand, consists of a single UAV that speaks with the ground and relays information to additional UAVs.
- **Radar cross-section:** Multi-UAV systems generate a tiny radar cross-section for military applications, which improves operational security.

Table 1 Comparison of single UAV system and drone swarm (multi UAV) system (Chen et al. 2015, Gupta et al. 2015, Aljehani and Inoue 2017)

Features	Single UAV system	Multi UAV system
Survivability	Poor	High
Scalability	Limited	High
Speed of mission	Slow	Fast
Autonomy	Low	High
Cost	High	Low
Communication needs	High	Low
Radar cross-sections	Large	Small
Impact of Failure	High	Low
Multitasks capability	Low	High
Reconfigurability	Low	High
Ad-Hoc networks	Cannot	Optimal
Complexity	Low	High
Heterogeneous	Inapplicable	Applicable
Bandwidth need	High	Medium
Antenna	Omni-directional	Directional
Failure to coordinate	Low	Present

- Impact of failure: A single UAV failure might have a domino effect, affecting the safety and success of the entire mission. But Multi UAV can successfully handle such errors, fault-tolerant control strategies (Li et al. 2021)
- Multitask capability: Multi-UAV systems are becoming more and more proficient at managing a variety of duties at once, such as environmental monitoring and surveillance, which greatly improves their operational efficiency. These systems use sophisticated algorithms to dynamically assign jobs based on real-time data, maximizing the efficiency of mission objectives and resource utilization (Zhang and Han 2021)
- Reconfigurability: Reconfigurability enables multi-UAV systems to dynamically modify their roles and formations in response to shifting mission needs and external circumstances. This flexibility guarantees that even in the face of unforeseen difficulties or mission modifications, the system can continue to operate at peak efficiency (Liu and Li 2020).
- Ad-Hoc Networks: Operational flexibility is limited by the direct communication requirement of single UAVs with ground stations. Ad hoc networks are used by multi-UAV systems to provide decentralized control and a wider operational range (Wei and Fang 2021).
- Complexity: When the mission criteria are clear, single UAV systems are easier to operate. Advanced coordination and control algorithms are necessary for multi-UAV systems, which are more intricate (Wang and Li 2021).
- Heterogeneous: Individual UAVs are uniform and constrained by their particular set of abilities. Multi-UAV systems can be heterogeneous, utilizing many UAV kinds to do a wider variety of tasks (Chen and Liu 2020).
- Bandwidth Need: Data transmission bandwidth needs are lower for single UAV systems. Multi-UAV systems need a lot of bandwidth in order to send data from several UAVs at once (Sun & Zhao 2021).
- Antenna: Standard antenna configurations are usually needed for single UAVs to communicate with base stations. Phased arrays are one type of advanced antenna technology that multi-UAV systems require for reliable inter-UAV communication (Xu & Zhang 2020)
- Failure to Coordinate: Single UAV systems do not have coordination problems. Robust coordination mechanisms are necessary because multi-UAV systems are susceptible to coordination failures (Li and Wang 2021).

4 Communication in UAV Swarm Networks

Effective communication amongst UAVs, whether distributed or centralized, is a basic prerequisite for establishing a cooperative mechanism for collective control in order to sustain swarm formation and cooperation. UAV-to-UAV and UAV-to-infrastructure communication are both included in swarm communication.

1. U-T-U communication: By establishing effective communication, all of the swarm's UAVs enable the sharing of information via radar or sensors. By building multi-hop

communication pathways with other UAVs, two UAVs can communicate with one another directly or indirectly.

2. U-T-I communication: In order to get real-time mission information, UAVs interact with a fixed controlling center, like a ground station. Direct communication is typically used in U-T-I interactions. The central control center is represented by infrastructure in the remaining sections of this essay (Chen et al. 2020b).

For intra-swarm communication, UAVs can implement one of the few topologies of a ring, star, mesh or fully connected. The UAVs exchange data in a loop, having a bi-directional link, an example of such topology is a ring. When a direct connection between any two UAVs is disrupted, the two UAVs back loop information to each other. All UAVs communicate through one UAV which is called a gateway UAV while in the star topology. The problem with this topology is the single point of failure. Mesh—The mesh topology connects every UAV in a swarm to more than one UAV. It bypasses the single point of error question with this topology. In a fully connected topology, all the UAVs in a swarm are linked between them.

There is a communication architecture that integrates the U-T-I and U-T-U communication. Put differently, the information sharing mechanism between UAVs or between UAVs and infrastructure is defined by the UAV swarm communication architecture. When taking U-T-U communication into account, the routing protocol is crucial to ensuring dependable end-to-end data delivery (Khan et al. 2017).

For UAV swarms to operate intelligently and collaborate autonomously, communication architecture is crucial. A central station that is capable of communicating with the entire UAV swarm is accessible during the initial phase of the swarm's development. As a result, UAV swarm communication architecture was added to the conventional single-UAV system using the centralized technique. The proposal is for a decentralized approach as the swarm gets larger.

Although the decentralized method is more intricately structured and organized, it lessens the swarm's reliance on central stations (Chen et al. 2020b). UAV swarm communication designs have been the subject of extensive study by numerous professionals and academics.

4.1 Communication Architecture

Communication architectures for networking UAVs in a swarm are categorized in two major categories which are centralized and decentralized communication architecture.

- (a) **Centralized communication architecture:** Decentralized UAV communication methods necessitate the use of a central node in order to link UAVs to remote pilots or other UAVs. Here, instead of interacting with one another, UAVs are directly linked to a ground station, satellite, or cellular base station. The cellular, satellite, and direct link network designs are examples of this centralized architecture, which has a number of downsides.
 - (i) **Direct Link Network Architecture:** A direct link network design uses a star topology to connect UAVs to a ground station as shown in Fig. 3a. This allows for a simple implementation that offers low latency and dependable data delivery for communication between the UAVs and the ground station. Unfortunately, this design needs expensive components and has a considerable latency in UAV-to-UAV communication inside a swarm.
 - (ii) **Satellite Network Architecture:** Compared to direct link architectures, satellite network architecture provides superior coverage by using satellites to link UAVs to ground stations as shown in Fig. 3b. This design is constrained by inadequate data transmission,

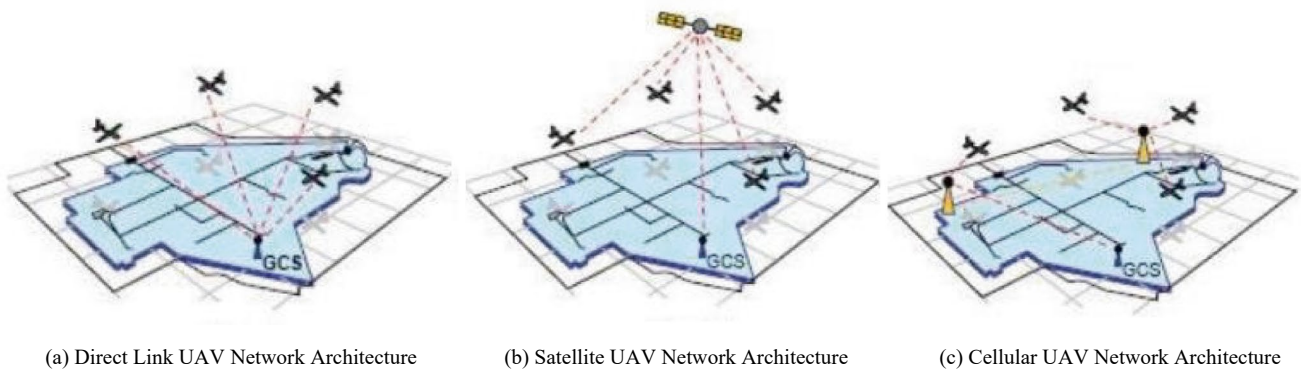


Fig. 3 Centralized Communication Architecture [adapted from Gupta et al. (2015)]

delayed communication, and limited satellite bandwidth, even if it covers a larger area.

- (iii) **Cellular Network Architecture:** The UAV network is distributed over cells, each of which is serviced by a base station, thanks to cellular network architecture as shown in Fig. 3c, which makes it possible to expand the network by adding more base stations and efficiently reuse bandwidth. This configuration offers dependable data delivery and strong connectivity, but it comes with a high cost for base station equipment, and it can have issues with frequency hand-offs when UAVs switch between cells.

- (b) **Decentralized UAV Communication Architectures/ Mobile Ad-hoc Networks:** A central node is not necessary in decentralized communication topologies. It is possible for two UAVs to converse without sending data back to the base station. This category includes three architectures: multi-group UAV network, multilayer UAV Ad-hoc network, and UAV ad-hoc network architecture, also referred to as Mobile Ad-hoc Network (MANET) or Mesh.

- (i) **Single-Group Swarm Ad-hoc Network/Mobile Ad-hoc Network (MANET) Architecture or Mesh:** In a UAV ad-hoc network, no pre-existing infrastructure is required for UAVs to communicate. Each UAV operates in a distributed peer-to-peer mode, participating in data forwarding for other UAVs within its transmission range using broadcasting, uni-casting, or multi-casting. When a UAV needs to communicate beyond its range, it uses intermediate nodes to relay messages, a process known as multi-hopping. Typically, one UAV equipped with two radios acts as a gateway to the ground station. This setup is feasible for micro-UAVs due to the use of low-cost, short-range, and lightweight radio frequency transceivers. However, these transceivers have limited transmission distances, so optimizing connectivity requires proper wireless standards and routing protocols, along with control algorithms to synchronize UAV mobility patterns. Ad-hoc network architecture (Fig. 4) is particularly effective for networking homogeneous, highly mobile UAVs in frequently changing network topologies.

An illustration of ring architecture is shown in Fig. 5a. UAVs with ring architecture use a bidirectional link to create a closed communication loop. The gateway node for the swarm might be any UAV. The target UAV can re-connect through the communication loop if a direct link between two nearby UAVs fails. The ring architecture is

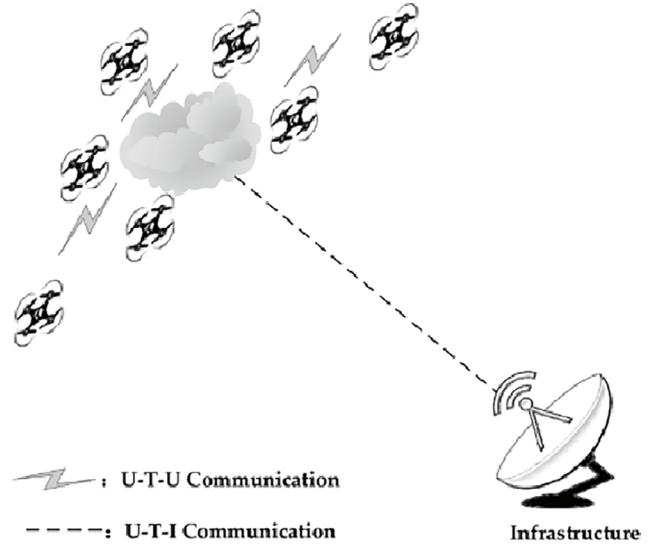


Fig. 4 Schematic representation of single-group swarm Ad-hoc Network [adapted from Chen et al. (2020)]

hence somewhat stable. Still, it is evident that this architecture is not scalable. The structure of communication is represented by a star architecture, as seen in Fig. 5b. Positioned at the center, the gateway UAV connects to the infrastructure and exchanges data with the complete UAV swarm. The inherent drawback of a SPOF in star construction is easily observed in Fig. 3b. In other words, the system as a whole will fail if the gateway node fails. The benefits of both ring and star topologies are combined in meshed architecture (Fig. 5c). It is observed that every UAV node in the swarm is equal with respect to both terminal nodes and routing functions. Any UAV can serve as a gateway node for the swarm, and the information stream between nodes can be implemented in a variety of ways. The industry standard for intra-swarm communication systems is presently mesh architecture.

- (ii) **Multi-Layer UAV Network Architecture:** The design of a multi-layer UAV ad hoc network consists of many UAV ad hoc networks, with the underlying assumption being that various UAVs are clustered together based on similarities in size or kind. The backbone, a communication route, connects all of these ad hoc networks. In Multi-Layer UAV Ad-hoc Network architecture (Fig. 6), just one backbone UAV is directly connected to the ground station, just like in UAV ad-hoc network architecture. Multi-group UAV Network architecture (Fig. 7) is an additional architecture that may be used to a network made up of many UAVs that are unable to establish a single Ad-hoc network.

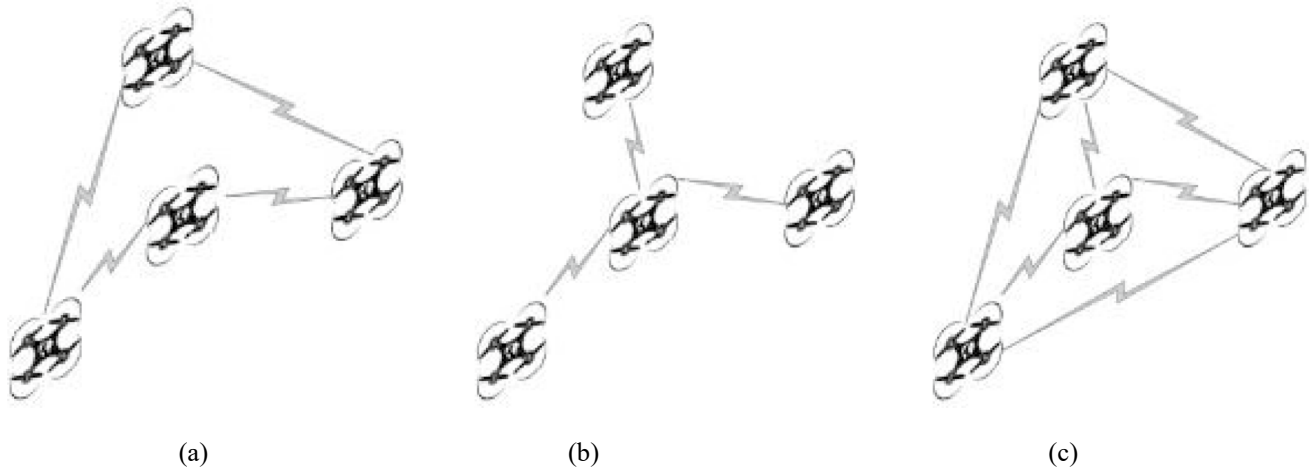


Fig. 5 Intra-swarm communication architecture: **a** ring architecture, **b** star architecture, **c** meshed architecture [adapted from Chen et al. (2020)]

(iii) **Multi-Group UAV Network Architecture:** Multi-Layer UAV Ad-hoc Network and Multi-group UAV Network design are comparable. Multiple UAV ad hoc networks comprise the architecture. Every ad-hoc network, in contrast to Multi-Layer UAV ad-hoc Network (Fig. 6), has a backbone UAV that connects to the ground station. Although the mobility patterns (speed and direction) of UAVs need to be controlled and kept within communication range of one another in order for Distributed UAV network architectures to be implemented, even though they are more robust and scalable than Centralized UAV network architectures (Chen et al. 2020b) (Table 2).

4.2 Wireless Communication Standards and Technology

Both wired and wireless connections can be used to establish communication between nodes in a network. Coaxial, fire, and twisted pair cables are among the physical connections used in the execution of a wired connection, commonly known as an Ethernet. Radio frequency technology is used in the air interface for wireless communication. When it comes to UAV networking, wireless connections are preferable over cable connections due to the mobility of UAVs. However, radio signals are susceptible to interference, low bandwidth, reduced security,

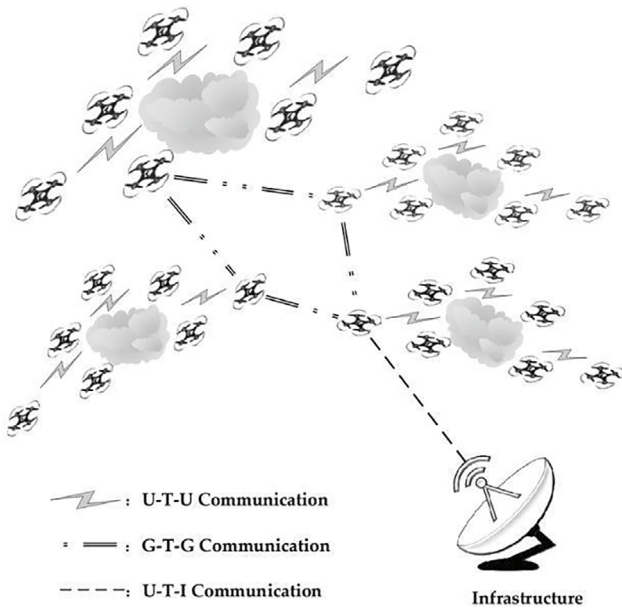


Fig. 6 Schematic of multi-layer swarm Ad hoc network [adapted from Chen et al. (2020)]

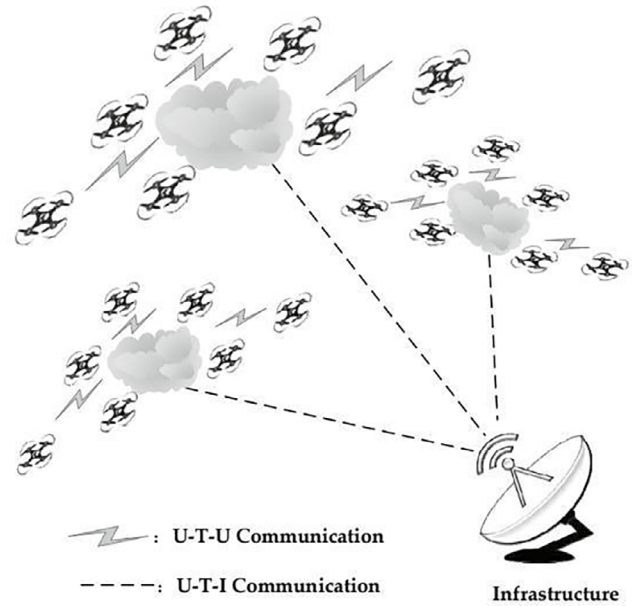


Fig. 7 Schematic of a multi-group swarm Ad hoc network [adapted from Chen et al. (2020)]

Table 2 Summary of UAV communication architecture [adapted from Chen et al. (2020)]

Features	Centralized communication architecture	Decentralized communication architecture		
		Single-group	Multi-group	Multi-layer
Multi-hop communication	×	√	√	√
UAVs relay traffic	×	√	√	√
Different types of UAVs	×	×	√	√
Self-configuration	×	√	×	√
Limited coverage	√	√	√	×
Single point of failure	√	×	√	×
Robustness	√	×	×	√

Note “√” = Supported, “×” = Not Supported

significant signal loss, and fading, which are limits of pure wireless connections or communication surfers. The foundation of their argument is the fact that mobile ad-hoc network architecture can establish a multi-hop wireless network without the need for pre-installed equipment and can establish connectivity outside of a line of sight (LoS).

Depending on their geographic coverage, mobile ad-hoc network communication can be categorized under Personal Area Networks (PAN), Local Area Networks (LAN), Metropolitan Area Networks (MAN), and Wide Area Networks (WAN).

With a data throughput of less than 10 Mbps, wireless personal area networks (PANs) offer a communication range of 1–10 m. PANs, or personal area networks, are wireless networks that use IEEE 802.15.4 to link electronic devices that are in close proximity to one another (e.g., connecting a personal computer to a printer, UAV to UAV communication). In order to minimize interference and reuse of bandwidth, the majority of radios with PAN are deployed in the 2.4 GHz ISM band. Wireless Local Area Networks are the type of networks that offer coverage that goes beyond PAN boundaries (WLAN). WLANs have a data transmission rate of up to 100Mbps and offer network coverage from 1 to 500 m. IEEE 802.11 and HiperLAN/2 technology standards are used in this network coverage, emphasize that problems with power consumption, mobility, and bandwidth constraints related to wireless environments should be taken into consideration while designing a WLAN. Furthermore, the same literature states that while designing or implementing a WLAN, ad hoc networking architecture should be taken into account.

To control the use and production of wireless communication devices, standards for communication were established. Hardware items for the Internet of Things (IoT) and machine-to-machine (M2M) communication can use any of the several wireless technology standards that are currently available.

4.2.1 IEEE 802.15.1: Bluetooth and Blue Low Energy (BLE)

For short-range data transmission, body area networks and personal area networks use the IEEE 802.15.1 family of protocols. Bluetooth is a low-end, ad-hoc, terrestrial wireless protocol that can be used for communication over short distances (100, 10 and 1 m). Bluetooth does not require devices to be in line of sight with one another because it operates on a broadcast communication model. Bluetooth transmissions are omnidirectional and wall-piercing, in contrast to infrared. Bluetooth's communication range is limited, though. Bluetooth transmissions are omnidirectional and wall-piercing, in contrast to infrared. Bluetooth's communication range is limited, though. A case study utilizing Bluetooth as a communication medium in a UAV research project can be located in Hyland et al. (2007).

4.2.2 IEEE 802.15.4: ZigBee

ZigBee is an open global mesh networking technology that allows hubs to be connected to one another via a variety of connections. ZigBee is an excellent solution for certain mechanical applications because it is inexpensive to implement. In addition to the previously indicated benefit, 128-piece AES encryption is used by the ZigBee standard. In the University of Pennsylvania's "Towards A Swarm of Agile Micro Quadrotors" research project (Kushleyev et al. 2013), Zigbee wireless technology was utilized to establish connections between UAVs and between UAVs and workstations.

4.2.3 IEEE 802.11: WiFi

The WiFi standards, 802.11b, 802.11 g, and 802.11n, are used to connect internet switches to devices such as PCs, tablets, and phones. The 2.4 GHz ISM, 2.4 UHF, and 5 GHz SHF ISM radio frequency bands are compatible with this set of standards.

4.2.4 IEEE 802.16: WiMax

Worldwide Interoperability for Microwave Access is referred to as WiMax. This wireless breakthrough allows

data to be transferred between indoor and outdoor areas at 30–40 megabits per second. A few cell carriers, most notably Sprint, used the standard in the past to provide their customers with distant information. Later, Sprint and many other carriers that used the standard switched to using faster LTE 4G network technology for data exchange.

5 Machine Learning Techniques in Swarm Drones

UAV swarms leverage principles from nature, particularly the behaviors of social insects like bees and ants. Machine learning (ML) techniques have become integral to enhancing the capabilities of swarm drones, enabling them to adapt, learn, and optimize their operations in dynamic environments. This section explores various ML techniques that empower these collaborative flying machines to navigate their surroundings, make intelligent decisions, and accomplish complex tasks (Fig. 8).

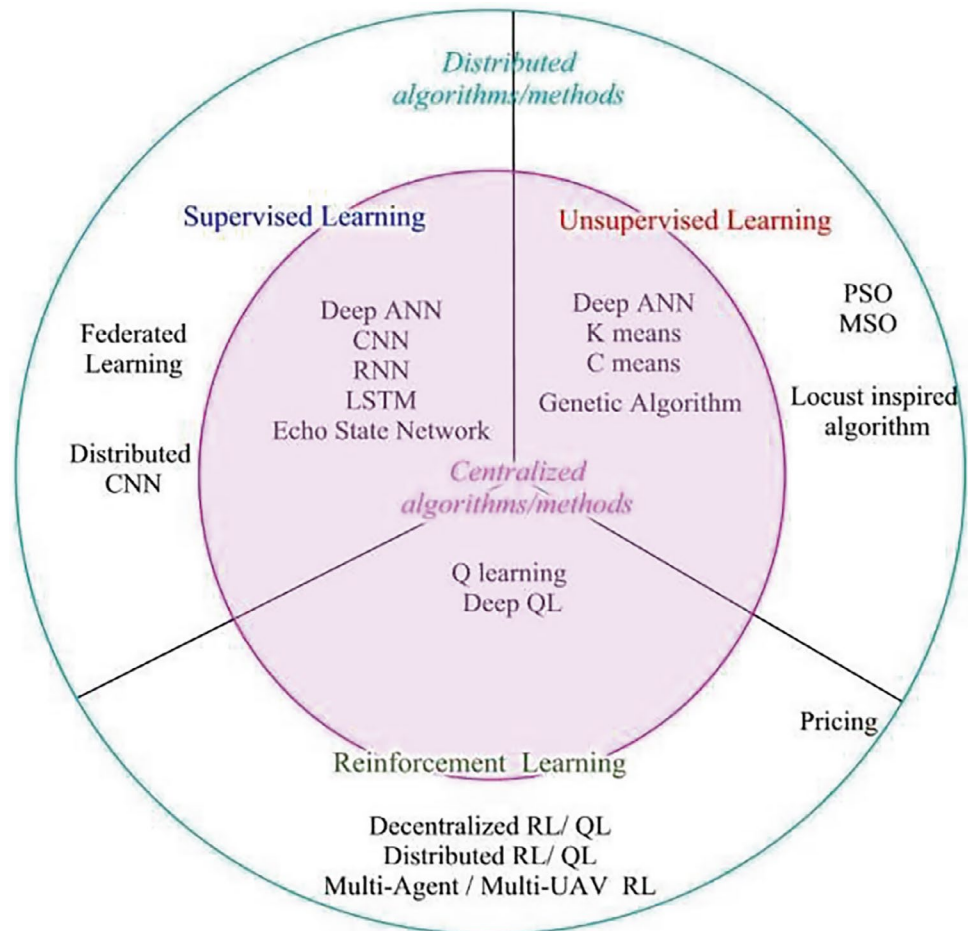
UAV technology has advanced significantly, leading to applications where flocks or swarms of UAVS can work

together to accomplish a global goal. This might be the case with UAVs that serve as cellular base stations (BSs), which should determine among themselves the best way to deploy in order to give ground users adequate coverage of a region. Additional examples cover a variety of fields in which unmanned aerial vehicles (UAVs) can be employed, including agriculture, transportation, security, and photography. All of these applications have one thing in common: they created a number of dilemmas that required decision-making, including UAV deployment, work distribution, and maintenance concerns (Azoulay et al. 2021).

(a) **Reinforcement Learning (RL):** It is a type of ML where agents learn by interacting with the environment and receiving rewards or penalties. In the context of swarm drones, RL helps in optimizing paths, improving coordination, and enhancing decision-making processes (Rahbari et al. 2021).

- **Q-Learning:** It is a model-free RL algorithm that helps drones in a swarm learn the best actions to take under different circumstances. Q-learning involves a Q-table where each state-action pair is

Fig. 8 A scheme with the details of several ML methods [adapted from Azoulay et al. (2021)]



stored and updated based on the rewards received. For example, drones can learn to avoid obstacles or optimize flight paths using Q-learning. Research has shown that Q-learning can be instrumental in multi-robot formation and path planning, significantly reducing convergence time and the number of convergence rounds compared to traditional algorithms (Fu and Kim 2023).

- **Deep Q-Networks (DQN):** It is developed by using Q-learning with deep neural networks for scaling up to large state spaces. Drones have been trained under more complex environments (for example, under surveillance and search-and-rescue), using DQN, leading to more efficient completion of tasks. The use of DQN has demonstrated exciting potential in several non-gaming applications like packet scheduling in IoT networks (Bonyadi and Michalewicz 2017).
- (b) **Swarm Intelligence Algorithms:** Swarm intelligence algorithms work by modeling the collective behavior of social insects. After all, we need these algorithms to be able to encode this type of decentralized control in drone swarms.
 - **PSO (Particle Swarm Optimization)** is a widely used algorithm for optimizing the trajectories of drones in a swarm in scenarios like area coverage and target tracking. PSO represents the social movement of birds flocking, fish schooling, particle odors where a cloud of swarm drones operate where particles form a flock freely following a leader as though a leader moves each particle individually in the cloud (Kalantari et al. 2016). The application of PSO to different drone-related scenarios includes optimizing the three-dimensional placement of drone base stations in wireless cellular networks (Zhao et al. 2022) and coverage optimization of wireless sensor networks using hybrid PSO and chaos optimization (Yang et al. 2017).
 - **Ant Colony Optimization (ACO)** is one of the nature inspired optimization methods which is simulated to solve various optimization problems based on the decision making process of the real ants. By using simulation of pheromone trails, drones in ACO can communicate the paths to each other, making it possible to discover optimal paths in complex environments. ACO algorithms are well known for their capabilities at preserving diversity in the elite sets, this is one of the reasons why they are deemed as excellent techniques for solving multi-modal optimization problems (Xue et al. 2019).

Since the performance of ACO can be improved using other methods for different applications, researchers extended

ACO algorithms. The fusion of ACO with the A* search algorithm, for example, has produced the the ACO-A* algorithm (Wang et al. 2020) and given promising results for 3D path planning in cluttered environments (Chun-jiang et al. 2020).

In the realm of path planning amidst traffic congestion, the ACO algorithm, in conjunction with Particle Swarm Optimization (PSO), is acknowledged for its rapid computational speed and optimization capabilities, rendering it a favored option for dynamic path planning (Chun-jiang et al. 2020).

(c) **Supervised and Unsupervised Learning**

Supervised and unsupervised learning techniques are employed for various tasks such as object detection, clustering, and pattern recognition.

In drone technology, **Supervised Learning** is used when you train your drones through labeled datasets to enable them to see and then respond to objects or obstacles. For instance, Convolutional Neural Networks (CNNs) is a technique that is widely used in image recognition and object detection (Luesutthiviboon 2023).

Recent work has demonstrated the ability of CNNs to excel at object recognition based on the appearance of the target in images, even in challenging, real-world environments where obstacles are present and in motion (Svanström et al. 2021). These networks are trained on datasets which contain images of drones and other objects like birds etc. to differentiate between different types of flying objects (Ajakwe et al. 2022). The application of CNNs enables real time and precise detection of drones in videos with the requirement of a static background improving surveillance efficiency. To this aim, other researchers have proposed different models and frameworks for drones such as the DRONET which is presented in an industrial facility surveillance approach in real-time (Radford 2015).

Unsupervised learning obstructs clustering and anomaly detection tasks. K-means clustering and Principal Component Analysis (PCA) are popularly used algorithms in unsupervised learning. Such approaches allow for systems, e.g., drones, to recognize patterns, cluster groups of objects of the same kind, and infer anomalies such as flight patterns (Caron et al. 2018). Nonetheless, clustering is a core idea in unsupervised learning through which data points are grouped according to their similarities. So far, it has been widely studied and employed in computer vision and image processing (Ji et al. 2019). Recent breakthroughs in unsupervised learning have also given rise to models that can classify and segment images with high performance in the absence of labeled data. For instance,

the Invariant Information Clustering model achieves the state-of-the-art in an unsupervised benchmark for clustering performance due to image classification and segmentation (Farrag et al. 2022).

(d) **Evolutionary Algorithms**

The process of optimization has its roots in Evolutionary algorithms, which are modeling the essence of natural selection, and they are optimizing problem-solving. For swarm drones, the optimization of the parameters such as flight path, energy consumption, and communication protocol, etc., are achieved by using **Genetic Algorithms (GA)**. These algorithms function by generations in which the best-performing solutions are chosen and combined to form the new solutions (Dorling et al. 2017). Specifically, Available Vehicle Routing Problems (VRPs) designed for drone delivery showed poor performance in finding optimal routes by ignoring battery state and payload weight issues, which resulted in suboptimal or even unfeasible routes for drones (Chen 2023). In response to these problems, with adaptive cross operator, researchers have proposed to improve the solving ability of GAs over conventional approaches (Hou et al. 2020). Besides, GAs have been coupled with other optimization methods as Particle Swarm Optimization (PSO) to create hybrid approaches for reducing energy consumption and delays in industrial Internet of Things (IIoT) applications (Asaamoning et al. 2021).

One of the most powerful techniques for achieving this complex of a behavior in drone swarms is **neuroevolution**, where neural network architectures are evolved using genetic algorithms. Drones swarms where any simple agent or drone can be networked together to send or receive messages with the system (Loquercio et al. 2018) Learning allowed the drones to optimize their neural network structures and navigate through more difficult paths, a concept shown through Neuroevolution methods that improve the drone performance over time (Liu et al. Aug. 2019).

The combination of neuroevolution, Convolutional Neural Networks, and Genetic algorithms has done the trick and the drones are benefiting from this hat trick in a myriad of situations from surveillance to object detection and classification. Artificial Intelligence (AI)-assisted machine learning and deep learning techniques are required to manage drone-based systems as reported in Pei et al. (2022); Genc et al. 2017, Yang et al. 2018). The learning-based approach in drone-based systems, employing online training, can exploit cloud/edge resources to learn on-the-fly, e.g., energy usage and network latency profiles to accommodate application requirements, such as computation-heavy object detection and tracking, in a system-wide manner.

6 Optimization Techniques in UAV Swarm Operations

The optimization algorithms are designed to off-load some of the missions and coordination tasks from each unit in the formation, so that each UAV/HUAV can contribute to the collective mission in an optimal way without overburdening any single unit. Dynamic load balancing and real-time scheduling methods are used to distribute computational tasks on the swarm with respect to the current workload and available resources of each drone. Priority of each mission and the capability required by them are used to prioritize the tasks by the resource allocation engine so that the no tasks get access to the spare vCPUs once the vCPU has provided the demand CPU time to complete it. Task scheduling algorithms decide how to schedule tasks for execution (e.g. using EDF or round-robin) so as to minimize latency and maximize throughput. More advanced techniques employ heuristic as well as machine learning approaches for workload distribution, predicting and adapting the distribution of tasks in a dynamic fashion, in a manner that the resource utilization rate is generally maximized. Such communication and coordination among swarm UAVs is facilitated by NECTAR (Networked Control Architecture) which complements in providing a unified architecture to carry out effective resource sharing and cooperative task execution, significantly increasing the operational effectiveness and robustness of the swarm in different applications.

Some of the key challenges faced during optimization of swarm operations are as follows:

- **Swarm Membership Management:** Secure communication between drones and cloud services is a must. In this, it provides a protocol for managing membership in the swarm by using several keys, so that the communication will be secure, and the attackers cannot eavesdrop on the critical information.
- **Fault Detection and Isolation:** Accurate fault detection and doing it in real-time requires extracting information from classified objects that could be helpful.
- **Drone Swarming:** The swarming of drones are planned into a global path plan to identify the near-optimal route the drones must be taking.
- **Obstacle Detection:** This is necessary to make sure your autonomous drone is capable of following the pre-planned path which should be free of collision, as this is what ensures the drone meets its mission without any failure or task interruption. Obstacle detection helps drones to be more aware of the environment to execute their flight safe and effective manners in a variety of locations.
- **Fully autonomous:** Essentially fully human-free systems, capable of making decisions based on great AI algorithms and a high standard of decisioning, allowing for instantaneous if not instantaneous decision-making

(no human in the loop). High degree techniques are required in controlling the UAVs to be independent. This is because they are in constant contact with central control units that limit their independent activity.

- **Scalability and flexibility:** effective integration of drone and ground control hardware, and software essential for scalable and adaptive future, (Javaid et al. 2023).
- **Communication Architecture:** It's critical to set up effective networks for UAV-to-UAV communication. High mobility, sporadic connectivity, and constrained transmission range are among the difficulties.
- **Resource Management:** The power, size, and weight limitations that UAVs encounter limit their capacity to carry out cooperative tasks. To solve these problems, effective resource management and data transmission are crucial.
- **Channel Characterization:** Propagation, shadowing, and fading effects must be overcome by UAV communication networks. Reliable communication requires accurate channel characterization.
- **Integration of Emerging Technologies:** Although their deployment poses difficulties, the integration of technologies such as 5G and 6G is required to improve UAV communication networks in order to increase speed, reliability, and coverage (Zafar and Khan 2016).

7 Routing Protocols and Swarm Intelligence

Swarm routing protocol is a research focus in the field of UAV swarm communication. However, the routing in UAV swarm communication networks is a new and important challenge due to the high mobility of UAVs, air links, limited resources, and QoS fluctuations (Chen et al. 2020b). Similarly the traditional Ad Hoc routing protocols cannot be applied for communication among UAV swarms without any advancement, hence it is still a challenge for researchers to construct routing protocols with respect to most of the mission scenarios and features of UAVs.

Route technology helps to configure routing protocols. More often, those described as routing protocols are actually various routing technologies combined or refined from these basic routing technologies. UAV swarm Ad hoc network based on standard Ad hoc network. Hence, after the analysis, improvement and fusion, it is feasible to use the traditional routing method in the UAV swarm Ad hoc network. Figure 9 presents the six major routing schemes used by UAV ad hoc networks. Store-carry-forward, greedy forward, path find, single-path, multi-path, and predictive routing are some of the static routing techniques.

- (1) Store-carry-forward technology: the current node stores and forwards the datagram till it locates the forwarding

node if at some point no relay node can be located. This method has the drawback of creating a significant delay, although it is appropriate for intermittent networks (Fig. 9a).

- (2) Greedy forward technology: The idea behind greedy forward technology is to choose the neighbor node that is closest to the destination to serve as the relay node until the datagram is transferred there (Fig. 9b). It can be applied in situations when there is a high level of UAV deployment. On the other hand, a failure will occur if a node is the closest to the target but there is no path to its neighboring node. In this instance, combining it with additional technologies will make the technology more dependable.
- (3) Path discover technology: The main method is by flooding the path with routing request (RREQ) responses, which maximizes accessibility. By using this method, the chance of communication breakdowns occurring when the present node loses its intended geographic location is decreased (Fig. 9c). However, it has the drawback of using up a lot of internet resources.
- (4) Single-path technology: defined by the data transmission process using a single path (Fig. 9d). Ideal for incredibly constrained bandwidth systems. However, its lack of robustness is a drawback. It is simple to lose packets because there isn't a backup route in case of network breakdown. Seldom is this method applied in settings involving UAV swarm communication.
- (5) Multi-path technology: The robustness of the link can be efficiently increased with multi-path propagation technology (Fig. 9e). Other links may take over if one fails. It works well in situations when a high link reliability is necessary. The network will contain a loop that will stop it when the multi-path intersections fail, which is a drawback.
- (6) Predictive routing technology: By using a node's current position, velocity, and direction as a guide, predictive routing technology selects the best hop node to follow (Fig. 9f). This technology is extensively utilized in a UAV swarm Ad hoc network because it may be applied to situations where nodes' positions vary quickly.

7.1 Classification of Routing Protocols

The benefits and drawbacks of various routing protocols (Fig. 10) and application scenarios are then covered.

1. **Topology-Based Routing Protocols:** Internet Protocol (IP) addresses are used by topology-based routing protocols to identify nodes, and they utilize connection information already in place to route packets through the

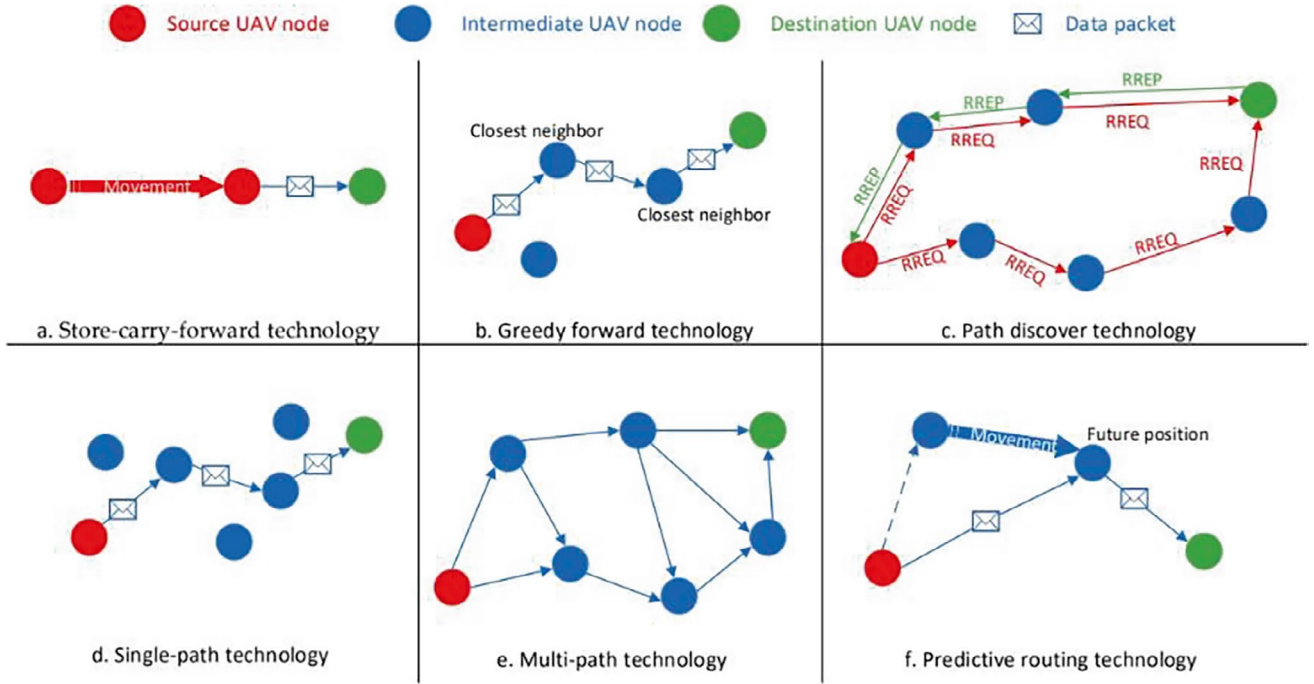


Fig. 9 The rationales for common routing technologies of UAV ad hoc network. **a** Store-carry-forward technology, **b** Greedy forward technology, **c** Path discover technology, **d** Single-path technology,

e Multi-path technology, **f** Predictive routing technology [adapted from Turnadereli & Bekmezci (2013)]

proper channels. Routing protocols based on topology can be classified into four categories: proactive, reactive, hybrid, and static routing protocols.

(a) **Static Routing Protocols:** The routing tables used by static routing protocols are fixed and cannot be changed. They are mostly applied in scenarios when there are no task changes and a stable topology. Traditional static routing protocols are not as flexible or resilient to changing conditions, which limits their applicability to UAV swarm systems. Three static routing protocols are as follows:

- (i) **Load Carry and Deliver Routing (LCAD):** Data packets are sent from the infrastructure to a UAV, which transports and delivers them to their intended location. LCAD aims to increase security and maximize network throughput. In order to meet batch data transmission and delay tolerance criteria, it reduces the number of hops. Furthermore, one drawback of LCAD is that it causes a noticeable delay when the UAVs' communication distance grows.
- (ii) **Data Centric Routing (DCR):** Multi-UAV systems require one-to-many data transmission for real-world applications. A "single-group swarm Ad hoc network," in which the gateway UAV is in charge of dispersing information to other nodes, is a good fit for DCR. The

"publish-subscribe" architecture is a common application that facilitates automatic connections between data publishers and subscribers while simultaneously gathering traffic. DCR is the preferred method when the system consists of a small number of UAVs with a predetermined flight route since it requires less support from other UAVs.

- (iii) **Multilevel Hierarchical Routing (MLHR):** To address the scalability issue of large-scale vehicle networks, it is moved from the mobile vehicle network. Depending on their size and purpose, UAV swarms can be grouped together, as the "multi-layer swarm ad hoc network" explains. That's why hierarchical structures are used in communication networks. For G-T-G communication, only gateway UAVs are accountable. By broadcasting data, the gateway UAV communicates with other UAV nodes in the group. This is seen visually in Fig. 11. To choose gateway UAVs based on geographic data, a clustering technique has been suggested in the present literature. Both the scalability issue and the dynamic network's stability are greatly enhanced by this routing paradigm.
- (iv) **Proactive Routing Protocols (PRPs):** PRPs regularly update routing tables to maintain an

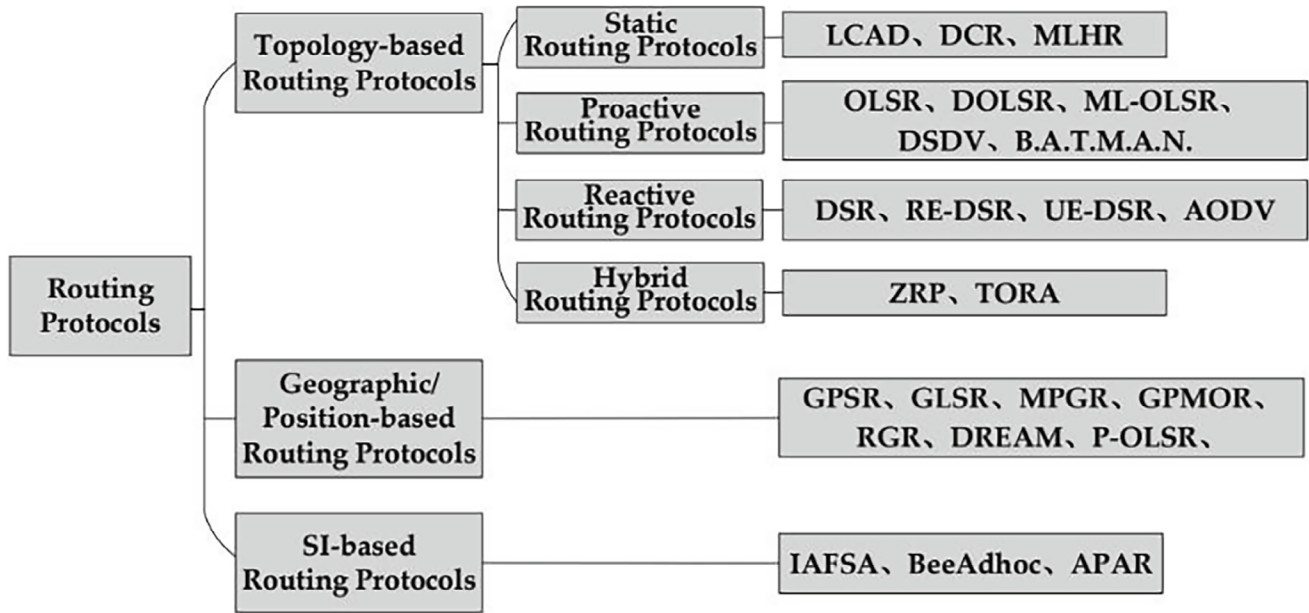


Fig. 10 Classification of routing protocols [adapted from Chen et al. (2020)]

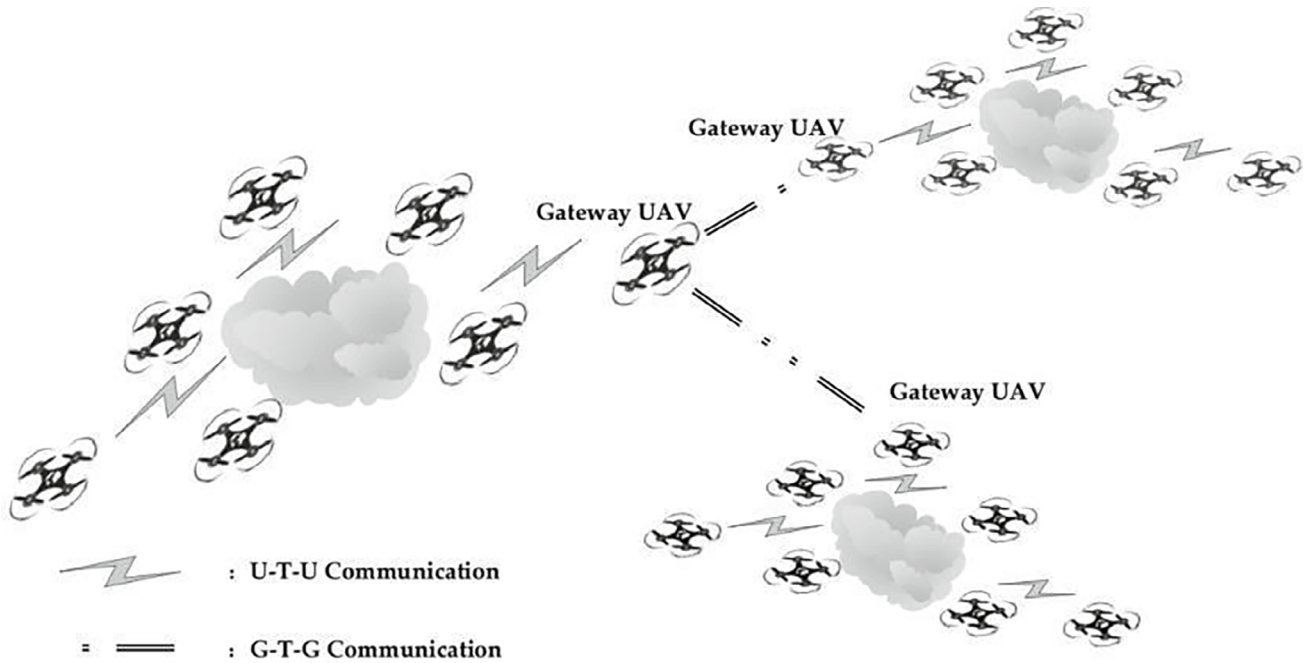


Fig. 11 Multilevel hierarchical routing in a UAV swarm Ad hoc network [adapted from Chen et al. (2020)]

accurate network topology and minimal route discovery delay, making them suitable for real-time applications. These protocols generate continuous message exchanges, leading to inefficiencies in highly dynamic networks like UAV swarms. Common PRPs include Optimized Link State Routing (OLSR) and Destination Sequenced Distance Vector (DSDV). OLSR

uses multiple point relays (MPR) to optimize overhead and is widely used despite challenges in route recalculations (Khan et al. 2023a).

- (v) DSDV involves frequent routing table updates, causing high bandwidth consumption in dynamic networks.
- (b) **Reactive Routing Protocols (RRPs):** RRP, or on-demand routing protocols, initiate route

discovery only when necessary, reducing control overhead. They suffer from high transmission delays due to the need for route discovery before data transmission. Notable RRP methods include Dynamic Source Routing (DSR) and Ad hoc On-Demand Distance Vector (AODV).

- (i) DSR is adaptable to self-organizing networks and suitable for UAV ad hoc networks due to its low infrastructure requirements.
- (ii) AODV combines elements of DSDV and DSR, minimizing overhead and utilizing route requests and replies for path discovery.
- (c) **Hybrid Routing Protocols (HRPs):** HRPs combine PRP and RRP methods, using proactive strategies within zones and reactive strategies between zones. Examples include Zone Routing Protocol (ZRP) and Temporarily Ordered Routing Algorithm (TORA).
 - (i) ZRP reduces routing overhead and transmission delays by using adaptive zone-based strategies.
 - (ii) TORA minimizes sensitivity to topology changes, quickly discovering new routes during link outages.
- 2. **Geographic/Position-Based Routing Protocols:** These protocols use nodes' geographical positions to maintain routing tables, reducing overhead in highly dynamic networks. Protocols like Greedy Perimeter Stateless Routing (GPSR) and its variants (e.g., GLSR, MPGR) improve routing reliability by selecting optimal paths based on location data.
- 3. **Swarm Intelligence-Based Routing Protocols:** Inspired by the behavior of natural swarms, these protocols use multi-agent systems for optimized collaborative tasks. They address communication range expansion and information security challenges in large UAV swarms, exemplified by the Improved Artificial Fish-Swarm Algorithm (IAFSA).

8 Non-terrestrial Networks and the 6G Era

The key to the evolution of 6G wireless communication technologies lies in integrating Non-Terrestrial Networks (NTNs), namely satellite systems, and High Altitude Platform Stations (HAPS) and Unmanned Aerial Vehicles (UAVs). Swarm UAV: Groups of coordinated, disciplined, and dedicated UAVs can effectively improve network resilience, performance, and coverage among others. Capacity expansion with swarm UAVs to address the connectivity problems, with the guarantee of ultra-high throughput and low latency, especially in presence of varying network density, bandwidth and traffic patterns (Mozaffari et al. 2016).

Swarm UAVs provide flexible relays to set up aerial base stations, particularly on occasion of insufficient or no terrestrial infrastructures, e.g., remote regions or emergency areas. The mesh network in sky is connected by swapping the position of UAVs deployed as aerial base stations, resulting in extended network coverage and improved capacity, since the load is burdened among multiple UAVs as shown in Lyu et al. (2017). Applications: UAV mounted mobile base stations, which can be strategically deployed located relative to ground terminals, leverage the mobility of UAVs. More flexible version than the terrestrial recovery as UAVs are easier to move around and cover the UAV system, while covering ground terminals.

Unmanned Aerial Vehicle (UAV) swarms have a unique advantage over traditional terrestrial base stations as they can move with freedom so as to cover areas on-demand and as per requirement. They can automatically calculate, analyze, and etc., to demonstrate their intelligent nature (Wang et al. 2022). As UAV swarms with a comprehensive sensor array cost-effectively enable high-precision positioning and are extremely adaptive, this creates an aerodynamic, low-cost and adaptive work structure that can lead to moment optimization and target tracking (Mozaffari et al. 2017).

In this way the employment of UAVs as mobile base stations is not restricted to network coverage; rather it can be widely used in IoT networks for low power communication (Xiao and Fu 2023). Also, UAV swarms can be useful in cooperative detection games which improve the security by mutual intruder detection and defense coordination, as suggested in Jung (2020). Millimeter-wave UAV swarm networks can establish non-line-of-sight links as well to provide better connectivity for users, especially in areas where physical obstacles are limiting connectivity (e.g., disaster-stricken scenarios (Shao-Horn et al. 2023)). Also control optimization methods can be applied to make the control energy-efficient in cellular-connected UAV swarm networks to improve the performance of the entire network (Peng et al. 2021). The use of UAV swarms for communications, especially in disaster recovery operations should be properly scheduled involving efficient task allocation methods for effective mission planning and execution (Gao et al. 2020). They can be used to optimize UAV swarms for different applications (wireless sensor networks, 5G mobile communications like massive MIMO) by using dynamic UAV positioning algorithms and resource allocation techniques (Saraswat et al. 2022).

6G is targeting 1 ms latency and 1 Tbps peak data rate to enable real-time services, such as autonomous driving and industrial automation. Swarm UAVs have unique advantages in promoting these goals because they establish a direct line-of-sight communication link, thereby avoiding signal blockages or interferences usually found in terrestrial networks (Khan et al. 2023b). With 6G technology with AI/

ML and IoT, blockchain integration is expected to open up ultra-reliable, intelligent, secure, and ubiquitous UAV networks, with support for near real-time communication even in ultra-dense connectivity configurations (Noumikos et al. 2023). 6G may also become part of new network architecture paradigms such as those that involve mobile nodes like UAVs—to move resources dynamically or enhance network resiliency, for example (Gupta et al. 2016). These tight requirements can be satisfied by swarm UAVs in 6G networks, representing the availability of the next generation of wireless communication systems. As an example, 6G technology, supported by direct, line-of-sight, communication links established between ground units and UAVs are about to change the infrastructure of various industries, because it will allow for real-time applications to work faster and more reliably than ever.

The deployment of swarm UAVs splits the network load and this allows NTNs (Network Tactical Networks) to be more robust and reliable, even in the case of failures through dynamic reconfiguration (Phadke 2024). In case of a failure of any single UAV, the rest of the UAVs in the swarm can change their positions and roles so that the network behavior will degenerate a little and the performance of the network will be only slightly affected. The ability to adapt and reconfigure in real-time is essential to keep a network running under these conditions (Phadke 2023).

UAV swarm cooperative networks have also been addressed with studies on improving performance with distributed computing strategies. The evolution of distributed computing techniques provide potential solutions for real-time coordination problems in large-scale UAV swarm networks (Zhang 2023). In deployment optimization, evolutionary game analysis was introduced to improve the adaptability and autonomy of UAV swarm deployment in communication-constrained scenarios. The second one is the use of a congestion game to simulate the interactions among UAV swarms and to steer the flying movements of the UAVs in order to enhance data collection and anti-interference performance (Umoga 2024).

In addition, with the help of AI and machine learning algorithms, swarm UAVs can improve the network resource management by making an optimal spectrum utilization and energy efficiency. By harnessing AI, these UAVs can forecast network demand trends and dynamically configure themselves, optimizing resource allocation in order to maintain an efficient and sustainable network performance (Zhang et al. 2023). This is the technique where AI helps intelligent routing of cognitive UAV swarms for emergency communication in the field of network communication. The work clearly demonstrated that a UAV swarm within a hierarchical aerial-ground network architecture combined with cognitive radio can be used to automatically deliver broadband wireless service to users connected into the network

during emergency scenarios, i.e., AI-driven intelligent routing possible to optimize the communication efficiency (Jacobsen et al. 2023).

9 Applications of Drone Networks

Drones, which are a crucial element of UAS, are mostly used in industries such as FedEx package delivery, agricultural management, national defense, border patrol, real estate, surveillance, infrastructure inspections, aerial imaging, search & rescue, disaster management, law enforcement and scientific research. These are essential for tasks like infrastructure inspections, post-disaster transportation, and as countermeasures against bad drones. With recent, successful applications such as autonomous blimp robots, multirotor drones for inspection, and hybrid solutions for infrastructure evaluations demonstrating their versatility across industry, the opportunities to leverage these airframes grow daily. The drone swarms provide superior efficiency compared to classical methods, making them an effective solution for infrastructure inspections.

9.1 Energy Industry Infrastructure Inspection

The energy industry is increasingly reliant on drones for inspecting energy infrastructure such as wind turbines, pipelines, oil storage tanks etc. Traditional inspection methods often require operations to stop and place inspectors in harm's way. Carried out while the infrastructure is still in operation, these checks can cause congestion and be dangerous, and drones now make them possible without these cost and risk burdens, improving asset utilization and health. Drones equipped with thermal sensors and high-resolution cameras can catch issues such as leaks or structural issues early, preventing more expensive repairs and downtime.

Rune Hylsberg Jacobsen et al. in Guven and Parlak (2022), proposed a system to perform safe inspections of critical infrastructure, such as bridges, power lines, and railroads by autonomous cooperative drones. It offers formation flying, obstacle avoidance, and AI-based fault detection. With automated fault identification, secured communication, coordinated swarming, autonomous navigation, and real-time telemetry monitoring, inspections are both secure and effective. The architecture of its system integrates numerous tiers of safety that allow the safe operation of the system. It starts with pre-planned missions, target identification, 3D reconstruction and the inspection plan process. Camera drones with high-definition visual cameras and onboard fault detection algorithms are used for capturing the damage of bridge structural elements and

railway systems. These include fault identification algorithms, motion path planning, obstacle avoidance, cable detection, and formation flying for an unmanned aerial system (UAS). It ultimately comes down to the integration of system components in the operations of a UAS, serving as the foundation to cloud services for validation with respect to infrastructure on ground. Validation: Drone testing (ex: powerline inspections, ROS communication, etc.)

But using drone swarms to inspect infrastructure brings additional hurdles—including how to coordinate several drones at once, and how data transfer was going to be secure. Some of the critical challenges faced are autonomous decision-making, cybersecurity, defect detection, obstacle avoidance, and scalability. The same applies to keeping costs down, maintaining and certifying the UAVs, along with integrating swarms over the top of existing infrastructure. Solving these problems requires more thought, more sophisticated algorithms, tighter security, and maybe a dose of extra regulation. This is the negative effect of drone swarms such as the increased expense and time associated with inspections, declines in inspection types, and a drop in safety. But the use of drone swarms comes with its advantages as well, including the reduction in inspection costs and time, inspection quality improvements, and safety enhancements. Because of this drones can be more accurate in covering the inspection target area even if it means a more difficult to reach location. The swarm can now examine several parts at once, which speeds up the process. Image fidelity for the inspection process is improved when drones can hover and capture high-detail images. This enables drone swarms to perform autonomous inspection of critical infrastructure without putting human lives at risk of accidents and injuries which is both cost effective and efficient as well as safe. As drone swarms are evermore deployed they need to continue to innovate in planning, algorithms, cybersecurity, and regulatory compliance to be successful.

9.2 Precision Agriculture

Precision farming, sometimes referred to as precision agriculture, is a contemporary farming technique that maximizes crop yield by managing it with the use of cutting-edge technology. Drones are transforming precision farming in agriculture by carrying out operations including mapping, spraying, and crop monitoring. Drones with multispectral sensors are able to assess crop health by identifying changes in plant reflectance that are imperceptible to the human eye. This optimizes the application of fertilizers and pesticides by enabling farmers to pinpoint stressed crop sections and treat them specifically. Drones, for instance, may precisely apply chemical sprays to fields, minimizing

waste and their negative effects on the environment while boosting crop yields. There are resolving issues such as energy consumption, confidentiality, authenticity, availability, and route de-confliction in swarm UAV networks for precision agriculture applications. To address these challenges, the paper (Stodola et al. 2020) suggests Drones as a Service (DAAS), a conceptual approach and architecture that combines blockchain, AI, and IoT technologies. The study employs various techniques, including machine learning algorithms for image recognition in agriculture, consensus algorithms for blockchain data validation, optimization algorithms for energy-efficient route planning of swarm drones, and computer vision-assisted systems for data collection and analysis. Blockchain technology is also used to secure data storage and transactions, and AI algorithms are utilized for route optimization and decision-making processes.

However, the incorporation of many technologies into swarm drones for precision agriculture presents significant challenges, necessitating substantial research and testing. Despite this drawback, the article provides a comprehensive strategy aimed at modernizing precision agriculture methods. It leverages cutting-edge technologies to deploy swarm drones with enhanced capabilities, efficiency, and security measures, reflecting the growing trend of using advanced technology to improve farming practices and increase sustainability in agriculture. Future research directions include optimizing the energy consumption of swarm drones, improving route planning algorithms, and enhancing data security and authenticity in precision agriculture applications.

9.3 Military Reconnaissance

The UAV reconnaissance model (Oubbati et al. 2019) streamlines the planning process, which greatly helps military operations. UAVs can be used to monitor urban surroundings and provide military troops with complete, real-time data about terrain and potential threats. Rapid optimization is made possible by the effective metaheuristic method, which is essential for time-sensitive missions, has low memory requirements, fast convergence, and a technique to avoid local optima. It is based on the concepts of simulated annealing. The algorithm strikes a balance between the computationally demanding solution evaluation process and speed. The incorporation of this system into the decision support system of the Czech Army has proven to be useful in terms of cutting down on planning times and improving mission effectiveness. The model will become even more useful in contemporary military reconnaissance and surveillance operations with additional improvements and real-world testing.

Furthermore, the approach is demonstrated to be effective (shorter optimization times) ensuring critical missions do not get postponed. Moreover, being able to make rapid judgments in complex urban environments can be critical in dynamic and fast-paced scenarios, such as in counter-terrorism, urban warfare, or support to civil authorities. The model represents a significant advance in military reconnaissance technology, combining autonomous decision-making with increased UAV swarm coordination. It also provides a very potent capability for enhancing urban situational awareness and effectiveness.

9.4 Urban Transportation Systems

In Urban areas, swarm networks of Unmanned Aerial Vehicles (UAVs) significantly help city transportation systems to be faster in emergency response, gross automation and demonstrated connectivity as they appear as intelligent transportation systems. These networks go about solving the high-mobility and interference issues encountered in metropolitan environments. Unmanned Aerial Vehicles (UAVs) are used for traffic management, rescue by guiding traffic, and optimal vehicle paths, and eye in the sky for surveillance. To ensure reliable routing and data delivery, recent research has proposed hybrid models, where the UAV swarm network and vehicular ad hoc network are merged (Green et al. 2019; Liu et al. 2020). This makes it easy to build sustainable transportation solutions.

9.5 Integration with IoT Frameworks for Environmental Monitoring Systems

Swarm Network is adding strong environmental monitoring capabilities. Small UAVs, using a variety of sensors, can monitor large urban areas, tracking environmental parameters and providing real-time, high resolution images of the city. These systems are applicable in intelligent monitoring to help in e.g. coordinated firefighting efforts or industrial air quality management (Liu et al. 2020). Spinning in conjunction with each other, UAV swarms function with ground sensors to optimize data harvesting, chosen for minimizing latency while improving decision-making processes and ensuring comprehensive and effective environmental monitoring.

9.6 Smart Surveillance

UAV swarm networks are instrumental in urban surveillance to provide automatic, scalable, 24/7 monitoring for securing surveillance to ensure human safety and security. Drones

equipped with HD cameras can locate unauthorized persons and also suspicious behavior of things such as violence or theft, and then relay the information to police and emergency personnel. Using state of the art recognition and detection algorithms, these networks help in improving the surveillance quality (Ishtiaq et al. 2021). Recent advancements like artificial intelligence (AI)-powered optimization of surveillance operations and reinforcement learning-based control frameworks for persistent surveillance, on the other hand, ensure real-time, reliable, and cost-effective surveillance.

9.7 Resource Management Using Edge Computing

The deployment of UAV swarm networks in an urban environment generates massive amounts of data that require efficient resource management. Edge computing solves these issues by processing data at the edge of the network, reducing latency and increasing the quality of service (QoS) (Qadir et al. 2022). Smart computing machine learning on edge computing frameworks enable UAVs to streamline processing decisions and real-time video analytics. These built-in elements ensure optimal utilization of available resources, enhances peak time processing and provides seamless coverage across a variety of urban use-case scenarios (perhaps with internet service, surveillance and disaster relief).

9.8 Disaster Management and Relief

UAV swarm networks are critical for disaster management and relief, allowing rapid situation awareness, damage detection and relief coordination. They help to ensure rapid and efficient disaster response by enhancing data sharing and communication among emergency responders. Highly effective and sophisticated UAVs are able to identify survivors, evaluate disaster-stricken regions, and even bring supplies to where they are most needed. Collaborated Unmanned Aerial Vehicle (UAV) operations will increase the resilience of urban infrastructure, and improve overall safety and longevity of cities (Grau et al. 2021).

A variety of applications are critical to the military, commercial, and civilian sectors, including aerial photography for remote sensing surveillance, border patrolling, wildlife monitoring, homeland security, forest fire monitoring, and nuclear power plant perimeter surveillance. The high-resolution camera on UAVs provide a cooperative aerial imaging experience, its cameras have the capability to capture the large area from different angles as needed and will be divided into separate or collective for more thorough coverage. Despite the aforementioned benefits, UAVs at low altitudes provide detailed, cost- and

time-saving achievable photos rather than conventional aircraft through integrated multimodal sensors, advanced photogrammetry and computer algorithms (Mozaffari et al. 2017). The combination of these technologies allows unmanned aerial vehicles (UAVs) to acquire and process terrain data, resulting in digital surface and elevation models required for accurate floodplain modeling and overland flow predictions. In addition, there are improvements in providing valid metadata that is indispensable for data retrieval when UAVs incorporate multispectral cameras that increase vegetation mapping accuracy (Nwankwo Constance Obiuto 2024). These attributes enhance the utility of these instruments from an environmental monitoring as well as security standpoints by complementing satellite data in its ability to work in cloud cover and because it fuses all together, it enhances the overall spatial, temporal, and spectral resolutions.

10 Future Directions in Drone Networks

Swarm drone research and development will likely take many different forms in the future as a result of the convergence of technology breakthroughs. Integration of artificial intelligence (AI) to enable swarm intelligence, which facilitates numerous drones to cooperate and display collective behavior among multiple drones, is one important field of development. (Semwal et al. 2023). Swarm drone systems have been transformed by the incorporation of AI algorithms, allowing for more autonomous and effective operations. (Myers et al. 2024).

The communication and coordination abilities of drone swarms have also been greatly enhanced by developments in wireless networking, such as multi-beam beamforming-based machine learning algorithms (Myers et al. 2024). Drone swarm operations are becoming increasingly popular in military operations for surveillance, target engagement, and engagement of adversaries (Hu et al. 2023). Additionally, research is concentrated on improving drone swarm navigation and control in difficult situations. One example is the increasing interest in creating swarm control systems based on models such as the Vicsek model to reduce operational complexity and improve applications in different domains (Varun et al. 2022).

Swarm drones are also being used for exploration in small spaces, with an emphasis on route optimization and safe operations via learning, optimization, and quick responses. Kumari et al. (2024) and Kim (2022), highlight the focus of research directions, which is to establish reliable identification and tracking systems for drones within a swarm. Swarm drone security and anomaly detection issues must also be resolved in the future. In order to provide a decentralized network architecture and improve security,

research is being done on the use of blockchain governance games for advanced drone swarm security (Marek et al. 2024). According to Semwal et al. (2023), it is imperative to develop effective testing procedures for swarm robotics and unmanned aerial vehicles.

In complicated environments, UAV trajectory optimization relies on lowering flight cost and journey time, especially in crowded cities where barriers and signal interference are common. Obstacles, both natural and man-made, can cause attenuation, multipath interference, and other problems for low-altitude planes. These difficulties impair mission autonomy, control, and planning, which lowers UAV performance in smart city applications including mobility, traffic control, and environmental monitoring. Making precise trajectory predictions for UAVs, safety modeling, and scheduling choices are essential for reducing flight duration and battery usage. In these intricate situations, metaheuristic algorithms present viable solutions for scheduling and UAV trajectory optimization. Although machine learning (ML), a branch of artificial intelligence, uses time-consuming and computationally expensive methods, it improves collaborative UAV operations. In order to handle these difficulties, collaborative UAVs with limited memory, high latency, and computational complexity require a resource-aware Federated Learning (FL) architecture. To provide an energy-efficient framework for smooth UAV collaboration, more research on FL and trajectory optimization is needed, which is still a big difficulty.

In order to improve cellular coverage in densely populated areas and provide self-sustaining communication infrastructure in the event of a natural disaster, UAV deployment and alternate association schemes are essential. Understanding the bandwidth, range, speed, cost, and power requirements for large-scale UAV-integrated cellular networks is crucial, as is making sure that low latency and high data rate ultra-reliable connections are maintained.

Collaborative UAVs that are self-organizing and automated improve data gathering, management, privacy protection, and path optimization. This allows for effective operations in a variety of fields, including disaster assistance and agriculture, without requiring regular instructions from a central base station. To enable efficient cooperation and data processing, self-organization in UAV swarms must overcome obstacles including collision and obstacle avoidance, which calls for sophisticated algorithms and additional study.

In complex contexts, task offloading frequently results in network congestion and performance delays. However, collaborative edge computing (CEC), which lowers computational costs while securely delivering data, can help to offset these problems. UAV-based task offloading improves QoS, lowers latency, and increases computing efficiency; nevertheless, due to certain obstacles, such as signal

coverage, UAV mobility and cooperation tactics must be optimized for maximum efficiency.

Control system, communication, and security measures are areas where swarm drone technology is expected to progress in the future. Large swarm maneuvers that involve more agents are the subject of research in order to allow swarms to collect data for the purpose of training Graphing Neural Networks (GNNs) to plan and control swarm agents automatically with little human intervention (Beck et al. 2022). In order to improve security in UAV swarm situations and safeguard against digital threats while maintaining data integrity, blockchain technology integration—such as Hyperledger Fabric—is also being investigated (Beck et al. 2022). Furthermore, the advancement of communication, guidance, and localization technologies is essential to enabling autonomous coordination among swarm drones, which enables them to assign tasks effectively and function flawlessly in a variety of applications, such as public entertainment, military combat, and environmental surveillance (Beck et al. 2022).

Overall, to improve the capabilities, effectiveness, and safety of drone swarms across a range of applications and industries, future trends in swarm drones will take a multidisciplinary approach that blends AI, wireless networking, swarm robotics, and security measures.

11 Conclusion

The evolution of UAV swarm technology has made way for applications of UAV swarms across military and civilian applications. Compared to a single UAV system, UAV swarms offer a number of advantages such as; better fault tolerance, scalability, and efficiency. Using DL algorithms, UAV swarms can improve in target tracking, Remote Surveillance, power management as well as the trajectory planning. Machine learning algorithms like reinforcement learning and further with them, federated learning, enabled a better maintenance of the swarm communication and coordination leading to more effective and autonomous operations.

The next steps in the development of swarm drone technology are probably going to focus on enhancing the control systems, communication protocols, and security measures. Cutting edge research is looking into large scale swarm maneuvers that can teach Graph Neural Networks (GNNs) to plan and control swarms on their own with minimal human interference. A blockchain technology such as Hyperledger Fabric is being considered for integration to secure the data and prevent digital threats from forged or tampered data. It also requires the development of communication, guidance and localization technology to allow drones to coordinate autonomously in swarms. This should

then enable effective task allocation and interoperability with different applications, ranging from public entertainment, military combat and environment surveillance. Swarm drones are also expected to be important for enhancing cellular coverage as well as the provision of a self-sufficient communication network in areas affected by disasters.

The bottom line is that future UAV swarms require an interdisciplinary research ranging from artificial intelligence, wireless networking, swarm robotics, to high-level security horrors. These integrated autonomous capabilities will improve the effectiveness, productivity and security of drone swarms, and will unlock a huge diversity of intelligent asset applications throughout many sectors.

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Security and Privacy Issues in Drone-Enabled IoT Networks

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Abstract

Even though there are many prospects for the Internet of Things (IoT) and drone technology integration across different industries, there are also many security and privacy concerns. The topics of data interception, unauthorized access, data breaches, physical security, and defenses against denial-of-service (DoS) assaults are examined in this article. It also covers the usage of machine learning in the identification of malicious activity. Investors may ensure safer and more efficient operations by enhancing the privacy and integrity of drone-enabled IoT networks by implementing strong security standards and utilizing cutting-edge technologies. Data that has been encrypted will only be accessible to the owner; without the required level of permissions, others will not be able to interfere and misuse the data.

Keywords

DoS · IoT · Drone technology

1 Introduction

The inclusion of Drones into the matrix of the internet and other innovative applications of technological progressions has broadened Horizonte's. They have the potential implemented in many fields like agriculture, surveillance, delivery services, and disaster management, all a testament to their transformation capabilities. Automatic Dependent Surveillance-Broadcast (ADS-B) is a prime example of this report, though not the only way in which such advanced technologies can be used to improve safety and efficiency—but application of such advances does not come without its challenges, especially concerning security and privacy. The mobility, connectivity, and data-gathering capacities that drones bring with them imply novel vulnerabilities that must be mitigated to guarantee the security of this data (integrity, confidentiality, and availability).

Drones enhanced by AI and machine learning capabilities are changing the game for industries like agriculture, surveillance, logistics, and environmental monitoring as they enable a new level of integration with IoT networks. However, more data means an increase in the need for AI and autonomy, as drones can manage and process data in real time and make autonomous decisions. To preserve both user trust and quality, secure and access controlled sensitive data to ensure data privacy, prevent data breaches, and align with data protection and privacy regulations. This section elaborates on these risks and challenges along with the recent strategies related to the same.

2 Interception of Data

In drone-enabled IoT networks, the interception of data is one of the main security problems. Drones ingest and send a lot of data to the cloud, some of which are generally sensitive. This data moves between Wi-Fi, and mobile

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networks, in addition to the unique drone common system. Channel Capturing: The channels can be captured by malicious agents.

2.1 Vulnerabilities and Threats

- **Eavesdropping:** Using insecure communication routes, attackers can listen in on data transfers. This may lead to the release of sensitive data, such as location data, video feeds, and sensor readings.
- **Man-in-the-Middle (MitM) Attacks:** These entail intercepting data being sent between the drone and other Internet of Things devices or the control station, with the potential to modify it. This could lead to inaccurate information, unapproved control, and compromised decision-making.
- **Signal jamming and spoofing:** An attacker can jam signals used for data transmission or spoof legitimate signals to confuse a drone or Internet of Things network.

2.2 Encryption

To mitigate the risks associated with the interception of data, encryption plays a crucial role in securing communications in drone-enabled IoT networks. Encryption ensures that even if data is intercepted, it cannot be read or altered without the proper decryption key.

2.2.1 Types of Encryptions

- **Symmetric Encryption:** employs the same key for decryption as well as encryption. Although it is effective, secure key distribution and management are necessary. AES (Advanced Encryption Standard) is one example.
- **Asymmetric Encryption:** makes use of the public and private keys. The encryption process uses the public key, whereas the decryption process uses the private key. Although this solution requires more computing power, it improves security. Rivest-Shamir-Adleman, or RSA, is a popular example.
- **End-to-End Encryption (E2EE):** guarantees that information is encrypted during transmission and should only be decoded by the intended receiver. A higher level of security for data in transit is offered by this method.

2.2.2 Implementation Challenges

- **Resource Constraints:** Drones and IoT devices often have limited processing power and energy resources, which can make implementing strong encryption protocols challenging.

- **Latency and Bandwidth:** Encryption can introduce additional latency and increase the bandwidth requirements for data transmission, potentially affecting the performance of real-time applications.
- **Key Management:** Secure generation, distribution, and storage of encryption keys are critical to maintaining the integrity of encrypted communications. Key management systems must be robust and resilient to attacks.

2.2.3 Best Practices to Secure the Data

- **Using Strong Encryption Algorithms:** Employing well-established and tested encryption standards like AES-256 and RSA-2048.
- **Regular Key Rotation:** Frequently changing encryption keys to minimize the risk of key compromise.
- **Multi-Factor Authentication (MFA):** Combining encryption with MFA for accessing drone systems and IoT networks adds an extra security layer.

3 Unauthorized Access and Authentication Drones

Occasionally appertained to as unmanned upstanding vehicles, have fully changed several sectors, including surveillance and agribusiness. Despite their benefits, the wide use of drones has sparked worries about security excrescencies, related to issues with authentication and unapproved access icing UAVs may only be covered and operated by authorized individuals is essential for guaranteeing sequestration, safety, and nonsupervisory compliance. Nonetheless, due to the constantly changing and decentralized characteristics of drone operations, conventional authentication ways occasionally fall suddenly. Administering stronger authentication mechanisms and contending unauthorized access are critical pretensions in this script. A groundbreaking frame known as guard (Secure and Effective Authentication for Unmanned Aerial Vehicles) has arisen as a signal of invention within the drone technology terrain in response to the growing security enterprises that unmanned upstanding vehicles (UAVs) defy. Drones are getting more and more common in numerous diligences, similar to structure examinations and delivery services, therefore it's critical to strengthen their security protocols. When it comes to diving into the complex web of vulnerability that mounts UAV processes, especially in the areas of unapproved access and verification, the guard offers a paradigm vault. In discrepancy to traditional authentication styles, which can number substantial computing costs and are unfit to accommodate the constantly changing geography of drone networks, guard provides an—encompassing and effective system of authentication. Unnaturally, the guard is designed to

reduce computational cargo and business, perfecting functional effectiveness without abstaining from security. Guard strengthens access control measures by utilizing slice-edge styles including multi-factor authentication and dynamic retreat proprieties, reducing the liability of hostile exploitation and unauthorized access. Also, interoperability is a crucial design consideration for a guard, which readily integrates to present Internet of Drones (IoD) architectures to guarantee comity across colorful drone bionetworks. In addition to promoting cooperation and information sharing, interoperability also makes standardizing security procedures easier, which strengthens the cohesiveness and adaptability of the UAV structure. Important features offered by Sentinel, a top platform for all-encompassing security and monitoring results, include several slice-edge technologies that guarantee strong security and perceptive analytics. Important characteristics correspond to the following.

3.1 Immediate Trouble Identification

Sentinel's real-time trouble Discovery point makes use of slice-edge machine literacy (ML) and artificial intelligence (AI) technologies for unexampled security checking and response. Sentinel creates a birth of typical exertion by continually assessing stoner and system gets. This allows it to identify variations that can point to possible pitfalls. Sentinel's connection with global trouble intelligence feeds complements its behavioral analysis by furnishing it with over-to-date information on recently arising pitfalls and attack vectors. Security brigades can snappily assay and take applicable action when a trouble is linked thanks to Sentinel's automatic warning system, which gives them comprehensive, contextual information. Through the insulation of impacted systems or accounts and the prosecution of written remediation scripts to contain and neutralize pitfalls, automated incident response capabilities further ameliorate security. Over time, Sentinel's tone-literacy algorithms get better by taking input into account.

3.2 Comprehensive Observation

Sentinel's comprehensive monitoring point ensures a comprehensive picture of an association's security geography by gauging across networks, endpoints, and all surroundings. Sentinel is suitable to completely cover and examine every element of the Information Technology structure, including mobile bias, distant pall services, and—demesne waiters and workstations Sentinel offers a single and coherent picture of security proceedings by combining data from several sources, making trouble recognition and response more precise. No matter where suspicious behavior arises

on the network, the platform's ongoing monitoring features guarantee that it's snappily detected. Sentinel's monitoring efficacy is further increased by its smooth integration with a variety of security tools and data sources, which provides in-depth analysis and—encompassing visibility into possible pitfalls and sins.

3.3 Authentication Challenges for UAVs

The issues of authenticating UAVs involve coordinating security needs with drone technology's functional conditions. Traditional systems for authentication can dodge considerable communication and computational charges, rendering them unhappy for the operations involving unmanned upstanding vehicles.

- *Biometric identification for UAVs*: This content investigates the effectiveness of biometric authentication in perfecting UAV security. Biometric identification, which relies on distinct biophysical or behavioral traits, provides an exceptionally high degree of assurance when certifying the identification of UAV drivers. colorful biometric modalities, including fingerprints, face recognition, and speech patterns, are considered for UAV identification
- *Two- Two-factor authentication (2FA) to Improve Security*: it's a critical element in the guard system, furnishing a redundant degree of protection to UAV authentication styles. By taking druggies to give two unique kinds of authentication, which include a standard word and a spontaneously produced one-time law texted to a handheld device, 2FA vastly improves authentication ways' robustness against unauthorized use attempts. This part explores the addition of 2FA inside UAV authentication ways, pressing its critical part in strengthening safety preventives.
- *Biometric authentication operations in unmanned upstanding vehicles*: Biometric authorization is transubstantiating (UAV) operations, furnishing a wide range of operations that ameliorate security, expedite procedures, and ameliorate the stoner experience. UAVs use distinct physiological traits or behavioral characteristics like as fingerprints, facial expressions, and voice patterns to authenticate regulators with unexampled perfection and thickness. This enhanced security medium guarantees that authorized people may pierce and manage the UAV, reducing the possibility of unauthorized operation and guarding critical conditioning. Likewise, biometric verification streamlines the authentication procedure by barring the need for tedious word-grounded approaches and shortening authentication times. The performing effectiveness not only increases the productivity of operations

but also adds to a better stoner experience. In the headway of expediting operations, biometric verification guarantees dependable verification of the UAV drivers, lowering the threat of identification of theft and assuring nonsupervisory compliance. UAVs use biometric authentication to operate safely and efficiently in a variety of areas, similar to monitoring, surveillance, visualization, and also delivery services, thus perfecting entire functional security as well as safety.

- *The operation of Two-Factor Authentication in unmanned upstanding vehicles:* 2FA is a significant enhancement in perfecting security measures for UAV operations. As its growing significance in a variety of sectors, strong authentication procedures are pivotal to precluding unauthorized access and possible troubles. The use of 2FA in UAV systems adds a redundant subcaste of safety by forcing drivers to give two different types of authentications before getting access to the driver interfaces. This two-factor authentication approach generally combines a regular word or Leg with a periodically generated one-time law transferred to a registered mobile device. By taking multiple authentication factors, 2FA dramatically minimizes the possibility of unauthorized access attempts while also strengthening the system's adaptability to bushwhackers. Also, 2FA improves security for operations without losing usability by using common authentication mechanisms while maintaining strict security norms. As the use of UAVs grows across diligence, the use of 2FA emerges as a critical element in icing the integrity and responsibility of UAV operations, promoting confidence in them in their security and sequestration measures.

4 Privacy Issues and Data Breaches in Drone-Driven IoT Networks Based on Machine Learning

The rapid advancements in drone technology have led to their widespread adoption in various IoT applications, such as package delivery, surveillance, and agricultural monitoring. As drones become integral to operations in sectors like agriculture, where they enhance productivity through linked technologies, the potential for associated security threats transforms from theoretical concerns to urgent realities that demand immediate attention and action. Moreover, real-world incidents, such as cases of GPS spoofing that compromise the operational integrity of drones, highlight the vulnerabilities that can be exploited in these systems, necessitating a serious reevaluation of security protocols to safeguard against malicious attacks (Khan et al. 2021). In one notable instance, attackers reportedly executed GPS spoofing to redirect drones, which resulted in unauthorized

access to sensitive areas and disruption of intended missions, emphasizing the critical need for robust security frameworks in drone-enabled IoT networks to prevent such exploitative manoeuvres (Khan et al. 2021).

Existing research has delved into the various types of attacks that can target drone-enabled IoT systems, including data interception, unauthorized access, and denial-of-service attacks (Khan et al. 2021). The ramifications of such attacks are not merely theoretical; they manifest in real-world scenarios that pose significant risks to both privacy and operational security, illustrating the necessity for comprehensive security measures to be integrated into the design and deployment of these technologies (Khan et al. 2021). In one example, attackers compromised an in-car WiFi system, which allowed them to access personal data from the car's network, thereby exemplifying how vulnerabilities in interconnected systems can lead to serious security breaches and data theft in the context of drone operations and IoT networks (Arshad et al. 2018).

To address these challenges, researchers have proposed a range of solution mechanisms, such as secure frameworks, privacy-preserving techniques, and robust authentication methods. These solutions aim to create more resilient IoT ecosystems that can withstand emerging threats while ensuring that user privacy is adequately protected, thus fostering greater trust in the use of drone technologies across various applications in daily life and industry (Nassi et al. 2019). However, despite ongoing efforts, the ad-hoc nature of IoT systems often complicates the implementation of these security measures, as devices may be resource-constrained and thus limited in their capacity to support sophisticated security management systems that monitor network vulnerabilities in real.

Drones enhanced by AI and machine learning capabilities are changing the game for industries like agriculture, surveillance, logistics, and environmental monitoring as they enable a new level of integration with IoT networks. But more data means an increase in the need of AI and autonomy, as drones can manage and process data in real-time and take autonomous decisions. To preserve both user trust and quality, secure and access controlled sensitive data to ensure data privacy, prevent data breaches and align with data protection and privacy regulations. This section elaborates on these risks and challenges along with the recent strategies related to the same.

4.1 Data Breaches and Privacy Issues—an Overview

Now they are integrating drones into IoT networks, and so we are seeing an explosion in the amount and significance of the data gathered and processed by drones. This data is

massive: everything from air-quality readings to traffic patterns, from personal identities to locations. While this phenomenon is not a new one, the increasing use of data mining handled by trained and supervised machine learning algorithms, based on large amounts of data, is causing data breaches and privacy violations.

Data breaches in which sensitive data is compromised by unauthorized people entering the secure environment can see several consequences e.g. identity theft, financial losses, and damage to your reputation. Lack of adequate security in a drone-enabled IoT network, could allow hackers to see video feed from drone, capture location information, access sensor data, etc.

In 2021, a security researcher demonstrated how a drone could be used to hijack a smart home network and gain unauthorized access to connected devices, including security cameras and door locks. This case study highlights the need for robust access control and encryption mechanisms to prevent such intrusions.

Another recent example occurred in 2022, when a swarm of drones was used to conduct a coordinated aerial surveillance operation over a sensitive government facility. This incident underscores the potential for drone-enabled IoT networks to be exploited for espionage and intelligence gathering, and the importance of implementing effective countermeasures to detect and mitigate such threats.

4.1.1 Key Risks and Vulnerabilities

- Risks Related to Data Collection and Storage

Drones produce great pieces of sensitive data. Such data usually resides on onboard systems before being sent to central servers or cloud storage for processing and analysis. Data could be intercepted at any step during the collection, storage, and transmission phases.

The reason onboard storage is vulnerable is that drones are stationary objects that can be touched or grabbed by bad actors. Moreover, unencrypted storage of data is simply a low-hanging fruit for data thieves. The transfer of data from drones to central locations exposes this information to risks if not transmitted using secure communication protocols. It is quite possible that this data can be intercepted by some attackers who can find very precious information.

- Communication Risks

Wireless communication plays a massive role in the navigation, data transfer, and communication with other devices in the IoT network for drones. These wireless channels are very insecure and can be taken over, and man-in-the-middle attacks can be advanced where data can be stolen or changed before it reaches its final destination (Vasconcelos et al. 2019).

Also, communication between drones if with unsecured or weakly secured protocols may reveal drones to cyberattacks against the jam, false identity, or listening (Vasconcelos et al. 2019). These vulnerabilities could allow attackers to specify an unauthorized destination server and link the server connection established by the drone controller to a malicious server outside the cell infrastructure, thus compromising the integrity and confidentiality of the data traffic between the drone to central servers/remote server or other IOT devices (Sharma et al. 2020).

- Machine Learning Model Risks

The machine learning models employed in drone-enabled IoT networks are also open to various attacks. An adversarial attack requires the model to subtly adjust the input data so that the ML model will contest the wrong results or divisions (Rovira-Sugranes and Razi 2022). The visual data arriving in the ML model of a drone, for example, contains a small variation, which can induce a state of confusion that results in consequences such as dangerous imprecision.

Also, data poisoning attacks can alter the training data that led to the formation of the ML model which can be used to break the ML models by training the model with inaccurate data. These models often work autonomously, therefore, the data they provide must be accurate and reliable for ensuring the safety of the drone-enabled IOT networks.

4.2 Mitigation Strategies

4.2.1 Encryption

At a slightly fundamental level, encryption is one of the ways (nowadays becoming the main way) to protect your data from unauthorized access. By encrypting data at rest (at rest on board the drone and in transit on the drone whilst ingesting), If the data is recorded or accessed by non-authorized entities, it is meaningless and unusable without the decryption keys. To protect the data collected from drones, advanced encryption standards (AES) and public key infrastructures (PKI) are utilized. By deploying end-to-end encryption, you can safeguard the data at rest, the data in motion, along with the data that is collected and processed.

4.2.2 Securing Communication Protocols

Secure communication protocols, such as HTTPS, TLS, and VPNs can protect data while it is sent and received. Those protocols make for encrypting and securing sent-out data over air so that it cannot be decoded or manipulated (Javaid et al. 2023). Also, it uses robust authentication mechanisms like digital certificates, and cryptographic keys to make sure only allowed devices and entities can talk to the communication channels.

4.2.3 Security Authentication and Access Control

When solid access control mechanisms are in place, sensitive data and systems are restricted to only those who are allowed access. RBAC (Role-based Access Control) can be the best way to provide each user in an org their set of permissions depending on their roles, which reduces the risk of unauthenticated accesses.

Multi-factor authentication requires the users to provide more than one type of verification to get into the systems, which serves as an extra layer of security. This greatly mitigates the chances of an attacker getting in, even if they have taken away one form of authentication (a password, for instance).

4.2.4 Secure ML Models

There are multiple ways to secure machine learning models—Protecting training data integrity to avoid data poisoning attacks. Another way of fixing this can be to validate and clean your data before using it for training. The implementation of strong adversarial training techniques makes ML models less vulnerable to adversarial attacks. Differential privacy techniques can be used to prevent ML models from accidentally exposing private information about the data used in training.

4.3 Regulatory and Compliance Considerations

4.3.1 Data Protection Regulations

Drone IoT network operating organizations are required to comply with data protection regulations such as the California Consumer Privacy Act (CCPA) and the General Data Protection Regulation (GDPR). These regulations provide exhaustive instructions on how to collect, store, process, and share data with a strong focus on safeguarding the privacy of each person. Compliance with these regulations necessitates the implementation of vigorous data security as mentioned in this article, regular privacy impact assessments, and transparency in every data processing operation. Failure to obey these regulations can lead to judicial fines and entanglement in an embarrassing mess.

4.3.2 Standards and Best Practices for Industry

Leading industry standards such as ISO/IEC 27001 and the NIST Cybersecurity Framework offer a suite of security controls and a roadmap to align effective cybersecurity and data protection controls with organizational priorities. By following these guidelines, organizations can develop a secure environment and protect their most sensitive information. In drone-enabled IoT hauntings, data protection is an pre-eminently essential factor, and organizations must comply with the best practices to safeguard information regular security audits, vulnerability assessments, and penetration testing are

life into a new and complex routine to uniquely identify and anonymize nefarious gaps. Keep current on potential threats and keep security up to date to keep your data safe.

4.4 Ground-Breaking Solutions and Technologies

4.4.1 Data Integrity and Security with Blockchain

Fortunately, blockchain technology provides an excellent opportunity to improve the integrity and security of data flowing through drone-enabled IoT networks. Blockchain creates a decentralized and tamper-proof record of data (Mohsan et al. 2023). This makes the data collected by drones more trustworthy and reliable.

The credibility contracts allow for automated and accurate enforcement of the terms of data privacy and security policies. A smart contract, for instance, can bartend who can access data collected by drones and how it will be used as per the coded rules and regulations.

4.4.2 Homomorphic Encryption

Is a type of encryption that allows you to process data while keeping it in the encrypted state. This way, vital data can remain encrypted also during computation, as a result, increases the level of security prevents data breaches, and respects the privacy of sensitive information. Processing and analyzing drone-collected data without ever exposing the raw data can be achieved via homomorphic encryption. The ability to perform secure and privacy-preserving data analytics while exposing only the required domain of the data to an untrusted environment is the objective that was chased behind every one of those works.

4.4.3 Decentralized Consortium (E.G., Secure Multi-Party Computation, or SMPC)

SMPC refers to a mathematical concept where multiple users can compute a function collectively over their inputs while keeping the inputs private. This is very useful when data needs to be shared and processed by multiple parties without disclosing the actual data. SMPC is an enabler for privacy-preserving collaboration among participants. This allow them to e.g. collectively analyze drone data without revealing their datasets and still protect the original elements of the information.

4.5 Emerging Trends

4.5.1 Zero-Knowledge Proofs (ZKPs)

These are cryptographic protocols used to prove to one party that the other knows a value without demonstrating this value much more than it is a fact. They can be more

complicated but this point is enough for this use case. Using ZKPs in a drone-enabled network would mean that we can validate the authenticity and integrity of data without needing to reveal the underlying data. One of these areas where data privacy and security concerns could be crucial is military or intelligence operations. The ZKPs can provide secure and privacy-preserving verification leading to trust and collaboration among multiple drone-based IoT networks.

In recent years, advancements in the ZKP protocol have made them more efficient and scalable. Due to this advancement, they became apt to be used on applications that need to be deployed in production environments. For IoT network-enabled drones, ZKP can be used to verify data transactions and enforce data consumption, as well as to support secure and privacy-preserving data sharing.

4.5.2 PETs (Privacy-Enhancing Technologies)

PETs are a set of tools and methods to provide data privacy and cybersecurity. Best Practice methods such as anonymization, encryption, access control, and data masking are a few of the comparative advantages of PET. This technology of protecting PETs through drone-based IoT also secures data from unauthorized access, prevents data breaches, and assists in regulatory and privacy compliance.

The recent advancements in PET are towards efficacy, scalability, and deployability. For instance, new anonymization techniques have emerged to provide better privacy guarantees preserving the utility of the data. Meanwhile, improvements in encryption algorithms, including quantum-resistant encryption, have improved the security and redundancy of data security practices, as well.

Generally, PETs allow the protection of data that drones collect, provide protection of communications (data transmission and storage), and make data analysis done in a privacy-preserving way. This might involve mechanisms to anonymize who is appearing in video footage captured by drones, or encrypting data during transmission and storage depending on your specific adversarial model. Also access control mechanisms which can restrict the access and use of sensitive data to only authorized entities, thus empowering data privacy and security.

4.5.3 Edge Computing for Privacy

Rather than transfer it back to a central server, edge computing processes data near the source or where the data is gathered in the first place (like the drone or a nearby edge device). The result is decreased latency, more effectively employed in real-time decisions, and less data going over the network, which in turn reduces privacy risks. This allows critical data to be analyzed and acted upon without being transmitted into areas of possible vulnerability.

It is especially useful in edge environments that are remote or bandwidth-constrained, such as with drones, allowing for the localization of compute processing securely and without relying on centralized infrastructure. Edge computing brings another layer of data privacy by confining the settings of what's processed through a network and allowing less data to be ever shared throughout networks. A drone may, for instance, receive sensor data and act on it locally, detecting anomalies or learning patterns that it then shares only aggregated or anonymized results with central servers or other drones.

With the advent of edge AI and edge analytics, edge computing solutions have greatly increased their capabilities and performance. Given the context, Edge AI's main objective is to deploy and run machine learning models directly into edge devices to analyze the data in real-time, and without exposing any privacy information. Edge analytics, however, offers sophisticated data processing and analysis at the edge, leading to nimbler and more secure data insights.

Edge computing offers promising data privacy, security, and efficiency advantages. Edge Computing mitigates the possibility of data breaches, improves real-time decision-making, and enables privacy-preserving data analysis by processing data locally. Edge computing can also work hand in hand with other privacy preservation technologies like encryption and anonymization to offer a complete and failsafe data privacy solution.

4.5.4 Privacy, AI and Machine Learning

With the rise in drone-enabled IoT networks, there is increasing activity concerning improving data privacy and security through leveraging AI and ML. For example, AI and ML can be leveraged to embed more sophisticated privacy-preserving techniques, detect and predict data breaches, and comply with privacy regulations (Rovira-Sugranes and Razi 2022). For this, you can use AI and ML to create anomaly detection systems that can detect uncommon data access or usage patterns early on, thereby preemptive intervention in case of potential data breaches.

Secondly, applications of AI/ML in privacy-preserving machine learning models like federated learning and differential privacy can be developed that encourage safe and privacy-respecting data analysis. The privacy compliance checks can be automated, data transactions can be checked automatically, guarantees rules and regulation base controls for each transaction and the above things can be easily achieved using AI and ML. AI-embedded smart contracts, for example, can validate that appropriate user consents are in place to collect and store data for a certain time, and that data is only used for specified purposes.

New AI and ML technologies, like explainable AI (XAI) and AI ethics, add even more to the capabilities and potential use cases for data privacy and security. Explainable AI is needed to give transparency and interpretability of models available in AI, and for people to be able to understand AI decisions and trust it. AI ethics ensures that these technologies are developed and used in a fair, accountable, and proper manner.

You googled the benefits of security and AI / ML is emerging. Organizations can utilize AI and ML technologies to build sophisticated privacy-preserving solutions to improve data breach detection and prevention, and always stay compliant with privacy regulations. Moreover, AI and ML are capable of providing a better data privacy management system, which can also support cooperation across heterogeneous drone-based IoT network data usage.

5 Physical Security Measures

Before getting into specifics of security, it's necessary to fete the UAV- IoT integration frame:

- (a) Data Networks UAVs are airborne bumps that establish connections with terrestrial Internet of effects networks by acquiring, information, and assessing data.
- (b) Communication Transmission During the flight, these aircraft will shoot an enormous quantum of data, including control signals, detector readings, and real-time film land.
- (c) Functional Flexibleness Functional Flexibleness allows the machines to operate in dynamic and changing surroundings; covering places that may not be respectable or accessible for Internet of effects bias stationed d on the ground (Wang et al. 2020).

Combining UAVs with IoT networks requires complex relations between the upstanding and terrestrial factors, which causes them trouble in several ways. The primary security issues are as follows:

• Physical Attacks

UAVs may face a variety of physical hazards, similar to deserter individuals trying to take down or destroy drones. These adversaries might want to disrupt UAV operations by interdicting them. There are many cases of UAV interception and kidnapping: This is the process by which vicious individualities use the drones' vulnerabilities to seize control of them, potentially resulting in the loss of important data or dislocations to operating procedures. Destruction

and Interference vicious actors can physically damage UAVs using munitions or electronic jammers, dismembering their operations.

• Security of Information and Confidentiality:

Unauthorized access and interception of data are possible when it comes to data transferred between UAVs and base stations. It's vital to guard the integrity and confidentiality of this information, especially for operations that involve sensitive or secret information. Significant enterprises are as follows:

Wiretapping: Data dispatches may be interdicted by unauthorized parties, potentially exposing private information.

Data tampering: When data is transmitted, vicious parties have the power to change it, venturing its correctness and responsibility.

Physically dangerous: These include purposeful conduct similar as shooting down or jamming unmanned upstanding vehicles (UAVs) to render them inactive.

• Security of Communication Pitfalls:

Bushwhackers can target the communication lines that connect UAVs to IoT networks in several ways, including:

A jamming attack: These involve snooping with the signals that the UAVs use to communicate, thereby stopping them.

Spoofing: Falsely manipulating the data to trick UAVs or ground stations into taking conduct that could lead to miscalculations.

5.1 Ways for Perfecting UAV- IoT System Security

Numerous sophisticated ways are used to overcome these issues. These tactics include data protection styles, secure communication protocols, and physical security measures.

5.1.1 Measures for Physical Security

Conserving the UAVs' physical security is the primary line of protection against violent attacks. Among the measures are

Hard structure and rudiments: Making use of strong, durable substances to defend against physical assaults.

Technologies that help tampering: Using detectors and announcement signals to identify and reply to tampering operations.

GPS shadowing: Construct virtual walls to limit unmanned upstanding vehicle (UAV) conditioning to specific regions and stop unwanted access to places that are sensitive as well.

5.1.2 Conserved Zone Data Secrecy Approach:

The intention of operating a defended zone is to establish an area of security around the UAV so that data dispatches are secured from bushwhackers. This approach can look like this:

Morphological defended zones: Using antenna radiation pattern design, the content area of UAV transmissions is formed to reduce the possibility of interception.

Dynamic ray shaping: This fashion lowers the possibility of surveillance by precisely directing signals toward intended listeners using innovative antenna technologies.

The defended zone's armature and structure are largely significant. To ensure that only intended druggies within the zone may pierce the data, the broadcast area can be limited, for case, by using a defined, concentrated ray().

Then is a simple illustration shown in Fig. 1 illustrating the Conserved Zone Data Secrecy Approach:

The UAV is in the middle of this illustration, encircled by a safe area. To avoid unwanted access or hindrance, all transmission signals within this zone are vindicated and translated.

Only honored outfit like ground stations and Internet of Effects bias that are believed to be suitable to communicate with the UAV is allowed to be inside the defended zone. These widgets can connect with the UAV via translated and vindicated signals as they're positioned inside the defended area.

Since the communication signals are vulnerable and unencrypted outside of the secured zone, it's more delicate for an unapproved outfit to intercept or access the UAV's

transmission. The possibility of illegal access or hindrance is reduced by designing the defended zone, icing the confidentiality and delicacy of the UAV's communication.

5.1.3 Installations for Safe Storage

Secure storehouse installations can be fitted to UAVs to guard data against physical attacks. These installations ensure that the data is safe indeed in the case that a UAV is taken down or compromised. Styles correspond to:

Data encryption: Sophisticated styles are used to cipher data stored on UAVs, performing in it unworkable without the decryption key.

Tone-destructive styles: If theft is linked, UAVs can be programmed to automatically destroy critical data in critical circumstances.

Tackle cinches: Data storehouse bias and important factors can be defended by using physically locked secure chambers.

Although integrating UAVs with IoT networks has revolutionary eventuality, there are also serious security pitfalls. A complete strategy that includes artificial noise, defended zones, physical security, and safe storehouse results is demanded to address these issues. We can cover UAV-IoT systems and ensure their safe and effective deployment in a variety of operations by enforcing these ways.

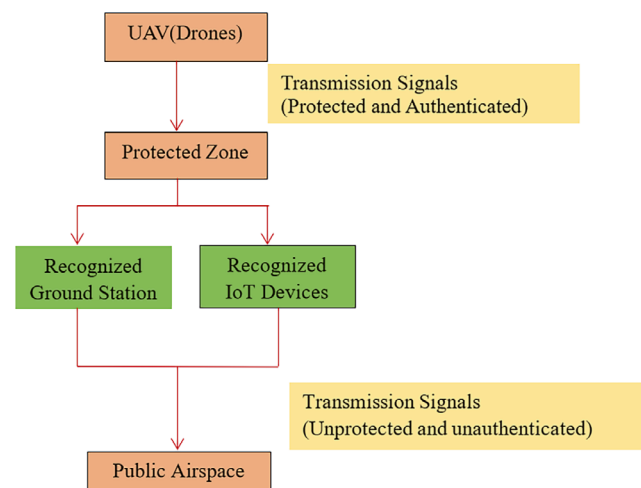


Fig. 1 Conserved zone data secrecy approach

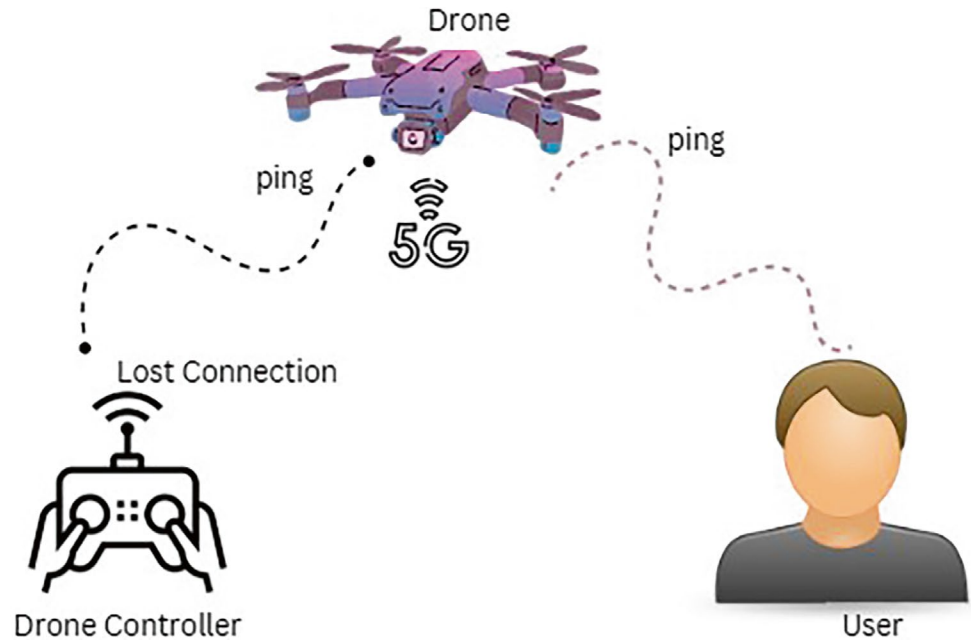
6 Defense Against DoS Attacks

6.1 Introduction to DoS Attacks

DoS attacks are intended to overwhelm a network or service, making it unavailable to its intended users. Some of such attacks can block the communication channels, disturb the computational resources, and also can affect the Machine Learning algorithm behavior in IoT networks with the entry of drones into this field (Zhijun, et al. 2020).

In drone-enabled IoT networks, many applications have been revolutionized with machine learning integration, including delivery systems, surveillance, and environmental monitoring. The higher levels of connectivity and dependence on digital communication in these systems render the systems vulnerable to cyber threats, with Denial of Service (DoS) attacks being an example. A DoS is an attempt to make a service unavailable to intended users by unfairly blasting resource-rich areas. Distributed Denial of Service (DDoS) is a commonly known example of this type of attack in which multiple devices attack the same computational resource. These attacks aim to interfere with the provision of services, which could have serious consequences for operations and finances.

Fig. 2 DoS attack on IoT-enabled drone



The annual Verizon Data Breach Investigation Report (Solutions 2015) reports that there has been a rise in DDoS attacks, particularly those associated with DDoS incidents. A cyberattack that exploits a large number of insecure IoT devices would be one example of this. An Internet of Things network constructed using the Mirai malware was the source of a significant attack on a security blog in September 2016. A few months later, the French web host OVH and the DNS service provider Dyn reported further Mirai-based attacks. As per (Bertino and Islam 2017), Mirai's approach is simple: it finds under-secured Internet of Things devices that can be accessed remotely by using the most commonly used credentials (Vasconcelos et al. 2019). Figure 2 shows a typical scenario of a DoS attack in IoT-enabled drones. Unnecessary requests limit the access of shared resources to legitimate users. That will overload the system and could prevent some or all valid requests from being filled. In this stage the adversary starts sending up multiple data packets to the drone and as a result, causes de-authentication of the network taking place between the ground controller and the drone.

The next section examines different DoS attacks and countermeasures, highlighting their applicability and significance in the context of drone-enabled IoT networks.

6.2 Types of DoS Attacks and Their Impact on Drone-Enabled IoT Networks

(a) **Jamming Attack:** The act of interfering with the radio frequencies that drones and their control systems use for communication is known as a jamming

attack. Because the attackers can also send signals of the same frequency, it can cause interference, making the drones to go out of control or not respond to important commands. In the worst case, drones might stop responding or crash, endangering people's safety and affecting operations. Figure 3 shows the common jamming attack scenario in drones. The attacker sends the jamming signal using the actual signal's frequency with the UAV's help to the base station. Thereafter the communication between the drone and base station is disrupted. The drone stops responding as signals are not allowed to reach the server thereafter. An autopilot mode gets activated once the drone loses contact with a base station. By manipulating the GPS signals, an attacker can easily initiate a GPS-spoofing assault and force a landing away from the intended destination when using auto-pilot mode.

- (b) **Physical Attacks:** Drones or their infrastructure getting damaged or maliciously tampered with is called physical attacks. For example, sensors or communication modules of drones may be compromised or the attack on drones may destroy hardware. This is what so-called swarms of drones can impose, rendering the UAS inefficient and hence damaging all sorts of missions.
- (c) **Protocol Exploitation:** Attackers can exploit the vulnerabilities within the communication protocols in drone-enabled IoT networks to execute targeted attacks. By exploiting the vulnerabilities in the communication protocols, for instance, attackers can insert malicious commands and modify the data exchanged between the drones and the control systems; thus, the attackers could endanger the services or gain access to sensitive data.

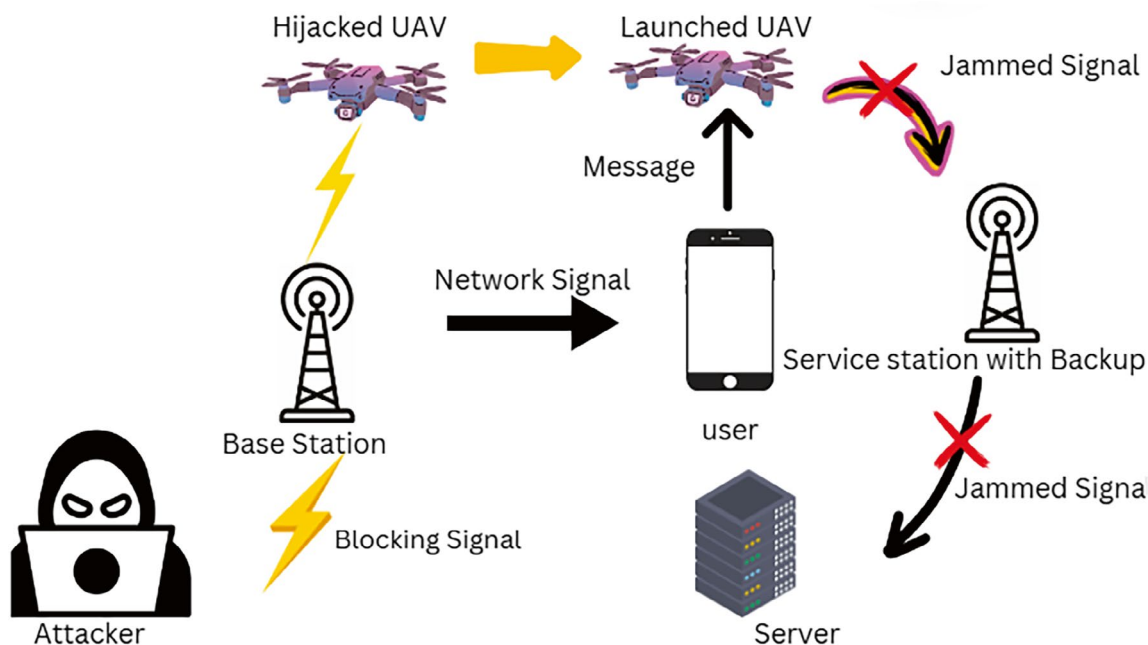


Fig. 3 Jamming attacks on drone

- (d) **Resource Exhaustion Attacks:** The idea here is to consume arbitrary system resources (for instance: CPU, memory, bandwidth) and the target will no longer be able to perform in the way it is supposed to. In such kinds of networks, static rates of flows or a high number of intervals attacks may be launched to the central servers or communication infrastructure against each other and this can cause an exhaustion of resources. What happens is these servers get swamped and cannot return requests in response to legitimate drone queries and this can lead to service outages or failures of servers.
- (e) **Firmware/Software Exploits:** Exploiting vulnerabilities in the firmware or software that could be running on the drones, or even in their control systems. For example, they could exploit buffer overflow vulnerabilities to execute arbitrary code on a flying drone, crash systems, or provide a wrong way to gain access to confidential data. In these complexities, the integrity and availability of the IoT networks under drone cover should not be compromised, while stringent measures against DoS attacks must be undertaken.

6.3 Strategies to Defend Against DoS

A multi-tiered approach that includes proactive measures to help prevent, detect, and respond to attacks is overburdening in shielding your network or website from Denial

of Service (DoS) attacks. Here we will cover some common defense mechanisms and we will map how these would be used in the context of IoT networks where drones are used.

6.3.1 Network-Based Defense Mechanisms

Firewalls and Intrusion Detection Systems (IDS)

Firewalls and IDS help in controlling and monitoring incoming and outbound network traffic. They are the foundation of network security and also help ease the impact of DoS attacks. Firewalls act as front-line soldiers in the security battle by blocking traffic that fails to meet the established security parameters as defined by the system administrator. Some DoS attacks may be targeted to less important systems and partially blocked (not a complete block, because of firewalls). IDS observes network traffic for signs of unusual or suspicious patterns that may indicate a DoS attack. IDS can be either anomaly-based (detect outliers in the traffic) or signature-based (match a set of patterns with known signatures of attacks).

Applications:

- (a) **Edge Firewalls:** Firewalls at the edge where drones connect to the Internet of Things.
- (b) **Distributed IDS:** Implementing IDS at distributed network nodes would allow earlier detection and hence modify DoS attacks to the extent.

Load Balancing and Traffic Filtering

These technologies are also designed for load balancing and traffic filtering methods, which spread traffic of the network across multiple servers or pathways to prevent bottlenecks if it is routed through a single point. Load Balancers; Distributing incoming traffic among different servers so that no single server gets overwhelmed. During DoS attacks, Load balancers can identify and route around traffic, thus preserving the service availability.

Rate Limiting and Black Listing are Traffic Filtering Techniques that can filter malicious traffic. Rate limiting restricts the rate of incoming requests and Blacklisting blocks known malicious IP addresses.

Applications:

- (a) Distributed Load Balancers: To generate functionality of distributing traffic across multiple deployed instances to evenly distribute traffic across application components and prevent hotspots that could make your service/application unusable.
- (b) Adaptive Filtering: Traditional traffic filtering mechanisms can be modified to use ML algorithms, which can automatically learn from the network evolve traffic characteristics, and effectively detect illegal traffic.

6.3.2 ML and AI-Based Defense Mechanisms

Anomaly Detection

Using ML-based Anomaly Detection, the Machine Learning algorithms are trained to monitor the thresholds and detect any normal deviations from the actual behavior, thus detecting the DoS attacks. Some common methods are supervised learning, where the developed models are trained on labeled datasets, or unsupervised learning where the models find patterns and anomalies in the materials that are unlabeled.

Applications:

- (a) Behavioral Analysis: ML algorithms analyze traffic patterns and drone behavior to detect anomalies that might reflect a DoS attack. This may feed from alerts that are triggered whenever an alarming change occurs in the communication pattern, e.g. a sudden high volume of traffic coming in is followed by silence.
- (b) Adaptive Learning: An active learning mechanism that continuously learns and adapts to new threats can improve anomaly detection accuracy, safeguarding the system against the changing DoS attack strategies.

Predictive Analytics

Predictive analytics is historical data combined with Machine Learning (ML) algorithms to anticipate likely DoS attacks,

providing a measure for proactive defences. Regression and time-series forecasting work as some of the methods to forecast types of future attacks on the base of historical data.

Applications:

- (a) Proactive Defense: Detecting an impending DoS attack before it strikes would allow the network administrators to take pre-emptive actions such as modifying existing firewall rules, scaling up the resources of susceptible nodes, etc.
- (b) Resource Optimization: Predictive analytics can also be used to optimize resource allocation to help ensure drones and IoT devices are as protected as they can be prevented from expected attacks.

6.3.3 Blockchain and Decentralized Security

Blockchain Technology

A blockchain is a useful technology that can be used to ensure that drone-based IoT networks are secure by providing an immutable, decentralized ledger. By decentralizing control, blockchain eliminates the kind of single points of failure that are attractive to attackers and reduces their ability to do a successful DoS attack. Blockchain has immutable records which make them tricky to play with, so network operations cannot be tempered with unauthorized data (data integrity).

Applications:

- (a) Secure Data Transmission: Blockchain can ensure a secure data transmission between drones and IoT devices, keeping data integrity in nature and avoiding entire tampering.
- (b) Decentralized Authentication: Using blockchain for decentralized authentication can prevent unauthorized access and reduce the risk of DoS attacks targeting authentication servers.

Smart Contracts

Smart contracts are smart because once you input the agreement they automatically execute themselves, the contract is written in code into the terms of the agreement, allowing transactions to take place in a secure and automated way. This can be quite powerful, for example, you could have Smart contracts to automatically create defenses (such as blocking IP addresses or reconfiguring the network, etc.) after detecting the threat detection range.

Applications:

- (a) Automated Mitigation: It can trigger mitigation mechanisms to automatically reallocate resources or reroute

the traffic so that continuity is ensured for the service offered even under DoS attacks.

- (b) **Transparency and Accountability:** Transparency of smart contracts makes all its actions completely over the network which improves the trust and availability of defense mechanisms.

6.3.4 Current Developments and Emerging Trends

Advances in AI and ML for Cybersecurity

These innovations in AI and ML have led to the process of developing advanced defense mechanisms for DoS attacks (Premkumar et al. 2022; Maruvala et al. 2024). This technology allows immediate identification and response and improves the security of drone-enabled IoT networks (Ceviz et al. 2306). The use of Explainable AI (XAI) techniques helps in making the ML-decision-making processes transparent, so it is less of a black-box and we can understand it clearly, and trust anomaly detection and predictive analytics. An example is the Federated Learning technique which allows for decentralized training of machine learning models on many devices, which provides for data privacy, data security, and overall improved accuracy detection of attacks.

Quantum Cryptography

Quantum Mechanics is the principle used by Quantum cryptography to implement encryption that is theoretically impossible to break, guaranteeing security against security through denial of services and other potential cyber threats (Said 2023). Quantum Key Distribution allows the secure exchange of cryptographic keys which is the basis for securing communication channels during DoS attacks.

Applications:

- (a) **Secure Communication:** Uses the secure communication channel between drones and IoT devices which can be tamper-proof against an outside attack.
- (b) **Future-Proof Security:** As quantum computing progresses, quantum cryptography will enable future-proof protection against evolving cyber threats, so that the security of drone-powered IoT networks will improve and can be taken forward.

unauthorized access to data. The first step in the procedure is precisely gathering comprehensive information about former security incidents from a variety of sources, including news stories, business databases, exploration journals, and commercial logs. This data includes a range of rudiments, similar to the details of the event, the specialized details of the drone, operating data, security protocols, and incident results. Processing is the coming step, in which the data is formalized, sanctified, and point- finagled to ameliorate prophetic power and guarantee thickness. A supervised literacy approach exercising machine literacy can be used to strengthen drone operations security against pitfalls similar to physical hindrance, breaches of information, and unauthorized access to data. The unsupervised ways like relating anomalies and clustering examine underpinning patterns and outliers in the data. Textual information similar to event reports and temporal trends can also be anatomized using time series analysis and natural language processing. Prophetic algorithms that estimate threats and recommend preventative conduct can be developed by relating typical attack vectors, drone models that are vulnerable, and temporal or geographical patterns. Before these models are included in real-time monitoring systems, they suffer a thorough confirmation process that makes use of cross-validation and performance criteria. Robust drone security is assured by this visionary strategy, which not only helps anticipate and help unborn security incidents but also continuously adapts and improves through feedback and fresh data.

7.2 Extraction of Features

Guard can easily adapt to a variety of structure configurations thanks to its scalability and inflexibility features, meeting the needs of businesses of all sizes. This implies that Sentinel can be easily modified to accommodate the requirements of businesses of any size. One of Sentinel's primary scalability advantages is its flexible adaptation capability, which enables it to handle massive amounts of data and scale either upward or downcast as needed. This guarantees that Sentinel will keep up its excellent performance and fulfill the association's growing needs. Sentinel can quickly identify and respond to attacks thanks to its elastic scalability and AI-invested surveillance capabilities. For large enterprises that need sophisticated problem detection and response capabilities, this feature is very beneficial.

7 Machine Learning Approaches

7.1 Collecting and Assessing Data

An expansive approach exercising machine literacy can be used to strengthen drone operations security against pitfalls similar to physical hindrance, breaches of information, and

7.3 Training the Models

One of the most important ways to guard IoT-grounded drone operations from security pitfalls is to train machine literacy models. These models, which make use of once

security incident data, can be trained using either supervised, unsupervised, or underpinning- -literacy ways, grounded on the features of the data and the particular security pretensions. The models are erected on labeled datasets in supervised literacy, where each case is designated by either trouble or typical gets. With this system, the model can be trained to identify and learn particular patterns that point to security breaches, like unusual flight patterns that diverge from anticipated bones, irregular login patterns that reveal unauthorized access attempts, or strange communication signals that point to jamming or spoofing. On the other hand, unsupervised literacy is applied when markers aren't handed and the ideal is to find retired structures in the data. Anomaly discovery and clustering are two ways that can be used to find odd patterns as well as actions that could indicate new pitfalls. Underpinning literacy is especially helpful in dynamic surroundings, where it can be used to pretend different assault scripts and educate models on the stylish ways to respond in real time to help implicit attacks. We're suitable to make strong defenses that ameliorate the robustness of drone systems that are connected to the Internet of Effects by methodically training these types of models to identify and reply to any security attacks.

7.4 Relating and Vaticinating Pitfalls

Enforcing trained machine literacy algorithms for real-time trouble identification and vaticination is an essential step in guarding drone operations that are grounded on the Internet of Effects. The functional structure of the drone incorporates these trained models to continuously dissect incoming aqueducts of data from several sources, such as detectors, communication networks, and drones. As possible security pitfalls arise, they can be honored by using the trained models on this real data. For illustration, the models can identify unusual communication signals that may indicate jamming or spoofing sweats, snappily identify unusual flying actions that diverge from destined routes, and describe illegal access attempts by examining login patterns. Likewise, by examining one data and seeing new trends, these models use prophetic analysis methodologies to read unborn security pitfalls. Pre-emptive safeguards can be put in place because of this visionary strategy, which makes it possible to anticipate troubles like changing hacking ways. In addition to perfecting rapid-fire peril identification, the nonstop real-time analysis creates a dynamic defense system that adapts to new trouble patterns and data, guaranteeing the uninterrupted safety of IoT-grounded drone operations. This technology offers a strong, adaptive defense against the intricate and snappily changing world of drone security pitfalls by exercising machine literacy (Baig et al. 2022).

7.5 Anomaly Discovery

Anomaly discovery entails recognizing patterns in information that don't acclimatize to awaited gets. In the case of UAVs, this involves tracking flight paths, communication signals, sensitive measures, and responses from drugies. Algorithms using machine literacy are especially well-suited for this purpose because they can learn from large datasets to distinguish between regular and aberrant patterns. Integrating ultramodern drone systems that employ anomaly discovery ways to give security and responsibility by automatically relating diversions from awaited gets during design and while the system is operating utilizing slice-edge machine literacy styles. For illustration, if an unmanned aircraft deviates from its intended flight route or makes erratic movements, this might indicate a kidnapping or navigation problem. Also, strange patterns in transmission signals may indicate jamming sweat or unauthorised entry to the drone's subsystems. Inharmonious detector measures, similar to rapid-fire temperature harpoons or irregular GPS data, may indicate an error or external intervention. Abnormal stoner conditioning, similar to odd command series or access tries, may indicate unauthorised use or phishing sweats (Diro and Kaisar 2023).

When an abnormality is set up, the system sends out announcements to inform drivers of the possible hazard. Also, responses that are automated can be carried out, similar to driving exigency procedures to operate the drone to a safe area, altering the flight route to avoid threats, or installing defensive measures for neutralizing the trouble. This real-time irregularity discovery and response capacity enables the drone to snappily identify and alleviate possible troubles while conserving functional integrity. The perpetration procedure starts with substantial data gathering and medication to guarantee that the machine literacy models admit superior inputs. Similar models suffer training on large datasets that include exemplifications of both typical and atypical gets, enabling them to separate between both of them. After training, the models are linked to the drone's internal systems and run constantly, analyzing current information and upgrading their information with fresh data to reply to changing troubled surroundings. This innovative fashion to identify anomalies not only improves the security of specific drones, but also adds to the overall security of drone processes in a variety of operations, including service and artificial use, disaster operation, and important structure observation. Drones are decreasingly robust and able to singly regulate and respond to possible troubles as they use machine literacy, guaranteeing their effective and secure operation in varied and dynamic surroundings.

7.6 Measures of Adaptive Security

Implementing artificial intelligence-grounded safety protocols into unmanned aircraft enables a flexible and adaptable response to the ever-changing outlook of troubles and sins. Drones may use important algorithms for underpinning learning to optimize their safety measures and communication tactics presently. This enhancement is fulfilled by continually analyzing responses from the drone's backgrounds and security event data. For illustration, if a drone gets signal blocking, the system may reply by transferring to a different communication frequency or using encryption styles learned from one hassle. This inflexibility is especially important in hostile or unlooked-for situations where standard, stationary safety preventives may fail. By understanding the real-time data, the drone can honor and respond to preliminarily unknown or unexpected hazards. Similarly, the automated literacy algorithms built into drone technologies are continuously improved and streamlined. Similarly, the automated literacy algorithms built into drone technologies are continuously improved and streamlined. Drones can handle and respond to security issues autonomously with minimal human commerce thanks to these adaptive security results. This is particularly relevant to operations involving the surveillance of critical infrastructure, military deployments, disaster relief efforts, and the provision of marketable services, as the operational environment is subject to abrupt and dramatic changes. By embracing machine literacy-grounded adaptive security, drones can further adapt and ensure the safe and successful completion of their operations. It is crucial to conduct a cross-domain investigation to identify relationships between security events caused by drone processes and other elements like the surrounding environment, regulatory compliance, and geopolitical risks.

8 Conclusion

Due to the sensitive nature of the data they manage and the vital roles they play, drone-enabled IoT networks must prioritize data security and privacy. Protecting these networks requires addressing data interception issues and putting strong encryption techniques in place. Despite certain obstacles, implementing best practices and constantly improving security measures can greatly reduce risks and improve the robustness of drone-enabled Internet of Things systems.

There are a lot of problems with deploying as well as the usage of ML in drone-enabled IoT networks and data breach and privacy concerns are some of the major issues in it. By encrypting data, only the owner of those data will have the key to access them, and no one else will be able

to intervene and misuse it without the necessary level of permissions.

In this context, newly emerging technologies, including blockchain, homomorphic encryption, and secure multi-party computation, have shown potential to better ensure data confidentiality and integrity within the context of privacy preservation within IoT networks empowered by drones. These technologies will be important as the challenges for data protection look only to get more complex in the burgeoning industry.

In the coming years, more in-depth research, innovation, and collaboration between industry, academia, and regulatory bodies will be necessary to create strong and efficient techniques to protect data privacy in drone-enabled IoT networks for machine learning. Maintaining continual situational awareness is crucial for organizations to use the full potential of this technology, and to protect sensitive information and public trust.

The countermeasures to deal with DoS attacks in drone-based IoT networks need a multi-layer security framework including conventional network security solutions along with the pro-active usage of AI and ML, blockchain technology, and also the evolutionary developments in quantum cryptography. Using these IoT security mechanisms, businesses can maintain the uptime, integrity, and security of drone-enabled networks and keep enabling the growth of this domain. The technology keeps growing over the years, and the innovations and advancements behind cyber security will be vital to keep these drone-enabled IoT networks future-proofed from possible cyber threats.

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Enhancing Capabilities and Security Features of Drone Networks Through Machine Learning: A Comprehensive Overview

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Abstract

The integration of machine learning algorithms into drone networks offers significant opportunities to enhance the processing power and security of Unmanned Aerial Vehicles (UAVs). This chapter provides a comprehensive overview of how leveraging IoT, and collaborative learning techniques can enable drones to operate autonomously, adapt to changing environments, and perform complex tasks with greater efficiency. By analyzing large datasets, drones can achieve improved situational awareness, predictive maintenance, and optimized mission planning through adaptive reinforcement learning and deep reinforcement learning. Techniques like federated learning, collaborative swarm intelligence, and ensemble learning contribute to the accuracy and effectiveness of drone operations, making them valuable in applications such as traffic management, disaster response, smart city infrastructure, and agricultural monitoring. The chapter also addresses critical issues such as privacy concerns, data security, regulatory compliance, and transparency in data usage. Additionally, it explores future directions, including the development of advanced machine learning algorithms, edge computing, interoperability, ethical frameworks, multi-domain integration, resilience, and human-drone interaction. By continuing to explore these areas, researchers and industry professionals can drive innovation and fully harness the potential of machine learning to advance drone networks for both civilian and military uses.

Keywords

Drone networks · IoT (Internet of Things) · Machine learning · Security · Features · Privacy concerns · Regulatory compliance · Autonomous systems

1 Introduction

Utilization of the Internet of Things (IoT) to increase connectivity and to enhance data-driven decision making in drone networks has been a topic of massive interest among researchers. Drones are autonomous devices that provide an economic solution to conventional vehicles that require human labor to navigate and capture various categories of data. A swarm of drones can be used in various facets of research and real life applications that can benefit normal day-to-day lives of both civilians and military personnel (Javaid et al. 2023). The integration of machine learning approaches in constructing a comprehensive and efficient data collection system of drones for further investigation leverages the potential of IoT in fields of research that are deemed impossible and hard to reach in real life. Machine Learning algorithms can further analyze the data retrieved to predict and synthesize information, thus improving the autonomous capabilities and efficiency of the drone networks.

A huge surge in the number of connected devices can be observed within the last decade, and projected numbers will rise to 75.44 billion by 2025 (Alam 2018). This huge boost in the number of devices is due to huge advancements in sensor technology and wireless communication. Utilization of this connectivity is a big opportunity to gather vital data and embed connectivity in everyday objects. Devices such as smartphones and gadgets and UAVs like drones all fall under wireless networks all over the world. UAV devices such as drones have been used in multiple civilian and military applications. Many such activities are related to military operations, such as medical supply delivery (Brik

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et al. 2020) and military intercommunication. A Smart Bangladesh Vision (2022) has been an esteemed plan for Bangladesh for futuristic technological advancements such as 5G internet, cashless transactions, increased internet speed and complete smartphone-based systems. In terms of smart cities, UAV devices can be used to deliver orders of merchandise and as taxi for future technological transpiration facilities (Mohamed et al. 2020). Smart agriculture and monitoring of environmental situations can be carried out by drones by taking sensor data from sites that require monitoring (Manju et al. 2021). Underwater object detection using Underwater Autonomous Vehicle (Dipta Gomes and Saif 2021) to study and observe marine life, can give a huge surge of movement in terms of maintaining ecological balance as well as detection of natural resources. Emergency services like medical supplies, express shipping of medicines in developing cities has been discussed by Ling et al. (Ling and Draghic 2019). Delivery of blood on an emergency basis in hospitals and medical services has been a prime concern of the research.

This chapter aims to explore the integration of machine learning into drone networks within IoT applications. The objectives include providing an overview of collaborative distributed machine learning techniques suitable for drone networks; Investigating security and privacy issues related to drone-enabled IoT networks and future directions of IoT based drone networks. The chapter is structured as follows: Sect. 2 discusses collaborative distributed machine learning techniques in drone networks. This chapter further explores the fields of federated learning, swarm intelligence, and ensemble learning. Section 3 provides the application of Adaptive Learning Techniques through reinforcement learning and Deep Reinforcement Learning. The section also discusses distributed machine learning frameworks like TensorFlow Federated Framework (TFF), PySyft, Apache Spark and Edge ML. Section 4 deals with the security and privacy issues in drone-enabled IoT networks, discussing security issues, threat landscape analysis, vulnerabilities and attack vectors and security measures and protocols for UAV drone networks.

2 Literature Review on Collaborative Learning in Drone Networks

A drone network consists of multiple devices working together to carry out a specific goal, and collaboration among devices needs to be established for higher accuracy and efficiency in carrying out various types of operations. Key learning principles through collaboration need to be integrated into the network for distributed data sharing and task allocation. A collaborative swarm of UAV ensures the task at hand is completed, ensuring higher robustness of the

network (Javaid et al. 2023). Data sharing between drones from the surrounding environment is extracted from sensors such as Light Detection and Ranging (LiDAR), camera and GPS to understand the nature of the task collectively.

Multi-UAV communication has been used in various civilian applications like service delivery in a smart city, traffic control, disaster management, and enhancing agricultural capabilities (Gupta et al. 2016). Drone collaborative learning enhances the convergence rate and final accuracy in neural network training, resulting in substantial time and battery savings for image recognition and semantic segmentation tasks (Mashhadi et al. 2024). One approach is Federated Learning, which we will now investigate.

2.1 Federated Learning

Federated Learning is a distributed machine learning model that takes into consideration a huge bulk of data residing in wireless devices like drones (Bonawitz et al. 2019) without physically storing any data. Autonomous devices like drones have certain resource limitations regarding battery life, limited computation, and efficient trajectory. As a result, federated learning (Bonawitz et al. 2019) gives a huge benefit to swarm collaboration of drones. In the case of severe and vital missions carried out by drones, these factors must be addressed. As a result, a distributed learning model proposed by Google called Federated Learning takes into consideration factors such as flight time, energy consumption, and speed in various altitudes and in dynamic surroundings (Brik et al. 2020).

A federated learning model is based on two parts and sections of participation, the first being clients, which are the drones, and the other being the FL (federated learning) server (Brik et al. 2020). A Federated Learning model consists of three sequential steps: first the training initialization by the FL Server, then the model training of the drones, and finally, the global model aggregation.

At first, the FL server initiates the training process by initializing the global model.

G_0 and broadcasting the model weight requirements to the surrounding clients for starting the local model training. All the existing drones available in a network are then trained using local datasets collected from their surroundings. The aggregation of the local models are then carried out by the FL server to compute the global model, ensuring complete data privacy. While updating the local models, the UAV device aims to find the most accurate weight with the lowest value of the loss function through iterative optimization using techniques like gradient descent.

Wireless devices such as drones send their locally trained model weights to the FL Server, which then accumulates the final weight of the global model using the

averaging (*FedAvg*) algorithm (McMahan et al. 2023). The aggregation step is regulated by the server, which selects a certain number of clients in one step, where each of the selected clients breaks their data into smaller segments called batches. With the help of a fixed computed learning rate, the local models are updated using the average gradient. The stochastic gradient descent (SGD) in the distributed network is then calculated to optimize the loss function. The global weight is then sent to the drones for updating the parameters of the models. To ensure complete privacy of the drone network and its underlying information and collected data, a decryption layer using the secure aggregation algorithm (Brik et al. 2020) is also constructed during the aggregation of local models. Figure 1 briefly discusses the conventional federated learning model.

2.1.1 Conventional FL Approaches

Powerful FL approaches have been developed from conventional FL approaches which will be discussed in this respective section. A conventional Federated Learning approach has been proposed by Chen et al. (2020) known as Collaborative FL where drone user equipment (UE) placed far from the network or disconnected from the server can permute the communication with the other drones and the base station (BS) by using Device-to-device communication (D2D). Here the drones share their trained data with their neighbors rather than the base server, thus ensuring more participating drones and more data to train the global model through aggregation.

The Multi-Hop FL model has been proposed by Pinyoanuntapong et al. (2020) where collaboration between drones is carried out using a multi-hop network. Here, UAVs not connected to the base server share their local models through routing policies to their neighboring drones. Here, the aggregation step is carried out before being passed to their neighboring User Equipment (UE) thus ensuring greater robustness and dependence on neighboring UAV devices.

A fog learning based FL has been proposed by Hosseinalipour et al. (2020) to overcome issues of conventional implementation of FL in fog networks. The topology of the drones is taken into consideration along with D2D communication between devices, inducing the heterogeneity of devices connected in the network. To optimize the model, poor resource devices and resource abundant devices are segregated. Whereas the other conventional FL methods consider only star topology between devices, the proposed fog learning is robust to all possible topology of drones in real time. Clusters of nodes of UAV are grouped and local models are shared in between the devices forming multi-layered network. Each layer passes downstream or upstream to other clustered layers with horizontal communication between grouped devices and vertical transmission of model parameters between clustered groups. The model further leverages reduced network traffic, less power consumption, efficient use of spectrum, more mobility of devices in the network, separation of weak and strong nodes with separated datasets and faster aggregation with fewer steps in summing all the local models.

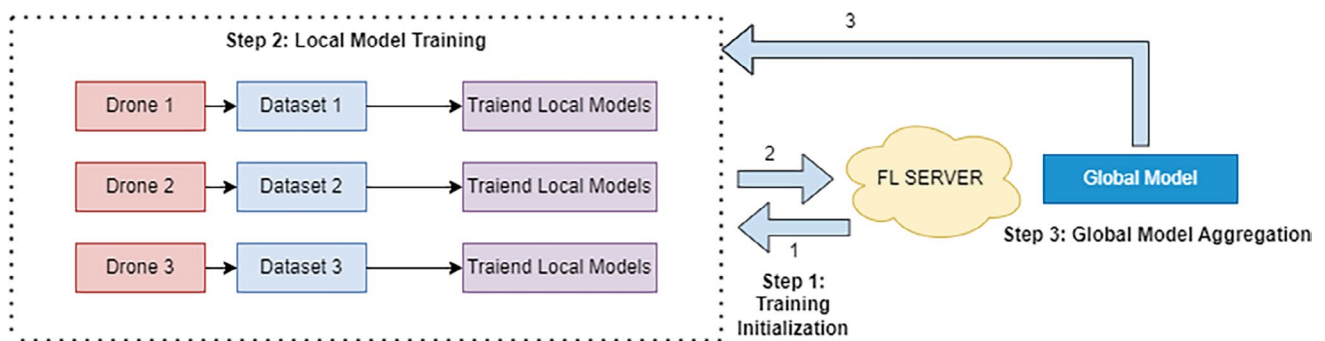


Fig. 1 Conventional federated learning model

Algorithm 1: Federated Learning in a Drone Swarm

Input: Initial global model G_0 , local datasets on drones

Output: Trained global model G

Step 1: Initialize Global Model:

FL Server initializes the global model G_0 with initial weights.

FL Server broadcasts the initial global model weights to all participating drones.

Step 2: Local Model Training:

for each drone i in the swarm **do**

Drone i receives the global model weights from the FL Server.

Drone i collects local data from its environment.

Drone i trains the local model using its local dataset.

Drone i optimizes the model's weights by minimizing the loss function.

Drone i sends the locally trained model weights W_i to the FL Server.

end for

Step 3: Global Model Aggregation:

FL Server aggregates the local model weights from all participating drones:

Compute the average of the local model weights: $W = \frac{1}{N} \sum_{i=1}^N W_i$

Update the global model G with the aggregated weights W .

Step 4: Update Drones with Global Model:

FL Server sends the updated global model G back to the drones.

Drones update their local models with the new global model weights.

Step 5: Iterate Until Convergence:

Repeat steps 2 to 4 until the global model converges or meets the desired accuracy.

Step 6: Ensure Data Privacy:

Implement secure aggregation algorithms to maintain privacy.

Use encryption mechanisms for secure communication between drones and the FL Server.

Controlling UAV in real time to overcome issues like wind dynamics and collisions between UAV devices (Shiri et al. 2020) pose a serious threat to efficiency of UAV in their autonomous features. A federated learning approach using Mean Field Game for controlling UAV (FL-MFG) method has been proposed by Shiri et al. (2020). The proposed method substantially decreases communication between UAVs in a network by introducing a Network based approximator function using Hamilton–Jacobi–Bellman (HJB) and Fokker–Planck–Kolmogorov (FPK) equations. The acceleration and position of the drones are determined by deriving the entire drone population state distribution without using conventional communication between the drones. To decrease the complexity, neural network-based model has been implemented, where both the NN-FPK and NN-HJB has been considered. The UAV control using the proposed FL-MFG method substantially decreases the travel time by 50% and travelling power consumption by 25%. A promising increase in collision avoidance of 50% projects a positive light in the longevity of the devices running in real time.

2.1.2 Case Studies on Federated Learning

A decentralized Federated Learning for UAV Networks (FDN-UN) model has been proposed by Qu et al. (2021) to overcome the need of cloud based conventional Machine Learning approaches integrated with Federated Learning. In place of a central server, the local models of the drones is trained and shared with the one-hop neighbor for global aggregation. It ensures a complete server-less approach, thus ensuring complete privacy of the data passed by the drones since the raw data is not passed to any central entity. As a result, the overall latency decreases, and power consumption also decreases to a minimal value since media such as images and videos need not to be updated to a server.

An energy-efficient Federated Learning model (E2FL) has been proposed by Pham et al. (2022) where a Seminal algorithm for UAV placement, location, energy consumption and latency time of data transmission is taken into consideration. Terrestrial communication between devices poses a challenge due to the physical existence of servers for control of trajectory of drones. Thus, aerial access

networks equipped with wireless powered communication (WPC) boost up the capability of drones flying for a longer duration of time and carrying out power intensive tasks. An exponential function is proposed to find the required harvested power consumption where optimization is carried out by sub dividing the optimization variables into three distinctive parts. A aerial FL server is deployed which aggregates the model updates of the FL user drones through an efficient iterative algorithm E2FL. The proposed algorithm shows energy increases with the increase of completion time of the drones from 50 to 90 and hovering altitude of the drones. With the increase of model size, the energy consumption also increases with high global accuracy rates compared with other benchmarking FL models with high accuracy learning rates.

A federated learning based multi-UAV system has been proposed by Zhang and Hanzo (2020) where image classification is carried out and where a Ground Fusion Center (GFC) is placed in an inaccessible area where recharging of devices cannot be carried out. The federated learning is carried out to reduce the communication overload between devices and UAV. The locally trained models are passed to the Ground Fusion Center (GFC) through a fading wireless channel. The global model is then formulated and passed back to the UAVs from the GFC for future iterations. To facilitate updates on global models, weighted zero-forcing (WZF) transmit precoding (TPC) was issued with a highest accuracy of 95% of image classification tasks.

In the research by Lim et al. (2021) a series of real-life applications of Federated Learning have been presented. The Gboard keyboard by Google for accurately predicting keystrokes has successfully been trained using FL. Sensitive data generated while texting is safely protected since the data is not shared with any central server. In the healthcare sector, the Nvidia Clara FL has been designed for distributed collaboration between hospitals. The personal information of the patients are kept confidential following federated learning as a means of communication between healthcare facilities. The WeBank FedAI uses Federated Learning for collaboration between banks. Thus, they had also able to store the private financial information of their clients through federated learning.

2.2 Swarm Intelligence

Even though FL approaches has proven a powerful mechanism in collaborative learning for UAV, but in case of large-scale co-ordination and further management of swarm of drones, biologically inspired algorithms called Swarm Intelligence needs to be addressed. In real life, living organisms that inhabit groups or swarms exhibit certain unique features and ways of coordinating. The coordination in

groups increases the overall vitality of the organisms in carrying out their day-to-day activities (Sun et al. 2020). Swarm intelligence is a biologically inspired algorithm for collaborative learning between IoT devices within a network. It refers to the act of behaving collectively for a group of decentralized, autonomous and self-organized objects within a network to carry out a specific task. Some classical swarm intelligence algorithms include particle swarm optimization (PSO) and Ant colony optimization (ACO) (Sun et al. 2020), which provide self-adaptation in complex environments. Certain advancements were carried out in Swarm Intelligence algorithms, such as Artificial Bee Colony (ABC) (Karaboga 2005), Bacterial Foraging Algorithm (BFO) (Passino and Passino 2002), and Butterfly Optimization Algorithm (BOA) (Arora and Singh 2019).

In case of UAV, the swarm Intelligence provides decentralized control of drones since UAV in swarms operate independently without relying on a central controller or server. Thus, the drones are able to adapt quickly and efficiently in dynamic environments, making the whole UAV network more robust and less prone to damage. On the other hand, swarm intelligence provides total autonomous behavior in drones without any external instruction. This is ensured since local interactions are put forward and, in return, feedback mechanisms are also implemented, thus making complex operations like search, surveillance, and exploration bring feasibility with UAV.

2.2.1 Review on Conventional Swarm Intelligence Algorithms

IoT is now a major area of research and technological advancements such as wireless sensor network (WSN) (Sun et al. 2020) have opened new doors of solving real-life problems. WSN has indeed increased connectivity among devices for inter-device communication and enhanced decision-making. A WSN network is composed of end points known as nodes which in case of a UAV Network are drones that help in gathering and processing data. Since there are limitations of power and processing capabilities of nodes in WSN, swarm intelligence algorithms come into action. The routing protocol of WSN is enhanced through algorithms such as Ant Colony Optimization (Sun et al. 2020) which helps to find the exact localization of nodes and selection of the cluster head (CH). The SI algorithm further helps in solving the NP-hard problem of finding the CH of a WSN and the error optimization problem of node localization.

Particle Swarm Intelligence: The particle swarm algorithm (PSO) is proposed by Kennedy and Eberhart (1995) which is a biologically inspired random search algorithm that takes into consideration the behavior of a swarm of animals such as birds or fish in cases of foraging. The algorithm is based on basic principles, that state that no

neighboring individuals should collide, the neighborhood will have the same speed flying in a flock, and the entire swarm will have a common target or destination. The fitness value of the members of the flock is calculated by an optimized function based on the direction and distance of flying. First, the position and velocity of particles are initialized randomly in the search space. The local best position of each particle along with the global best position, is initialized. Each particle is evaluated using a fitness function, which is the optimized function based on the current position. The particle velocity and position are updated based on the following Eq. (1). Here, v_{id} is the velocity of the i th particle in the d th dimension, x_{id} is the position of the i th particle in the d th dimension, $pbest_{id}$ is the best-known position of the i th particle, $gbest_d$ is the best-known global position in the d th dimension, ω is the inertia weight

that controls the trade-off between exploration and exploitation, c_1 and c_2 are acceleration coefficients (cognitive and social components, respectively) and r_1 and r_2 are random values between 0 and 1. The best-known position of each particle is updated based on the fitness value; if the fitness value is higher than the current fitness value of the particle, then the new best position of the particle is updated. The global best position of all particles are updated based on the best fitness value of all particles aggregated. Finally, the process is terminated until the best solution is met, thus bringing feasibility with UAV.

$$\begin{aligned} \underline{v}_{id} = & \omega * v_{id} + c_1 * r_1 * (pbest_{id} - x_{id}) \\ & + c_2 * r_2 * (gbest_{id} - x_{id}), x_{id} = x_{id} + v_{id} \end{aligned} \quad (1)$$

Algorithm 2 Particle Swarm Optimization (PSO)

Input: Inertia weight ω , acceleration coefficients c_1 , c_2 , random values r_1 , r_2

Output: Optimal position and velocity of particles

Step 1: Initialization:

Randomly initialize the position x_{id} and velocity v_{id} of each particle in the swarm.
Initialize the local best position $pbest_{id}$ and global best position $gbest_d$.

Step 2: Fitness Evaluation:

for each particle i in the swarm **do**
 Evaluate the fitness of each particle based on its current position.
 if current fitness of particle i is better than $pbest_{id}$ **then**
 Update $pbest_{id}$ to the current position.
 end if
 if current fitness of particle i is better than $gbest_d$ **then**
 Update $gbest_d$ to the current position.
 end if
end for

Step 3: Update Velocity and Position:

for each particle i in the swarm **do**
 Update the velocity v_{id} using the equation:
 $\underline{v}_{id} = \omega * v_{id} + c_1 * r_1 * (pbest_{id} - x_{id}) + c_2 * r_2 * (gbest_{id} - x_{id})$
 Update the position x_{id} using the equation:
 $x_{id} = x_{id} + v_{id}$
end for

Step 4: Termination:

Repeat steps 2-3 until the stopping criteria (e.g., maximum number of iterations or convergence) are met.

Ant Colony Optimization (ACO): The Ant colony Optimization (ACO) proposed by Dorigo and Caro (1999) is a biologically inspired meta heuristic approach based on behavior of ants in the case of foraging to find the shortest optimal path in a specific search space. While ants leave a trail of pheromones on their path in case of food collection, which is traceable for other members of the ant colony, the ants select the path with a higher presence of pheromones. Thus, the whole colony will be able to trace the best path to reach its prime destination, which is the location of the food. The first step of ACO is the initialization of pheromone levels on edges between nodes representing the paths and ants in a network represented in a graph. Nodes in the graph will be the waypoints the ant need to visit, in our case UAV, during a specific mission. An optimization function is defined based on total flight time, maximizing area coverage, minimizing energy consumption, or maximizing the number of tasks completed. UAVs probabilistically choose their next waypoint based on factors such as distance to the waypoint, amount of pheromone on the edge, and the specific mission objective. Additional factors, such as battery level, weight of the payload and mission priorities, are taken into consideration. Pheromones of the edges are updated based on the quality of the solutions found by the UAVs. Stronger pheromone trails should be left on edges corresponding to shorter flight paths or more favorable routes. The process is terminated based on reaching the final target. By customizing the ACO algorithm to address the unique characteristics of UAV operations, it can be an effective tool for optimizing tasks such as route planning, area coverage, and resource allocation in UAV missions. At each step of the algorithm, ants probabilistically select

the next move based on a combination of pheromone levels and heuristic information. The probability P_{ij}^k of ant k moving from node i to node j is typically calculated using the Eq. (2) (Dorigo and Caro 1999) where τ_{ij} is the amount of pheromone on the edge between nodes i and j , η_{ij} is a heuristic value, such as the inverse of the distance between nodes i and j . and α and β are parameters that control the relative importance of pheromone vs. heuristic information.

$$P_{ij}^k = \frac{\tau_{ij}^\alpha \cdot \eta_{ij}^\beta}{\sum_{l \in \text{allowed moves}} \tau_{il}^\alpha \cdot \eta_{il}^\beta} \quad (2)$$

After all ants have completed their tours, the pheromone levels on edges are updated.

The amount of pheromone $\Delta\tau_{ij}^k$ deposited by ant k on edge (i, j) is typically calculated using Eq. 3 where Q is a constant representing the total amount of pheromone deposited by each ant and L_k is the length of the tour constructed by ant k . The objective function defines what constitutes a good or bad solution, guiding the ants towards finding better solutions over time. This function is problem-dependent and might not have a fixed representation.

$$\Delta\tau_{ij}^k = \frac{Q}{L_k} \quad (3)$$

Additionally, pheromone evaporation is applied to prevent pheromone buildup and encourage exploration. The update rule typically looks like the Eq. 4, where ρ is the evaporation rate, which is a value between 0 and 1.

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \sum_k \Delta\tau_{ij}^k \quad (4)$$

Algorithm 3 Ant Colony Optimization (ACO)**Input:** Pheromone levels τ_{ij} , heuristic information η_{ij} , evaporation rate ρ **Output:** Optimal path found by ants**Step 1:** Initialization:Initialize pheromone levels τ_{ij} on all edges.

Initialize all ants at their starting positions in the graph.

Step 2: Path Construction:**for** each ant k **do****for** each node i visited by ant k **do**Select the next node j to move to base on the probability:

$$p_{ij}^k = \frac{\tau_{ij}^\alpha \cdot \eta_{ij}^\beta}{\sum_{l \in \text{allowed moves}} \tau_{il}^\alpha \cdot \eta_{il}^\beta}$$

end for**end for****Step 3:** Pheromone Update:**for** each ant k **do**Deposit pheromone $\Delta\tau_{ij}^k$ on the edges used by the ant, calculated as:

$$\Delta\tau_{ij}^k = \frac{Q}{L_k}, \text{ where } L_k \text{ is the length of the tour constructed by ant } k$$

end for

Apply pheromone evaporation on all edges:

$$\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \sum_k \Delta\tau_{ij}^k$$

Step 4: Termination:

Repeat steps 2-3 until the stopping criteria (e.g., maximum number of iterations or convergence) are met.

2.3 Ensemble Learning

Ensemble Learning is a machine learning approach which combines multiple machine learning base models called 'weak learners' to create a single optimum model with higher efficiency (Zhang and Ma 2012). The ensemble models enhance diversity of the learning of IoT devices to improve overall performance of the model increasing robustness and generalization. The main types of ensemble learning are bagging, boosting and stacking. Since conventional machine learning models have limitations and accuracy vastly dominated by errors, thus ensemble network provides a better alternative since strength of multiple base models are considered. Some of the common disadvantages of using sole machine learning approach includes high bias, high variance and low accuracy for distributed systems (Mienye and Sun 2022). Selection of strongest subset base models is proposed by Caruana et al. (2004). The selection

algorithm starts with an empty ensemble. Then a base classifier is selected from a list of classifiers based on performance metric using a validation set. The step is repeated for n number of iterations for n numbers of base models.

2.3.1 Review on Conventional Ensemble Learning Methods

Bagging is a type of ensemble learning which is also called Bootstrap Aggregating proposed by Breiman (1996), where multiple versions of a model is trained on different subsets of the training set of data for higher accuracy of classification and regression (Nguyen et al. 2021). Bagging provides a wider scope of improvement since it reduces the variance of a model without increasing the bias (Mienye and Sun 2022). Some popular bagging algorithms include Random Forest and Extra Trees classifier. Initially multiple subsets of original data is created based on random sampling. Weak Learner or the base model within the ensemble is trained

Table 1 Comparison between bagging and boosting

Feature	Bagging	Boosting
Objective	Reduce variance	Reduce bias
Proposed by	Breiman (1996)	Schapire (1990)
Training	Parallel, independent	Sequential, dependent
Data sampling	Random with replacement	Entire data-set, adjusted weights
Aggregation	Voting or averaging	Weighted combination
Overfitting	Lower risk	Higher risk
Complexity	Simpler	More complex
Speed	Faster	Slower
Variance	Reduced	Not reduced
Bias	Not reduced	Reduced
Robustness	More robust to noise	Less robust to noise
Algorithms	Random forest, extra trees	AdaBoost, XGBoost
Use cases	High-variance models	High-bias models

separately where in most cases decision trees are considered as base models. Prediction of all base models is combined to find the optimum result.

Boosting is another ensemble learning approach proposed by Schapire (1990) that considers weak learning models and transforms the base models to strong classifiers. Boosting builds models sequentially, with each new model attempting to correct the errors made by the previous ones (Mienye and Sun 2022). All models are then combined to construct a strong predictor for classification tasks. Specifically boosting techniques use the input data to train a weak learner then use the computed predictions from the weak learner to select misclassified training samples and use the adjusted training set that contains the misclassified instances from the previous training round to train the next weak learner. After a certain number of base learners is acquired the iterative learning process is continued and the base learners are then weighted collectively. Reducing bias is the main goal of boosting rather than variance. Consequently, it improves base learners that have a high bias and low variance like decision stumps which are decision trees with a single internal node. Samples that are misclassified are given more weight which makes the 326 base learners concentrate on them. Therefore, in the event that the base classifier exhibits bias towards particular samples the algorithm rectifies the bias by assigning greater weight to those samples. But because the weights assigned

to noisy samples are typically much larger than the weights assigned to the other samples boosting is not appropriate for learning noisy data due to this iterative learning approach. This forces the algorithm to concentrate too much on the noisy samples which leads to over-fitting. Popular boosting algorithms include Adaboost (Freund and Schapire 1999), Gradient Boosting (Breiman 1998), XGBoost (Chen and Guestrin 2016), LightGBM (Ke et al. 2017) and CatBoost (Prokhorenkova et al. 2018). Stacking, also known as generalization, is an ensemble technique proposed by Wolpert (1992) which trains a different machine learning model to combine the predictions of two or more ensemble members. Stacking is used when efficient machine learning models are trained by a separate machine learning model. A comparison of Bagging and Boosting ensemble learning approach has been presented in (Table 1).

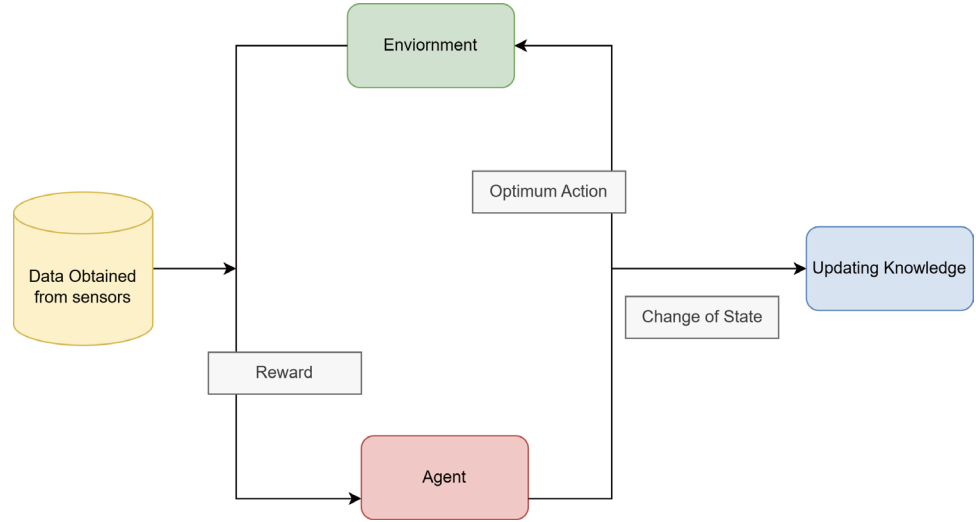
In the research by Nguyen et al. (2021) comparison among ensemble machine learning approaches has been discussed for predicting soil erosion in watersheds. Among the comparisons made, it was concluded that Gradient Boosting Machine (GBM) from an enhanced boosting ensemble learning, and Random Forest were the top performing models for detecting soil erosion. In respect to measuring depth of soil erosion, GBM outperforms Random Forest.

A new ensemble learning approach was proposed by Goudos and Athanasiadou (2019) for predicting signal strength for UAV cellular communication. The new model has been constructed with five base learners Support Vector Machine, Artificial Neural Network, Least Square Boosting (LSBoost), Gaussian Process (GP) and bagging model. The research had been carried out in an urban environment, with higher efficiency compared to the traditional machine learning approaches. The proposed model has been evaluated using the mean absolute error (MAE), root-mean square error (RMSE) and mean absolute percentage error (MAPE). The proposed ensemble method obtained the best results with an RMSE of 6.26 dB, an MAE of 3.54 dB, and an MAPE of 3.92%.

3 Literature Review on Adaptive Learning Techniques of Drone Networks

Drones are UAV devices that adapt to their surroundings and environmental obstacles. Adaptive gives UAV to learn from their surroundings and adjust the behavior in real-time ensuring an optimum performance and navigation. Real time processing of data is carried out by integrating sensor data like cameras, Light Detection and Ranging (LiDAR) and Global Positioning System (GPS) that continuously retrieve data in real time from the environment (Mashhadi et al. 2024). A comprehensive understanding of

Fig. 2 Simple reinforcement learning model



the surroundings can be achieved by combining data from multiple sensors. Reinforcement Learning is one of the approaches which is used to learn optimal actions through trial and error, and receiving rewards or penalties based on their performance.

The research presented by Motetti et al. (2024) introduces an adaptive inference policy method which guides towards efficient pose estimation of nano-drones such as Crazyfire 2.1 nano-drone. Multiple policy-based algorithms such as Output based policy (OP) and two Auxiliary based policies were introduced to increase the accuracy and reduce computation complexity in pose estimation of drones. In the research it was found that the OP policy was far more efficient than the Auxiliary policies for a static Convolutions Neural Network model (CNN). For another random CNN model, the auxiliary based policy method proved promising. The CNN model based on auxiliary based policy showed a maximum latency reduction of 9.2% and the best Mean Absolute Error (MAE) of 0.98.

3.1 Reinforcement Learning (RL)

For adaptive learning for UAV networks, Reinforcement learning has been introduced to learn in a dynamic environment. Reinforcement Learning is a type of machine learning approach where an agent, that learns to take a specific decision based on their surroundings to explore a dynamic environment (Pham et al. 2018). The learning mechanism is based on a cumulative reward based on the performance of learning. An environment is everything an agent or learner interacts with, represented as the current state s of an agent at a specific point of time. All possible moves for an agent is known as action a and the feedback of the action carried out by an agent is known as reward r . Policy p is a set

of rules followed by an agent to select the next action that need to be carried out based on the current state. A value function V is used to determine the expected cumulative reward attained by an agent following a specific policy. The expected return (cumulative reward) starting from state (s) and following policy p is represented in Eq. (5).

$$V^p(s) = E_p \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) | s_0 = s \right] \quad (5)$$

The Q-Function Q is an estimate for probable rewards for a given set of State- and Action S - A pair, using a particular policy. The first step of reinforcement learning is the initialization, which involves an agent to be initiated with a certain policy through randomization. The Q function is initialized to a certain random initial value during the initialization phase. The Interaction is the second step where an agent observes the current state s_t and takes an action based on the policy it has been assigned at. The agent then executes the action a_t in the environment it is in. Following an action taken by an agent, the environment changes to a new state s_{t+1} and then a reward r_t is generated based on the performance of the agent. The next step that follows is the Learning stage, where the agent updates its knowledge based on the reward and new state. This involves updating the policy and the Q-Function. The overall process is repeated for agent to be allowed to receive rewards through interaction with the environment, receiving rewards and updating its knowledge. The Eq. 6 describes the updating of the Q-Function ($Q^p(s, a)$) which is the expected return starting from state s , taking action a , and then following policy p (Fig. 2).

$$Q^p(s, a) = E_p \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t, s_{t+1}) | s_0 = s, a_0 = a \right] \quad (6)$$

3.2 Applications of RL in Leveraging Adaptive Nature in Drones

Accurate landing of UAV like drones has been presented by Vandadari et al. (2018) using a Reinforcement Learning based controller using Least Square Policy Iteration (LSPI) algorithm. The methods also utilized Parallel Tracking and mapping (PATM) based mechanism which gives the drones the ability to track the position and orientation of the camera sensors in real time through creating a map of the environment. The proposed landing algorithm was trained and tested on a Parrot AR Drone 2.0. The training was carried out on a drone simulator and the final testing was carried out on the actual drone, where model information of the drone was not required. Effective use of an RL-based controller was successfully presented in the research in both simulated and real-world environments.

An adaptive Data processing framework has been proposed by Motetti et al. (2024) for sensing dynamic urban environment (ADDSEN) using a RL approach. A knowledge sharing middle ware was constructed using knowledge sharing (POKS) helping in a distributed sharing of data across drone swarms. The system dynamically adjust the broadcast rate of data sharing between UAV drones based on knowledge loss rates. The proposed cyber-physical sensing framework provides effective aggregation and analysis of data through distributed knowledge sharing. The cooperative dissemination strategy further leverages an optimum storage capacity with efficient energy allocation, maximizing overall performance of the drones. The results of the experiment showed that swarm of drones exhibit proper urban sensing tasks with higher knowledge transmission rates, lower knowledge loss rates and more stability in the performance of the drones compared to conventional drone broadcast protocols. To evaluate the proposed framework a swarm of Matrices 100 drones equipped with Zenmuse $\times$ 3 camera was used in traffic monitoring tasks. The CO₂ sensor is used to measure the air pollution rate and prescience of CO₂ in the environment. The results for ADDSEN enabled drone swarm showed a promising mean result of 560 parts per million (ppm).

Adaptive path planning of drones in urban environment has been proposed by Wu et al. (Wu and Low 2021) to conduct coordinated operations of drones. Activities such as air mobility to increase the efficiency and safety of the drones for transportation activities has been discussed. For the path planning method, the airspace of the drones are divided into small cube like structures using the “AirMax” framework. The framework allows the drones to move freely in selected spaces maintaining a safe distance from obstacles. Optimized path points are derived for a swarm of drones with flight safety and efficiency. For changes in the dynamic environment an adaptive path re planning

(APReP) algorithm has been proposed to prioritize tasks along with selection of optimum paths in real time. A Dynamic Rapidly Exploring Random Tree DRRT algorithm was developed for efficient path planning ensuring proper obstacle avoidance in dynamic environment. The ARPep algorithm showed promising computing time of path planning of 0.021 s. In different dynamic changes in the environment, a T-test has been conducted using a confidence coefficient of 0.05 which showed that the ARPep based algorithm performed better than conventional methods.

A drone navigation framework based on RL has been proposed by Pham et al. (2018) using a simple PID controller for positioning on a Parrot AR Drone 2.0 with a Motion Capture System. A simulation using MATLAB was conducted on the drone with a learning rate of 0.1 and discount rate of 0.9. With the increase of episodes the number of states required in first episode was 100 while on the last episode it only took 8 steps, thus increasing the navigation capacity of the drone.

A novel multistage RL model has been proposed by Xiao et al. (2023) known as Adaptive Curriculum embedded multistage learning (ACEMSL) to carry out target search in a dynamic environment. The model proved efficient with constraint observations to navigate effectively in a 3D environment, focusing on knowledge transfer of policies of real-life swarm of drones in real time. Each UAV device is first fed the exact location of the target in a sparse environment. In the experimental results it was found that the algorithm is evaluated by computing the Success Rate (SR) and Time to Reach (TTR). It was found that the Collaborate Target Search (CTS) has obtained more success with TTR of 64% and SR of 44% for two drones and TTR of 48% and SR of 28% for one drone.

A novel Handover (HO) mechanism for drones for a cellular network of drone-User Equipment (UE) has been proposed by Chen et al. (2020). The reference signal received power (RSRP) data has been optimized in this research. In terms of results the minimum serving cell of the RSRP is higher than -85 dB and 28 dB signal to noise Ratio which gives a promising result with regards to connectivity (Table 2).

3.2.1 Deep Reinforcement Learning

Deep Reinforcement Learning is an advancement of RL which combines RL methods with deep learning to process high dimensional data. Some notable Deep RL methods include Deep Q-Networks and policy gradient methods. Deep Q-Learning uses a Neural Network to find the value of the Q-Function and uses experience replay and target networks to train the models more effectively. On the other hand, Policy Gradient Methods optimizes the policy within a RL network using gradient ascent (Munoz et al. 2019).

In the research by Munoz et al. (2019) a flying method of drones is proposed based on the deep RL method. The

Table 2 Summary of reinforcement learning applications

Study	Technique	Findings
Vankadari et al. (2018)	RL-based Controller using LSPI and PATM	Accurate landing of UAVs using an RL-based controller trained on a simulator and tested on a Parrot AR Drone 2.0. Demonstrated success in both simulated and real environments
Motetti et al. (2024)	ADDSEN framework with RL and Middleware	Effective urban sensing with drone swarms, showing high knowledge transmission rates, low knowledge loss, and stable performance. CO ₂ sensor results showed 560 ppm
Wu and Low (2021)	APReP and DRRT algorithms for Path Planning	Efficient path planning in dynamic urban environments with a planning time of 0.021 s. Better performance compared to conventional methods in T-tests
Pham et al. (2018)	RL-based Navigation with PID Controller	Improved navigation capacity of UAVs on a Parrot AR Drone 2.0. Increased efficiency with fewer steps needed in later episodes
Xiao et al. (2023)	ACEMSL for target search	Effective target search in 3D environments with higher success rates (SR) and faster time to reach (TTR) for UAV swarms
Chen et al. (2020b)	Handover mechanism for drone networking	Optimized RSRP data for drone-User Equipment (UE). Higher minimum serving cell (−85 dB) and improved connectivity with 28 dB SNR
Munoz et al. (2019)	Deep RL with Double Deep Q Network (DDQN)	Highest success rate in obstacle-rich environments. Achieved maximum reward of 98% in urban delivery task simulations
Zanol et al. (2019)	Dynamic Trajectories using RL	Efficient dynamic trajectories for multiple drones with average rewards of 0.44. Evaluated using Normalized Cumulative Average cost and SOC simulation

method is based on the double deep Q Network (DDQN) which propagates a drone within an environment rich in obstacles. The method initiates an action based on maximum Q-Value. A simulation of the drones is carried out in ‘AirSim’ environment, based on an urban environment for delivery tasks. In terms of results, three destinations were selected, and the highest rate of success was achieved for the CNN based deep RL model with a maximum reward of 98%.

In the research work by Zanol et al. (2019) dynamic trajectories of multiple drones were predicted using RL for a dynamic environment. The problem discussed in the paper has been represented by the Markov Decision Process, where each drone learns from the environment independently to carry out optimal action. A fixed staging point is selected where the drones are assigned to patrol a defined area. A return of battery recharging is returned, which is modeled using a Gaussian variable. To calculate the efficiency and frequency of the data captured, a two-states Markov chain is used. The proposed model has been evaluated using the normalized Cumulative Average cost and State of Change (SOC) simulation with average rewards of 0.44.

4 Literature Review on Distributed Machine Learning Framework and Tools

To train models composed of multiple nodes or devices such as UAV, multiple frameworks of Distributed Machine Learning (DML) (2024) has been introduced in the last couple of years. The main approach includes a scalable solution of training dataset of large magnitude minimizing computation

complexity. DML framework plays a vital role in implementing abstract concepts of drone networks facilitation high performance computing. Some notable DML frameworks used for drones are discussed in the later part of this section. A summary of all the mentioned frameworks is given in Table 3.

4.1 TensorFlow Federated (TFF) Framework

TensorFlow Federated (TFF) Framework is an open-source framework proposed by Google (Clarke and Moses 2014) for various machine learning and other computational tasks where data security is of prime concern in distributed systems. Some key features of TFF model includes:

1. Federate Learning and computations:

Collaborative Model Training: TFF facilitates multiple devices to collaboratively train a machine learning model without sharing any personal information or raw data. As a result, each device computes updates to the model based on local data and finally through aggregation updated centrally.

- Collaborative Model Training: TFF facilitates multiple devices to collaboratively train a machine learning model without sharing any personal information or raw data. As a result, each device computes updates to the model based on local data and finally through aggregation updated centrally.
- Privacy Preservation: Due to decentralization of data, the preservation of privacy withholding regulatory requirements are maintained within the framework.

Table 3 Comparative analysis of machine learning frameworks for UAV applications

Feature	TensorFlow Federated (TFF) Framework (Clarke and Moses 2014)	PySyft (2023)	Apache Spark (2024)	EdgeML (2024)
Primary focus	Federated learning and computations	Privacy preserving machine learning	Big data processing	Machine learning on edge devices
Collaborative model training	Yes, without sharing personal data	Yes, with no central data access	No direct support for collaborative training	Yes, on edge devices
Privacy preservation	Yes, through decentralized data	Yes, with differential privacy and encrypted computations	Limited, mainly through data processing techniques	Yes, by performing computation on the device
Custom computations	Supported	Supported	Supported	Supported
APIs	Federated Learning API, Federated Core API	Extends PyTorch APIs	Various APIs for data processing, ML, streaming, etc	Lightweight ML algorithms APIs
Real-time data processing	Limited, more focused on federated learning	Limited, focused on privacy	Yes, through Spark Streaming	Yes
Machine learning support	Integrates with TensorFlow models	Extends PyTorch models	Yes, through MLlib	Lightweight ML models
Encryption techniques	Not considered	Secure Multi-Party computation, homomorphic encryption	Not considered	Not considered
Key components	Federated learning API, federated core API	Virtual workers, pointer tensors, plans and protocols	Spark core, spark SQL, spark streaming, MLlib, GraphX	ProtoNN, Bonsai, FastGRNN
Scalability	Scalable across multiple devices	Scalable across multiple devices	Scalable for big data processing	Scalable across edge devices
Use with UAVs	Collaborative training, privacy-preserving computations	Privacy-preserving distributed learning	Processing and analyzing large datasets generated by UAVs	Real-time decision-making on edge devices

- Custom computations: Along with Federated Learning, TFF also supports general computations, thus giving the scope of further implementation of various other algorithms in a federated setting.
- Flexible Execution: The API within the framework enables flexible experimentation and research, allowing both computational logics to be implemented and FL Learning process.

2. Two Main APIs of the Framework:

- Federated Learning API: FL API is a high-level API used for integrating existing TensorFlow models with the Federated Learning process. This API is a good option for standard FL use cases.
- Federated Core API: Federated Core API is a low-level API meant for custom built Federated Algorithms, giving more flexibility and control over advanced use cases and research flow (Table 3).

4.2 PySyft

PySyft is an open-source library (OpenMined: PySyft Blog Posts. 2023) that extends the capability of PyTorch and

other machine learning frameworks in order to expand the capability of preserving privacy in machine learning models. Direct access to a external source of machine learning data is not given to researchers, ensuring data privacy and security. Below are some salient features of PySyft.

1. **PySyft Support for FL:** PySyft enables the FL models to be trained on distributed data, across devices and swarms of drones, thus ensuring no existence of centralization of data.
2. **Differential Privacy:** Differential Privacy is a technique to minimize the exposure of the input data from output data, by adding noise to the gradients during the training phase, which helps to protect individual data from exposure.
3. **Encrypted Computations:** Secured Multi Party Computation (SMPC) is integrated in the environment, which allows multiple UAV devices to carry out computational tasks, ensuring the input data is kept private among the parties.
4. **Homomorphic Encryption:** Encryption mechanism which does not require decryption to enable computation within the environment.

PySyft allows tensors and models to be sent and operated in remote computations thus adding another layer of privacy to the overall environment. Since the data used is protected through maintaining data within the owner, thus no central server is required. The PySyft framework is composed of multiple key parts:

1. **Visual Workers:** PySyft introduces the concept of Visual Workers, which are remote devices that can hold data and models and perform computations in place of the user.
2. **Pointer Tensors:** Pointer tensors are abstract data, that is sent to remote workers which allows users to manipulate and use the data remotely without changing or transferring the actual data.
3. **Plans and protocol:** Plans are sequence of operations that is sent to the worker to be executed, and protocol enables multiple parties to securely complete a Computation.

PySyft helps in creating UAV applications with much enhanced privacy and security, particularly if the handled data is sensitive and requires extra set of protection in collection and processing.

4.3 Apache Spark

Apache Spark is an open-source project under the Apache Software Foundation (2024) used for big data processing for streaming, SQL, machine learning and graph processing. The framework has some benefits that include speed, easiness and advanced techniques of data processing. The Architecture of the framework is discussed below:

- **Driver Program:** The central control of the application is the driver program that communicates with cluster manager and oversees execution of the tasks.
- **Cluster Manager:** Cluster Managers are used to oversee clustered data, and some notable cluster managers include Hadoop, Apache Mesos and custom cluster managers.
- **Worker:** Workers of Executors are nodes present in the clusters which executes assigned tasks. While executing, the workers run an executor which helps in the overall running of subset tasks.

The components of Apache Spark are given below:

- **Spark Core:** The Spark Core lies as the foundation of the platform providing the basic functionalities of the system.
- **Spark SQL:** Querying of the data within the framework is carried out by Spark SQL, where Data-Frames are used to store distributed collections of data organized in columns.

- **Spark Streaming:** For real time data processing, Spark Streaming are used to process real time data integrating mini-batches and performing resilient distributed dataset (RDD) transformations on data.
- **Machine Learning Library (MLlib):** MLlib is a machine learning library that include all conventional machine learning tools and algorithms through a scalable solution.
- **GraphX:** GraphX is a component of the Apache Spark to process Graph data, allowing user to create, manipulate and query graphs.

UAV devices like drones generate huge sums of data, thus processing of the data efficiently is an important aspect of carrying out designated tasks. In this regards, Apache Spark is utilized to analyze and process the generated data efficiently.

4.4 Edge ML

EdgeML is an open-source project (EdgeML 2024) that focuses on bringing machine learning capabilities to edge devices like drones. The primary goal of EdgeML is to enable efficient and scalable machine learning (ML) on resource-constrained devices on Internet of Things (IoT) devices. By leveraging lightweight ML models and algorithms, EdgeML ensures that edge devices can perform complex data processing and inference tasks without relying on cloud infrastructure.

1. Key Features of EdgeML:

- **Lightweight Algorithms:** EdgeML provides a suite of ML algorithms optimized for low-power and resource-limited environments. These algorithms are designed to perform with minimal computational and memory overhead.
- **Efficient Inference:** The project emphasizes real-time inference capabilities, enabling edge devices to make predictions and decisions promptly.
- **Scalability:** EdgeML's solutions are scalable across a variety of edge devices, from simple microcontrollers to more powerful edge servers.
- **Open-Source:** As an open-source initiative, EdgeML encourages collaboration and contributions from the research and developer community.

2. Components of EdgeML:

- **ProtoNN:** A prototype-based neural network designed for classification tasks. It is highly efficient in terms of both memory and computation.
- **Bonsai:** A tree-based algorithm for classification and regression that strikes a balance between accuracy and resource usage.

- **FastGRNN:** An optimized version of recurrent neural networks for sequence modeling, offering a good trade-off between accuracy and efficiency.

A significant increase of productivity of UAV operations can be carried out through EdgeML since the framework allows real-time decision-making capabilities in drones with limited connectivity with special on-boarding processing of sensor data with immediate insights and required actions.

5 Comparative Analysis of Key Studies

For successful collaborative Learning in Drone networks, a list of key studies has been presented in this research. Data transmission among UAV devices helps in carrying out major civilian and military applications. A list of key research on Federated Learning has been elaborated for further analysis. The research by Chen et al. (2020) shows Device-to-Device communicative method has been used to share the trained data with neighboring devices, leveraging the participation of the drones, placed far from the network. As a result, the overall involvement of the drones increases substantially, causing data availability without incorporating any base server. Despite increased participation, this may cause increased overhead of communication and latency in multi-hop scenario. To overcome the limitation, a Multi-Hop Federated Learning method has been proposed by Pinyoanuntapong et al. (2020), where a multi-hop communicative approach has been followed, where UAVs shares local models to their networks through routing policies before aggregation. This method increases robustness to the Federated Learning approach, further increasing reliance on neighboring UAVs. However, the approach introduces a complex model and increases delay due to multi-hop routing. To address the delay in network traffic, a Fog based Federated Learning approach has been proposed by Hosseinalipour et al. (2020), considering power consumption and efficient spectrum use. The model segregates poor and resource abundant devices, resulting in enhanced robustness to various network topology. It further decreases the latency of devices communications among neighboring devices. On the other hand, due to increased segregation of devices, the overall complexity in cluster nodes and layers management increased. To reduce the complexity of node management, the FL-MFG based method by Shiri et al. (2020), facilitating network based approximator function as communicative method. Since navigation of drones has been a challenge in terms of simplicity, the FL-MFG method decreases the communication of devices through deriving state distribution. The

proposed method reduces the travel time by 50%, power consumption by 25% and improves collision avoidance by 50%. The method even though successful in controlling the mobility of UAV devices but lacks in working successfully in real time situations due to increased complexity in facilitating a neural network in real time. A Decentralized FL for UAV Network (FDN-UN) by Qu et al. (2021) approach integrates a server-less method though local model training and one-hop neighbor sharing. This method enhances complete privacy, causes reduced latency and minimal power consumption. Potential issues with scalability and synchronization can still be observed in this method. The energy efficient FL (E2FL) method by Pham et al. (2022a) considers UAV placement, location, energy consumption, and latency further optimizing navigation for aerial networks and thus ensuing increased flight duration, and power-intensive task capability. In this research, a increased model size is achieved, in increases further energy consumption. For multi-UAV navigation in real time systems, a FL-based Multi-UAV System has been proposed by Zhang and Hanzo (2020) in carrying out image classification task with a Ground Fusion Center (GFC) using WZF TPC. A high accuracy on image classification of 95% and reduced communication overload has been achieved by the method, but due to dependency of a Ground Fusion Center for node aggregation and updates, causes a decreased capability of carryout long ranged complex activities.

In terms of collaborative Learning, a biologically inspired algorithm called Swarm Intelligence for collaborative learning has been discussed in the chapter. Swarm intelligence is equipped with decentralized, autonomous and self-organized mechanisms for carrying out precise assigned tasks using conventional algorithms such as particle swarm optimization (PSO) and ant colony optimization (ACO) (Passino and Passino 2002), providing self-adaptation in complex environments. Some advanced algorithms such as Artificial Bee Colony (ABC) (Arora and Singh 2019), Bacterial Foraging Algorithm (BFO) (Lim et al. 2021), and Butterfly Optimization Algorithm (BOA) (Kennedy and Eberhart 1995) has also been introduced in this chapter.

6 Practical Challenges in Integrating Machine Learning Techniques of Drone Enabled IoT Networks

1. **Hardware Limitation:** Drones are inherently constrained by their size and weight, which limits the hardware they can carry. This includes limitations on processing power, memory, storage, and sensor capabilities (Gupta et al. 2016). These constraints can impact

the drone's ability to perform complex tasks, process data efficiently, and store large amounts of information. Potential solutions include:

- a. **Custom Hardware Design:** Develop lightweight, energy-efficient processors and sensors specifically designed for drones. These can be optimized for the unique needs of drone operations, such as high computational power in a compact form.
 - b. **Modular Systems:** Implement modular hardware systems that allow drones to carry only the necessary components for specific missions, thereby reducing weight and power consumption.
2. **Power Consumption:** Drones have limited battery life, which restricts the duration and range of their operations. High power consumption by onboard systems, such as cameras, sensors, and processors, can further reduce flight time. Potential solutions include:
- a. **Energy Harvesting:** Incorporate energy-harvesting technologies, such as solar panels, to extend battery life during flights (Do et al. 2022).
 - b. **Power-Efficient Algorithms:** Develop and implement power-efficient algorithms (Pham et al. 2022) for data processing and communication to reduce the overall energy consumption of onboard systems.
 - c. **Optimized Flight Paths:** Use algorithms to calculate the most energy efficient flight paths, minimizing the power required for propulsion (Pham et al. 2022).
3. **Real-Time Processing:** Real-time data processing is critical for many drop applications, such as surveillance, search and rescue, and environmental monitoring. However, limited onboard computational resources can make it challenging to process data in real time. Potential solutions include:
- a. **Edge Computing:** Implement edge computing to process data locally on the drone (Shi et al. 2016), reducing the need to transmit large amounts of data to central servers and enabling faster decision-making.
 - b. **Distributed Computing:** Use a network of drones to distribute the processing load (Javaid et al. 2023), allowing more complex tasks to be handled collaboratively by multiple drones.
4. **Environmental Factors:** Drones must operate in various environmental conditions, such as extreme temperatures, high winds, rain, and varying altitudes. These factors can impact the drone's stability, sensor accuracy, and overall mission success (Motetti et al. 2024). Potential solutions include:
- a. **Robust Design:** Design drones with robust and durable components that can withstand harsh environmental conditions, such as temperature-resistant materials and waterproofing.
 - b. **Adaptive Control Systems:** Implement adaptive control systems that can adjust the drone's flight

parameters in response to changing environmental conditions, such as wind speed and direction.

With a detailed understanding of machine learning techniques in enhancing collaborative and adaptive communication in drone enabled IoT networks, new domains of security and privacy in data transmission needs to be addressed. It is now of utmost importance to examine how these advancements intersect with security and privacy concerns. As machine learning increasingly drives the capabilities of drones, ensuring that these systems are secure, and that privacy is protected becomes vital for the safe and reliable operation of drone networks. In the following sections, we will explore these critical aspects, highlighting the challenges and strategies necessary to safeguard the technological innovations discussed thus far.

7 Security and Privacy Issues in Drone-Enabled IoT Networks

Drone-enabled IoT networks are increasingly being used in various fields, such as agriculture, surveillance, delivery services, disaster management, etc. (Besada et al. 2019). Different important e-commerce services, such as Pathao, Parcel, Courier, and Foodpanda, in third world countries, especially Bangladesh, are being involved day by day in our regular lives (Wali Ullah and Islam 2017). They may use the service of drones, where traffic jams are the greatest concern. Traffic jam scenarios in Bangladesh and in other densely populated countries need to be sketched out. Around 3.2 men-hour are being lost on a daily basis in Bangladesh (Tribune 2023). On the other hand, Bangladesh and other natural calamities regularly hit. Post incident services become tough to provide, even for defense personnel. So if we provide different services, even the relief provided by drones, then many potential services can be brought out. While these networks offer significant benefits in terms of efficiency and capabilities, they also introduce a range of security and privacy issues. If we consider computer security in a basic sense, then three types of issues need to be considered, (a) authentication, (b) authorization level, and (c) encryption level issues (Qadir and Quadri 2016). Here are some key concerns.

7.1 Security Issues

7.1.1 Data Transmission Vulnerabilities

1. **Interception and Eavesdropping:** Drones often transmit data wirelessly, making them susceptible to interception by unauthorized entities. The encryption of data is

crucial to preventing eavesdropping. Different kinds of feasible encryption standards (Qadir and Quadri 2016) need to be considered that are feasible with respect to power consumption and CPU consumption. Because power is one of the greatest concerns for all kinds of drones. Solar power and battery power also need to be considered.

2. **Jamming and Spoofing:** Attackers can use jamming techniques to disrupt the communication between drones and control stations (Miljanovic et al. 2022). Spoofing attacks can trick drones into following malicious commands by mimicking legitimate signals. They can also cause disasters in power consumption. They might cause computational power even by keeping the drones busy with unnecessary rather than principal works.

7.1.2 Unauthorized Access

1. **Hacking:** Though it is very difficult and not so easy work, if we maintain standard security guidelines, drones can be hacked to gain unauthorized access to the drone's controls and data. This can lead to the theft of sensitive information or malicious control of the drone.
2. **Weak Authentication Mechanisms:** Inadequate authentication protocols can allow attackers to gain control over the drones or the network, highlighting the need for robust authentication methods. Manufacturers and users should place high emphasis on these issues. Possible two-factor authentication will need to be considered. Modern authentication methods need to be abide by duly (Papathanasaki et al. 2022).

7.1.3 Malware and Software Vulnerabilities

1. **Malicious Software:** Computer systems, drones, and their control systems can be targeted by malware, which can disrupt operations or allow attackers to gain control (Papathanasaki et al. 2022). Firmware and system software need to be customized in such a way that no new applications or software can be installed prior to confirmation from the system admin level. Besides that, regular, periodic, authentic, and effective anti-virus systems need to be rolled out on a daily basis. Researchers may pay attention to the operating system level to avoid any kind of so that such drone security features will be scalable at the minimum cost.
2. **Software Exploits:** Exploits in the drone's software or the IoT platform can be used to gain unauthorized access or cause disruptions. Regular updates and patches are necessary to mitigate these risks.

7.1.4 Physical Security Threats

1. **Theft and Tampering:** Drones are susceptible to physical theft or tampering, which can lead to loss of data or

physical damage to the infrastructure. The data and core software may be incorporated in a system like a black box so that they cannot be hijacked until they fully physically damaged (Pfleege 1997).

2. **Damage and Sabotage:** Attackers can physically damage drones to disrupt services or cause accidents. Devices and drones need to be manufactured in such a way that, even if the drones (Papathanasaki et al. 2022).

7.2 Privacy Issues

7.2.1 Surveillance and Monitoring

1. **Unauthorized Surveillance:** Drones equipped with cameras and sensors can be used for unauthorized surveillance, infringing on individuals' privacy. Hence, they need to be properly used prior to proper authentication and authorization for IOT devices (Papathanasaki et al. 2022).
2. **Data Collection:** The extensive data collection capabilities of drones can lead to concerns over the privacy of individuals being monitored, especially if the data is collected without consent. Such kind of data is not only few data, but they are big data. So different data policies and data patterns need to explore with a view to proper handling with the drones (Ling and Draghic 2019).

7.2.2 Data Storage and Access

1. **Sensitive Information:** Drones collect and store a large amount of data, including potentially sensitive information. If this data is not adequately protected, it can be accessed and misused. Database administrators should be trustworthy. There needs to be different levels of security for these sensitive data.
2. **Data Sharing:** The sharing of data collected by drones among different entities can lead to privacy breaches if not properly controlled and anonymous. Hence sharing of data among different drones shall have to follow standard guidelines and rules and regulations to deal such kind of issues.

7.2.3 Regulatory and Compliance Challenges

1. **Legal Compliance:** Ensuring that drone operations comply with privacy laws and regulations is challenging, especially when drones operate across different jurisdictions (Clarke and Moses 2014). Air law, navigation law, civil aviation laws need to be fixed and roll out for all kinds of drones.
2. **Accountability and Transparency:** Lack of transparency in how data collected by drones is used and who has access to it can lead to trust issues and potential privacy violations. National and international act, ordinances need to be strictly maintained (Papathanasaki et al. 2022).

7.3 Ethical Issues

Ethical considerations are crucial when deploying drone technology, especially as they relate to data privacy, ownership, and potential misuse. An in-dept discussion on Data Ownership and privacy is given below:

1. Data Ownership and Privacy:

- **Personal Data Collection:** Drones gather a lot of information, from pictures and videos to various sensor readings. This can include sensitive personal details, such as images of people, locations, and activities. The question of ownership is complex. Is the data owned by the drone operator, the company that made the drone, or the individual captured in the data? This lack of clarity can lead to disputes and misuse.
- **Informed Consent:** Getting explicit permission from everyone who might be recorded by a drone is nearly impossible, especially in public places where people might not even be aware they are being monitored. People have a right to know when and how their data is being collected, but with drones, this can be difficult to manage. How do we balance privacy with the benefits of drone technology?
- **Anonymization and De-Identification:** Efforts to anonymize or de-identify data (such as blurring faces) are important for privacy. However, even with these measures, there's a risk that data could be re-identified or pieced together to reveal identities. The possibility that anonymous data might still lead to privacy breaches raises ethical concerns about how we handle and protect data.

2. Surveillance and Misuse of Data:

- **Mass Surveillance:** Using drones for large-scale surveillance can be controversial. While they can help with security and law enforcement, there's a risk of infringing on people's civil liberties, such as their right to privacy and freedom from excessive government monitoring.
- **Potential for Abuse:**
 - **Personal Misuse:** Drones could be used for harmful purposes, such as stalking or harassment. This misuse could invade personal spaces and cause distress to individuals.
 - **Unauthorized Surveillance:** There's also the risk that drones could be used to spy on people without their knowledge, raising ethical questions about the right to privacy and personal security.
- **Data Misuse by Third Parties:**
 - **Hacking Risks:** Data collected by drones could be stolen or hacked by unauthorized individuals or groups. If this data is sensitive, its misuse could have serious consequences.

- **Unregulated Use:** Companies or entities that access drone data might use it in ways that are not regulated or transparent, leading to potential abuses of that data.

7.4 Mitigation Strategies

To address these security and privacy issues, several mitigation strategies can be employed:

- a. **Robust Encryption:** Implement strong encryption methods for data transmission and storage to protect against interception and unauthorized access. In this case skilled human resources in collaboration with professional and academic researchers need to be involved.
- b. **Secure Authentication:** Use multi-factor authentication and other secure authentication methods to ensure that only authorized individuals can access the drone network. Open problems will be having to be discussed, and they will be have to be fixed with to view to ensuring secure authentication.
- c. **Data Anonymization Techniques:** To protect individual identities and sensitive information the following data anonymization techniques (Svaigen et al. 2022) provides a higher leverage towards protecting the privacy in drone IoT enabled networks. The following techniques are given below:
 - **Data Masking:** Replacing sensitive data with fictional yet structurally similar data.
 - **Generalization:** Reducing the granularity of data, such as reporting age ranges instead of specific ages.
 - **K-Anonymity:** Ensuring that each data entry is indistinguishable from at least $k-1$ other entries.
 - **Differential Privacy:** Introducing random noise to the data to protect individual entries while preserving overall patterns.
A method of implementing K-anonymity in data collected from drone surveillance to ensure that no individual's location or activities to be singled out has been proposed by Neves et al. (2023).
- d. **Regular Software Updates:** Keep all drone software and IoT platforms up to date with the latest security patches and updates to mitigate vulnerabilities. Service from authentic third parties may be taken to ensure such updates. Online journals and log book need to be checked whether all are being performed or not with a zero tolerance.
- e. **Physical Security Measures:** Implement physical security measures to protect drones from theft and tampering. Different types of hardware security dongles may play a significant role in this regard.

- f. **Privacy by Design:** Incorporate privacy considerations into the design and operation of drone systems, ensuring data minimization and anonymization where possible. System designers and architects will keep them into their heads before final launch.
- g. **Regulatory Compliance:** Ensure compliance with relevant laws and regulations regarding data protection and privacy and maintain transparency with stakeholders about data use. Again, it needs to be stated that standard guidelines need to be strictly followed. Regulatory compliance has been reformed with time into set of requirements in different parts of the world. The following compliance laws play a vital role in preserving privacy concerns all over the world.
 - **General Data Protection Regulation (GDPR)** (Regulation (EU) 2016): Ensuring data processing practices adhere to EU privacy standards.
 - **California Consumer Privacy Act (CCPA)** (California Office of the Attorney General: California Consumer Privacy Act (CCPA) 2018): Complying with state-level regulations on data collection and sharing.
 - **Industry-Specific Standards:** Adhering to standards such as NIST SP 800–53 (Standards 2011) for security and privacy controls.

By addressing these security and privacy issues, the potential risks associated with drone-enabled IoT networks can be significantly reduced, allowing for safer and more effective deployment of this technology.

7.4.1 Real-World Challenges and Solutions

Drones collect vast amounts of data for various purposes, and this data collection raises significant privacy concerns, especially when personal or sensitive information is involved. The challenge is to ensure that while valuable data is gathered, it does not infringe upon individual privacy rights or lead to unauthorized use of personal information. The following provides a detailed description of each challenge and corresponding solution related to privacy and security in drone networks.

1. **Balancing the Need for Data Collection with Privacy Concerns:** Privacy by Design (Rest et al. 2014) is an approach that integrates privacy considerations into the design and operation of systems from the outset. This principle ensures that privacy is not an afterthought but a fundamental aspect of system architecture.
2. **Ensuring Data Security in Areas with Limited Infrastructure:** In remote or inaccessible areas, the infrastructure for secure data transmission and storage may be lacking. This poses a risk as sensitive data collected by drones could be exposed or intercepted due

to inadequate security measures. Edge Computing (Shi et al. 2016) involves processing data locally on the drone rather than sending it to a centralized cloud server. This approach addresses several issues related to limited infrastructure and data security.

3. **Keeping Up with Evolving Regulatory Landscapes:** Regulations governing data protection, privacy, and drone operations are continuously evolving. Keeping track of and complying with these changes can be challenging for organizations using drones, especially those operating across multiple jurisdictions. A Compliance Monitoring Framework (Lonzetta and Hayajneh 2018) is a structured approach to ensuring that systems and operations remain in compliance with current regulations and standards.

7.5 Threat Landscape Analysis

A threat landscape analysis for drone-enabled IoT networks involves identifying and understanding the potential security threats, vulnerabilities, and attack vectors that can compromise the integrity, confidentiality, and availability of these systems. Here's a detailed examination:

7.5.1 Potential Security Threats

1. **Interception and Eavesdropping:**
 - **Threat:** Unauthorized entities intercept communication between drones and control systems.
 - **Impact:** loss of sensitive data, compromised operational integrity. Active intruders not only eavesdrop but also make very critical changes in the system. Passive intruders only eavesdrop the system prior to not changes any change in the system. They do it very carefully, and sometimes passive intrusion is more harmful than active intrusion. In this regard both the concepts of active intruder and passive intruder should be kept in mind (Pfleege 1997).
2. **Jamming and Spoofing:**
 - **Threat:** Disrupting communication signals or sending fake signals to drones.
 - **Impact:** loss of control over drones, possible crashes, or diversion of drones to unintended locations.
3. **Hacking and Unauthorized Access:**
 - **Threat:** Gaining unauthorized access to drone control systems or networks.
 - **Impact:** Theft of data, malicious control of drones, disruption of services.
4. **Malware and Software Exploits:**
 - **Threat:** Infection of drone systems with malware or exploitation of software vulnerabilities.
 - **Impact:** Unauthorized control, data breaches, service interruptions.

5. Physical Attacks:

- **Threat:** Theft, tampering, or physical destruction of drones.
- **Impact:** Loss of drones, data breaches, operational disruption.

6. Insider Threats:

- **Threat:** Malicious actions by individuals with authorized access.
- **Impact:** Data theft, operational sabotage, unauthorized information disclosure (Pfleeger 1997).

7.5.2 Vulnerabilities and Attack Vectors

a. Communication Protocols:

- **Vulnerability:** Use of insecure communication protocols.
- **Attack Vector:** Eavesdropping and man-in-the-middle attacks.

b. Weak Authentication Mechanisms:

- **Vulnerability:** Insufficient authentication processes.
- **Attack Vector:** Credential theft and brute-force attacks.

c. Unpatched Software:

- **Vulnerability:** Outdated software with known vulnerabilities.
- **Attack Vector:** Exploitation through known exploits and malware.

d. Insecure Data Storage:

- **Vulnerability:** Storage of data without encryption or proper access controls.
- **Attack Vector:** Data theft via direct access or hacking.

e. Inadequate Physical Security:

- **Vulnerability:** Lack of physical security measures for drones.
- **Attack Vector:** Theft or tampering by physically accessing drones.

f. Sensor Exploitation:

- **Vulnerability:** Exploitable weaknesses in sensors and data input mechanisms.
- **Attack Vector:** Feeding drones false data to misguide operations.

g. Network Vulnerabilities:

- **Vulnerability:** Weaknesses in the supporting network infrastructure.
- **Attack Vector:** Network-based attacks like DDoS, jamming, and spoofing (Pfleeger 1997).

7.5.3 Detailed Discussion on Attack Vectors

1. Eavesdropping and Man-In-The-Middle Attacks:

- **How:** Attackers intercept communication channels using tools like software-defined radios (SDRs) to capture data packets.
- **Mitigation:** Employ strong encryption (e.g., AES-256) for all communications and use secure channels (e.g., TLS).

2. Jamming and Spoofing:

- **How:** Attackers use jamming devices to disrupt GPS signals or use software to spoof legitimate signals.
- **Mitigation:** Use anti-jamming techniques, frequency hopping, and cryptography verification of signals.

3. Credential Theft and Brute-Force Attacks:

- **How:** Attackers use phishing, key logging, or brute-force methods to obtain access credentials.
- **Mitigation:** Implement strong password policies, two-factor authentication (2FA), and regular credential updates.

4. Malware Infections:

- **How:** Through phishing emails, infected USB devices, or exploiting software vulnerabilities.
- **Mitigation:** Regularly update and patch software, use robust anti-malware solutions, and conduct security audits.

5. Physical Attacks:

- **How:** Physically accessing or capturing drones during operation or storage.
- **Mitigation:** Use tamper-evident seals, GPS tracking, and secure storage facilities.

6. Sensor Data Manipulation:

- **How:** Feeding false data to sensors, such as GPS spoofing or environmental sensors.
- **Mitigation:** Implement sensor fusion techniques and validate sensor data through multiple sources.

7. Network-Based Attacks:

- **How:** DDoS attacks to overwhelm network resources or exploiting network protocol vulnerabilities.
- **Mitigation:** Use network security tools like firewalls, intrusion detection/prevention systems (IDS/IPS), and implement redundancy.

Drone-enabled IoT networks are vulnerable to a variety of security threats and attack vectors. Understanding these threats and implementing robust security measures, including encryption, secure authentication, regular software updates, physical security, and network protections, is essential for safeguarding these networks. Regular security assessments and staying updated with the latest security trends are also critical to mitigate emerging threats.

7.5.4 Consequences of Security Breaches Due to Hacking

1. Military Context:

- **National Security Threats:** If drones used by the military are hacked, the implications could be severe. Unauthorized access to military drones could lead to the exposure of classified information, compromise of strategic operations, or even hostile takeovers of drone controls. Such breaches could be exploited for espionage, sabotage, or even direct attacks on military assets or personnel.

- **Terrorism and Espionage:** Hacked drones could be used to carry out terrorist attacks, disrupt communications, or gather intelligence. The use of military drones for surveillance, reconnaissance, and combat means that any compromise could have devastating consequences for national security.

2. Healthcare Context:

- **Patient Safety:** In healthcare, drones might be used to transport medical supplies, deliver vital health services, or monitor patients remotely. A security breach could disrupt these services, potentially leading to harm or even loss of life.
- **Confidentiality of Medical Data:** Hacked drones could leak sensitive patient information, violating privacy and leading to legal and ethical issues. Unauthorized access to health data could also lead to identity theft or discrimination if the information falls into the wrong hands.

7.6 Examination of Potential Security Threats in Drone-Enabled IoT Networks

1. Physical Threats:

- **Theft or Damage:** Drones can be stolen or damaged, leading to loss of both hardware and sensitive data.
- **Interference:** Signal jamming or GPS spoofing can disrupt drone operations, causing navigation errors or loss of control.

2. Cyber Threats:

- **Unauthorized Access:** Hackers can gain unauthorized access to drones or their control systems, leading to potential hijacking or misuse.
- **Data Breaches:** Sensitive data collected by drones can be intercepted or stolen during transmission.
- **Malware Attacks:** Drones can be infected with malware, which can then spread to other devices in the IoT network.

3. Network Threats:

- **Denial of Service (DoS) Attacks:** Attackers can flood the network with traffic, rendering the drone and IoT systems inoperable.
- **Man-in-the-Middle (MitM) Attacks:** Intercepting communications between drones and control systems to alter or steal data.

7.7 Discussion on Vulnerabilities and Attack Vectors

1. Communication Channels:

- **Wireless Communication Protocols:** Weaknesses in protocols like Wi-Fi, Bluetooth, or proprietary connections can be exploited.

- **Data Transmission:** Unencrypted data transmission can be intercepted, leading to data theft or manipulation.

2. Software and Firmware:

- **Outdated Software:** Unpatched vulnerabilities in drone software and firmware can be exploited by attackers.
- **Insecure APIs:** Application Programming Interfaces (APIs) that are not secure can provide entry points for attacks.

3. Hardware Components:

- **Sensor Spoofing:** Sensors on drones can be tricked into providing false data, leading to incorrect decisions or actions.
- **Inadequate Encryption:** Lack of encryption in hardware components can expose sensitive data to interception.

7.8 Security Measures and Protocols

7.8.1 Strategies for Mitigating Security Risks and Protecting Drone-Assisted IoT Networks

1. Robust Authentication and Authorization:

- **Multi-Factor Authentication (MFA):** Implementing MFA to ensure that only authorized users can access drone systems.
 - **Strengthening Access Controls:** Implementing MFA adds an extra layer of security by requiring users to verify their identity through multiple methods (e.g., password, bio-metric verification, or a security token). This makes it more difficult for unauthorized users to gain access to drone systems, even if they manage to steal a password.
 - **Reducing the Risk of Unauthorized Access:** MFA can significantly reduce the likelihood of breaches by making it harder for hackers to gain control over drones. It ensures that even if one authentication factor is compromised, additional layers protect the system.
- **Role-Based Access Control (RBAC):** Restricting access based on user roles to minimize the risk of unauthorized access.

2. Data Encryption:

- **End-to-End Encryption (E2EE):** Ensuring that data is encrypted during transmission and at rest to protect it from interception and theft.
- **Public Key Infrastructure (PKI):** Utilizing PKI for secure communication and authentication between drones and control systems.

3. Regular Software and Firmware Updates:

- **Patch Management:** Regularly updating software and firmware to fix known vulnerabilities and enhance security features.

- **Automated Updates:** Implementing automated update mechanisms to ensure that all devices are consistently protected.

4. Advanced Encryption Techniques:

- **Protecting Data in Transit and Storage:** Encrypting the data that drones collect, store, and transmit ensures that even if the data is intercepted, it cannot be easily deciphered by unauthorized parties. This includes encryption of communications between the drone and control stations, as well as any data stored on the drone itself.
- **Secure Communication Channels:** Using end-to-end encryption ensures that data remains secure throughout its journey from the drone to its destination. This is particularly important in military and healthcare contexts where data sensitivity is high.

7.8.2 Implementation of Security Measures and Protocols for Ensuring Data Integrity and Confidentiality

1. Secure Communication Protocols:

- **VPNs and Secure Tunnels:** Using Virtual Private Networks (VPNs) and secure tunneling protocols to protect data transmission.
- **TLS/SSL:** Implementing Transport Layer Security (TLS) or Secure Sockets Layer (SSL) to encrypt communications.

2. Intrusion Detection and Prevention Systems (IDPS):

- **Anomaly Detection:** Using IDPS to detect unusual activities or potential intrusions in real-time.
- **Response Mechanisms:** Establishing protocols for responding to detected threats, such as automatic shutdown or isolation of compromised drones.

3. Physical Security Measures:

- **Tamper-Resistant Hardware:** Designing drones with tamper-resistant hardware to prevent unauthorized physical access.
- **Secure Storage:** Ensuring that drones and their data storage components are securely stored when not in use.

4. Redundancy and Fail over Mechanisms:

- **Backup Systems:** Implementing redundant systems and fail over mechanisms to ensure continuous operation in case of a failure or attack.
- **Data Backup:** Regularly backing up data to secure locations to prevent data loss in case of a breach or hardware failure.

By addressing both the potential threats and implementing robust security measures, drone-assisted IoT networks can be significantly fortified against a wide range of security risks, ensuring data integrity and confidentiality.

8 Future Directions

Some key areas for future exploration include:

1. **Advanced machine Learning:** Continued research into advanced machine learning algorithms, such as deep learning, reinforcement learning, and federated learning, can further enhance the autonomy and decision-making capabilities of drones. These algorithms can enable drones to learn from large datasets, adapt to changing environments, and optimize their performance in real-time.
2. **Edge Computing and On-Device Processing:** Exploring edge computing and on-device processing capabilities can enable drones to perform complex computations locally, reducing latency and dependence on cloud-based services. This can enhance the efficiency and responsiveness of drone networks, especially in scenarios where real-time decision-making is critical.
3. **Interoperability and Standardization:** Developing interoperable and standardized protocols for communication and data exchange among drones from different manufacturers can improve collaboration and coordination in heterogeneous drone networks. This can facilitate seamless integration of drones into existing infrastructure and enable more efficient resource sharing.
4. **Ethical and Legal Frameworks:** Establishing ethical guidelines and legal frameworks for the use of drones in various applications is essential to ensuring responsible and ethical deployment of this technology. Addressing privacy concerns, data security, and accountability issues will be crucial in building trust and acceptance of drone networks in society.
5. **Resilience and robustness:** Enhancing the resilience and robustness of drone networks against cyber-attacks, signal jamming, and physical threats is essential for ensuring the reliability and security of operations. Developing adaptive security mechanisms and redundancy strategies can mitigate risks and safeguard drone networks in hostile environments.
6. **Human-Drone Interaction:** Investigating human-drone interaction mechanisms, user interfaces, and communication protocols can improve the usability and acceptance of drones in various user scenarios. Designing intuitive interfaces and user-friendly controls can empower users to interact effectively with drone networks and leverage their capabilities.

9 Conclusion

With the advancement of IoT technologies and the integration of Machine Learning Techniques for drone-enabled networks through a secured and private environment,

opened new fields of research domains. Applications ranging from agriculture and infrastructure planning to disaster management and surveillance monitored by UAV devices have greatly influenced in improving the day-to-day activities in human life. This chapter explores the various machine learning technologies and addresses the security challenges associated with drone networks. As drone networks evolve, the Integration of machine learning into these systems becomes increasingly important. Leveraging machine learning, along with IoT and collaborative learning methods, can significantly boost the autonomy, efficiency, and adaptability of drones across a wide range of applications. This combination not only enhances drone capabilities but also strengthens their security features, showcasing the promising future of these technologies. Incorporating machine learning into drone networks brings new opportunities for data-driven decision-making, real-time analysis, and predictive capabilities. This allows drones to gather essential information, process data effectively, and enhance their operational performance across various scenarios. By analyzing the vast data collected, drones can achieve better situational awareness, predictive maintenance, and improved mission planning. Additionally, the collaborative aspect of drone networks, supported by machine learning algorithms, promotes distributed data sharing, task allocation, and swarm intelligence. This collaborative strategy increases the accuracy, resilience, and efficiency of drone operations, making them more capable in areas such as traffic control, disaster response, and agricultural monitoring. Future advancements in machine learning for drone networks offer significant potential for enhancing drone capabilities and security features. Collaborative efforts among academic institutions, industry leaders, and regulatory bodies will play a crucial role in shaping the future of drone technology and unlocking its full potential for both civilian and military uses. By leveraging the opportunities presented by machine learning, drone networks can develop into advanced, intelligent systems that support safer, more efficient, and sustainable operations across various industries. The ongoing effort to improve drone networks through machine learning is set to transform how drones are utilized in today's world.

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Unlocking the Potential of Machine Learning for Enhanced Security in Drone-Enabled IoT Networks

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Abstract

The integration of drones into Internet of Things (IoT) ecosystems has revolutionized various domains, offering unparalleled mobility and adaptability for data collection, monitoring, and surveillance. However, drones are susceptible to security vulnerabilities, stemming from network infrastructure weaknesses, software vulnerabilities, and potential malicious intent. Unsecured drones are vulnerable to hacking, data theft, and manipulation, posing risks to public safety and privacy. Protecting against these susceptibilities is paramount to ensure the safe and responsible integration of drones into society. Incorporating sophisticated Machine Learning (ML) techniques into drone networks presents a promising solution to mitigate security risks effectively. ML algorithms analyse real-time data to detect anomalies and predict threats, enabling prompt responses to potential risks. Machine Learning powered drones can autonomously adapt security measures to evolving threats, enhancing encryption protocols to resist tampering and interception. The chapter explores the recent privacy and security concerns surrounding drone networks and discusses the integration of ML-based security solutions to alleviate risks. Challenges and opportunities in ML-driven drone security are also explored, emphasizing the importance of ensuring safe and secure drone operations in the evolving landscape of IoT ecosystems.

Keywords

Internet of drones · UAV · Security · Attacks · Countermeasures · Machine learning

1 Introduction

To concept, picture a world where drones, those shiny flying gizmos we encounter in the movies or on television are no longer novelties for enjoyment by enthusiasts, but rather essential devices in the daily lives of the populace. Far from merely being recreational tools used to produce spectacular aerial photography, or commercial vehicles for transporting packages and other shipments ordered over the Internet, these Unmanned Aerial Vehicles—UAVs—called drones today, have numerous applications in various fields, ranging from commercial to military and scientific. They have transformed into highly relevant tools in many sectors including farming, emergency response, medicine, security and surveillance. The IoT ecosystem encompassing drones, sensors, microcontrollers for data exchange and Ground Control Stations (GCS) communicating over the Internet known as the Internet of Drones (IoD) has boosted these applications and opened up new opportunities (Lakshman and Ebenezer 2021; Boccadoro et al. 2021; Abualigah et al. 2021). However, this kind of reliance has consequential implications on issues to do with privacy and security (Fig. 1).

Drones, while revolutionary in their capabilities, are prone to security vulnerabilities due to several factors. These include vulnerabilities in the network infrastructure, drone systems, and IoT devices, as well as malicious intent or inadvertent actions by individuals or organizations (Majeed et al. 2021; Tsao et al. 2022). Ensuring the safety of drone operations is paramount. Unsecured drones are susceptible to hacking or unauthorized access. Protecting the privacy of individuals and organizations is essential. Drones

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Fig. 1 Application domains of internet of drones

equipped with cameras and sensors can collect sensitive data. Without adequate security measures, this data could be intercepted or exploited, leading to privacy breaches or potential misuse of personal information. Hackers can intercept or jam the signals between the drone and its controller, disrupting control or stealing sensitive information. Attackers may tamper with the drone's firmware to modify its behaviour or install malicious software, posing significant security risks to the drone and its operators. GPS spoofing can lead to misdirection or a complete takeover of the drone's flight path. Likewise, different kinds of attacks on drone-enabled IoT networks can be anticipated (McClure and Krentz 2022; Yaacoub et al. 2020). Thus, ensuring that drone operations cannot be breached and penetrated by unauthorized personnel is a major necessity to protect these flying machines and the people benefitting from them.

Security risks can be effectively mitigated by incorporating sophisticated Machine learning techniques into the Drone networks making it more protected and secure. Machine Learning algorithms can analyse vast amounts of data collected from drones in real-time to detect anomalies, identify patterns, and predict potential security threats. Moreover, Machine Learning powered drones can adaptively adjust their security measures based on evolving threats to autonomously respond to security incidents. The integration of Machine learning based counter and preventive measures for drone security will enable the safe integration of drones into various domains.

This chapter discusses the recent privacy and security concerns influencing Drone networks. A taxonomy of different types of attacks is provided with the need to address those attacks. Various approaches and techniques

of Machine learning will be explored to alleviate vulnerabilities in Drone security emphasizing the importance of integrating Machine Learning based security solutions into operational drone ecosystems. Further, the challenges and opportunities in integrating Machine Learning into drone security will be discussed.

2 Overview of Layers in the Internet of Drones

With most drones, owners and operators communicate and control drones using applications in smart devices such as smartphones, software developed for web use, or specially designed applications. It includes services such as delivery of packages, security, mapping, and agriculture, with all these services being supported by drones in their respective tasks. Furthermore, APIs featured within the drone ecosystem allow for third-party services to not only interface with, but also function within the ecosystem itself, thereby adding utility and diversifying the service offerings. The basics of the Internet of Drones as in its physical architecture and its main components are detailed as follows. It offers a clear concept of how each layer works and how each component interfaces with the other components to provide the IoD environment (Boccardo et al. 2021; Abdelmaboud 2021).

Service Layer. Service provision by drones is an organized procedure whereby the organization manages the flow of services offered concerning the timetable of services, the use of resources and service succession among other factors. Business Intelligence involves identifying parts and patterns from information which is received from drones for

decision making and with the use of applications such as machine learning for pattern recognition, anomalous observations, and predictions. Security services ensure accuracy, integrated Autonomous Swarm Networks (ASNs), coherency and integrity of data which in turn use the approved encrypted, authenticated, and authorized method.

Network Layer. Communication networks facilitate the interfacing of drones with ground control through distinct networks like 5G, LTE, Wi-Fi and satellite networking systems. Networks of these networks are controlled to address the access of resources, moderate bandwidths and guarantee correct data passing, including the principles of network slicing and administration of Quality of Service (QoS). It also deals with coordination between networks like cellular, satellite and local networks to schedule or coordinate their operations, functions etc., for smooth communication.

Control Layer. Mobile airborne platforms are operated remotely from the ground through Ground Control Centre (GCC) or with GCS that provide directions, security and additional command and control facilities. Air Traffic Control (ATC) is responsible for ensuring that drones in the skies are given proper routing.

Data Layer. Data layer is the foundational component for managing, processing, and analyzing the huge amount of data produced by drones and their associated systems. The storage solutions suggested are centralized storage facilities, cloud-based storage and Edge computing storage. The policy can help adopt relevant data quality management policies such as data governance and data stewardship because of the importance of data and its storage.

Device Layer. Physical UAVs or drones combine different sensors, cameras, and communication modules and are applied to various uses. The ship's own sensors, for example, GPS, LiDAR and cameras, gather data from the physical operating environment excluding interactions with people. Motors, servos, and robotic arms which are in essence, actuators, are commonly used to operate drones, making them move and act in a predetermined way.

3 Security and Privacy Requirements in IoD

This section covers recent emerging privacy and security requirements for drone networks. Documenting these concerns also reveals their relevance to the protection and proper functioning of the drone system. It is essential to understand the threat space of Unmanned Aircraft systems (UAS) which encompasses fundamental nodes: operators and drones, the interaction fields between the human operators, the drone, and the drone environment, which shows allowable communication zones. Figure 2 depicts the UAS Attack-Surface Illustration Template inspired by Best (2020). It outlines and distinguishes the concept of 'attack surfaces' which is defined as areas accessible to attack from 'attack vectors,' which are means of access to these areas. It also embeds more nodes, like the navigation software, and controller, showing possible attack vectors.

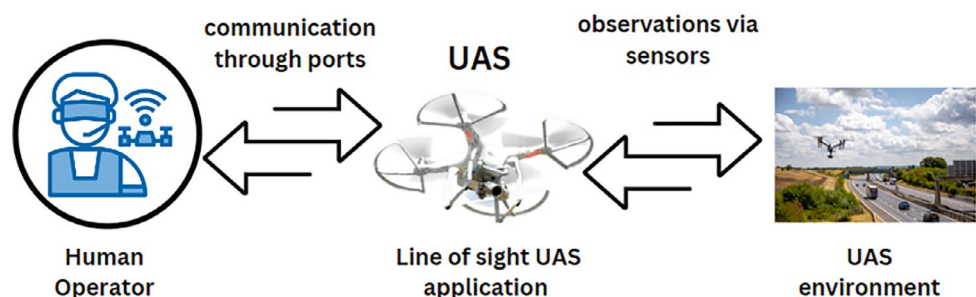
For the confidential, integral, and compliant protection of the information gathered and transmitted by IoD, it is necessary to protect the data from being exposed to any unauthorized person who may attempt to delete or alter it among other interactions. The Federal Aviation Administration (FAA) requirements must be extended to cover other security agencies and supporting aviation entities that deal with drones' manufacturers and operators. These include privacy advocates/activist groups, drone operators and manufacturers as well as aviation organizations supporting catapulting.

Critical security requirements (Yang et al. 2022; Samanth and Balachandra 2022; Ayamga et al. 2021; Nikooghadam et al. 2021; Yahuza et al. 2021; Choudhary et al. 2018) for secure operations of drones should include,

Authorised Access. Drones need to ensure that only the persons with a right of operation will be provided with access to the components of the UAV system among which there are GCS and aircraft. With regards to the maturity of the GCS, it is evident that there is a requirement to

Fig. 2 UAS attack-surface illustration template

UAS Attack-Surface Illustration Template



implement optimal authentication procedures, as well as compulsory access control measures that would prevent penetration into the system by other unauthorized personnel. As the two engage in their constant communication, both the operator and the drone must verify their identities with each other at all times. They also apply distance bounding protocol for proximity security. This helps in combating spoofing attacks because unless the spoofer is in the proximity of the legitimate party, he or she cannot open the attack practically.

Availability. To meet the required performance levels in operational space and time throughout its period of usage, elements of the UAV system should be available. Some of the procedures like the anomaly-based intrusion detection systems must be implemented to differentiate between normal messaging and messaging that is undertaken due to Denial of Service. Moreover, there are ways by which one could introduce redundancies into the system which would allow the flight control system to cope with specific malfunctions or alterations to the selected components of the system. It remains important to efficiently handle the updating processes to prevent the availability of the system during operation from being affected.

Information Confidentiality. It is undesirable for the telemetric, and control information to be communicated by some unauthorized parties. Hence, there is a need to use proper encryption systems to encrypt the data link. This makes it possible that an error such as supplying wrong account details cannot easily occur during data communication.

Information Integrity. The GPS signals and the control signals received through telemetric information by the drone systems, need to be authenticated to ensure the signals are genuine and have not been altered. The authenticated encryption cryptographic primitives should be employed so that both the integrity and confidentiality of the information be preserved.

System Integrity. The authenticity of components of the software and the drone hardware or any part of the entire system needs to be ensured. For instance, memory curtaining can offer assurances that only trusted code is executing, sealed storage can be employed to check on the content integrity and remote attestation can be helpful in ensuring that offered assurances remain valid. Infestation prevention can also be prevented by the use of the following detection programs- Intrusion Detection System, Anti-viruses, Firewall and strict policies on the use of external storage devices.

Accountability of Actions. From the aspect of non-repudiation to consolidate accountability and individual control on operators, there should be effects like digital signatures initiated for authenticating the operators and linking

them with the actions provided to them. There should also be time-lined logs that contain all the necessary details of each event and action as well as the system modifications.

Security Architecture. A systematic integration of several levels of security services working within the elements of the UAV system should be proposed to fulfil the requirements of a full set of security solutions for UAV security. These services are a shield and slip through threats that are well-known and other threats that are in emerging stages.

Physical Security Services. The physical operating components such as the control modules, buses, and computers are protected against intentional or accidental destruction. This involves optimizing the supply chain management of the UAV systems and securing gadgets not to be tampered with.

Software-Based Security Services. Several security services for software-based security are required such as Anti-malware service, Access control service, services for creating a secure environment for updates, Firewall service, and Intrusion Detection service.

4 Taxonomy of Attacks on the Internet of Drones

The IoD lacks a well-defined framework for identifying its critical infrastructure. Common targets of attacks typically include various components of the IoD, such as the devices themselves, the networks these devices operate on, and the connections between them. These threats significantly contribute to security breaches, resulting in substantial losses of resources, trust, and availability. These vulnerabilities have made it increasingly easier for attackers to compromise systems. The main categories of attacks (Yaacoub et al. 2020; Yang et al. 2022; Choudhary et al. 2018; Derhab et al. 2023; Alrefaei et al. 2022; Guo et al. 2020; Global 2021; Dey et al. 2018) are listed as follows.

Communication Attacks in IoD

- **Intercepting Communications.** These attacks enable attackers to intercept data exchanges between IoD devices and sensors. One specific type, Rogue Access Points masquerades as a legitimate access point to nearby devices gaining permission to control traffic flow. Preventive measures include employing strong encryption, HTTPS, and PKI-based authentication to secure communications.
- **Man-in-the-Middle Attack.** This attack takes place between a drone and the GCS. In this attack, an attacker intercepts and possibly alters the communication between drones and their control systems. By

intercepting and altering messages between the drone and GCS, the attacker can eavesdrop on communications or inject malicious commands to control the drone.

- **Identity Spoofing.** Attackers often target the identities of legitimate users of the IoD to breach security through impersonation. An attacker can impersonate a legitimate customer to gain access to the IoD system. However, a malicious drone can mimic the identity of an authorized drone, thus communicating with other drones or the GCS. To mitigate these risks, some possible solutions include employing encrypted or so-called one-time-use pseudo-IDs. These measures can help enhance security by making it more difficult for attackers to exploit identity-related vulnerabilities.
- **Unauthorised Intrusions.** This occurs in cases where malicious actors hack into IoD servers and services, potentially resulting in the leakage of critical information.
- **Replay Exploits.** These attacks involve intercepting and capturing a valid data transmission, then retransmitting it to deceive the recipient into performing an unauthorized action. For instance, an attacker can capture control commands sent to drones and can replay them to repeat actions. As a result, this can lead to a crash of drones or other unintended behaviour.
- **Information Tampering.** This is accomplished by injecting falsehoods into messages through various forms of data alteration during transmission. These alterations include administrative or misrepresented changes, fabrication, substitutions, and data injections, all of which can provide users with false information.
- **Unauthorised Eavesdropping.** Since the targeted communications are real-time, interception could lead to the exposure of private information. This risk is exacerbated by the lack of authentication and encryption, leaving the information vulnerable to interception and disclosure.
- **Flooding Attacks.** It is a form of Denial-of-service (DoS) attack where a barrage of random commands to the drone's receiver system is sent by attackers, thereby hindering it from executing authentic commands. This attack might consume the victim's computation power and possibly discharge its battery. They jam the communication channel consuming the bandwidth by periodically sending random commands or control signals.

Signal Manipulation Attacks in IoD

- **GPS Spoofing.** GPS is utilized for coordinating the location of drones and planning their movement. An attacker could impersonate the GPS satellite in transmitting incorrect GPS signals to the GPS receiver and this

could make the GPS receiver compute the wrong geographic coordinates. Attackers can interfere with GPS signals or generate spoofing signals, leading to unsynchronized events and mid-air collisions. This way, any malicious person could get hold of the drone and change its flight path. To mitigate these risks, measures such as using authenticated and encrypted signals and constantly assessing the strength of GPS signals should be implemented.

- **GPS Jamming.** This can be achieved by limiting the GPS reception or by interfering with the signals so that the drone may not receive accurate positioning data and will be forced to crash or deviate from its intended course.
- **Signal Jamming.** Signals are easily jammed, and the attacker interferes with communication between drones or between a drone and the ground control station, which could disable the communication and therefore put the system out of service. It could also disrupt drone operations as important command messages could not reach the drone, leading to crashes or collisions.
- **Signal Spoofing.** In this, the attacker sends false signals to deceive a receiver into accepting false data as legitimate. It is an intelligent form of interference making the drones believe that they are in a false location. As signals are unencrypted and unauthenticated, the transmission systems cannot confirm their authenticity, and hence spoofed signals from the drones to GCS or vice versa could negatively disrupt the drone operations.

Data Attacks in IoD

- **Manipulation of Captured Footage.** Manipulation of captured footage involves altering recorded video or images captured from drones. This attack primarily targets low-altitude drones because these drones rely on visual feedback from their attached cameras for navigation and manoeuvring to avoid obstacles. Advanced knowledge of the flight control system allows the creation of a different actual video with a fake one, and the attacker may transmit only the manipulated information received.
- **Third-Party Trust Violations.** In IoD communications, third parties can be unreliable. This involves breaches of trust with external entities accessing or managing drone-related data, services, or systems selling or disclosing information.
- **False Sensor Data Injection.** Injecting hostile data into the drone's sensors can disrupt its functionality. Sensors such as infrared, radar, and electro-optical ones can be manipulated. This attack can lead to misinformed decision-making and compromised drone operations.

Device Attacks in IoD

- **Physical Sabotage.** These attacks target the hardware components of IoD devices with the intention of damaging or destroying them. It involves deliberate actions to physically damage or disrupt the operation of drones or their associated hardware components. As drones are valuable assets in IoD applications, protecting them against physical sabotage is essential to ensure their proper operation and security.
- **Deployment of Counterfeit Devices.** Deployment of counterfeit devices introduces unauthorized drones in the field. Counterfeit devices imitate real ones, and they compromise activity information that is sensitive during important operations.
- **Internal Threats with Key Loggers.** Keylogging software can be internally planted, such as in the IoD version, and then used to forward sensitive information to attackers.
- **Malicious Firmware Updates.** During a mission or in its process, IoD firmware or software can be installed with malicious code, compromising entire sessions of IoD operations.
- **Manipulation of Flight Controller.** If an attacker can unduly influence the parameters of the flight controller by feeding in incorrect data or by installing a virus on the system, then the drone is vulnerable to disruption or complete compromise.

Cloud Attacks in IoD

- **Cloud Data Breaches.** Data on the cloud could be compromised due to various attacks. Denial of Service, cloud service misconfiguration, data breaches, and data manipulation are a few examples of types of attacks targeting cloud data.
- **Model Manipulation.** Adversarial attacks on machine learning models hosted in the cloud can lead to altered drone behaviour or compromised decision-making in IoD applications (Fig. 3).

5 Security Countermeasures: A Machine Learning Perspective

The Internet of Drones is vulnerable to many types of attacks, and machine learning is a powerful tool for mitigating these threats. Understanding the various countermeasures available for security is crucial for protecting against both cyber and physical threats in the IoD ecosystem. This section categorizes and briefs different strategies to mitigate these threats unlocking the potential of Machine Learning. We focus on a few prevalent threats and their machine

learning-based mitigation strategies offering a clearer understanding of how machine learning can effectively address these critical issues.

5.1 Preventive Countermeasures

Countermeasures utilize Artificial Intelligence to detect probable cyber-attacks and prevent them in a bid to safeguard vulnerabilities in the drone system.

Firewalls

Firewalls can be used to protect drone control systems and communication channels, preventing unauthorized access and data theft (Shafik., Wasswa., S. Mojtaba Matinkhah., and Fawad Shokoor. 2023). Standard firewalls are good at ensuring simple security policies are obeyed at the network layer by analysing packets to discern whether they should either allow or deny them as per predefined rules. However, they face several challenges:

- **Sophisticated Attack Detection.** Current firewalls are based on the principles of traditional firewalls and can only detect exact matches of well-known signatures, which are extremely disadvantageous in dealing with new and fast-changing attacks.
- **High Traffic Volumes.** Large quantities of data flow through the Internet can put a load on previous generations' firewalls rendering it slow or even causing datagram loss.
- **False Positives.** It might be difficult to keep balanced accuracy in threat detection as creating numerous rules, which produce low rates of false positives, might be problematic or not give proper segregation between legitimate traffic and threats.

Machine Learning Solutions for Firewalls

Anomaly Detection. Machine learning algorithms can be iteratively trained to study network traffic flow over a period of time and be able to recognize unusual activities that may be causing a network breakdown. This is convenient for identifying novel attacks and new and viral behaviour that static rules may not pick out. Some algorithms and approaches used are as follows:

- **Variational Autoencoders (VAEs).** These are generative models that learn the pattern of activities typical in the network and thus identify activities that deviate from the normal standards. VAEs are a good choice for anomaly detection when no supervision signal is available for training.

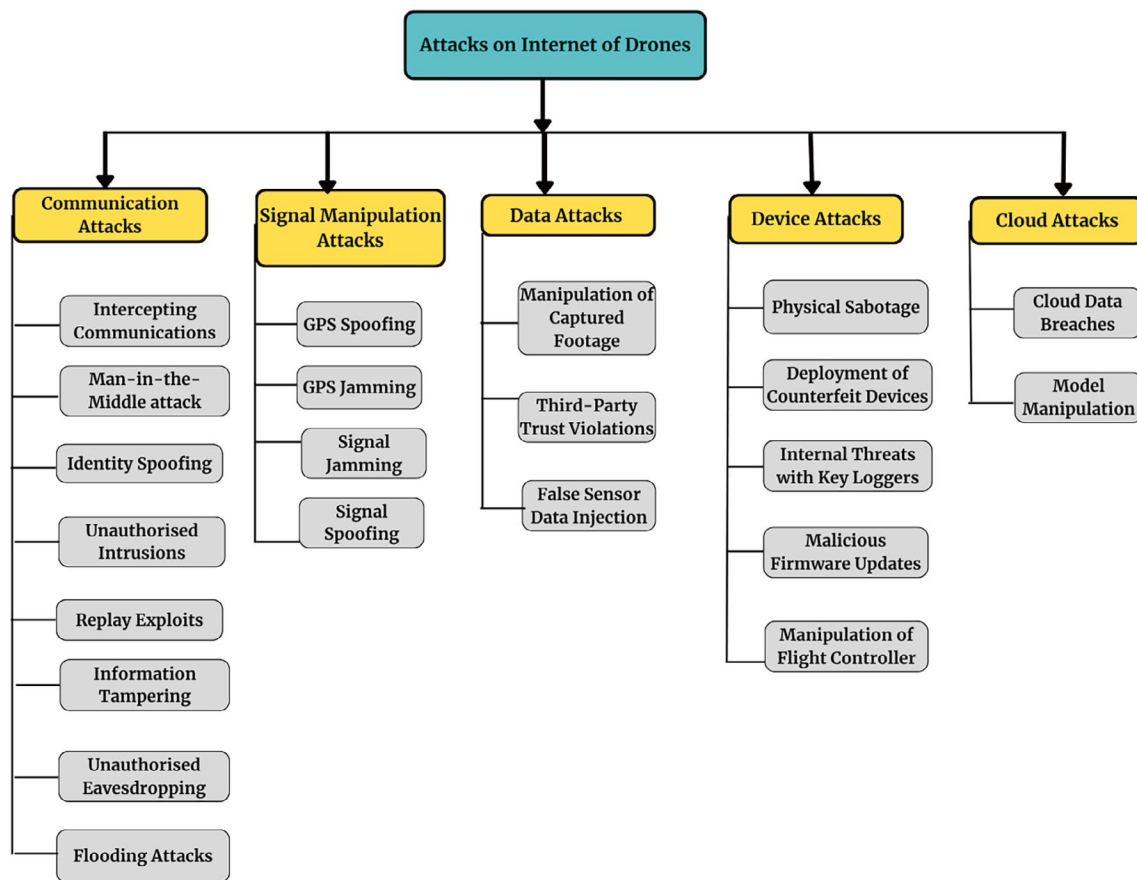


Fig. 3 A taxonomy of attacks on the internet of drones

- **Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) Networks.** These are used for sequential data analysis as they recognize temporal orders in the traffic patterns.
- **Random Forests and Support Vector Machines (SVMs).** These are conventional methods of supervised machine learning that can predict/control traffic flow as either normal/malicious given prior knowledge.
- **Behavioural Analysis.** One of the ways in which the functionality of the networks can be enhanced is by training the ML models to understand possible anomalous-user and anomalous device behaviour to proactively detect possible attacks in those networks.
- **Real-time Decision Making.** Continuous analysis of incoming data makes it possible to change the filtering rules of an ordinary firewall in real time, and the use of an ML model allows for a proactive response to emerging threats.

5.2 Detective Countermeasures

Detective steps are necessary for monitoring security risks in prevention or interruption mode or for investigation after an

incident. Below are the specific strategies for detective measures and how Machine learning can be applied to counter:

Intrusion Detection Systems (IDS)

Intrusion Detection Systems are developed to check for relevant activity and its assessment to determine if it represents a threat to a network (Rahman et al. 2022). AI and ML help in enhancing IDS in aspects of the detection rate of invasions and the time taken to detect them. Key algorithms and techniques (Shrestha et al. 1549; Ramadan et al. 2633; Moustafa and Alireza Jolfaei 2020; Shaw 2020; Hoang et al. 2019) include,

Signature-Based Detection

- **Pattern Matching.** To identify a known attack pattern, there are commonly employed methods like regex matching.

Anomaly-Based Detection

- **Supervised Learning Algorithms.** These algorithms need tagged sets for learning. SVMs can be used on

trained models to identify and categorize network traffic as either benign or a security threat. Statistical tools like Random Forests can be employed to reveal anomalies because they can be trained on labelled data to come up with decision rules. Neural Network models such as CNN, RNN, LSTM, BiLSTM, GRU are good examples of deep learning models to learn features from network traffic since they can accurately learn many complicated patterns.

- **Unsupervised Learning Algorithms.** These do not need any labeled data to work and aim at detecting unseen threats. Clustering Algorithms such as the K-Means, DBSCAN and the hierarchical approach involve grouping typical behavior patterns and the ones deviating can be viewed as a possible threat. Further, Generative architectures like Autoencoders, Restricted Boltzmann Machine (RBM), Deep Belief Networks can learn automatically from unlabelled raw data to accomplish tasks. The major concepts that encompass the use of Principal Component Analysis (PCA) are data compression and outlier detection.

Log Analysis

This involves analysis of logs that are captured as system logs, network logs etc., for clues of security events (DShar 2022; Minn et al. 2024).

Log Parsing and Preprocessing

- **Regular Expressions.** Specific data about user activities, specific application usage and error history are obtained from logs with the help of regular expressions. This, in turn, aids in preparing the reports and compiling the data for the next steps of analysis and processing.
- **NLP Techniques.** Logs are unstructured in nature; NLP procedures aid them in guessing structure for comprehension. For example, SVM or the neural networks that are used for NLP can be used to classify log entries, extract important features from text and learn textual patterns.
- **Frequent Itemset Mining.** Examples of algorithms are FP-Growth that are used to find the often-reoccurring patterns within logs.
- **Sequential Pattern Mining.** Finds established temporal aggregates or states that result from regular combinations of events.

Machine Learning for Log Analysis

- **Supervised Learning.** Models such as Logistic Regression can be used which are designed for binary

classification to detect log entries that are suspicious. Random Forests can model complex patterns from logs and the instances that violate them can be found. Similar to XGBoost, Gradient Boosting Machines (GBM) can be used for increased accuracy in the identification of anomalous behaviours in logs.

- **Unsupervised Learning.** Clustering algorithms could be employed to find abnormalities from the logs. Autoencoders can be used to detect anomalous patterns in logs by analyzing compressed input distributions and identifying high reconstruction errors.
- **Deep Learning Techniques.** Log data is a text format and CNN can be effective in measuring spatial patterns. RNNs and LSTMs are very useful for examining the sequences of log events to identify temporal differences.

5.3 Anti-Jamming Techniques

Jamming is the interference of the frequencies of drones that are engaged in various operations. It alters the work of the drone and may lead to the loss of drone security and safety. Thus, it is effective to employ anti-jamming strategies to ensure drone systems and their subsequent components communicate in a jam-free environment. There are two options for the implementation of the anti-jamming techniques—frequency hopping and channel hopping are widely used in drone communication systems. It is used in the manner that the system constantly changes the specified frequency band to the other thus, creating difficulties for a jammer to block communication signals. A drawback that exists in the antijamming mechanisms is that the jamming signal characteristics and its impact need to be studied and the protocol settings to be shifted by the communicating parties need to occur within a short time frame.

Machine Learning Integration. Jamming detection and prevention can be effectively done through the application of machine learning algorithms. Supervised machine learning models like decision trees, SVM, and artificial neural networks develop learning models that can be utilized to classify normal and jammed signals. For unlearned scenarios, unsupervised learning algorithms such as clustering techniques can be employed to cluster normal signal patterns. Signal patterns can be analyzed in real-time and appropriate responses can be made to counteract any attempts at jamming. Significant deviations from these clusters can indicate the presence of jamming. Q-learning, a reinforcement learning approach can be utilized to dynamically select frequencies based on the current state of the communication environment to avoid jammed frequencies through maximizing cumulative rewards over a series of time events. Game theory has also often been used to model jamming attack scenarios (Anti-jamming communications in UAV

swarms 2019, Shamaileh et al. 2021, Šimon et al. 2022, Lin et al. 2019, Anti-jamming strategy based on game theory in single-channel UAV communication network 2021).

5.4 Anti-GPS Spoofing Techniques

GPS spoofing involves the transmission of false GPS signals to cheat GPS enabled drones, leading to potentially unsafe situations and infrastructure. It is one of the most prevalent types of attack along with jamming. Various strategies like monitoring the GPS signal level for step change, rate of change, and recording time shifts to detect alarming variations.

Machine Learning Integration. Supervised machine learning algorithms can be employed to analyze the parameters of GPS signals to predict deviations through learned models. For instance, ML algorithms can be trained to detect abnormal time delays in receiving GPS signals which may suggest a spoofing attack. GPS signals can be matched against known patterns of authentic signals to detect any discrepancies to detect spoofing attacks. ML algorithms can be utilized to learn unique signatures and signal strength of authentic GPS signals and any deviation from these signatures and signal strength can trigger an alert for a potential spoofing attack. Deep learning models can be utilized to recognize patterns in signal strength that are indicative of spoofing attacks. Temporal models like RNN can be utilized to detect deviations from the expected trajectories (Agyapong et al. 2021, Nayfeh 2023, Wei et al. 2022, Umer et al. 2024; Sun et al. 2023).

5.5 Denial of Service (DoS) Attacks' Mitigation

DoS attack is the most common and easy type of attack an adversary can use to induce into the IoD to disrupt the normal functionality of drones. Flooding attacks including network flooding and data flooding attacks can be induced by overloading the network and processing power of drones and eventually leading to many vulnerabilities. Threshold and rule-based methods are utilized to mark deviations from predefined thresholds.

Machine Learning Integration. Machine learning can enhance DoS mitigation techniques by analysing network traffic patterns and identifying anomalies indicative of DoS attacks. For example, supervised machine learning algorithms such as Random Forest, SVM, and Gradient Boosting can learn to distinguish between legitimate and malicious traffic based on features such as packet size, frequency, source IP address and communication patterns.

Autoencoders can be implemented to learn normal communication patterns and identify anomalies through reconstruction errors. Game theory is another popular strategy for mitigating DoS attacks by modeling the strategic interactions between attackers and defenders to develop optimal defense strategies. Dynamic adaptive strategies to adjust traffic rates, security policies and others based on the congestions, compromised channels, and current threat environment, can be implemented using Reinforcement learning strategies (Ouiazane et al. 2022, 2023, Gaber et al. 2021).

5.6 Collision Avoidance Methods

Drones operate in shared airspace where the risk of collision with other drones, obstacles, and even manned aircraft is high. Effective collision avoidance methods are essential for ensuring safety. Obstacle detection and avoidance systems such as LiDAR, Ultrasonic sensors and infrared sensors, GPS and communication protocols aid in collision avoidance. Integration of AI and ML algorithms aids in efficient avoidance of collision through Autonomous navigation, path planning and decision making (Ouahouah et al. 2021, Hsu and Gau 2020, Yan, Xiang, and Wang 2020, Wang and Chen 2014, Puente-Castro et al. 2022, Cheng et al. 2021, An et al. 2024, Roghair et al. 2021).

- **Vision based Approaches.** Vision based approaches work by images and videos captured through drones. Object detection and image segmentation algorithms such as CNNs, Mask R-CNN can be utilized with the capabilities of YOLO for effective object detection and classification.
- **Path Planning.** Path planning algorithms help in optimal navigation decisions and learning efficient routes while avoiding collisions. Algorithms such as A*, Rapidly-exploring Random Trees, genetic algorithms and neural networks are being popularly used in path planning for finding the optimal route. Further, Reinforcement learning algorithms can be utilized where drones learn optimal paths by receiving rewards or penalties based on their actions. The application of Swarm Intelligence ensures coordinated movements and allows multiple drones to operate collaboratively and autonomously, enhancing their ability to avoid collisions.
- **Trajectory Prediction.** The future trajectory of drones can be predicted using algorithms like LSTM and RNN which uphold temporal dependencies for better predictions. Hybrid models integrating kinematics models with machine learning models can improve detection of collision-free trajectory future paths.

6 Case Study on Drone Operation Attacks in the Ukraine-Russia Conflict (2022-Present)

The ongoing Ukraine-Russia conflict has been marked by the extensive use of drones for various military purposes, including surveillance, artillery targeting, and direct strikes. The conflict has also seen a significant cyber warfare component, with both sides employing sophisticated techniques to disrupt each other's drone operations. Below are some real-life case studies that illustrate how cyber warfare has impacted drone operations in this conflict (Brantly 2023, Ilyas 2024, Rickli and Mantellassi 2024).

6.1 Ukrainian Drone Operations Disrupted by Russian Electronic Warfare

During the conflict, Ukrainian forces have relied heavily on drones for intelligence, surveillance, and reconnaissance (ISR) missions. These drones, ranging from commercial off-the-shelf models to military-grade UAVs, have played a crucial role in monitoring Russian troop movements and directing artillery strikes.

Attack Details

- **Russian Electronic Warfare (EW) Tactics.** Russia has employed a variety of electronic warfare tactics to disrupt Ukrainian drone operations. These tactics include GPS jamming, which interferes with the navigation systems of drones, and GPS spoofing, which provides false coordinates to the drone, causing it to lose its way or crash.
- **Impact on Ukrainian Operations.** Ukrainian forces have reported that their drones frequently lose GPS signals, making it difficult to complete missions or return to base. In some cases, drones have been lost due to these disruptions. The electronic warfare capabilities of Russian forces have forced Ukraine to adapt its tactics, including flying drones at lower altitudes to avoid detection and relying more on autonomous navigation systems.

Machine Learning Countermeasures

- **Adaptive Navigation Systems.** In response to these challenges, Ukraine has worked on developing machine learning-enhanced navigation systems that can operate with limited or no GPS input. These systems use onboard sensors and machine learning algorithms to predict the drone's position based on its flight path, allowing it to continue its mission even when GPS signals are unreliable.

- **Signal Anomaly Detection.** Machine learning models have been trained to detect anomalies in the signals received by drones, such as sudden changes in GPS coordinates that may indicate spoofing. By recognizing these anomalies, the drone can switch to alternative navigation methods or alert operators to potential threats.

6.2 Russian Drone Strikes and Cyber-Physical Attacks on Ukrainian Infrastructure

Russia has used drones not only for reconnaissance but also for conducting direct attacks on Ukrainian infrastructure. These drones are often equipped with explosives or serve as loitering munitions (commonly known as "kamikaze drones") to strike critical targets such as energy facilities, communication hubs, and military installations.

Attack Details

- **Cyber-Physical Attacks.** Russian forces have combined drone strikes with cyber attacks to maximize the impact on Ukrainian infrastructure. For instance, a drone strike on a power grid might be accompanied by a cyber attack on the grid's control systems, making it more difficult for Ukrainian forces to respond and recover. In one instance, Russian drones targeted Ukrainian communication networks, while a simultaneous cyber attack disrupted the command and control systems of Ukrainian forces.
- **Coordinated Multi-Domain Attacks.** These coordinated attacks demonstrate the integration of cyber and kinetic operations, where drones serve as both the weapon and the means of delivery for cyber payloads. This combination increases the effectiveness of the attack, creating a more complex and challenging environment for Ukrainian defenders.

Machine Learning Countermeasures

- **AI-Driven Threat Detection.** Ukrainian forces have started to deploy machine learning systems that can detect and respond to these coordinated attacks. These systems analyse patterns in the data from drone operations and cyber networks to identify potential multi-domain threats. For example, an unexpected increase in drone activity near a critical infrastructure site might trigger a pre-emptive cyber defence response.
- **Resilient Network Architectures.** Machine learning is also being used to develop more resilient communication networks that can withstand both physical and cyber attacks. These networks use machine learning algorithms to automatically reroute traffic, isolate compromised segments, and maintain connectivity even under heavy attack.

6.3 The Role of Commercial Drones and Open-Source Intelligence (OSINT) in Ukraine

In addition to military-grade drones, both Ukrainian and Russian forces have used commercial drones for various purposes, including battlefield reconnaissance and artillery spotting. These drones are often equipped with consumer-grade cameras and sensors, making them accessible and easy to deploy.

Attack Details

- **Interception and Hijacking of Commercial Drones.** Russian forces have attempted to intercept and hijack commercial drones used by Ukrainian forces. This is often done through signal interception, where the communication link between the drone and its operator is compromised. Once intercepted, the drone can be redirected, grounded, or even used against its original operator.
- **Exploitation of OSINT.** Both sides have also used open-source intelligence (OSINT) to track drone operations. By monitoring social media, online forums, and other publicly available information, adversaries can gather intelligence on the location and timing of drone flights. This information is then used to plan countermeasures, including cyber attacks on the drone operators' infrastructure.

Machine Learning Countermeasures

- **Encryption and Secure Communication.** Machine learning is being applied to develop more secure communication protocols for commercial drones. These protocols use machine learning to detect and prevent unauthorized access to the communication link, making it more difficult for adversaries to intercept or hijack the drone.
- **Real-Time Data Analysis.** Machine learning algorithms are also being used to analyse OSINT data in real-time, allowing Ukrainian forces to anticipate and counteract potential threats. By identifying patterns in the information being shared online, machine learning models can predict when and where cyber attacks might occur, enabling pre-emptive defensive measures.

The Ukraine-Russia conflict has demonstrated the critical importance of cyber warfare in modern drone operations. The integration of machine learning into both offensive and defensive strategies has become essential for maintaining the effectiveness and resilience of drone fleets in contested environments. These case studies highlight the

ongoing evolution of drone warfare, where cyber and physical domains are increasingly interconnected, and machine learning plays a pivotal role in shaping the outcomes of conflicts.

7 Open Issues and Challenges with Integration of Machine Learning in IoD Security

Integrating machine learning in the Internet of Drones offers numerous advantages for enhancing security and privacy, but it also introduces several open issues and challenges. Here are some key open issues and challenges.

Many of the IDSs of the IoD today are only designed to detect wireless communication interference even when there are other risks at hand. Further studies to extend IDS coverage to the ground control systems, drones, and cloud servers should be conducted to improve the general security of the systems. It is suggested that IoD's relevance should be focused in relation to specific datasets and testing methodologies that are particular to IoD contexts. Furthermore, it may be possible to strengthen IDS through the study of adversarial machine learning techniques, which create more robust strategies that cannot be easily tricked by fake alerts.

There are various complexities and interference factors associated with IoD which make it mandatory to verify the formal security protocols and techniques. Formal methods allow for checking whether proposed solutions are genuinely secure, especially in civilian contexts and offer a more robust approach to security engineering that has to graduate from arguing that 'my solution works' to proving that an approach is secure instead. This verification ensures that the implemented protocols are secure within any form of threat point and hence, offers better security assurance.

Malware threat level is constantly increasing as the newest danger for drones. Malware has no specialized anti-virus checkers for drone operating systems. The study should continue with the aim of comparing the different techniques such as static analysis, dynamic analysis, signature-based detection, and categorised into supervised and unsupervised learning. This knowledge of the potential malware threats to drones goes a long way in explaining the need for effective preventative detection measures as a way of ensuring that drone systems running critical operations and services are not compromised and or rendered ineffective.

The usage of AI and ML in wireless networks and IoD systems improves the new delivering approaches for real-time decisions, anticipation and autopilot which gives good performance, security and reliability as all decisions are not fully relayed on human interface. Nevertheless, there is still much to add like the capacity of the AI system to operate efficiently in contexts that are dynamic, constant real-time

data analysis capability, precise navigation and sound cybersecurity for dealing with new threats which are possibly more complex and more lethal. However, some potential issues in the deployment of such a sophisticated system include interoperability to other systems in case a complex structure is adopted, getting rid of the errors in the interfaces as much as possible and training for any residual complexity which remains as further research questions.

Drones typically have limited computational power, memory, and battery life. Running complex ML models directly on these devices can drain resources quickly, leading to reduced flight time and overall performance. Lightweight models or edge computing solutions are often required, but these may compromise the accuracy and capabilities of the ML algorithms. Different drones may be equipped with varying hardware and software, leading to challenges in ensuring that ML models can operate consistently across different platforms. Standardization of protocols and interfaces is crucial for interoperability but is often lacking in rapidly evolving drone technologies.

In many drone applications, such as object detection or obstacle avoidance, ML models need to process data and make decisions in real time. Delays in processing can lead to catastrophic failures, especially in dynamic environments. Achieving low-latency processing while maintaining accuracy is a significant challenge. Maintaining and updating ML models in a distributed drone network can be complex. Models may need to be updated regularly to adapt to new data or security threats. Ensuring that all drones receive and implement these updates without disruption to their operations requires careful planning and robust communication protocols.

These advancements and considerations will play an important role in creating a safer and more efficient Internet of Drones, which will allow the use of drones in various applications and integrate them at a high level of security risks and compliance.

8 Emerging Technologies and Future Threats

The security landscape for drone-enabled IoT networks is poised to face significant changes as emerging technologies continue to evolve. These advancements, while offering enhanced capabilities, also introduce new vulnerabilities and threats that must be anticipated and mitigated. In this section, we explore how these evolving technologies could impact security and the potential role of machine learning in addressing these challenges.

Quantum computing is one of the most anticipated technological advancements with the potential to revolutionize various fields, including cryptography. For drone-enabled

IoT networks, this could mean that sensitive data transmitted over the network could be easily intercepted and decrypted by adversaries with quantum capabilities. Machine learning could play a crucial role in developing quantum-resistant algorithms by modelling and simulating quantum attacks, thereby enabling the design of more secure encryption schemes.

As IoT networks expand, the shift towards edge and fog computing is becoming more prominent. These paradigms distribute computational tasks closer to the data source, reducing latency and bandwidth usage. However, they also introduce new security challenges, such as the increased attack surface and the need for secure communication between distributed nodes. Machine learning can enhance the security of edge and fog computing by enabling decentralized threat detection and response mechanisms. For instance, federated learning can be employed to train models across distributed nodes without sharing raw data, preserving privacy while improving the collective defence against emerging threats.

The concept of swarm robotics, where multiple drones operate in a coordinated manner to achieve complex tasks, is gaining traction in various applications. However, the coordination and communication between swarm members are vulnerable to various attacks, such as jamming, spoofing, and hijacking. Securing swarm robotics will require advanced machine learning techniques capable of ensuring reliable and secure communication within the swarm, even in the presence of adversarial interference. Machine learning can also be used to model and simulate potential attack scenarios, helping to develop resilient swarm algorithms that can maintain operational integrity under hostile conditions.

The future of drone-enabled IoT networks is intertwined with the evolution of emerging technologies, each bringing its own set of opportunities and threats. As these technologies continue to develop, the security landscape will become increasingly complex, necessitating the integration of advanced machine learning solutions to stay ahead of potential threats. By anticipating these challenges and investing in cutting-edge research, we can unlock the full potential of machine learning for enhanced security in this rapidly evolving domain.

9 Conclusion

The integration of drones into our daily lives through the Internet of Drones presents a promising future filled with opportunities across various sectors. However, with this technological advancement comes the challenge of ensuring the security and privacy of drone operations. Recent advancements in cyber and physical attacks have

highlighted the vulnerabilities inherent in drone networks, necessitating a proactive approach to mitigate risks.

In the introductory chapter, we explored the evolving landscape of drone technology and the critical role played by the IoD in facilitating seamless communication and coordination among drones. Furthermore, we delved into the architecture of the IoD, illustrating the interconnected layers that enable drones to function efficiently within a networked environment. We also explored the various security requirements that must be met to ensure the integrity, confidentiality, availability, and privacy of drone operations. We deliberated the security concerns surrounding drone networks, emphasizing the need to safeguard both the drones themselves and the data they collect.

Additionally, we discussed the classification of attacks against IoD based on different criteria, highlighting the nature of attacks, their target assets and compromised security requirements. This comprehensive classification framework provides insights into the diverse range of threats facing drone networks and helps guide the development of effective countermeasures.

The integration of machine learning into the security frameworks of the Internet of Drones is indispensable. It not only enhances the detection and mitigation of sophisticated threats but also enables adaptive and resilient security mechanisms that evolve with emerging challenges. While traditional methods provide foundational protection, machine learning's ability to analyse large volumes of data in real time and recognize complex patterns ensures that IoD systems remain robust against an ever-evolving threat landscape.

As drone technology advances, so must the security measures that protect it. A holistic approach, combining preventive, detective, and responsive strategies powered by machine learning, offers the best defense against the diverse and dynamic risks facing the IoD. By continuously refining these techniques and staying ahead of potential adversaries, we can ensure the safe, reliable, and efficient operation of drones, which are becoming increasingly integral to various sectors of society.

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Machine Learning Models for Drone Security: Cognitive Versus Cyber Intelligence for Safety Operations

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Abstract

The legitimacy, appropriation, and authorization of data during end-to-end communication between participating nodes in a network determines the intelligence and integrity of a real-time cyber-physical system. Given the persistent and increasing violations of the airspace by drones worldwide, the fundamental intelligence and security architectures governing drone usage operations must be empirically evaluated. This chapter examined the significant contributions of artificial intelligence models in developing trustworthy, reliable, intelligent, and secure drone systems to ascertain cyberspace, intelligence space, and airspace security. Furthermore, it explored the role of converging zero trust architecture, existing blockchain technology, and explainable AI in addressing drone security and safety issues like drone ownership authentication, drone package delivery verification, drone operation authorization, and jurisdiction accountability.

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Keywords

Artificial intelligence · Autonomous vehicles · Blockchain · Cyber-attacks · Drones · Explainable AI · Networks · Security · UAV · Zero trust

1 Introduction

There is a rising global deployment of unmanned aerial vehicles (UAVs) (also known as drones) for priority logistics, to expand network connectivity, and to access difficult terrains with 360-degree technological coverage (Javed et al. 2022). Thanks to the green logistics, speedy delivery, scalable flexibility, cost-effectiveness, etc. associated with UAV usage, made possible by disruptive technologies (Ajakwe et al. 2023a). However, this smart use of UAVs has several security and safety issues that cut across cybersecurity, cognition, and space intelligence (Ajakwe et al. 2023a). To be precise, nowadays, drones have become a powerful arsenal and strategic tool for carrying out information warfare to achieve a competitive advantage, which can be in the form of space, psychological, electronic, and cyber (SPEC) warfare (Kallenborn 2022). Thus, as autonomous intelligent vehicles, UAVs can be used as autonomous weapons to spread unsolicited, unverified, and unauthorized information to a wide spectrum of society without gauge (Kallenborn 2020). These associated perceived risks and dangers are indications of out-of-control innovation intelligence vs safety consciousness, with reports of airspace violations, cross-border infringement, and private security breaches despite the plethora of counter-security architecture installations and regulations (Li et al. 2022). Furthermore, this abusive use of UAVs for unsolicited intrusion into private properties, creating airspace apprehension, and invasion of key public infrastructures threatens the survivability and sustainability of the emerging drone transportation system as an alternative priority-based aerial logistics freight carrier. By extension, it questions the

trustworthiness of UAV operations, undermines its technology diffusion acceptability, casts doubts on the veracity of its feedback, deflates its cognitive garrulity, and unsettles the reliability of the underlying technologies in guaranteeing holistic sanity, safety, and maximum security of the national space (cyber, cataclysmic, and cognition). Hence, there is a need for an altruistic assessment of both cognitive and security architectures controlling the operational characterizations of these autonomous designs to douse their provocative applications.

According to Federica (Laricchia 2022), the Federal Aviation Administration (FAA) issued 331,573 remote pilot certifications as of July 2023, and 869,472 drones were registered in the U.S. 60% of these UAVs were for leisure activities, and the remaining 40% were for business uses. This shows an astronomical increase and demand for UAV networks (Castrillo et al. 2022). UAVs have, however, become a clear target for cyberattacks because of their expanding interest and use in both civil and military applications (Keller 2017). Strict cyber compliance by actors/stakeholders in the intelligence, network, and security domains does not necessarily translate into cybersecurity intelligence (cyber-intelligence), evidenced by the rapidity of quanta of threats emanating from modernization and technological advancement (Ajakwe et al. 2023a). Since innovation drives integration, to succeed, proactive and synergistic counter-defense designs must develop at the speed of the pre-empted threats to knit them in the bud and restore public confidence in the potency of national security, viz-a-viz intelligentization, impenetrability, interoperability, and mobility-friendliness of the associated networks that govern their day-to-day activities, especially in an era of digital frenzies such as cashless economy, metaverse, bitcoin, deep fakes, digital twins, etc. Over the years, more research efforts have been given to physical security and cybersecurity (network and data-centric consciousness), with less attention given to the underlying permutations and cogitations that produce the eventual outcome that drives most decisions (Ihekoronye et al. 2022). With artificial intelligence (AI) being at the heart of the final decision-making process of these interconnected systems based on the available data, a shift from network-centric (physical security) to data-centric (cybersecurity) to decision-centric (AI security) designs of networks is necessary (Ajakwe et al. 2024a). Hence, addressing issues such as near-instant availability, edge-friendly distributed operations, true interoperability and scalability, and tamper-proof-flexible security becomes imperative.

Furthermore, data is now a force multiplier and access to verified and untampered (viable) transformed data (called information) emanating from multiple sources (velocity and mobility) can define the fate of an entity

(Ajakwe et al. 2024b). Hence, there is a need to protect the data used by the AI models and the AI models' decision-making process. Nowadays, several attacks on key government buildings carried out with the aid of UAVs, to beat installed reactive surveillance and defense apparatus (Zhang et al. 2020), are done by manipulating the AI models' data (source). Generally, cyber-attackers leverage sublime and surreptitious loopholes in the connected intelligence and interaction of disruptive technologies on data to their advantage and wreak havoc in a swift, unsuspecting manner (Tlili et al. 2022). It is therefore pertinent to x-ray empirically the sophistications of UAV physical security, cybersecurity, and AI security architectures about cyber-attacks inherent in such networks. This is necessary to juxtapose the extent of technological integration of recent AI models into UAVs to match up with exuding vicious attacks on UAV networks.

Hence, this chapter seeks to answer the question: What factors govern the dynamic landscape of UAV networks as potential attacks? What are the different attacks on UAV networks and their potential effects? To what extent have AI models helped to mitigate these attacks? Beyond the AI model, what role do blockchain technology and zero-trust architecture play in strengthening the security and sustainability of UAV networks, by protecting the cognitive behavioral tendencies and decision-making capability of the underlying AI models? To answer these questions and address emerging issues in drone security, developing convergence-based innovations that meet users' expectations, are ethically friendly, and foster peaceful, just, and inclusive societies, free from fear and violence in line with the Sustainable Development Goals (SDGs) approved by the United Nations (UN) (Wu et al. 2018).

Therefore, the chapter explored how to develop a robust, responsive, and resilient drone security framework with transparency features that do not deliver advantages to adversaries by integrating the potentials of zero trust architecture (ZTA) (Sedjelmaci and Ansari 2023) to maintain standardized and enterprise resource-based security policy and governance, blockchain technology (BTC) (Chen et al. 2024) for tamper-proof decentralization data storage and authentication, and explainable AI (XAI) (Haque et al. 2403) algorithms for reliable and transparent interpretation of the discovered insights necessary for context-aware decision-making in drone networks and operations. The specific contributions of this chapter are:

1. We explored the dynamic landscape of drone networks as a potential source of cyberattacks.
2. Then, we examined the growth index of cyber-attacks on drone networks with advancements in drone technology.

3. Next, we checked the extent to which AI models have fared in matching up with growing incidences of attacks on UAV networks and curtailing them.
4. Finally, we highlighted the role of blockchain technology and zero-trust architecture in strengthening the existing drone network security.

2 Dynamic Landscape of Drone Application and Network Security Implications

Drones have swiftly transitioned from specialized military applications to widespread usage in a variety of commercial and civilian sectors. Their quick adoption is a result of their adaptability and effectiveness in a variety of domains, including disaster management, agriculture, logistics, and surveillance. However, drone networks are now positioned as possible targets for cyberattacks due to the significant cybersecurity challenges brought forth by their increasing ubiquity. Drones themselves, ground control stations (GCS), and the communication links that allow data transfer between these entities are some of the interrelated parts that make up drone networks. Malicious actors can take advantage of the distinct vulnerabilities that each component possesses. Due to their dependence on advanced sensors, wireless communication channels, and autonomous operating systems, drone networks are more vulnerable to cyber threats due to their expanded attack surface.

Researchers highlighted the expanding landscape of UAV-related cyberattacks. To defend drone networks from a variety of threats, Wang et al. (2021) emphasized the increasing security issues and the necessity for strong countermeasures (Wang et al. 2021). Similarly, Sun et al. (2024) demonstrated how cutting-edge technology, such as AI-driven anomaly detection, may greatly improve cyber threat identification and mitigation in drone networks. Cyberattacks are becoming more sophisticated and frequent as UAV technology develops, putting military and civilian activities at serious risk. These threats have increased due to advancements in UAV systems, including better connectivity, increased capabilities, and wider usage. In their discussion of blockchain technology's role in protecting UAV communications, Tan et al. (2022) explain how decentralized security measures can offer a tamper-proof record for UAV transactions (Tan et al. 2022). Moreover, there is a need for the incorporation of zero-trust architecture as a means of putting strict access restrictions and ongoing verification processes in place to shield UAV networks against illegal access and misuse.

It is essential to understand the dynamic landscape of UAV networks, which is crucial for developing effective cybersecurity strategies. This entails identifying the possible sources of cyberattacks and understanding how

technological advancements both exacerbate vulnerabilities and provide defensive measures. Through the utilization of cutting-edge technologies like blockchain, artificial intelligence, and zero-trust architectures, strong security frameworks that minimize risks and guarantee the safe functioning of unmanned aerial vehicles (UAVs) in a world growing more linked can be established. By exploring the dynamic landscape of UAV networks as a potential source of cyberattacks, this section provides insight into the constantly changing issues and offers solutions to protect UAV networks from the ever-present risk of cyberattacks.

2.1 Drone Technological Dynamics and Derelictions

Miniature size, swift maneuverability, cost-effectiveness, multipurpose use cases, and adaptability characteristics of drones make them potential tools for state and non-state actors in perpetrating heinous attacks (Ajakwe et al. 2023b). The FAA had issued 331,573 remote pilot certifications as of July 2023, and 869,472 drones had been registered in the country (Satell 2023) (M. S. 2023). 60% of these UAVs were for leisure activities, and the remaining 40% were for business uses. UAVs can be used for various purposes, such as search and rescue (SAR) operations, network relays, construction projects, product dynamic delivery, and remote monitoring of the environment, traffic, or people (Panetta 2019). For example, SAR missions involve time-critical communication, requiring instantaneous video transmission to the ground control station (GCS). As a result, the communication requirements of UAVs often adapt to their specialized uses. Figure 1 highlights the categories of drones based on application.

Particularly, the capacity to function autonomously and collaboratively without requiring human input drives the growth of UAV applications. Currently, drones are used for a wide range of tasks, including military operations (such as search and rescue missions, target tracking missions, reconnaissance, etc.) (Ajakwe et al. 2021), environmental protection investigations (Radoglou-Grammatikis et al. 2020), agricultural missions (Ajakwe et al. 2024a), and many more uses when their advantages are appropriate. The capabilities of drones are undergoing rapid development by many military and paramilitary organizations to reduce soldier deaths as an affordable option other than using human aircraft. This sublime shift indicates an awakened interest in tactical and pragmatic security intelligence against cyberspace warfare from internal and external aggressors.

2.1.1 Indicators of Growth in Drone Security

The rapid evolution of drones has transformed several industries. The capacities and functionalities of UAVs have

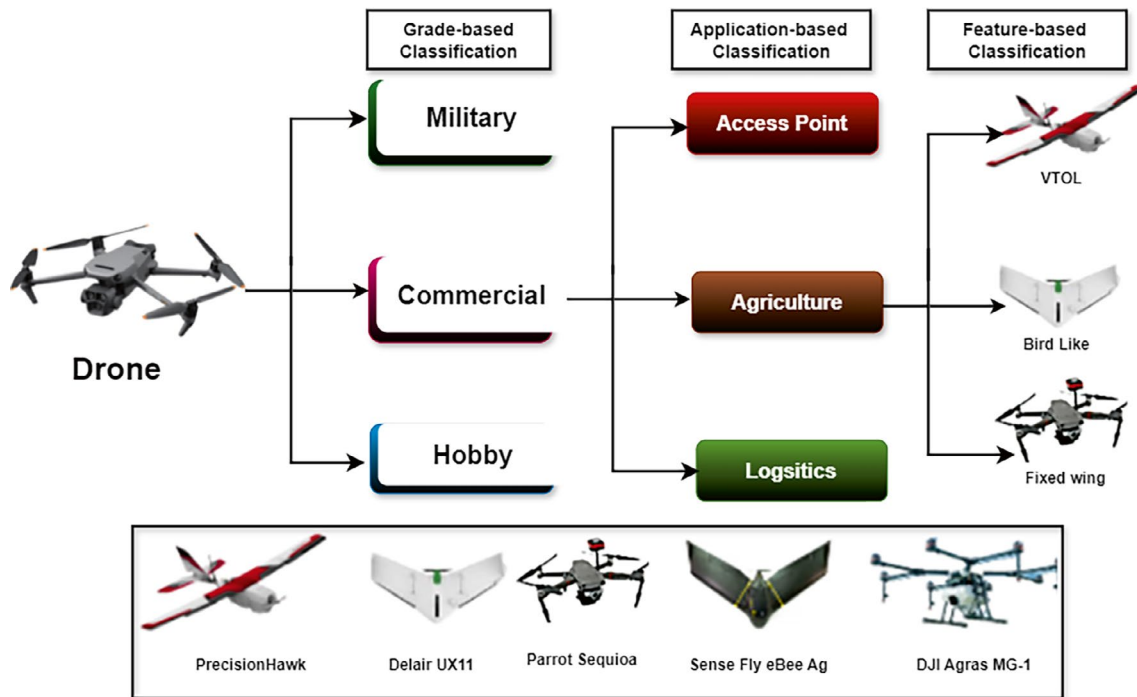


Fig. 1 Categories of drones and application areas highlighting some of the drones used in agriculture

been greatly enhanced by this technological innovation, making them vital tools in contemporary operations. Sun et al. (2024) stated that there is a significant increase in the potential and applicability of drones due to recent advances in cutting-edge machine-learning techniques (Sun et al. 2024). The technological factors triggering the sporadic growth in drones can be summarized into drone components, drone communication systems, and drone-cloud/edge computing infrastructure.

- **Enhanced HW/SW Component Capabilities:** With their sophisticated sensors and high-definition cameras, modern UAVs are more useful for carrying out various activities. Integrating machine learning (ML) algorithms as the underlying cognitive engines in drones increases their atomicity, makes them function independently, and decreases the need for constant human intervention, thereby boosting their operational effectiveness.
- **Expanded Communication Systems:** Real-time data processing and decision-making are made easier by the deployment of 5G networks, which improve data transmission speed and reliability between UAVs and UAV-GCS communication. To facilitate more intricate and coordinated operations, UAVs are becoming more and more integrated into broader network systems, including the Internet of Drones (IoD).
- **Efficient Cloud and Edge Computing:** UAVs frequently use cloud services for data processing and

storage via cloud integration, giving them access to powerful analytics and scalability. Also, edge data processing lowers latency and improves the ability to make decisions in real-time. With the presence of federated learning (FL), explainable AI (XAI), Tiny ML, and edge AI (EAI), assisted by improved crypto-graphic hashing via tamperproof security architecture such as blockchain technology, drone security issues are receiving more research attention.

2.1.2 Indexes of Gross Drone Violations

The increasing sophistication and dynamic application of drone systems make them increasingly appealing targets for cyberattacks, as seen in the plethora of drone-related violations globally. Unintentionally, the attack surface has grown as a result of the development of UAV technology, giving adversaries more opportunities to target and exploit vulnerabilities.

Table 1 presents the statistics of UAVs sighted in different cities in the United States between 2020 and 2024, as reported by the FAA (2024).

The 823 abusive usage of drones out of the 8048 sighted drones represents a significant-10.2% $\left(\frac{823}{8048} * 100\right)$ violation of airspace, intelligence space, and cyberspace using drones. Though there is a slight decline $\left(\frac{37}{823} * 100 = 62.9\%\right)$ in abusive drone usage between 2020 and 2023, the 37 drone violations reported in the first quarter of 2024 indicate a 76.4% $\left(\frac{37}{823} * 17quarters\right)$ rise in drone

Table 1 Indices of UAV-related violations, 2020–2024

Year	Number of sighted UAV	Abusive UAV usage
2020	1633	232
2021	2585	225
2022	1819	183
2023	1685	146
2024	326 (Jan.–Mar.)	37 (Jan.–Mar)
Total	8048	823

violations in a quarter. This raises concerns about the intelligence and security mechanisms that guarantee safe drone operations because 10% increase in drone-related violations portends great risks and questions the veracity and reliability of the underlying cognitive models.

Beyond the physical violation of airspace by drones, there is an increase in the frequency and sophistication of drone-related cyberattacks. According to recent reports, cyberattacks on UAV systems have significantly increased, and the attackers are using more advanced methods. According to the Cybersecurity and Infrastructure Security Agency (CISA 2024a), UAV-related cyber incidents have increased annually, especially in critical infrastructure and defense sectors (CISA 2024a). Attackers target UAV systems with advanced persistent threats (APTs) and zero-day exploits, making it difficult to detect and mitigate these attacks (Ceviz et al. 2023). Also, there are diverse attack vectors. Technological advancements have increased the attack surface, which has led to the exploitation of a variety of attack vectors, including GPS spoofing, hijacking, ransomware, and denial of service (DoS) attacks (Ceviz et al. 2023). Finally, there is a likelihood of the emergence of new threats. As UAV technology advances, new types of cyber threats—like supply chain attacks and insider threats—are emerging. During software development or manufacture, attacks on the supply chain create vulnerabilities that can be exploited later (Belikovetsky et al. 2017). Authorized users may maliciously or carelessly penetrate UAV systems, resulting in data breaches and disruptions to operations.

As expected, drones can wirelessly interact with one another, a base station, and/or a ground controller. Consequently, wireless connectivity and distributed control systems of drones predispose UAV networks to constant active interference attacks and espionage. A closer look at drone network architecture helps to provide clues to unraveling loopholes exploited by intruders to breach its security.

2.2 Drone Network Architecture Dynamics

Drone network architecture is different from the conventional Vehicular Ad hoc Networks (VANETs). Drones use

Flying Ad hoc Network (FANET); a subset of Mobile Ad hoc Network (MANET). FANETs stand out from other ad hoc network types due to their unique features in response to the high mobility of UAVs (Ceviz et al. 2023). One of the most distinguishing features of UAVs' FANET is their ever-evolving topology, which allows the building of a high-speed (above 720 km/h) and extensive communication ecosystem (terrestrial, aerial, and space networks), thereby promoting an innovation known as drone transportation system (Ajakwe et al. 2023a, 2023c). The UAV network architecture provides the fundamental framework for connected intelligent and semantic interactions and communications between different UAVs (UAV-to-UAV), UAV-to-ground-based control stations (UAV-to-GCS), and other interconnected system components (UAV-to-Satellite). Conventionally, a drone is an autonomous node equipped with sensors, cameras, GPS modules, computing power, and navigational formats that form the backbone of UAV networks. Figure 2 highlights the decentralized nature of communication in drone security networks.

During operations, the UAVs are monitored and controlled in real-time by the GCS, which also acts as the primary command post. The network architecture relies heavily on the communication channels that connect the GCS and the UAVs. The communication links allow data transmission between UAVs and GCS through wired (e.g., Ethernet, fiber optics) and wireless (e.g., radio frequencies, Wi-Fi, cellular).

According to Laghari et al. (2023), in the communication technique, these links may make use of several methods, including satellite communication via the tower infrastructure and drone controlling system, cell phones, Wi-Fi, mesh networks, and satellite communication (Laghari et al. 2023). For UAV networks, low latency is essential, particularly when performing quick tasks like screening and emergency response. The data processing systems provide real-time data analysis, mission planning, and decision-making; these are located at GCS and onboard UAVs. Nowadays, FANETs have advanced and unique characteristics of improved protocols for faster cooperativeness of nodes in the network in terms of packet delivery ratio (PDR) and energy efficiency (Ceviz et al. 2023; Bekmezci et al. 2016; Tropea et al. 2023). Thanks to the development of AI-driven cognitive software-defined radios (SDR) (Group R 2024) with improved mesh network capacity for self-healing and self-forming networks, self-situation-aware navigation in drones is possible. Due to these outstanding characteristics, UAVs are now the focus of several cyberattacks (Ahamed Ahanger et al. 2022).

Thus, in recent years it has been increasingly challenging to ascertain the reliability of UAV networks and, by extension the safety of drone operations as seen in Sect. 2.1.2. Malicious users run the danger of exploiting

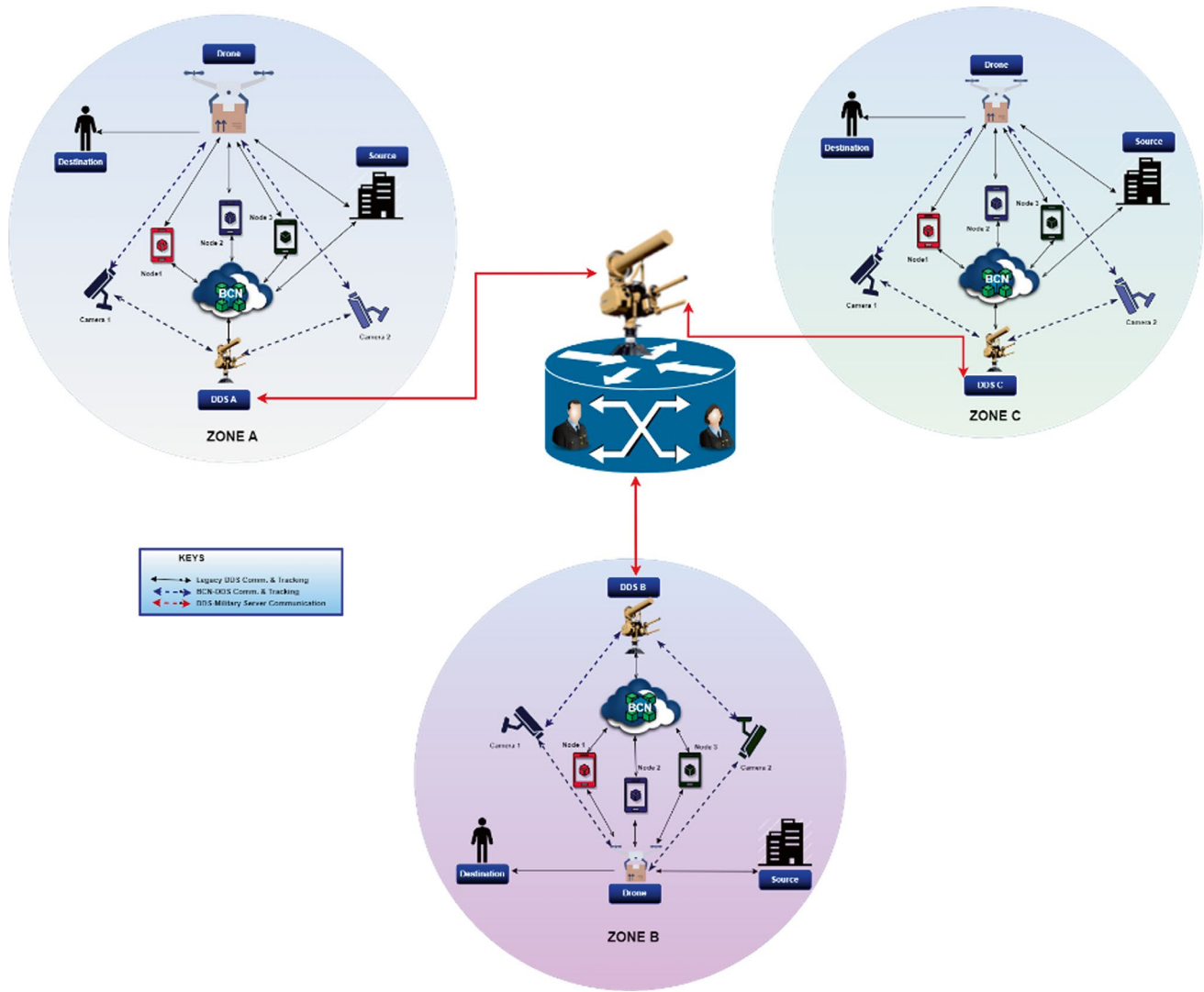


Fig. 2 A decentralized collaborative blockchain-assisted network for tamperproof UAV communication authentication and authorization highlighting the interactions between edge devices and aerial and terrestrial networks (Ajakwe 2023)

potential vulnerabilities and threats in the drone network as the network's applications grow and its technology evolves to cause harm swiftly and undetected. As UAVs continue to evolve, enhancing cybersecurity procedures is essential to protecting UAVs and their networks from potential threats. To achieve this, there is a need for a comprehensive exploration of the myriad of cyberattacks on drone networks to gain insights into their characteristics.

3 Taxonomy of Drone Security and Safety Attacks

Drone security attacks are categorized into two broad groups; physical and logical attacks (Ihekoronye et al. 2022). Based on the characteristics of the attack, drone

attack nomenclature can be grouped as cracking/jamming, exploitative, infectious, information gathering, and interceptive attacks (Ajakwe et al. 2023a). On the other hand, considering the intention of the intruder on the communication network, attacks on drones can be either active or passive (Ajakwe et al. 2022a). However, a broader categorization of UAV attacks can be established based on the vulnerabilities associated with FANETs, as an integral part of the ground base station. According to Ceviz et al. (2023), the taxonomy of cyberattacks on drone networks can be grouped into three (3) categories: UAV-component attack, UAV-communication layer attack, and UAV-communication architecture (Ceviz et al. 2023). Table 2 presents a detailed categorization of these attacks. From Table 2, it is obvious that virtually all aspects of the drone network are prone to cyberattacks, making it difficult to develop a one-for-all

Table 2 Categorization of drone network attacks

S. No.	Aspect of drone	Targeted area	Sub-target	Specific name of attack
1	HW/SW component	Hardware-based (HW)	✓ Battery attacks	Depletion of battery (DoB)
			✓ HW-DoS attacks	Denial of sleep (DoSp)
		Software-based (SW)	✓ Supply chain attacks	
			✓ Side-channel attacks	
			✓ Backdoor attacks	
			✓ Malware attacks	
		Sensor-based	✓ Zero-day attacks	Active spoofing attacks
			✓ GPS-spoofing attacks	Passive spoofing attacks
2	Communication architecture	UAV-to-UAV	✓ Data tampering	
			✓ De-authentication attacks	
		UAV-to-others	✓ Sybil attacks	
			✓ Jamming attacks	
			✓ Drone-in-the middle	
		(GCS, satellite)	✓ Spoofing attacks	Steppingstone assault
3	Communication layer	Network layer attacks	✓ Traffic attacks	Sybil, Greyhole (Dropping)
			✓ Topology attacks	Sinkhole, blackhole, wormhole
		Physical layer attacks	✓ Resources attack (replay), jamming attacks, eavesdropping attacks	Rushing, network reconnaissance
		Transport layer attacks	✓ UDP flooding, SYN flooding, session hijack-Ing	

intelligence model that can curtail all facets of drone security vulnerabilities.

1. **Communication Layer Vulnerabilities:** Cyber-attacks on the drone communication layer cut across the network (Sachdeva et al. 2023), physical (Zhi et al. 2020), and transport (Airlangga and Liu 2023) layers. Network layer attacks, otherwise called routing attacks (Sachdeva et al. 2023) are the most pronounced because it is targeted at the traffic and topology of the drone. Each of the routing attacks falsely claims ownership of the drone by diverting it through signal manipulation. For instance, since wireless communication plays a major role in UAV control, navigation, and data transfer, it renders them susceptible to eavesdropping and interception by adversaries with access to communication routes that are either poorly encrypted or unprotected (Zhi et al. 2020). Thus, eavesdropping occurs when an interception of a message by an unauthorized party occurs without the attacker disrupting the ongoing conversation. By breaching the network's privacy, the attacker can secretly listen in to the communication (Tlili et al. 2022). Furthermore, the attacker can escalate the attack by introducing fabricated messages and deleting or altering the intercepted message.

2. **Communication Architecture Vulnerabilities:** Cyber-attacks targeted at drone communication architectures aim at making drones collide with each other, especially during coordinated self-navigating swarming operations. For instance, in a Drone-in-the-Middle (DitM) attack (Deebak and Hwang 2023), the attacker secretly modifies and sends the navigational messages that are sent and received between the drone and its connected navigational control system. This leads to session hijacking, DNS spoofing, and packet sniffing. The integration of UAVs with cloud services for data processing and storage exposes them to vulnerabilities related to the cloud. Loss of control or data breaches might result from compromised cloud services. Additionally, weaknesses in IoT devices can be leveraged by UAVs that integrate with them, resulting in wider compromises of networks (Vajravelu et al. 2023). Common vulnerabilities in this category include denial of service, distributed denial of service, data theft, hijacking/unauthorized takeover of drones, forceful redirection/rerouting of drones to undefined destinations, etc.

3. **HW/SW Components Vulnerabilities:** Attacks on drone components encompass hardware, software, and sensor-based attacks (Ceviz et al. 2023) with different features and objectives. For instance, attacks on drone batteries

and the power system aim at either depleting the battery power (DoB) faster than the usual time (Desnitsky and Kotenko 2021) or denying it from entering sleeping mode (DoSp) (Shakhov and Koo 2018). Firmware is an essential part of UAV operation because it is the low-level software that manages the hardware of the drones. By taking advantage of firmware vulnerabilities, malicious operations such as data theft and unauthorized control can be carried out (Sharma and Mehra 2023). Firmware vulnerabilities enable unauthorized control or disablement. Physical UAV access may result in data theft or hardware modifications. Malware insertion is one common example of a firmware attack. An attacker can introduce malware that can result in unpredictable behavior or allow backdoor access. For instance, on an operator's computer, they could be introduced into UAV systems through malware downloads, corrupted data, or other attack vectors (Niyonsaba et al. 2023). Also, UAV physical access poses a serious cybersecurity risk. When unauthorized individuals physically access UAVs, they can alter their hardware or take out confidential information, such as sensor information (Feng et al. 2020), endangering the UAV's security and functionality. Risks to data privacy may exist for some software and firmware used in UAS operations, which could lead to data theft or unauthorized use of the UAVs (CISA 2024b).

The staggering statistics from Table 2 indicate a spiral growth index in the diversities of cyber-attacks and vulnerabilities associated with drone operations, which corroborates with the reported increase in drone-related incidents and violations (see Table 1) with huge economic impact across different sectors. In a recent report by Ponemon Institute, the average cost of data breach associated with drone operational outages and recovery attempts was estimated at \$4.5 million; a 12% rise between 2020 and 2024 (Institute 2024). This amount represents all the expenses related to a data breach, including costs for discovery, containment, recovery, notification, and legal, regulatory, and reputational costs. To be precise, nowadays, drones have become a powerful arsenal and strategic tool for carrying out information warfare to achieve a competitive advantage, which can be in the form of space, psychological, electronic, and cyber (SPEC) warfare (Kallenborn 2022). Thus, as autonomous intelligent vehicles, UAVs can be used as autonomous weapons to spread unsolicited, unverified, and unauthorized information to a wide spectrum of society without gauge (Kallenborn 2020). These associated perceived risks and dangers are indications of out-of-control innovation intelligence vs safety consciousness, with reports of airspace violations, cross-border infringement, and private security breaches despite the plethora of counter-security architecture installations and regulations

(Li et al. 2022). Thus, cyber-intelligence strategies developed to address these myriad attacks cannot have outdated behaviors, lopsided characteristics, and biased cognition tendencies in decision-making to achieve safety operations and prevent huge economic losses that threaten the extinction of drones.

4 Intelligence Models for Drone Security

As discussed earlier (see), the accelerated growth of drone capability and perceived improved safety operation is attributed to advancement in the underlying technologies that govern its decision-making process. Specifically, the intelligence of AI algorithms and other associated technologies influences the smart behavior of these aerial autonomous robots for positioning, timing, and navigation. Thus, AI models for ascertaining drone security cut across models for maintaining cognitive space security, airspace security, and cyberspace security (Ajakwe et al. 2023a).

4.1 Counter-Drone Defense Models

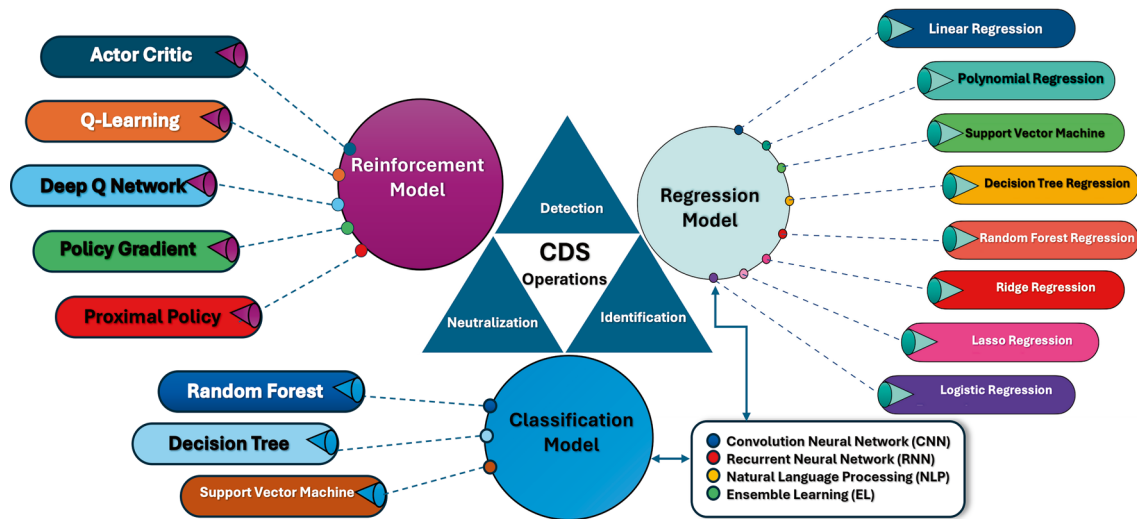
Counter-drone defense (CDS) systems, otherwise called anti-drone systems, are designed to gauge the airspace from malicious and abusive deployment of drones. To do this, CDS carries out three intertwined activities: detection, identification, and neutralization using AI models to process the acquired signal information emanating from the drone sensors (Okechukwu Ajakwe et al. 2022). Each of the CDS activities is carried out using different techniques depending on the sensor data characteristics and is greatly impacted by factors such as sunlight intensity, noise, rain, snow, fog, etc. According to Ajakwe et al. (2022b), AI models deployed in CDS can be categorized as single or hybrid approaches, with the latter having better performance due to multiple sensor data fusion, albeit with increased system complexity (Ajakwe et al. 2022b). Table 3 provides succinct details of hybrid-based CDS techniques for curtailing drone airspace insecurity.

Most of the AI models that perform these complex tasks are AI conventional models, which can be classified into regression, classification, classification-regression, and reinforcement learning models as seen in Fig. 3.

1. **Regression Model:** Based on gathered historical sensor data, regression models are used to analyze time series drone operational data (Ajakwe et al. 2020). The relationship between the variables in the data determines the sort of regression model that is used. Examples of such models include Linear, Ridge, Lasso Regression, Decision Tree (DT), Support Vector Regression (SVR),

Table 3 Advanced CDS strategies for airspace security (Ajakwe et al. 2023a)

Technique	Potentials	Limitations
Vision-RF (Aledhari et al. 2021)	Integrates the weather-resistant properties of RF with the optical camera's identification power. Forensic applications find it quite useful in precisely identifying things affixed to drones	Narrow use-case, unavailability of real-world datasets
Vision-Acoustic (Svanström et al. 2020)	Environmentally robust and climate-friendly drone detection system. Assure precise visual depiction of drones that have been detected minimizes problems with signal interference	Complex and non-uniform sensor fusion issues, high system complexity, and maintenance costs
Vision-LIDAR/Laser (Ajakwe et al. 2023c)	Supports precise identification of drones and other aerial objects, as well as high-dimensional 3D point cloud mapping. Ensure extensive coverage. They are inexpensive to install and have compact deployment	Affected by physical obstacles. High system cost and complexity issues
Vision-RADAR (Park et al. 2021)	Wide-area coverage and multidirectional scanning of multiple drones. Brief explanation of the target drone tolerant to noise and intricate land-scaping	High deployment and maintenance costs. High system complexity. affected by physical obstacles
MultipleRF, RFID (Basak et al. 2021a)	Identify several drones. Implementation costs are lower than using radar technology. Can transmit sensitive data belonging to an unauthorized operator	Cannot recognize attached objects to drones. Deficient detection capacity in terms of range and accuracy. Susceptible to interference

**Fig. 3** Categories of underlying AI models for airspace security in drone operations

K-nearest neighbor, Gradient Boosting, Bayesian Linear Regression, and Feedforward Neural Network.

2. **Classification Model:** These models use drone data to classify and identify drones based on their unique characteristics (Ajakwe et al. 2021). Examples are Random Forest (RF), Decision Trees, and Support Vector Machine (SVM).
3. **Regression and Classification Model:** These include deep learning (DL) models (such as CNN and RNN), Natural language Processing (NLP) models, and Ensemble Learning (EL) models (Ihekoronye et al. 2022). CNN and RNN are used to analyze remote sensor and satellite communication data of drone operations.

While CNNs are very good for object detection and classification, RNNs are good for regression and time series tasks. NLP is used to analyze textual semantic reports of drones in response to situational changes in the environment to discover possible patterns. EL models combine different ML models to achieve more reliable predictions. Common examples of classification and regression models used in CDS are YOLO (Ajakwe et al. 2023c), LSTM, Resnet, MobileNet, etc.

4. **Reinforcement Learning Model:** Reinforcement Learning (RL) models are used for tasks such as localization, navigation, and mapping of an environment. Typical RL models include Actor-Critic, Q-Learning,

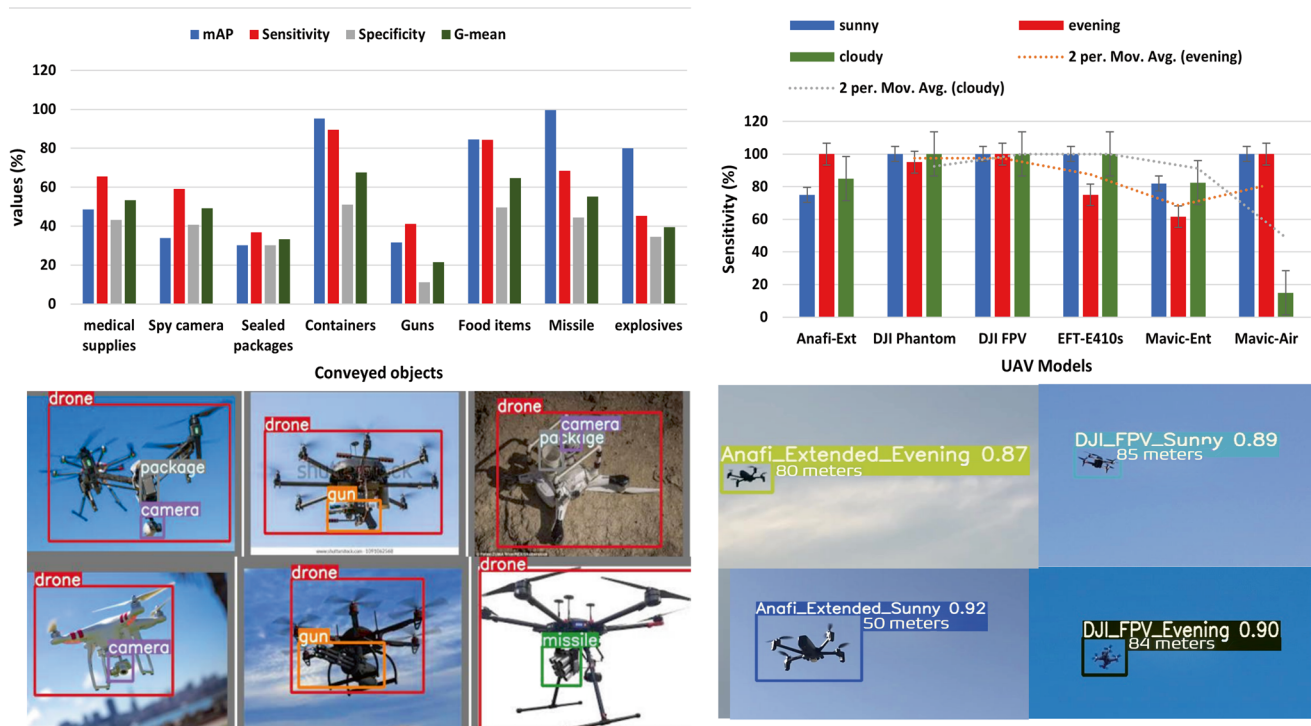


Fig. 4 Multiple drones and attached object detection and identification using a hybrid DL model, highlighting **a** Drone detection model performance and **b** Attached object detection result, at different weather condition (Ajakwe et al. 2023c)

Deep Q Network, Policy Gradient, Proximal Policy Optimization, etc.

The perceived threat of a drone in the airspace is not only the drone's function but also a combination of the harmful status of the object it is conveying (Ajakwe et al. 2022b). To guarantee the safety of industrial facilities, authors (Ajakwe et al. 2023c) developed a hybrid (vision-laser) DL model using Visiodect dataset (Ajakwe et al. 2022) that can detect and estimate the range of multiple classes of drones under different atmospheric conditions. The graph in Fig. 4 summarizes the drone detection and identification performance of the drone and the attached objects.

The result achieved in this study validates the possibility of determining the harmful status of a drone in operation, which is necessary for safety consciousness. However, despite this result, cognitive AI models used in CDS are still confronted with the problem of simultaneous detection, identification, and incapacitation of several perceived malicious drones deployed in the airspace. This is necessary for swift and proactive action to curtail abusive drone usage and drone-related violations. Furthermore, the lack of robust legislation against the unethical use of drones inadvertently affects the development of scalable AI models for improved drone airspace security. Finally, a study carried out by Ajakwe et al. (2023a) shows that out of 48,053 applications of newer AI models in different domains, CDS has

only 972 which is a 0.02% ($\frac{972}{48053} * 100$) deployment rate (Ajakwe et al. 2023a). This indicates a paucity in the integration of important AI models in CDS to achieve reliable drone security and safety. Little wonder, there is an increase in airspace drone-related violations.

4.2 Cognitive Navigation Models

Safe drone operation is a function of swift and smart navigation based on semantic analysis and awareness of the objects in their immediate environment, thereby achieving precise collision avoidance. As discussed in Sect. 3, most cyber-attacks on drone security are targeted at the entire drone system, with routing protocol attacks being the most pronounced. Software Defined Radios (SDR) are used to strengthen drone digital communication against code injections, spoofing, exploitation, and other threats, as drone networks are the main target of drone security threats (Romesburg et al. 2021). Furthermore, to improve security and safety, the majority of RF-based CDS monitor and limit drone mobility using SDR platforms (Chiper et al. 2022). With most of the signal processing handled by software under the direction of an AI algorithm, SDR's goal is to use a single, straightforward hardware configuration for a variety of radio applications. This makes sure that the software code is as close to the antenna as feasible,

Fig. 5 Categories of SDRs with different detection performance ranges for safe drone navigation and communication



enabling the AI-enabled software to handle signals from common hardware configurations for a variety of radio applications. Thus, among other benefits, SDR provides scalability, larger frequency coverage (Chiper et al. 2022), flexible deployment and interoperability, and cost-effectiveness. Common examples of SDR for safe and secure drone mobility are shown in Fig. 5.

An ML-enabled multi-sensor-SDR's hardware and software design integrates many wireless technologies, including mesh networking, cognitive radios (CR), and frequency hopping spread spectrum (FHSS), to provide public safety and UAV intelligence surveillance and reconnaissance (ISR). Enhancing situation analysis and environment-aware navigation, as well as improving the scenario-specific and dynamic neutralization capacity of CDS systems, this gives drones access to self-healing networks. This helps protect against UAVs being maliciously deployed to interfere with aerial autonomous vehicle mobility, which puts UAS applications' sustainability at risk. FHSS employs randomized frequency shifts several times per second to safeguard vital data, while CR enables the effective distribution of unused frequency channels, leading to enhanced quality of service, less interference, and raised spectrum efficiency.

A variety of input parameters and algorithms, including multiple signal classification (MUSIC) (Basak et al. 2021a), Raytheon Associative/Array Processor (RAP) (Basak et al. 2021a), received signal strength (RSS) (Gelman et al. 2019), RF jamming (Fang et al. 2018), etc., are used to achieve cognitive intelligence (such as GNU radio signal analysis, processing, and prediction) in SDR platforms, albeit with accompanying limitations. Nowadays, ML-enabled SDRs are developed with improved performance capacity to address the inherent constraints of conventional methodologies. Most of the ML-based algorithms

are based on radio frequency technologies. Common AI model technology deployed in SDRs includes RF, SVM, KNN, XGBoost, Ensemble Learning (Nemer et al. 2021), Deep Recurrent Neural Networks (Basak et al. 2021b), etc. Undoubtedly, the integration of AI algorithms as the underlying cognitive engine in SDRs has improved its functional capacities. These days SDRs have advanced mesh capacity for the self-forming network (SFN) or self-healing network (SHN), allowing long-range and self-navigating strategies (SNS) in drones to improve the safety of UAV intelligence deployment for logistics, surveillance, and reconnaissance.

SDRs with state-of-the-art software, SHN with multiple nodes capacity, and increased throughput via unlimited hops are commonly produced by RDTech (Group R 2024) and include RDL-2 (100 km), RDL-3 (200 km), RD-SR/MR/LR/XLR (omnidirectional signal tracking with ranges of 15, 30, 100, and 200 km), RD-HUB (10 km), RD-MR(30 km), RD-LR, RD-L-2 (70 km), RD-LR, RDL-3 (100 km), RD-XLR, RDL-3 (200 km), and RD-Edge (see Fig. 5). Though the results of these ML-enabled SDRs are worthwhile, there are several inherent limitations such as limited detection range, implicit algorithms interpretation, etc., which still raise concerns about the safety operations of drones. It makes sense that one significant drawback of RF-based technology is its inability to identify and destroy autonomous drones while they are flying along a predetermined route without actively communicating with a ground-based operator.

4.3 Cyberspace Intelligence Models

Several cyber-cognitive and security designs have been created to counter (prevent) and restrict (curtail) the impact of cybersecurity assaults in FANETs to preserve

confidentiality, integrity, and availability (CIA) in drone networks and ensure their safety. The goal of these security architectures is to increase the Internet of Drones' (IoD) network resilience as a crucial component of the Internet of Everything (IoE) to build a dependable and sustainable Internet of Value (IoV) ecosystem. Generally, drone cyber-cognitive security models are divided into preventive, defensive, and hybrid intelligence mechanisms.

1. **Intrusion Prevention Intelligence (IPI):** Techniques for prevention are very efficient against known and frequent attacks. However, they are likely to perform badly due to the advent of new assaults and insider threats that are not familiar to these architectures. Techniques for prevention work well. Authentication methods (Pu and Li 2020), physical-layer security (Maeng et al. 2022; Fu et al. 2022), sensor-based techniques (Ajakwe et al. 2022c; Davidson et al. 2016), blockchain technology (Tan et al. 2022; Ajakwe et al. 2023d; Saraswat et al. 2022), etc. are common UAV cyber-attack prevention strategies.
2. **Intrusion Detection Intelligence (IDI):** Intrusion detection systems (IDS) are extensively used in UAV networks for cyberattack detection (Ihekoronye et al. 2022; Silva et al. 2023) to detect possible assaults before the system's CIA is compromised. These systems serve as a supplement to preventative techniques. For drone networks, the necessity of a hybrid intrusion prevention system (IPS) as opposed to a traditional IDS has become important lately (Ajakwe et al. 2022d; Abro et al. 2022).
3. **Hybrid Intelligence:** This combines the features of IPI and IDI techniques to achieve robust protection against malicious tendencies on drone security. Hybrid intelligence schemes aim to curtail active and passive cyber-attacks on drone operations. Authors (Ajakwe et al. 2022a) developed a PIDS model using the RNN model

to mitigate cyber-attacks on different types of drones with different sensors. The performance of the model is shown in Fig. 6.

Thus, AI algorithms are the fundamental inference engine in prevention as well as detection intelligence schemes to ensure optimal performance, improve their cyber-resilience potentials, and find and analyze cyber-attack and intrusion patterns efficiently. To comply with the energy-conscious design of UAVs, researchers have devised AI algorithms that reduce cyberattacks. These models are lightweight and require less processing complexity. Table 4 lists the commonly adopted techniques and AI models to achieve cyber-resilience in drone operations, as found in the literature.

The security intelligence techniques in Table 4 promised significant improvements in curtailing drone security breaches and accelerating safe drone operation. Still, it is far from the desired expectation in matching up with the evolving complex technology landscape. Regrettably, traditional AI models typically lack transparency and adhere to "black-box" semantics, which restricts the models' capacity to be applied to real-world use cases and complicates the interpretation of their conclusions. Because of this, the security and safety issues surrounding UAVs for aerial mobility and logistics can be linked to these underlying AI models' incorrect or false attack detection, missed navigation obstacle detection, incorrect scenario and interference classification, and incorrect interpretation of results (Ajakwe et al. 2023a, 2024b).

From the foregoing analysis, it can be deduced that increased intelligence capability is necessary to outsmart vulnerabilities and support sustainable drone deployment. Hence, a transparent cognitive algorithm, tamperproof security approach, and trustworthy architecture are essential for ensuring drone application survival. These factors also

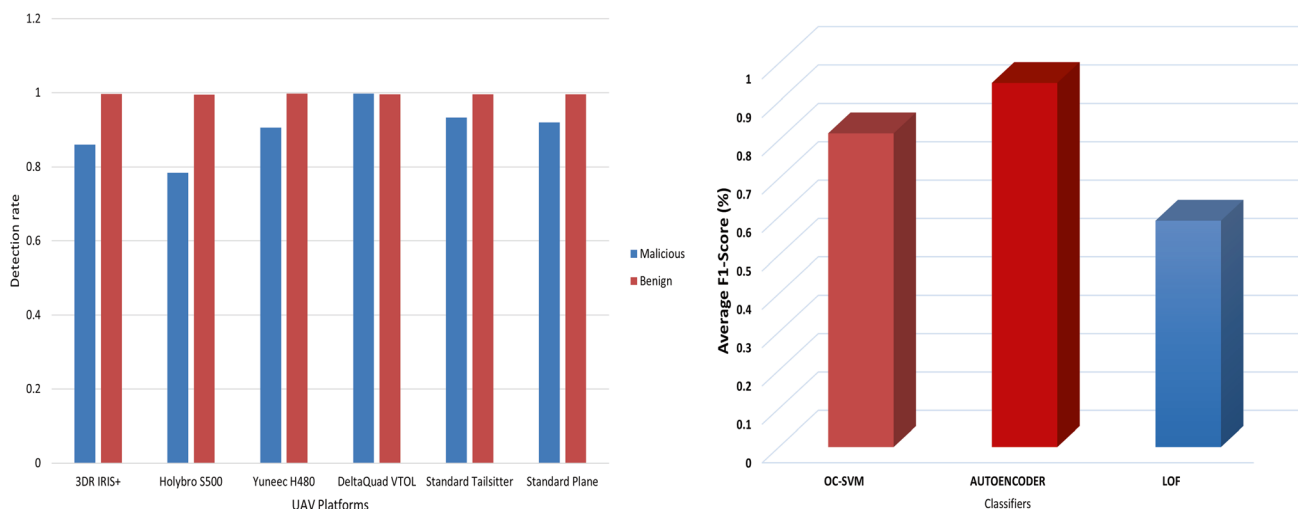


Fig. 6 Performance of Hybrid intelligence model for multiple drone sensor attack mitigation (Ajakwe et al. 2022a)

Table 4 Techniques and AI models for cyber-resilience in drone operations

Security mechanism	Technique/AI model	Cyber-attack addressed
Intrusion prevention intelligence (IPI)	Mutual authentication protocol Hussain et al. (2022)	Eavesdropping, DiTM
	Key agreement Yu et al. (2023)	Desynchronization, DiTM, leakage, impersonation, offline password guessing
	Authentication Scheme (Mallikarachchi et al. 2023)	Brute force, DoS, DiTM, impersonation, replay
Intrusion detection intelligence (IDI)	Machine Learning (ML) (Chulerttiyawong and Jamalipour 2023, Shrestha et al. 2021)	Blackhole, grayhole, GPS SPOOFING, GPS jamming, DoS, DDoS, Brute force, Botnet, web attack, infiltration, GPS spoofing, time delay, sybil
	Deep learning (DL) (Ramadan et al. 2021)	Brute force, Botnet, Dos, port scanning, XSS, backdoor, Dos, DiTM, injection, password, scanning
	Federated learning (FL) (Silva et al. 2023)	GPS spoofing, GPS jamming
Hybrid intelligence	Deep learning (DL) (Ajakwe et al. 2022a)	GPS spoofing, GPS jamming

include smartness-security intelligence to prevent cyberattacks, safety-security intelligence for quick navigation, situational-awareness intelligence for scenario-specific landing, and software-centric and storage intelligence for dependable decision-making in package delivery. Applying zero-trust architecture (ZTA) (Nkoro et al. 2024), blockchain technology (BTC) (Ajakwe et al. 2023d), and explainable AI (XAI) (Haque et al. 2403) have shown to be effective tools for tackling security problems that appear difficult in a variety of domains. Thus, the future security intelligence models for safe-conscious and smart-driven drone operations must leverage the unique potentials of these disruptive technologies to enhance their decision-making.

5 Future Security for Drone Safety Operations

AI, ML, and BTC have become powerful tools in the fight against UAV network cyber threats. Research on AI cyber-security is receiving a lot of attention, as seen by the Web of Science WoS indicators analysis for the last five years, which guarantees the safety of UAV assets. For instance, (AI) algorithms can greatly increase drones' autonomy and functionality without operator assistance, optimize their flight routes, react to changes in the environment, and make decisions based on data processing from sensors and cameras Veprytska and Kharchenko (2024). Increased autonomy leads to increased exposure to attacks that demand a swift and smart approach to handling emerging scenarios. Therefore, future security intelligence architecture for ensuring safe drone operation has to be hypersensitive to dynamic scenarios, event-triggered, and time-triggered. This is necessary to meet the resource constraint peculiarity of drone systems and address identified and emerging

security and safety issues. Developing a robust, responsive, and resilient security intelligence with transparency features that do not deliver advantages to adversaries is critical. This can be achieved by integrating the potentials of ZTA to maintain standardized and enterprise resource-based security policy and governance, BTC for tamper-proof decentralized data storage and authentication, and XAI algorithms for reliable and transparent interpretation of the discovered insights necessary for context-aware decision-making in UAV networks and operations. Figure 7 is a schematic diagram of a typical drone security intelligence framework for safe drone operation.

5.1 Sheer Explainable Drone Model Decision

To address the intelligence failures associated with UAVs due to programmatic conformity and ethnocentric bias (Ajakwe et al. 2022e), the objective function of the XAI paradigm (Trivedi et al. 2024) is to search for ethical AI, valid AI, and interpretable AI to ensure self-explanatory, reliable, unbiased, and transparent results. XAI shares 0.5% adoption in UAVs with other AI algorithms, according to authors (Ajakwe et al. 2022e). While still in its infancy (particularly for UAVs), XAI is expected to garner over \$20 billion in global investments by 2023 (Trivedi et al. 2024) for technological dissemination and adoption across many applications. Based on projections, the XAI market was valued at \$6.2 billion in 2023. By 2028 (MarketsandMarkets: Explainable AI Markets 2024), it is expected to expand by 20.9%. From the Industry 1.0 to the Industry 5.0 revolution (Trivedi et al. 2024), the industrial use of XAI as the underlying cognitive engine in cyber-physical systems has undergone systematic vertical integration to enhance human-centric processes. XAI models use SHapley

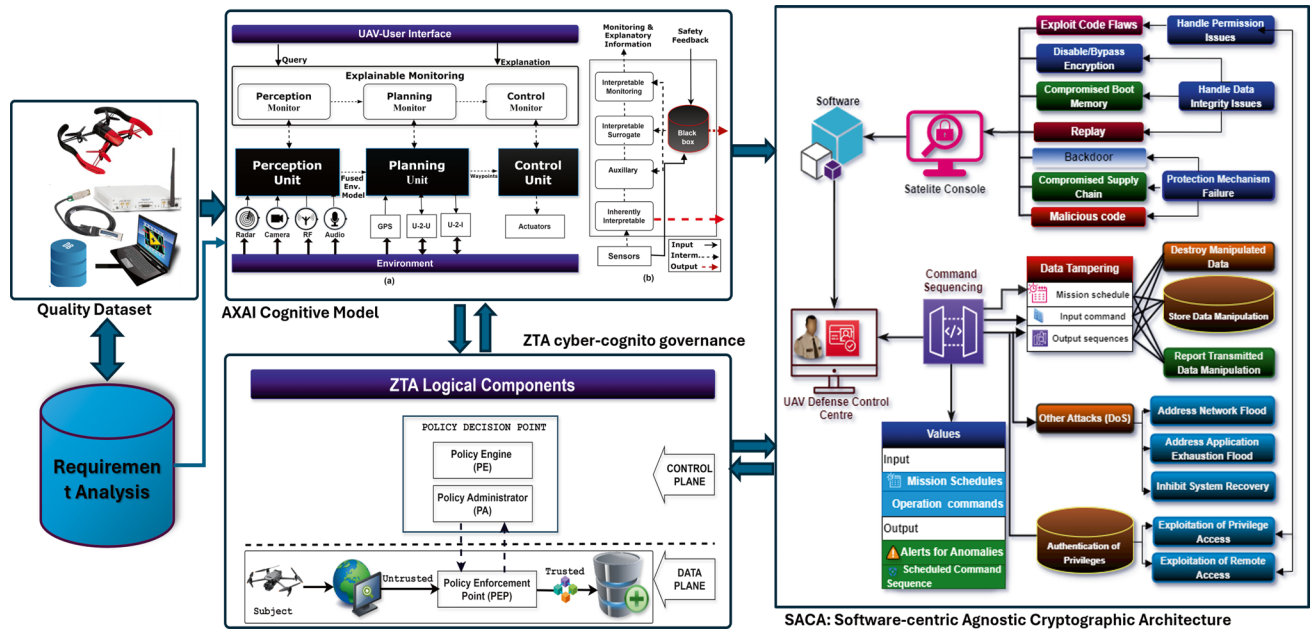


Fig. 7 Future security intelligence for safe drone operations highlighting the convergence of explainable AI, Blockchain technology, zero-trust architecture, and Quality dataset for reliable decision-making

Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) to achieve transparent decision-making and trustworthy model prediction interpretation. With these local and global explanation techniques, XAI models provide concise clarity of the connections between drone sensors' input–output data pairs and unmask the logic behind their operations which is necessary for tackling intrusions. Thus, with XAI, trustworthy and reliable drone autonomy is anticipated which is predicated on certain fundamental pillars and use case requirements.

AI risk management (2023) from the National Institute of Standards and Technology (NIST) states that an AI model must be (i) equitable with harmful bias management, (ii) privacy-enhanced, (iii) explainable and interpretable, (iv) accountable and transparent, (v) secure and resilient, (vi) safe, and (vii) valid and reliable to be considered trustworthy. Designing a future security intelligence cognitive model that meets these requirements demands that transparency mechanisms must be in place at the data, model, and human agency levels to prevent manipulative tendencies. As seen in Fig. 7, an agile XAI cognitive intelligence model that will serve as the underlying inference engine in future drone systems should be able to provide:

1. **Explainable Perception Monitoring:** Multi-sensor signal processing is carried out to extract UAV navigation and positioning information. These fused environmental parameters and semantics are valuable resources for determining the location of the UAV in the environment and its user intention.

2. **Explainable Planning Monitoring:** This interactive interface observes in real-time the UAV communications within a drone network which is used to plan the strategy for UAV navigation, authenticate, and validate the instance of UAV operation in the airspace. Observing the network's behavioral characteristics allows an unauthorized UAV to be detected and curtailed accordingly.
3. **Explainable Control Monitoring:** Intelligent actuators and cognitive radios are designed to allow interoperable hardware–software control of the UAV movement and its admissibility in any geographical space.

With the explicit interpretation and explanation of all cognition-based drone activities by the XAI model, cyber-attacks at the components, communication layer, and communication layer of the drone can be curtailed. Thus, an XAI-enabled cognition model guarantees an inclusive security and safety-conscious drone operation.

5.2 Soundproof Drone User Authentication via Blockchain

For both internal and external security, safeguarding the engine room—the intelligence center that sets navigational guidelines and UAV decision-making principles—is essential. To ensure optimal security against malicious UAV usage, an aerial security design must combine scalability, speed, and security in a way that is both inclusive and secure, enabling real-time (timely), scenario-specific

(tactile), traceable (tractable), trustworthy (transparent), and tamper-proof operations that are in line with the Sustainable Development Goals. Blockchain technology has the potential to completely transform data security in UAV user and package authentication by offering a secure, decentralized, and unhackable platform for sending and storing sensitive data (Ajakwe et al. 2023a; Zhang et al. 2020). The integration of BTC to enhance the security of drone networks is premised on the emergence of newer BTC consensus algorithms with reduced gas cost, computational complexity, and offline transaction capacity (Ajakwe et al. 2023d). Thus, BTC can facilitate soundproof drone security without sacrificing the speed of user authentication, authorization, and communication availability in FANETs as demonstrated by the works of Ajakwe et al. 2023d; Liu et al. 2024).

To address the multivariate landscape of cyber-attacks on drone networks (as highlighted in Table) requires a cross-layer, lightweight, and software-driven security architecture as shown in Fig. 7. This is critical to address the blockchain-trio-dilemma, SSS (speed, security, and scalability), and meet the lightweight resource requirement of drones. Nine (9) essential requirements are needed to integrate BTC to improve the safety of drone operations namely, (i) immutable data integrity of BTC ledger, (ii) interoperability with existing legacy system, (iii) intelligent automation of smart contract, (iv) identifiable supply chain (for logistics use case), (c) inclusive data privacy protection, (vi) informative monitoring and interpretation, (vii) innocuous scalability, (viii) intrinsic regulatory compliance, and (ix) incremental cost-effectiveness. Meeting these requirements by future drone security models empowered by blockchain demands a collaborative involvement of stakeholders in the system development life cycle.

BTC integration into drone operation offers:

1. **Secured Data Exchange and Communication:** The combination of 6G and BTC technologies can simplify pervasive control over UAV communication availability while enhancing network capacity and security to ensure effective task completion. To guarantee data consistency and entity authenticity, authors (Xie et al. 2024) proposed a ubiquitous UAV task security management architecture that combines BTC and 6G for FANETs using a distributed control network (DCN), rapid consensus mechanism (RCS), and a UAV emergency consensus mechanism.
2. **Seamless Drone Authentication:** Drone devices are vulnerable to privacy breaches when their private keys are stored on Physical Unclonable Functions (PUFs). Blockchain enables decentralized UAV user identity storage, lightweight unified management, and concurrent dynamic private key updates. A tamper-proof hierarchical blockchain-based dynamic secret key strategy was developed by Chen et al., (Chen et al. 2024) for use with

swarms of unmanned aerial vehicles (UAVs). This strategy removes the need for drones to store any secret information, thereby mitigating the vulnerabilities associated with key leakages that could result in physical attacks.

3. **Safe Drone Navigation Authorization:** Blockchain ensures data integrity for cooperative UAV swarm navigation through data sharing (Liu et al. 2024) and access control through decentralized ledger security (storage security). By using BTC, the adaptability of UAV systems to network fluctuations resulting from possible cyber attacks by masqueraders can be reduced. This is achieved by assessing the signal's properties and features, modifying the trust propagation, and revising methods as necessary.

As highlighted in Fig. 7, blockchain technology combined with artificial intelligence provides a holistic solution to UAV cybersecurity. The integration of blockchain's transparent and secure data management with AI's predictive analytics and adaptive learning powers allows for strong protection against cyberattacks. In addition to improving UAV security posture, this cooperation guarantees the reliability and trustworthiness of UAV operations in crucial applications. In drone communication, AI can optimize the Blockchain's architecture to make it more effective, safe, and energy efficient. The Blockchain provides AI-based solutions with data immutability and a trust mechanism, which makes them more transparent, reliable, and comprehensible (Bendiab et al. 2023).

5.3 Streamlined Drone Governance via Zero Trust Policy

According to Stafford et al. (2020), Zero Trust (ZT) represents a family of emerging cybersecurity concepts that shift defenses away from fixed, network-based perimeters and toward users, assets, and resources (Stafford 2020). ZT Architecture, or ZTA, plans industrial workflows and enterprise infrastructure using ZT principles and policies. It operates under the premise that no implicit trust is given to user accounts or assets based only on their location on the Internet or Local Area Network, or on who owns the asset (personal or enterprise-based). An enterprise resource is created as a result, and permission and authentication are handled separately. The NIST special report (Stafford 2020) states that ZTA concentrates on safeguarding resources (assets, workflows, services, network accounts, etc.) rather than network segments because the network location is no longer seen as a crucial part of the resource's security. ZTA is an answer to enterprise network trends such as cloud-based assets that are situated outside of the enterprise network's boundaries, remote users, and the link your device

(interoperability) concept. This is comparable to the properties of drone FANETs (see Sect. 2.2).

ZTA is a collection of guiding principles for system design, workflow, and operations rather than a single architecture that may be applied to enhance the categorization or sensitivity of any security posture. As a result, ZTA can be implemented as a ground-breaking solution to deal with the increasing security threats presented by the wide-scale use of drones. This is because it has the potential to fundamentally rethink traditional security models by encouraging continuous and comprehensive verification of all entities within a network environment and doing away with implicit trust. With the ZTA core tenets of “never trust” and “always verify” (Hasan et al. 2023), explicit authorization is sought before resource access is granted for any drone operation. This shift is essential in the context of UAV operations, where a more flexible and robust security architecture is needed due to the constantly shifting and often unpredictable nature of flight paths and interactions. The integration of ZTA in UAV security is still in its early phases, despite the obvious need for it. This is because the process is being introduced incrementally based on risk considerations, process modifications, and the use-case of technology solutions. One specific issue with using ZTA for UAV operations is creating ongoing authentication systems that can adapt to the evolving UAV operational landscape. Figure 7 highlights how ZTA can be adopted in drone systems to streamline their activities and achieve operational efficiency.

Conventional security strategies have proven to be inadequate. This can be as a result of their incapacity to adapt or their inability to deal with the whole gamut of potential risks presented by UAVs. The potential of the zero-trust cybersecurity paradigm for UAV resource protection to assure UAV data security, user identification, access management, operations, endpoint validation, and host environment authentication is demonstrated by a few use scenarios of ZTA adoption in UAV security. ZTA was utilized by Yang et al. (2021) to create a fast UAV user-identity authentication system for reliable and extremely tractable UAV swarm cooperation. This was done to mitigate vulnerabilities like identity theft, unlawful listening, tampering, reply to attacks, man-in-the-middle, or DiTM, and forgery attacks. Similarly, (Dong et al. 2022) adopted ZTA for continuous authentication of UAV delivery using a multi-factor authentication scheme empowered by blockchain technology. Furthermore, authors (Haque et al. 2403) combined XAI and ZTA to enhance drone security through a rigorous and continuous process of authenticating all the network entities and communications. Finally, (Sedjelmaci and Ansari 2023) proposed a ZTA-empowered IDS that combines security rules and ML algorithms (RL) for cooperative detection and curtailment of cyber-attacks on IoD.

These case studies are possibilities of ZTA in addressing drone security issues.

However, to integrate ZTA into drone systems, certain key considerations have to be adhered to in line with ZTA core principles.

- Drone data sources and computing services must be considered resources.
- All drone-related communication is secured, regardless of network location.
- Access to individual enterprise resources is granted on a per-session basis.
- Access to drone resources is determined by dynamic policy—including the observable state of client identity, application/service, and the requesting asset—and may include other behavioral and environmental attributes.
- The enterprise monitors and measures the integrity and security posture of all owned and associated assets.
- All resource authentication and authorization are dynamic and strictly enforced before access is allowed.
- The enterprise collects as much information as possible about the current state of assets, network infrastructure, and communications and uses it to improve its security posture.

Depending on the security requirements and security needs, adopting the ZTA cybersecurity paradigm to expedite safety-driven UAVs, increased identity governance, micro-segmentation, and/or network infrastructure-software-defined parameter-based techniques can be done. These basic considerations suggest that ZTA to maintain security governance, BTC for tamper-proof decentralized data storage, and XAI for transparent and dependable interpretation of the discovered insights for context-aware decision-making in drone networks can be sufficient to address the various systemic vulnerabilities, security attacks, and safety issues associated with the use of drone for logistics.

6 Conclusion

Due to technological developments in underlying UAV technologies, such as greater payload capacities, longer flight periods, situational awareness cognition, etc., there is increase in the use of drones. However, different security and safety concerns impede this scalable adoption, endangering the drones’ practicality, usability, and continued existence. This chapter looked at security concerns that affect the effectiveness of drone operations. In addressing the associated risks and facilitating safe and scalable drone deployment, the chapter explored the role of explainable AI algorithms for transparent context-aware cognition intelligence, blockchain technology for tamper-proof security

control intelligence, and zero-trust architecture for trustworthy core governance cooperation intelligence. It also covered important conditions and factors to take into account when incorporating these intelligence paradigms into drone system operations.

By integrating XAI, BTC, and ZTA into drone operations, security issues such as airspace security, cybersecurity, data privacy, payload security, infoswarm, etc., can be maximally addressed. Also, drone safety-related issues such as inadvertent landings, airspace accidents, collisions, disruptive tendencies of key infrastructures, etc., can be significantly curtailed. However, technology is irrelevant when it comes to enhancing the security and safety of UAV mobility and networks through the integration of explainable AI, blockchain, and zero trust. To accelerate the potential of smart mobility and intelligent logistic ecosystems via UAV, many use-case requirements must be taken into account for the scalable deployment of these new paradigms.

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Cost–Benefit Analysis of Sensing and Data Collection with Drones for IoT Applications

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Abstract

Drones for IoT applications have emerged as a transformative approach for efficient sensing and data collection. This chapter presents a comprehensive cost–benefit analysis of utilizing drones equipped with various sensors—environmental, imaging, and communication—for data gathering in IoT networks. By leveraging real-time and batch data collection methods, drones offer significant advantages in connectivity and data transmission. Initial acquisition costs of drones range from \$500 to \$50,000 depending on capabilities, with annual maintenance expenses averaging \$2,000 per unit. Operational costs, influenced by energy consumption and personnel training, add an additional \$1,500 per drone per year. Compliance costs, including licensing, permits, and insurance, contribute approximately \$1,000 annually. On the benefit side, drones significantly enhance operational efficiency and speed, enabling rapid deployment and real-time data availability. The ability to access remote and hazardous areas increases versatility, while high-resolution data capture improves accuracy and reliability. Economic analysis models, such as Cost–Benefit Analysis (CBA), Return on Investment (ROI), and Life Cycle Cost Analysis (LCCA), are applied to evaluate financial viability. Studies indicate that drones can reduce operational costs by 30–50% and improve

data quality by 20–40% compared to traditional methods. Challenges in this analysis include data uncertainty, technological risks, and ethical considerations. However, advancements in drone technology, AI integration, and improved data analytics are mitigating these issues, leading to cost reductions and enhanced benefits. Comparative analysis with ground-based and satellite-based methods shows that drones offer superior flexibility and cost-effectiveness, making them a preferred choice for modern IoT applications. Future trends suggest continued growth and adoption, driven by innovations and evolving regulatory frameworks, underscoring the strategic importance of integrating drones into IoT networks for optimized outcomes.

Keywords

Cost–benefit analysis · Data collection · Drones · Efficiency · IoT · Sensors

1 Introduction

Drones for IoT applications have emerged as a transformative technology, enabling advanced sensing and data collection across various sectors. This introduction delves into a cost–benefit analysis of utilizing drones for these purposes, focusing on the economic and operational impacts, technological advancements, and strategic implementation. The integration of drones with IoT networks has the potential to revolutionize data collection, offering unprecedented efficiency, accessibility, and precision (Mehta et al. 2023; Li et al. 2019; Tiwari et al. 2023). In terms of sensing and data collection, drones are equipped with a diverse array of sensors that cater to different needs. Environmental sensors measure parameters such as temperature, humidity, and

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air quality, crucial for applications in agriculture and environmental monitoring. Imaging sensors, including high-resolution cameras and thermal imagers, provide detailed visual data for sectors like construction, disaster management, and security (Gallacher 2016a; Weerasooriya et al. 2024). Communication sensors facilitate real-time data transmission, enabling drones to interact seamlessly with IoT networks. Data collection methods vary, with real-time data collection allowing for immediate analysis and decision-making, while batch data collection enables drones to gather large volumes of data for subsequent processing (Wu et al. 2021).

Integrating drones with IoT networks involves sophisticated connectivity protocols and data transmission techniques. Protocols such as MQTT (Message Queuing Telemetry Transport) and CoAP (Constrained Application Protocol) ensure reliable and efficient communication between drones and IoT devices. Data transmission techniques, including 4G/5G and satellite links, support continuous connectivity, essential for real-time applications. The cost factors associated with drone operations encompass acquisition, maintenance, operational, and compliance costs (Gallacher 2016b; Mazhar et al. 2024; Araújo et al. 2021). Initial purchase costs vary based on the drone's capabilities, ranging from a few hundred to several thousand dollars. Maintenance costs, including regular servicing and repairs, are essential to ensure optimal performance and longevity. Operational costs include energy consumption and battery replacements, with drones typically requiring frequent recharges or battery swaps to maintain flight times. Personnel training and labour costs are significant, as skilled operators are needed to handle the complex functionalities of drones (Dutta and Mitra, 2021; Wang et al. 2022a).

Compliance and regulatory costs are critical considerations in drone operations. Obtaining licenses and permits, which vary by region, is mandatory for legal operation. Additionally, insurance and liability costs protect against potential accidents and damages, safeguarding both operators and stakeholders. The benefits of using drones for IoT applications are substantial. In terms of efficiency and speed, drones offer rapid deployment and data collection capabilities, significantly reducing the time required for traditional methods (Feroz and Abu Dabous 2021; Bochie et al. 2021). Real-time data availability enhances decision-making processes, enabling timely responses to dynamic situations. Accessibility and flexibility are notable advantages, as drones can reach remote or hazardous areas that are challenging or dangerous for humans. Their versatility allows them to be used across various industries, from agriculture to infrastructure inspection (Wang et al. 2022b; Dulia and Shihab 2024).

Data quality and precision are enhanced through high-resolution data capture, providing detailed and accurate information for analysis. This improved accuracy and reliability contribute to better outcomes in applications such as precision agriculture, where precise data on soil conditions and crop health can lead to optimized resource use and increased yields. Economic analysis models, including Cost–Benefit Analysis (CBA), Return on Investment (ROI), and Life Cycle Cost Analysis (LCCA), provide frameworks for evaluating the financial viability of drone operations (Vierhauser et al. 2022; Ruan et al. 2019). CBA involves identifying and quantifying the costs and benefits, providing a comprehensive understanding of the economic impact. ROI calculations help determine the profitability of drone projects, comparing the returns with traditional methods. LCCA offers insights into the long-term costs and benefits, considering the entire lifecycle of drone-enabled IoT systems (Mishra 2024).

Quantitative analysis of costs and benefits involves collecting and analysing data using statistical tools and techniques. Case studies and hypothetical scenarios illustrate the practical implications, offering generalized metrics for assessment. However, challenges in cost–benefit analysis include data uncertainty and variability, technological and operational risks, and ethical and legal considerations (Kabashkin 2023). Mitigation strategies for these challenges involve robust data collection methodologies, contingency planning, and adherence to legal and ethical standards. Technological advancements in drone technology and data analytics are driving down costs and enhancing benefits. Innovations such as autonomous drones and AI integration improve operational efficiency and data quality. Advances in sensor technology and machine learning enable more precise data collection and analysis, supporting predictive analytics and decision-making (Mahamkali et al. 2023; Rejeb et al. 2021).

Comparative analysis with alternative methods, such as ground-based sensing technologies and satellite-based data collection, highlights the situational advantages and limitations of drones. While ground-based methods are cost-effective for localized data collection, drones offer superior coverage and flexibility. Satellite data provides large-scale insights but lacks the granularity of drone-collected data (Kooistra and Poppe, 2015; Mishra and Mishra, 2024). Traditional manual data collection, though labour-intensive, can be supplemented by drone operations to achieve higher efficiency and accuracy. Future trends in drone technology and IoT integration indicate growing adoption and market expansion. Emerging technologies, such as next-generation sensors and advanced IoT networks, promise to enhance the capabilities and applications of drones. Policy and regulatory developments are expected to shape the landscape,

with evolving legal frameworks and standards ensuring safe and effective drone operations (Kalatzis et al. 2018).

Strategic recommendations for optimizing cost–benefit outcomes include best practices for cost management, such as efficient resource allocation and regular maintenance. Strategies for maximizing benefits involve leveraging the latest technologies and continuous training for operators. Adoption and implementation strategies encompass steps for integrating drones into IoT networks, emphasizing training and capacity-building measures to ensure successful deployment and operation (Boškov et al. 2023). The cost–benefit analysis of sensing and data collection with drones for IoT applications underscores the significant advantages and considerations of this technology. With careful planning and strategic implementation, drones can offer transformative benefits across various sectors, driving innovation and efficiency in data collection and analysis (ur Rehman et al. 2017; Ali et al. 2021).

Based on the literature review, the economic analysis of AI-based Internet of Drones (IoD) is characterized by a number of significant research gaps, such as the absence of comprehensive cost–benefit analyses, insufficient data, scalability and generalizability, dynamic pricing and revenue models, regulatory impact assessments, environmental and social impact, standardization and interoperability, cybersecurity costs and benefits, long-term economic sustainability, and comparative analyses. By addressing these research gaps, we will gain a more comprehensive understanding of the economic implications, which will inform business decisions, policy-making, and future research directions.

2 Sensing and Data Collection with Drones

The integration of drones into the Internet of Things (IoT) ecosystem has revolutionized the landscape of data acquisition and environmental monitoring. This section provides a comprehensive examination of the methodologies and technologies involved in drone-enabled sensing and data collection. It elucidates the diverse types of sensors utilized in drones, which are pivotal in enhancing the precision, efficiency, and scope of data collection processes (Fernandes et al. 2022).

2.1 Types of Sensors Used in Drones

Drones, equipped with a variety of sensors, are capable of capturing and transmitting a wide array of data types. The choice of sensors is critical, as it determines the nature and quality of the data collected, thereby impacting the overall

effectiveness of the IoT applications. The primary categories of sensors deployed in drones include environmental sensors, imaging sensors, and communication sensors.

2.1.1 Environmental Sensors

Environmental sensors on drones are designed to monitor and measure various physical and chemical parameters of the environment. These sensors play a crucial role in applications such as weather forecasting, pollution monitoring, and agricultural management.

- *Temperature and Humidity Sensors:* These sensors measure ambient temperature and humidity levels, providing essential data for weather prediction models and climate studies. In agriculture, they assist in monitoring micro-climatic conditions, which can influence crop health and yield.
- *Gas Sensors:* Equipped to detect and quantify the concentration of various gases (e.g., CO₂, NO_x, and O₃), these sensors are invaluable in air quality monitoring and pollution tracking. They help in identifying sources of emissions and in assessing compliance with environmental regulations (Kiropoulos et al. 2021).
- *Pressure Sensors:* These sensors measure atmospheric pressure, which is critical for weather forecasting and for adjusting the flight parameters of drones to maintain stability and performance at varying altitudes.
- *Particulate Matter Sensors:* Used to detect and measure airborne particles such as dust, pollen, and smoke, these sensors are essential for assessing air quality and for health-related studies in urban environments (Shao et al. 2022; Kim et al. 2020).

2.1.2 Imaging Sensors

Imaging sensors on drones capture visual data, which can be processed to extract valuable information across various domains, including surveillance, mapping, and precision agriculture.

- *Optical Cameras:* These include standard RGB cameras that capture high-resolution images and videos. They are widely used in applications ranging from infrastructure inspection to wildlife monitoring, providing detailed visual documentation.
- *Multispectral and Hyperspectral Cameras:* These cameras capture data across multiple wavelengths beyond the visible spectrum. Multispectral sensors are particularly useful in agriculture for assessing crop health, soil properties, and water stress by analysing plant reflectance in different spectral bands. Hyperspectral cameras offer even finer spectral resolution, enabling more detailed material identification and classification.

- *Thermal Cameras:* These sensors detect infrared radiation and produce thermal images, which reveal temperature variations. They are utilized in applications such as search and rescue, fire detection, and building inspections to identify heat leaks and structural issues.
- *LiDAR (Light Detection and Ranging):* LiDAR sensors emit laser pulses and measure the time it takes for the pulses to return after hitting a surface. This technology generates precise 3D maps of the terrain and built environments, aiding in topographical mapping, forest management, and urban planning (Mustapha et al. 2021).

2.1.3 Communication Sensors

Communication sensors are integral to the operation of drones within the IoT framework, enabling real-time data transmission and interaction with other devices and systems.

- *Radio Frequency (RF) Transceivers:* These sensors facilitate wireless communication between the drone and ground control stations. They support data telemetry, remote control, and live video feed transmission, ensuring that the drone's operations can be monitored and adjusted in real-time.
- *Wi-Fi and Bluetooth Modules:* Used for short-range data transfer, these modules enable the drone to connect to local networks and IoT devices. They are particularly useful in indoor environments or areas with dense IoT infrastructure.
- *Cellular Modules (4G/5G):* These modules provide long-range communication capabilities, allowing drones to transmit data over cellular networks. This is essential for beyond visual line of sight (BVLOS) operations, enabling drones to operate over extended distances and in remote areas where direct RF communication is impractical (Walter et al. 2023).
- *Satellite Communication:* In areas with no cellular coverage, satellite communication modules ensure uninterrupted data transmission. This capability is crucial for drones operating in remote or maritime environments, supporting applications such as disaster response and environmental monitoring.

The diverse array of sensors employed in drones facilitates comprehensive and versatile data collection, underpinning the effectiveness of IoT applications. Each type of sensor contributes uniquely to the overall data acquisition process, enhancing our ability to monitor, analyse, and interact with the physical world in unprecedented ways. (Fernando et al. 2019; Tiwari et al. 2011).

2.2 Data Collection Methods

The methods employed for data collection using drones are pivotal in determining the efficiency and effectiveness of the gathered data for IoT applications. The two primary approaches are real-time data collection and batch data collection, each with distinct operational frameworks, advantages, and limitations.

2.2.1 Real-Time Data Collection

Real-time data collection involves the immediate transmission of data from the drone to the ground control station or a central processing unit as the data is being gathered. This method is facilitated by various communication technologies, such as cellular networks, satellite links, or specialized radio frequencies, which ensure that the data is available instantaneously for analysis and decision-making processes (Zaman et al. 2023).

Advantages:

- *Immediate Access to Data:* Real-time collection allows for prompt access to data, which is crucial for time-sensitive applications such as disaster management, search and rescue operations, and real-time environmental monitoring.
- *Dynamic Response:* The capability to receive and analyse data in real time enables dynamic response and real-time adjustments to the drone's flight path and sensing parameters based on the ongoing observations.
- *Reduced Data Loss Risk:* Continuous data transmission minimizes the risk of data loss that could occur due to drone malfunctions or crashes before landing (Khang et al. 2023).

Challenges:

- *Connectivity Issues:* Dependence on robust and uninterrupted communication networks can be a significant limitation, especially in remote or hostile environments where signal coverage is weak or non-existent.
- *High Energy Consumption:* Real-time data transmission consumes considerable power, which can reduce the drone's operational time and necessitate frequent battery changes or recharges.
- *Bandwidth Limitations:* The need for high bandwidth to transmit large volumes of data in real time can be a constraint, particularly for high-resolution imagery or video streams.

2.2.2 Batch Data Collection

Batch data collection entails storing the collected data onboard the drone during the mission and transmitting it to the ground control station or a central server after the mission has been completed. This method relies on the drone's onboard storage capabilities and is typically used when real-time transmission is not feasible.

Advantages:

- *Operational Simplicity:* Batch data collection is less dependent on continuous communication links, making it suitable for operations in areas with poor connectivity or in urban environments where signal interference is prevalent.
- *Energy Efficiency:* Since real-time transmission consumes significant energy, batch data collection is more energy-efficient, allowing for extended flight times and larger data sets to be gathered.
- *Higher Data Quality:* By storing data onboard, drones can collect higher resolution data without compression, which is often necessary for real-time transmission due to bandwidth limitations (Pidchosa and Hunda 2022).

Challenges:

- *Data Loss Risk:* The risk of data loss is higher if the drone crashes or malfunctions before data transmission to the ground station can occur.
- *Delayed Data Access:* The primary disadvantage is the delayed availability of data, which can be critical for applications requiring immediate analysis and response.
- *Storage Limitations:* The amount of data that can be collected is constrained by the onboard storage capacity, necessitating careful planning and optimization of data collection parameters (Gebresenbet et al. 2023).

2.3 Integration with IoT Networks

The integration of drones with IoT networks involves seamless connectivity and efficient data transmission techniques to ensure that the data collected by drones is effectively utilized within the broader IoT ecosystem. This integration requires robust connectivity protocols and advanced data transmission techniques.

2.3.1 Connectivity Protocols

Connectivity protocols are essential for ensuring reliable and efficient communication between drones and IoT networks. These protocols govern how data is transmitted, received, and processed within the network.

Types of Connectivity Protocols:

- *Wi-Fi:* Widely used for short-range, high-bandwidth data transmission, suitable for operations in areas with existing Wi-Fi infrastructure.
- *Cellular Networks (4G/5G):* Provide broader coverage and higher data transfer rates, essential for real-time applications and operations in urban or suburban areas.
- *LPWAN (Low Power Wide Area Network):* Such as LoRaWAN or Sigfox, suitable for long-range communication with low power consumption, ideal for rural or remote areas.
- *Satellite Communication:* Used for operations in extremely remote areas where terrestrial communication networks are unavailable, albeit with higher latency and cost (Alawad and Kraemer, 2022).

Considerations for Protocol Selection:

- *Range and Coverage:* The operational environment dictates the choice of protocol, with LPWAN and satellite communication being preferable for remote areas.
- *Data Rate Requirements:* High-resolution data or real-time video streams necessitate protocols with higher bandwidth, such as Wi-Fi or cellular networks.
- *Energy Consumption:* Low power consumption is critical for extending drone flight times, influencing the selection of protocols like LPWAN for energy-efficient communication.

2.3.2 Data Transmission Techniques

Data transmission techniques encompass the methods and technologies used to transfer data from drones to IoT networks efficiently and reliably. These techniques must address various challenges such as data volume, transmission distance, and energy constraints.

Techniques for Efficient Data Transmission:

- *Data Compression:* Reducing the size of the data before transmission to minimize bandwidth usage and speed up the transfer process. Techniques include lossless compression for sensitive data and lossy compression for less critical information.
- *Edge Computing:* Processing data locally on the drone or at a nearby edge server to reduce the amount of data that needs to be transmitted, thus saving bandwidth and reducing latency.
- *Adaptive Transmission:* Utilizing dynamic adjustment of transmission parameters based on the available network conditions to optimize data flow and maintain communication quality.

- *Multi-hop Communication*: Relaying data through intermediate nodes (e.g., other drones or ground stations) to extend the communication range and improve reliability in challenging environments (Rajamohan, 2023).

Challenges in Data Transmission:

- *Latency*: Ensuring low latency is essential for real-time applications, requiring high-speed communication networks and efficient data processing techniques.
- *Interference*: Managing interference from other devices and networks, particularly in crowded urban environments, is crucial for maintaining data integrity and transmission reliability.
- *Security*: Implementing robust encryption and authentication mechanisms to protect data during transmission from unauthorized access and tampering.

The methods of data collection and the integration of drones with IoT networks are critical components of modern IoT applications. Understanding the nuances of real-time versus batch data collection and the intricacies of connectivity protocols and data transmission techniques is essential for optimizing drone operations and maximizing the utility of collected data within the IoT ecosystem (Suhail et al. 2021).

3 Cost Factors in Drone Operations

Cost factors in drone operations are critical considerations for organizations and individuals looking to integrate drones into their IoT applications. Understanding these costs can aid in making informed decisions that balance operational efficiency with budget constraints. The primary cost factors include acquisition and maintenance costs, which encompass initial purchase, regular maintenance, and repairs.

3.1 Acquisition and Maintenance Costs

Acquisition and maintenance costs are pivotal in determining the overall financial feasibility of deploying drones for IoT applications. These costs can be categorized into initial purchase costs and ongoing costs associated with maintenance and repairs. Both categories play a significant role in the long-term sustainability and reliability of drone operations.

3.1.1 Initial Purchase Cost

The initial purchase cost of a drone is the upfront expense incurred when acquiring the drone hardware. This cost

varies significantly based on several factors, including the drone's specifications, capabilities, and the brand. High-end drones equipped with advanced sensors, extended flight times, and robust data collection capabilities can be substantially more expensive than basic models (Alanezi et al. 2022).

Moreover, the intended use of the drone heavily influences the initial purchase cost. Drones designed for specialized applications, such as high-resolution aerial photography, thermal imaging, or LiDAR mapping, often come with higher price tags due to the sophisticated technology they incorporate. For instance, drones utilized in agricultural monitoring might require multispectral cameras, which are more costly than standard optical cameras.

Additionally, purchasing decisions must consider the cost of ancillary equipment and software. This includes ground control stations, spare batteries, propellers, and software licenses for data processing and analysis. These supplementary items, although sometimes overlooked, can add significantly to the initial expenditure (Mishra and Mishra 2023; Wang et al. 2021).

3.1.2 Regular Maintenance and Repairs

Regular maintenance and repairs represent ongoing costs that are essential for ensuring the optimal performance and longevity of drones. These costs can be divided into routine maintenance expenses and repair costs arising from unexpected damages or wear and tear. Routine maintenance involves periodic checks and servicing of the drone's components to prevent potential issues and ensure safe operation. This includes inspecting and replacing propellers, calibrating sensors, updating firmware, and checking battery health. Regular maintenance schedules vary based on the drone model and usage frequency but are crucial for avoiding costly repairs and prolonging the drone's lifespan (Liu 2024).

Repair costs, on the other hand, can be more unpredictable. They arise from incidents such as crashes, component failures, or environmental damage. Depending on the severity of the damage, repair costs can range from minor fixes, like replacing a damaged propeller, to major repairs involving significant parts of the drone. Advanced drones with sophisticated components may incur higher repair costs due to the specialized nature of the parts and the technical expertise required for repairs (Hosseinzadeh et al. 2022). Insurance is another aspect that impacts the overall maintenance and repair costs. Many operators choose to ensure their drones to mitigate financial losses from accidents or damages. The cost of insurance premiums must be factored into the overall maintenance budget, along with any deductibles or coverage limitations (Krichen et al. 2023).

3.2 Operational Costs

3.2.1 Energy Consumption and Battery Costs

Energy consumption constitutes a significant operational cost in drone operations, particularly in the context of IoT applications where consistent and reliable data collection is paramount. Drones, by design, rely heavily on battery power, and the efficiency of these batteries directly impacts operational longevity and cost-effectiveness. The type and capacity of the battery, its charge–discharge cycles, and the drone's power management system are critical considerations (Maretto et al. 2023; Silva et al. 2022).

Lithium-polymer (LiPo) batteries are commonly used due to their high energy density and discharge rates, but they are also costly and require regular replacement due to their limited lifecycle. The expense of maintaining a fleet of drones with adequately charged batteries can be substantial. Furthermore, the energy consumption of the drone's various components—such as sensors, communication modules, and onboard processors—adds to the total power requirement, necessitating careful energy budget planning (Kumar et al. 2023).

The operational cost related to energy consumption can be mitigated through advancements in battery technology, such as the development of more efficient energy storage systems and energy-harvesting technologies. Additionally, implementing optimized flight paths and power management algorithms can reduce energy consumption, thereby lowering costs. Understanding and managing these energy requirements is crucial for the sustainability and economic viability of drone operations in IoT applications (Neumann et al. 2018).

3.2.2 Personnel Training and Labour

The successful deployment and operation of drones for IoT applications demand skilled personnel, which incurs significant training and labour costs. Operators must be proficient in piloting drones, often requiring certifications and ongoing education to stay abreast of evolving technologies and regulations. The complexity of drone operations necessitates comprehensive training programs that cover not only the mechanics of flight but also the integration and troubleshooting of IoT systems.

Labour costs extend beyond the initial training phase. Routine maintenance, data management, and analysis are labour-intensive activities that require specialized skills. Technicians must regularly inspect and repair drones to ensure optimal performance and safety. Moreover, the data collected by drones often needs to be processed and analysed, necessitating expertise in data science and IoT analytics (Khosravi et al. 2021).

The investment in personnel training and labour can be considerable, but it is essential for ensuring operational efficiency and safety. Cost-effective strategies include leveraging simulation-based training, which reduces the need for extensive hands-on flight training, and utilizing automated maintenance and diagnostic tools to decrease manual labour requirements. Developing a workforce with the necessary skills is fundamental to the sustainable and effective operation of drone fleets in IoT ecosystems (Thomopoulos et al. 2021; Jarašūnienė et al. 2023).

3.3 Compliance and Regulatory Costs

3.3.1 Licensing and Permits

Operating drones within the legal framework necessitates obtaining various licenses and permits, which contribute to the overall cost of drone operations. Regulatory requirements vary by country and often include airworthiness certifications, operator licenses, and specific mission permits. Compliance with these regulations ensures that drone operations are conducted safely and within the legal boundaries, but it also imposes financial burdens (Tzanidakis et al. 2023).

The process of acquiring the necessary licenses and permits can be time-consuming and costly, involving fees for application processing, periodic renewals, and possible inspections. Additionally, staying compliant with evolving regulations requires continuous monitoring and adaptation, which can incur further administrative costs.

Organizations must factor these costs into their budget planning and may benefit from establishing dedicated compliance teams to manage regulatory affairs efficiently. The expense associated with licensing and permits is an essential aspect of the cost–benefit analysis, as non-compliance can result in severe penalties, operational disruptions, and reputational damage (Kavga et al. 2020).

3.3.2 Insurance and Liability

Insurance is another critical component of the cost structure in drone operations. The inherent risks associated with drone flights—such as accidents, technical failures, and potential damages to property or individuals—necessitate comprehensive insurance coverage. Policies typically cover liability for third-party damages, physical damage to the drone, and other operational risks (Shedole and Madhu 2019). Insurance premiums are influenced by various factors, including the type of drone, its operational environment, the value of the equipment, and the scope of its usage. High-risk operations, such as flights in densely populated areas or critical infrastructure inspections, tend to attract higher premiums due to the increased likelihood of incidents.

Liability considerations also extend to data security and privacy concerns, especially in IoT applications where sensitive data is collected. Ensuring robust data protection measures and compliance with privacy regulations can mitigate these risks but may increase costs. The financial impact of insurance and liability is significant but necessary for safeguarding against potential losses and ensuring the continuity of operations. Proactively managing these risks through comprehensive insurance policies and adherence to safety protocols is essential for maintaining operational integrity and financial stability in drone-based IoT systems (Zaman 2023; Morrone et al. 2022).

The cost of drone-based sensing and data collection in the Internet of Things (IoT) can vary significantly across various regions as a result of factors such as:

1. Regulations: The regulations and permits for drones are subject to change in various countries or states.
2. Labor costs: Distinction between the labor costs of data analysts and trained pilots.
3. Equipment expenses: Maintenance and repair expenses, as well as fluctuations in the prices of drones and sensors.
4. Infrastructure: The quality and availability of infrastructure, including data transfer networks, charging stations, and landing sites.
5. Environmental factors: Airspace restrictions, terrain, and weather conditions that influence drone operations.

The following is a preliminary estimate of the costs in various regions:

1. North America:
 - Drone hardware: \$5,000 to \$20,000
 - Software and sensors: \$3,000 to \$10,000
 - Labor costs are between \$50 and \$150 per hour for pilots and analysts.
 - The total cost of a project is between \$10,000 and \$50,000.
2. Europe:
 - Drone hardware: €4,000–€15,000 (approximately \$4,500–\$16,700 USD)
 - Software and sensors: €2,000 to €8,000 (approximately \$2,200 to \$9,000 USD)
 - Labor costs (pilot, analyst): €30 to €100 per hour (approximately \$33 to \$110 USD)
 - The total cost of each undertaking is between €8,000 and €35,000 (approximately \$9,000 and \$39,000 USD).
3. Asia-Pacific:
 - Drone hardware: \$3,000–\$12,000
 - Software and sensors: \$2,000 to \$6,000—Labor costs (pilot, analyst): \$20 to \$80 per hour

- Total cost: \$6,000 to \$25,000 per project
4. South America:
 - Drone hardware: \$4,000–\$15,000
 - Software and sensors: \$2,500 to \$8,000
 - Labor costs (pilot, analyst): \$30 to \$100 per hour
 - Total cost: \$9,000 to \$35,000 per project
 5. Africa:
 - Drone hardware: \$3,500–\$12,000
 - Software and sensors: \$2,000–\$6,000
 - Labor costs (pilot, analyst): \$25–\$80 per hour
 - Total cost: \$7,000–\$25,000 per project

These estimates are approximate and may fluctuate based on the specific requirements, project scope, and location within a region.

Example:

Table 1 provides an example of the total annual cost analysis for drone operations considering various cost factors such as acquisition and maintenance, operational costs and compliance and regulatory costs.

4 Benefit Factors in Drone Operations

The integration of drones in various fields has significantly augmented operational efficiency and efficacy. This section delves into the multifaceted benefits offered by drone operations, elucidating how they manifest in enhancing efficiency and speed, particularly focusing on rapid deployment and data collection, as well as the real-time availability of data.

4.1 Efficiency and Speed

Efficiency and speed stand as quintessential pillars defining the efficacy of any operational process. Drones, equipped with cutting-edge technology and manoeuvrability, inherently possess the capability to revolutionize conventional paradigms, offering unparalleled efficiency and speed in diverse applications.

4.1.1 Rapid Deployment and Data Collection

One of the most conspicuous advantages of employing drones in IoT applications is their ability to facilitate rapid deployment and data collection. Traditional methods often entail substantial time investments in setting up equipment and mobilizing personnel, leading to delays in data acquisition (Ojha et al. 2023). Conversely, drones, characterized by their aerial mobility and swift deployment capabilities, circumvent such logistical constraints, ensuring prompt initiation of data collection endeavours.

Table 1 Total annual cost analysis for drone operations

Cost factor	Cost description	Cost calculation	Amount (\$)
<i>Acquisition and maintenance costs</i>			
Initial purchase cost	Cost of purchasing drones	\$50,000 per drone (× 5 drones)	\$250,000
Regular maintenance and repairs	Annual maintenance cost per drone	\$5,000 per drone (× 5 drones)	\$25,000
<i>Operational costs</i>			
Energy consumption and battery costs	Energy cost for charging batteries	\$500 per month (× 12 months)	\$6,000
	Replacement batteries cost	\$200 per battery (× 10 batteries)	\$2,000
Personnel training and labour	Training cost per operator	\$2,000 per operator (× 3 operators)	\$6,000
	Personnel labour cost per flight hour	\$50 per hour (× 300 h)	\$15,000
<i>Compliance and regulatory costs</i>			
Licensing and permits	Licensing fee for drone operation	\$1,000 per year	\$1,000
Insurance and liability	Insurance premium for drone operations	\$5,000 per year	\$5,000
	Liability coverage for potential damages	\$10,000 per year	\$10,000
<i>Total annual cost</i>			\$320,000

Moreover, the agility and adaptability of drones enable them to traverse diverse terrains and navigate through challenging environments with ease. This flexibility not only expedites the data collection process but also extends the scope of applications to encompass remote or inaccessible regions, thereby broadening the spectrum of data acquisition possibilities (Sharabatiet al. 2023).

Furthermore, the autonomous functionalities integrated into modern drone systems streamline the operational workflow, eliminating the need for constant manual intervention. Automated flight routes and pre-defined data collection parameters enable drones to execute missions with precision and consistency, minimizing human error and optimizing resource utilization.

4.1.2 Real-Time Data Availability

In addition to expediting the data collection phase, drones facilitate real-time data availability, thereby enhancing the responsiveness and agility of decision-making processes. By leveraging onboard sensors and telemetry systems, drones capture and transmit data instantaneously, enabling stakeholders to access crucial information without latency.

Real-time data streaming empowers decision-makers to monitor dynamic scenarios and respond proactively to emerging trends or anomalies. Whether in agricultural monitoring, infrastructure inspection, or disaster management, the ability to access timely insights fosters informed decision-making and facilitates proactive intervention strategies.

Furthermore, the integration of edge computing capabilities within drone platforms enables on-the-fly data processing, facilitating rapid analysis and interpretation of acquired data streams. This not only reduces dependence on centralized processing resources but also mitigates bandwidth constraints associated with transmitting

raw data over extended distances (Dudukcu et al. 2023; Sziroczak et al. 2022).

4.2 Accessibility and Flexibility

4.2.1 Reach in Remote or Hazardous Areas

One of the paramount advantages of employing drones in data collection is their unparalleled reach in remote or hazardous areas (as shown in Fig. 1). Traditional methods often face insurmountable challenges when accessing such environments due to geographical barriers or safety concerns. However, drones transcend these limitations by effortlessly navigating through rugged terrains, dense forests, or hazardous zones, thus enabling data collection from otherwise inaccessible areas. This capability not only enhances the scope of data acquisition but also mitigates risks to human personnel, thereby fostering safety and efficiency in operations (Aretoulaki et al. 2023).

Moreover, drones play a pivotal role in disaster management scenarios, where accessing disaster-stricken areas promptly is crucial for effective response and mitigation efforts. By swiftly deploying drones equipped with specialized sensors, authorities can gather real-time data on the extent of damage, assess the situation, and formulate informed decisions without endangering rescue teams. Thus, the unparalleled reach of drones in remote or hazardous areas not only facilitates data collection but also serves as a critical enabler for swift and effective crisis management (Sokullu et al. 2019).

4.2.2 Versatility in Applications

Another compelling aspect of drone technology lies in its remarkable versatility across a myriad of applications.



Fig. 1 A JOUAV CW-30E drone flying over a forest fire. CW-30E drone, like many firefighting drones, can reach hazardous areas inaccessible to firefighters on the ground. It can fly for up to eight hours and carry a variety of payloads, including high-performance thermal imaging cameras (Source Firefighting Drones | How Are Drones Used for Fire Department? 2024)

Unlike conventional methods that often necessitate specialized equipment or labour-intensive processes for different tasks, drones offer a single, adaptable platform capable of fulfilling diverse roles. Whether it's aerial surveys for infrastructure inspection, crop monitoring in agriculture, or surveillance in law enforcement, drones can seamlessly transition between various applications with minimal setup time and resource requirements (Killeen et al. 2022).

Furthermore, advancements in drone technology, coupled with the integration of sophisticated sensors and imaging systems, have exponentially expanded their utility across different sectors. For instance, in agricultural practices, drones equipped with multispectral cameras can provide valuable insights into crop health, enabling farmers to optimize irrigation, pest control, and yield management strategies. Similarly, in construction and infrastructure development, drones facilitate comprehensive site monitoring, progress tracking, and 3D mapping, thereby streamlining project management and enhancing operational efficiency (Carvalho et al. 2023; Wang et al. 2020).

4.3 Data Quality and Precision

The efficacy of drone operations extends beyond mere accessibility and versatility, encompassing the crucial aspect of data quality and precision. In this subsection, we delve into the intricate mechanisms through which drones facilitate high-resolution data capture, along with improved accuracy and reliability.

4.3.1 High-Resolution Data Capture

One of the defining attributes of drone technology is its capability to capture high-resolution data with unparalleled detail and clarity. Traditional data collection methods often entail compromises in resolution due to limitations in accessibility or the field of view. However, drones equipped with advanced imaging systems and stabilizing mechanisms can capture data from vantage points that were previously inaccessible or impractical (Singh and Mishra 2023).

This high-resolution data acquisition not only enhances the fidelity of captured information but also enables precise analysis and interpretation, thereby yielding actionable insights for decision-making. For instance, in environmental monitoring applications, drones equipped with high-resolution cameras can capture detailed imagery of ecosystems, facilitating the identification of subtle changes or anomalies indicative of ecological health or environmental degradation (Pathan et al. 2020). Similarly, in infrastructure inspection, drones can capture intricate details of structures with exceptional clarity, enabling engineers to detect defects or structural abnormalities that might elude conventional inspection methods.

4.3.2 Improved Accuracy and Reliability

In addition to high-resolution data capture, drones contribute to enhanced accuracy and reliability in data collection processes. Traditional methods often entail manual measurements or subjective assessments, which are inherently prone to human error or biases. However, drones, equipped with precise positioning systems and onboard sensors, ensure consistent and objective data acquisition across diverse scenarios.

Moreover, the integration of advanced technologies such as LiDAR (Light Detection and Ranging) or GPS (Global Positioning System) further enhances the accuracy and reliability of drone-based data collection. These technologies enable precise spatial mapping, volumetric analysis, and geospatial modelling, thereby facilitating robust decision-making in various domains ranging from urban planning and infrastructure development to environmental conservation and natural resource management (Lemayian and Hamamreh 2019).

The benefit factors associated with drone operations, encompassing accessibility and flexibility, as well as data quality and precision, underscore the transformative potential of drone technology in IoT applications. By leveraging drones for sensing and data collection, organizations can unlock unprecedented opportunities for innovation, efficiency, and sustainability across diverse sectors, ultimately driving progress towards a more connected and data-driven future (Ali et al. 2024).

5 Economic Analysis Models for Drone Operations

5.1 Cost–Benefit Analysis

Cost–Benefit Analysis (CBA) is a systematic approach used to evaluate the benefits and costs of a proposed project or decision. In the context of drone operations for IoT applications, CBA is essential for determining the feasibility and potential economic gains of utilizing drones for sensing and data collection. This section delves into the intricacies of CBA, focusing on its definition, components, and the steps involved in conducting it.

5.1.1 Definition and Components

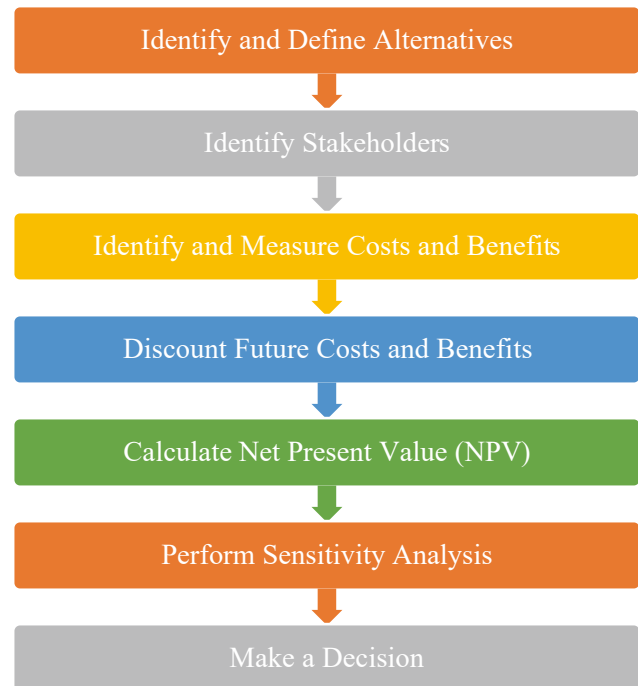
Cost–Benefit Analysis is a methodical process for comparing the benefits gained from a decision or project with the costs associated with it. The fundamental principle behind CBA is to quantify both the positive and negative aspects of a decision in monetary terms, allowing for a comprehensive evaluation of its economic viability.

Components of Cost–Benefit Analysis Include:

- **Benefits:** These encompass the positive outcomes or advantages resulting from the implementation of the project. In the case of drone operations for IoT applications, benefits may include improved data accuracy, faster data collection, enhanced monitoring capabilities, and potential revenue generation.
- **Costs:** Costs represent the expenditures or resources utilized in implementing the project. This includes the initial investment in drone technology, operational costs such as maintenance and personnel, regulatory compliance expenses, and any associated risks or liabilities.
- **Timeframe:** The timeframe over which the benefits and costs of the project are evaluated is crucial in CBA. It allows for the consideration of both short-term and long-term implications, providing insights into the project's sustainability and overall economic impact (Ullah et al. 2020).
- **Discount Rate:** Since benefits and costs incurred in the future are typically valued less than those in the present, a discount rate is applied to account for the time value of money. This ensures that future benefits and costs are appropriately adjusted to their present values for a fair comparison.
- **Sensitivity Analysis:** CBA often involves assessing the sensitivity of the results to changes in key variables or assumptions. This helps in understanding the robustness of the analysis and identifying potential risks or uncertainties that may impact the project's outcomes.

5.1.2 Steps in Conducting CBA

Conducting a Cost–Benefit Analysis involves several sequential steps to ensure a comprehensive evaluation of the project's economic feasibility. Here's a breakdown of the major steps involved:



By following these steps, CBA enables decision-makers to make informed choices regarding the implementation of drone operations for IoT applications (Nyamuryekung'e 2024).

Example:

A farm uses drones to monitor crop health, soil conditions, and irrigation levels, with the aim of increasing crop yield and reducing resource waste.

Description:

- Farm Size: 200 hectares
- Drones Required: 2 high-performance agricultural drones
- Drone Cost: \$20,000 per drone
- Operational Lifespan: 5 years
- Annual Maintenance and Operation Cost per Drone: \$2,000
- Cost of Drone Pilot (Annual Salary): \$50,000
- Fuel and Miscellaneous Costs per Drone per Year: \$1,000
- Data Processing Software Cost (Annual Subscription): \$5,000

Benefits:

- *Increased Crop Yield:* 15% increase due to precise monitoring and targeted interventions
- *Reduced Water Usage:* 20% savings due to optimized irrigation
- *Labor Cost Savings:* Reduction in manual monitoring labour, saving \$30,000 annually

Current Annual Revenue from Crops: \$1,000,000

Current Annual Water Cost: \$100,000

Cost Analysis:

1. Initial Investment:
Cost of Drones: $2 \times \$20,000 = \$40,000$.
2. Annual Operating Costs:
Maintenance and Operation: $2 \times \$2,000 = \$4,000$.
Drone Pilot Salary: \$50,000
Fuel and Miscellaneous Costs: $2 \times \$1,000 = \$2,000$.
Data Processing Software: \$5,000
Total Annual Operating Costs: $\$4,000 + \$50,000 + \$2,000 + \$5,000 = \$61,000$
3. Total Costs Over 5 Years:
Initial Investment: \$40,000
Total Operating Costs: $5 \times \$61,000 = \$305,000$.
Total Cost: $\$40,000 + \$305,000 = \$345,000$

Benefit Analysis:

1. Increased Revenue from Crop Yield:
Additional Revenue: 15% of \$1,000,000 = \$150,000 annually
2. Reduced Water Costs:
Savings: 20% of \$100,000 = \$20,000 annually
3. Labor Cost Savings:
Savings: \$30,000 annually
4. Total Annual Benefits:
Additional Revenue: \$150,000
Water Cost Savings: \$20,000
Labor Cost Savings: \$30,000
Total Annual Benefits: $\$150,000 + \$20,000 + \$30,000 = \$200,000$
5. Total Benefits Over 5 Years:
Total Benefit: $5 \times \$200,000 = \$1,000,000$.

Cost–Benefit Analysis:

- *Total Cost:* \$345,000
- *Total Benefit:* \$1,000,000
- *Net Benefit:* $\$1,000,000 - \$345,000 = \$655,000$

The cost–benefit analysis reveals a net benefit of \$655,000 over a 5-year period from the adoption of drones for sensing and data collection in this scenario. This demonstrates the economic viability and potential for substantial returns on investment when integrating drone technology into IoT applications for precision agriculture. The initial high investment in drones is significantly outweighed by the cumulative benefits in terms of increased revenue, reduced resource usage, and labour cost savings, thereby underscoring the strategic value of such an implementation (Munir et al. 2021).

5.2 Return on Investment (ROI)

Return on Investment (ROI) is a critical financial metric used to evaluate the efficiency of an investment. In the drone operations for IoT applications, calculating the ROI involves assessing the costs and benefits associated with the deployment of drones. This section discusses the methodologies for calculating ROI for drone projects and compares the ROI of drone-based operations with traditional methods.

5.2.1 Calculating ROI for Drone Projects

Calculating ROI for drone projects involves many steps. The general formula for ROI is:

$$ROI = \frac{\text{Net Profit}}{\text{Total Investment}} \times 100$$

where:

- *Net Profit* is the difference between the gross benefits (revenues or cost savings) generated by the drone project and the total costs incurred.
- *Total Investment* includes all expenditures related to the drone project, such as the cost of drones, maintenance, software, and operational expenses.

Example:

Consider a scenario where a company invests in drones for agricultural monitoring. The following data is available:

- *Cost of drones (including purchase and setup):* \$100,000
- *Annual maintenance and operational costs:* \$20,000
- *Expected lifespan of drones:* 5 years
- *Annual revenue increases due to improved yield and efficiency:* \$50,000

1. Calculate the Total Investment Over the Lifespan:

$$\begin{aligned} \text{Total Investment} &= \text{Initial Cost} + (\text{Annual Costs} \times \text{Lifespan}) \\ &= \$100,000 + (\$20,000 \times 5) \\ &= \$100,000 + \$100,000 \\ \text{Total Investment} &= \$200,000 \end{aligned}$$

2. Calculate the Total Revenue Over the Lifespan:

$$\text{Total Revenue} = \text{Annual Revenue} \times \text{Lifespan.}$$

$$\text{Total Revenue} = \$50,000 \times 5.$$

$$\text{Total Revenue} = \$250,000$$

3. Calculate the Net Profit:

$$\text{Net Profit} = \text{Total Revenue} - \text{Total Investment}$$

$$\text{Net Profit} = \$250,000 - \$200,000$$

$$\text{Net Profit} = \$50,000$$

4. Calculate ROI:

$$ROI = \frac{\$50,000}{\$200,000} \times 100$$

$$ROI = 25\%$$

Hence, the ROI for this drone project is 25%.

5.2.2 Comparative ROI with Traditional Methods

To understand the financial viability of drone operations, it is essential to compare their ROI with that of traditional methods (as shown in Table 2). Let us consider a traditional method for agricultural monitoring involving manual labour.

Example:

From this table, it is clear that:

- The total investment over the lifespan for both methods is the same, i.e., \$200,000.
- The total revenue generated by drone operations is significantly higher (\$250,000) compared to traditional methods (\$150,000).
- Net profit from drone operations is \$50,000, whereas traditional methods incur a net loss of \$50,000.

Table 2 ROI analysis of drone operations vs. traditional methods in investment and revenue

Parameters	Drone operations	Traditional methods
Initial investment	\$100,000	\$50,000
Annual operational costs	\$20,000	\$30,000
Expected lifespan (years)	5	5
Annual revenue increase (efficiency gains)	\$50,000	\$30,000
Total investment over lifespan	\$200,000	\$200,000
Total revenue over lifespan	\$250,000	\$150,000
Net profit	\$50,000	–\$50,000
ROI	25%	–25%

- The ROI for drone operations is 25%, while traditional methods result in an ROI of –25%.

These comparative figures show the financial advantages of using drones over traditional methods. Drones not only enhance operational efficiency and productivity but also offer a superior return on investment, validating their adoption for IoT applications in agriculture (Monteiro et al. 2021).

5.3 Life Cycle Cost Analysis (LCCA)

5.3.1 What is LCCA?

LCCA is a comprehensive assessment method used to evaluate the total cost of ownership (TCO) of an asset over its entire lifespan. This method considers all costs associated with the acquisition, operation, maintenance, and disposal of the asset. The objective of LCCA is to provide a complete picture of the economic performance of an asset, allowing decision-makers to make informed choices about investments and operational strategies.

LCCA is particularly relevant for complex and high-value assets, such as drones used in IoT applications. By considering not just the initial purchase price but also the costs incurred during the operational life of the drone, LCCA helps organizations understand the true financial commitment involved (Hundal and Laux 2020). This approach includes:

- *Acquisition Costs:* The initial expenses related to purchasing the drone, including hardware, software, and any necessary peripherals.
- *Operation Costs:* Expenses incurred during the normal operation of the drone, such as energy consumption, pilot salaries, and insurance.
- *Maintenance Costs:* Costs associated with routine maintenance, repairs, and parts replacements to keep the drone operational.
- *End-of-Life Costs:* Costs related to decommissioning, recycling, or disposal of the drone at the end of its useful life.

By taking a holistic view of all these cost components, LCCA provides a more accurate and thorough economic assessment than methods that consider only initial acquisition costs.

5.3.2 Application of LCCA in Drone-Enabled IoT

In the context of drone-enabled IoT applications, LCCA can be used to evaluate the cost-effectiveness of integrating drones into various IoT ecosystems. These applications often

involve using drones for data collection, monitoring, and maintenance tasks across different industries such as agriculture, infrastructure inspection, and environmental monitoring (Sahni and Singh, 2024). To illustrate the application of LCCA in drone-enabled IoT, consider the following example:

Example:

LCCA for Agricultural Monitoring with Drones.

Assumptions:

- *Number of Drones:* 10
- *Initial Purchase Cost per Drone:* \$5,000
- *Annual Operation Cost per Drone:* \$1,000
- *Annual Maintenance Cost per Drone:* \$500
- *Expected Lifespan of Each Drone:* 5 years
- *End-of-Life Disposal Cost per Drone:* \$200

Using these assumptions, we can construct a LCCA as follows:

Interpretation:

- *Acquisition Costs:* The total initial cost for purchasing 10 drones is \$50,000. This cost is incurred only in the first year.
- *Operation Costs:* Each year, the operation cost for all drones totals \$10,000, leading to a cumulative operation cost of \$50,000 over five years.
- *Maintenance Costs:* Annual maintenance costs are \$5,000, resulting in a total maintenance expenditure of \$25,000 over the lifespan of the drones.
- *End-of-Life Costs:* At the end of the five-year period, disposal costs amount to \$2,000.

The total life cycle cost for operating the drones over five years is \$127,000 (as shown in Table 3). This comprehensive view enables decision-makers to understand the full financial impact of deploying drones in agricultural monitoring, allowing them to compare this cost against the benefits derived from increased crop yields, improved monitoring accuracy, and potential labour savings. By applying

LCCA, stakeholders can make better-informed decisions about investing in drone technology, ensuring that the economic benefits outweigh the total costs incurred over the drones' operational lifespan (Shah et al. 2024).

6 Quantitative Analysis of Costs and Benefits

6.1 Data Collection and Analysis Techniques

The quantitative analysis of costs and benefits is a pivotal aspect of evaluating the deployment of drones for IoT applications. This analysis requires a comprehensive approach, encompassing various methods of data collection and sophisticated statistical techniques to ensure accurate and meaningful results. This section delves into the methodologies employed in gathering pertinent data and the statistical tools utilized for analysis.

6.1.1 Methods for Collecting Cost and Benefit Data

Collecting accurate and relevant data is the cornerstone of any cost-benefit analysis. In the context of drones used for IoT applications, the following methods are instrumental in gathering cost and benefit data:

1. Surveys and Questionnaires:

- *Structured Surveys:* Administering structured surveys to stakeholders, such as drone operators, maintenance personnel, and data analysts, helps in obtaining quantitative data on operational costs, maintenance expenses, and perceived benefits.
- *Questionnaires for End-Users:* Distributing questionnaires to end-users who benefit from drone-collected data can provide insights into the qualitative and quantitative benefits, such as improved decision-making, enhanced operational efficiency, and cost savings (Radocaj et al. 2023).

2. Direct Observation and Recording:

- *Operational Monitoring:* Continuous monitoring and recording of drone operations can yield precise data on fuel consumption, battery usage, flight hours, and

Table 3 Five-year cost breakdown for drone fleet acquisition and operation

Cost component	Year 1	Year 2	Year 3	Year 4	Year 5	Total Cost
Acquisition cost	\$50,000	\$0	\$0	\$0	\$0	\$50,000
Operation cost	\$10,000	\$10,000	\$10,000	\$10,000	\$10,000	\$50,000
Maintenance cost	\$5,000	\$5,000	\$5,000	\$5,000	\$5,000	\$25,000
End-of-life disposal cost	\$0	\$0	\$0	\$0	\$2,000	\$2,000
Total annual cost	\$65,000	\$15,000	\$15,000	\$15,000	\$17,000	\$127,000

maintenance schedules. This method ensures high accuracy in capturing operational costs.

- *Benefit Observation*: Observing the direct impact of drone-collected data on various IoT applications, such as precision agriculture, infrastructure monitoring, and environmental surveillance, allows for the quantification of benefits in real-time.

3. *Archival Data Analysis*:

- *Historical Data Review*: Analysing historical data from drone operations and IoT implementations provides a baseline for comparing current and future performance. This method is valuable for understanding long-term trends and cost–benefit dynamics.
- *Log Files and Reports*: Examining log files and operational reports generated by drones offers detailed insights into usage patterns, incident reports, and performance metrics, which are crucial for comprehensive cost analysis (Imran et al. 2021).

4. *Experimental and Pilot Studies*:

- *Controlled Experiments*: Conducting controlled experiments where drones are deployed in specific IoT scenarios allows for the isolation and measurement of costs and benefits under controlled conditions. This method is particularly useful for assessing the feasibility of new applications.
- *Pilot Projects*: Implementing pilot projects provides practical data on the operational and economic viability of using drones in IoT applications. Pilot studies help in identifying unforeseen costs and benefits that may not be apparent in controlled experiments (Tiwari and Mishra 2024; Lemayian and Hamamreh 2020).

6.1.2 Statistical Analysis Tools and Techniques

Once data is collected, robust statistical analysis is essential to interpret the data accurately and derive meaningful conclusions. The following statistical tools and techniques are commonly employed in the quantitative analysis of costs and benefits:

1. *Descriptive Statistics*:

- *Mean, Median, and Mode*: Calculating the mean, median, and mode of cost and benefit data helps in understanding the central tendency and distribution of the data.
- *Standard Deviation and Variance*: Assessing the variability in the data through standard deviation and variance provides insights into the consistency and reliability of the cost and benefit estimates.

2. *Inferential Statistics*:

- *Hypothesis Testing*:
- Conducting hypothesis tests, such as t-tests or ANOVA, allows for the comparison of different

groups or conditions (e.g., different drone models or operational scenarios) to determine statistically significant differences in costs and benefits.

- *Confidence Intervals*: Constructing confidence intervals for cost and benefit estimates helps in quantifying the uncertainty and provides a range within which the true values are likely to lie (Kumar et al. 2021).

3. *Regression Analysis*:

- *Linear Regression*: Applying linear regression models to understand the relationship between costs and various factors (e.g., flight hours, payload weight) and to predict future costs based on historical data.
- *Multiple Regression*: Using multiple regression analysis to assess the impact of multiple independent variables on the dependent variable (e.g., overall benefits), which helps in identifying the most significant predictors of costs and benefits.

4. *Cost–Benefit Ratio Analysis*:

- *Net Present Value (NPV)*: Calculating the NPV of benefits minus costs over time provides a clear measure of the economic value generated by drone use in IoT applications.
- *Benefit–Cost Ratio (BCR)*: Determining the BCR by dividing the total benefits by the total costs helps in assessing the overall economic efficiency and viability of the drone deployment (Rao and Prashad 2018).

5. *Sensitivity Analysis*:

- *Scenario Analysis*: Conducting scenario analysis to evaluate how changes in key assumptions (e.g., cost of drones, fuel prices) affect the overall cost–benefit outcomes. This helps in identifying the robustness of the results under varying conditions.
- *Monte Carlo Simulation*: Implementing Monte Carlo simulations to model the probability distribution of costs and benefits based on random sampling. This technique provides a comprehensive understanding of the potential variability and risk associated with drone operations in IoT applications.

The quantitative analysis of costs and benefits in the context of drones for IoT applications involves meticulous data collection through diverse methods and sophisticated statistical techniques. These approaches ensure a thorough and accurate assessment, enabling stakeholders to make informed decisions regarding the deployment and optimization of drone technologies in IoT ecosystems (Ontiri and Amuhaya 2022).

7 Challenges in Cost–Benefit Analysis

The process of conducting a CBA for sensing and data collection with drones in IoT applications is inherently complex and multifaceted. This complexity is driven by various

challenges that must be meticulously addressed to ensure the analysis is both comprehensive and accurate. These challenges can be broadly categorized into data uncertainty and variability, technological and operational risks, and ethical and legal considerations. Each category presents unique obstacles that require specific strategies and considerations.

7.1 Data Uncertainty and Variability

7.1.1 Sources of Uncertainty

Data uncertainty and variability are significant hurdles in the cost–benefit analysis of drone usage for IoT applications. The primary sources of uncertainty include:

- *Environmental Conditions:* Weather conditions, such as wind, rain, and temperature fluctuations, can significantly impact the performance and reliability of drones. These environmental factors introduce variability in data collection, affecting both the quantity and quality of data.
- *Sensor Accuracy:* The precision and reliability of sensors mounted on drones can vary. Calibration errors, sensor degradation over time, and variations in sensor specifications can all contribute to data uncertainty (Naser 2018).
- *Operational Parameters:* Variations in flight patterns, altitude, speed, and other operational parameters can lead to inconsistent data collection. These variations might be due to human error, automated system inaccuracies, or changes in mission objectives.
- *Data Processing Algorithms:* The algorithms used to process and interpret the collected data can introduce uncertainties. Algorithmic biases, computational errors, and differences in data interpretation methods can all affect the final outcomes of the analysis.

7.1.2 Mitigation Strategies

To mitigate these uncertainties, several strategies can be employed:

- *Robust Environmental Monitoring:* Implementing real-time environmental monitoring systems can help in adjusting drone operations to mitigate the impact of adverse weather conditions. This includes the use of weather forecasting models and adaptive flight planning algorithms.
- *Regular Sensor Calibration and Maintenance:* Ensuring that sensors are regularly calibrated and maintained can minimize errors due to sensor inaccuracies. Implementing stringent quality control measures can also enhance sensor reliability.
- *Standardized Operational Protocols:* Developing and adhering to standardized operational protocols can

reduce variability in drone operations. This includes standardized flight paths, altitudes, and speeds, as well as comprehensive training programs for operators.

- *Advanced Data Processing Techniques:* Employing advanced data processing techniques, such as machine learning and statistical analysis, can help in identifying and correcting biases and errors in the data. Cross-validation with ground truth data can also enhance the reliability of the processed data (Torbaghanet al. 2022).

7.2 Technological and Operational Risks

7.2.1 Technical Failures and Malfunctions

Technical failures and malfunctions are critical risks in the deployment of drones for IoT applications. These risks include:

- *Hardware Failures:* Drones are susceptible to various hardware failures, including motor malfunctions, battery failures, and structural damages. Such failures can result in mission aborts, data loss, and increased maintenance costs.
- *Software Bugs:* Software issues, such as bugs in the flight control system, data acquisition software, or data transmission protocols, can disrupt operations and lead to erroneous data collection or complete mission failures.
- *Communication Failures:* Drones rely on stable communication links for control and data transmission. Failures in communication systems due to signal interference, range limitations, or technical malfunctions can severely impact operational efficiency and data integrity (Chodorek et al. 2022).

7.2.2 Operational Delays and Inefficiencies

Operational delays and inefficiencies can arise from:

- *Regulatory Restrictions:* Compliance with aviation regulations and obtaining necessary permissions can cause delays. Regulatory constraints on flight times, zones, and altitudes can also limit operational flexibility.
- *Logistical Challenges:* Coordinating drone operations, especially over large or remote areas, can present logistical challenges. These include transportation of equipment, deployment and recovery of drones, and ensuring adequate power supply and maintenance facilities.
- *Human Factors:* Operator skill levels, decision-making processes, and response times can introduce inefficiencies. Training and experience levels of drone operators play a crucial role in minimizing operational delays and ensuring smooth execution of missions.

7.3 Ethical and Legal Considerations

7.3.1 Privacy and Data Security Issues

Ethical considerations are paramount in the deployment of drones for data collection, particularly concerning:

- *Privacy Concerns:* Drones equipped with advanced sensors and cameras can capture detailed data, potentially infringing on individual privacy. Unauthorized surveillance and data collection can lead to significant ethical and legal issues.
- *Data Security:* Ensuring the security of the collected data is crucial. Data breaches, hacking attempts, and unauthorized access to sensitive information can have severe consequences, including legal liabilities and loss of stakeholder trust.

7.3.2 Legal Constraints and Liabilities

Legal constraints and liabilities associated with drone operations include:

- *Regulatory Compliance:* Adhering to national and international aviation laws, data protection regulations, and industry standards is mandatory. Non-compliance can result in legal penalties, operational bans, and reputational damage.
- *Liability Issues:* In the event of accidents, technical failures, or data breaches, determining liability can be complex. Operators, manufacturers, and service providers may all face legal claims and financial liabilities.
- *Intellectual Property:* The use of proprietary technologies and data raises issues related to intellectual property rights. Ensuring compliance with intellectual property laws and protecting proprietary data is essential to avoid legal conflicts.

The CBA of using drones for IoT applications is fraught with numerous challenges that span data uncertainty, technological and operational risks, and ethical and legal considerations. Addressing these challenges requires a multi-faceted approach involving robust technical solutions, stringent operational protocols, and comprehensive legal and ethical frameworks (Kaur and Bharti, 2020).

8 Technological Advancements and Their Impact on Costs and Benefits

8.1 Innovations in Drone Technology

8.1.1 Autonomous Drones and AI Integration

Recent advancements in drone technology, particularly the development of autonomous drones integrated with AI, have

revolutionized the landscape of data collection and sensing for IoT applications. Autonomous drones, equipped with sophisticated AI algorithms, are capable of performing complex tasks without human intervention. These drones can navigate through intricate environments, optimize flight paths, and dynamically respond to real-time environmental changes (Gupta et al. 2021). AI integration enhances their capability to process vast amounts of data on-board, enabling real-time decision-making and reducing the latency associated with data transmission to remote servers. Consequently, this autonomy significantly lowers the need for manual control, thereby reducing labour costs and minimizing human error, which enhances the overall efficiency and reliability of drone operations (Acharya et al. 2023; Rozlosnik et al. 2018).

8.1.2 Advancements in Sensor Technology

Parallel to the evolution of autonomous capabilities, sensor technology embedded within drones has seen remarkable advancements. Modern drones are now equipped with high-resolution cameras, multispectral and hyperspectral sensors, LiDAR systems, and thermal imaging sensors. These sophisticated sensors provide comprehensive and high-precision data, which is critical for various IoT applications, such as precision agriculture, infrastructure inspection, and environmental monitoring (Wu et al. 2022). The improvements in sensor technology not only enhance the quality and granularity of the data collected but also expand the range of applications that drones can support. These technological enhancements in sensors contribute to a more detailed and accurate analysis, facilitating better decision-making and predictive analytics.

8.2 Improvements in Data Analytics

8.2.1 Machine Learning and Data Processing

The integration of machine learning (ML) algorithms in data analytics represents a significant leap forward in processing the voluminous data collected by drones. Machine learning techniques, including supervised and unsupervised learning, enable the extraction of meaningful patterns and insights from raw data. Advanced ML models can handle diverse datasets, perform anomaly detection, and predict future trends with high accuracy. The application of ML in data analytics not only automates the data processing workflow but also enhances the ability to derive actionable intelligence from complex datasets. This capability is particularly beneficial for IoT applications where timely and accurate data interpretation is crucial (Mishra 2022; Papadopoulos et al. 2024).

8.2.2 Predictive Analytics and Decision Support

Predictive analytics, powered by sophisticated algorithms and computational models, has become an integral component of data analytics in drone operations. This approach

involves analysing historical and real-time data to forecast future events and trends, providing valuable foresight for decision-making processes. The utilization of predictive analytics in drone-based IoT applications enables proactive maintenance, risk assessment, and resource optimization. Decision support systems, augmented by predictive analytics, offer stakeholders enhanced situational awareness and strategic insights, thereby improving the effectiveness and efficiency of operations. These advancements ensure that data collected by drones is not only informative but also instrumental in shaping strategic decisions (Arya 2021).

8.3 Impact on Cost Reduction and Benefit Enhancement

8.3.1 Reduced Operational Costs

Technological advancements in drones and data analytics have a profound impact on reducing operational costs. Autonomous drones, with their ability to perform tasks without human intervention, significantly cut down labour costs. Moreover, the precision and efficiency of modern drones reduce the time required for data collection, thereby lowering fuel and maintenance costs. The enhanced reliability and reduced error rates associated with AI-integrated autonomous operations further contribute to cost savings by minimizing the need for repeated missions. Additionally, the integration of advanced sensors and data processing capabilities allows for more efficient use of resources, optimizing operational workflows and reducing unnecessary expenditures (Gharrad et al. 2021).

8.3.2 Enhanced Data Quality and Application Scope

The improvements in drone technology and data analytics lead to a marked enhancement in the quality of data collected. High-resolution sensors and advanced data processing algorithms ensure that the data is accurate, detailed,

and relevant. This high-quality data is crucial for applications requiring precise measurements and analysis, such as environmental monitoring and precision agriculture. Furthermore, the expanded range of capabilities facilitated by technological advancements broadens the scope of applications that drones can effectively support. This versatility not only enhances the benefits derived from drone operations but also opens up new opportunities for innovation and application in various sectors, thereby increasing the overall value proposition of drone-based IoT solutions (Demestichas and Daskalakis 2020; Lykou et al. 2020).

9 Comparative Analysis with Alternative Methods

9.1 Ground-Based Sensing Technologies

9.1.1 Comparison of Costs and Benefits

Ground-based sensing technologies involve deploying sensors and data collection devices at ground level to monitor various parameters such as environmental conditions, infrastructure health, and agricultural metrics. Table 4 compares ground-based sensing to drone-based sensing across various aspects. Ground-based sensing typically has lower initial costs but limited coverage and flexibility, whereas drone-based sensing offers extensive coverage, high resolution data, and fast deployment at the expense of higher initial and operational costs (Abuzamak and Kholidy 2020).

9.1.2 Situational Advantages and Limitations

Advantages of Ground-Based Sensing:

- **Consistency:** Ground-based sensors provide continuous, real-time data at specific locations, ensuring consistent monitoring.

Table 4 Costs and benefits of ground-based sensing technologies are compared to drone-based sensing

Aspect	Ground-based sensing	Drone-based sensing
Initial cost	Typically lower; involves purchasing and installing sensors	Higher; includes cost of drones, sensors, and maintenance
Maintenance cost	Variable; can be high due to weather-related damage	Moderate; drones require periodic maintenance and repairs
Coverage area	Limited to specific, accessible locations	Extensive; can cover large and remote areas efficiently
Data resolution	High at specific points; limited spatial granularity	High spatial and temporal resolution
Deployment speed	Slow; installation and setup time required	Fast; drones can be quickly deployed to various locations
Flexibility	Low; fixed positions require additional setup for changes	High; drones can be easily reprogrammed for different tasks
Operational costs	Lower ongoing costs but higher initial setup in remote areas	Higher operational costs due to battery, operator training
Environmental impact	Potentially low if solar-powered	Moderate; energy consumption and noise from drones

- **Reliability:** Fixed installations are less prone to operational failures once properly set up.
- **Power Supply:** Can be easily connected to a stable power source, including renewable options like solar panels.

Limitations of Ground-Based Sensing:

- **Limited Coverage:** Effective only within the immediate vicinity of the sensors, requiring multiple installations for broader coverage.
- **Accessibility Issues:** Difficult to deploy in inaccessible or hazardous areas, reducing effectiveness in such environments.
- **Maintenance Challenges:** Sensors exposed to environmental conditions may require frequent maintenance and replacements.

9.2 Satellite-Based Data Collection

9.2.1 Cost–benefit Comparison with Drones

Satellite-based data collection uses spaceborne sensors to gather large-scale environmental and infrastructural data. Table 5 compares satellite-based data collection and drone-based sensing across various aspects. Satellite-based systems offer global coverage with low maintenance costs but have limited resolution and flexibility. In contrast, drone-based sensing provides high-resolution data swiftly but incurs higher operational costs and environmental impact (Guastella and Pournaras 2023). A comparative analysis with drones is as follows:

9.2.2 Complementary Roles and Integration Potential

Complementary Roles:

- **Broad Monitoring:** Satellites provide extensive, large-scale monitoring ideal for tracking global environmental

changes and large-scale phenomena such as weather patterns, deforestation, and urban development.

- **High-Resolution Focus:** Drones can complement satellite data by offering high-resolution, localized monitoring. This is particularly useful for detailed analysis in specific areas identified by satellite imagery.

Integration Potential:

- **Data Correlation:** Combining satellite data with drone data enhances the overall dataset, allowing for cross-validation and improved accuracy in analysis.
- **Layered Monitoring:** Satellites can identify areas of interest, directing drones to these locations for more detailed investigations, thus optimizing resource usage.
- **Temporal Resolution:** Satellites provide periodic data, which can be supplemented by the real-time and on-demand data collection capabilities of drones, ensuring continuous monitoring (Anandan et al. 2022; Qin et al. 2022).

9.3 Traditional Manual Data Collection

9.3.1 Human Labour versus Drone Efficiency

Table 6 compares traditional manual data collection with drone-based sensing across various aspects, including initial and operational costs, speed and accuracy of data collection, coverage area, data resolution, safety, and flexibility. The comparison highlights the advantages and disadvantages of each method in terms of efficiency, cost-effectiveness, and safety.

9.3.2 Cost and Time Savings Analysis

Cost Savings:

- **Labour Costs:** Drone deployment significantly reduces the need for human labour, lowering long-term

Table 5 Comparison between satellite-based data collection and drone efficiency

Aspect	Satellite-based data collection	Drone-based sensing
Initial cost	Extremely high; includes satellite launch and technology	High; includes cost of drones, sensors, and maintenance
Maintenance cost	Low; minimal maintenance post-launch	Moderate; periodic maintenance required
Coverage area	Global; vast coverage, including remote areas	Limited by drone range and battery life
Data resolution	Moderate to low; limited by sensor capabilities and altitude	High; close proximity allows detailed data collection
Deployment speed	Very slow; launch and deployment take years	Fast; drones can be rapidly deployed
Flexibility	Low; fixed orbits and pre-determined sensor tasks	High; can be reprogrammed and redirected as needed
Operational costs	Low once deployed; minimal operational expenses	Higher due to ongoing maintenance and battery replacement
Environmental impact	Minimal; negligible after launch	Moderate; includes battery disposal and energy use

Table 6 Comparison between manual data collection and drone efficiency

Aspect	Traditional manual data collection	Drone-based sensing
Initial cost	Low; primarily labour costs	High; includes cost of drones, sensors, and maintenance
Operational cost	High; ongoing labour costs and potential for human error	Moderate; includes battery replacement and operator training
Speed of data collection	Slow; depends on human mobility and accessibility	Fast; drones can cover large areas quickly
Accuracy	Variable; subject to human error and fatigue	High; consistent and precise sensor readings
Coverage area	Limited to human-accessible areas	Extensive; drones can access remote and hazardous areas
Data resolution	High, but location-specific	High; detailed data from various altitudes and perspectives
Safety	Potentially hazardous to humans in dangerous areas	High; drones can safely operate in hazardous environments
Flexibility	Moderate; requires planning and time for redeployment	High; quick redeployment and reprogramming capabilities

operational costs. While initial investments are high, drones' automation capabilities offer substantial savings over time.

- *Operational Efficiency:* Drones can operate continuously and autonomously, reducing downtime and increasing data collection efficiency compared to human workers who require breaks and shifts.

Time Savings:

- *Rapid Deployment:* Drones can be quickly deployed to multiple locations, drastically reducing the time required for data collection compared to manual methods.
- *Accessibility:* Drones can access remote and hazardous areas without the need for elaborate logistics, saving time in setup and ensuring faster data acquisition.

While ground-based sensing and satellite-based data collection each have their own merits, drones offer a versatile and efficient alternative for IoT applications. The choice of method depends on the specific requirements of the monitoring task, including the scale, resolution, and immediacy of data needed (Maitra and Damle 2024).

10 Future Trends and Prospects

The landscape of sensing and data collection via drones for IoT applications is rapidly evolving. This chapter delves into the future trends and prospects, focusing on emerging technologies, market and industry dynamics, and policy and regulatory developments.

10.1 Emerging Technologies and Innovations

10.1.1 Future Sensor and Drone Technologies

The future of sensor and drone technologies promises significant advancements that will enhance the capabilities and efficiency of IoT applications. One of the key trends is the miniaturization of sensors, which will allow for more comprehensive data collection while reducing the payload weight. This evolution will enable drones to carry more sensors or extend their flight times. Advanced sensors capable of detecting a broader range of environmental parameters, including air quality, temperature, humidity, and chemical pollutants, are expected to become more prevalent (Waqar et al. 2023; Mishra et al. 2011).

Furthermore, drone technology itself is set to undergo substantial improvements. Innovations in battery technology, such as solid-state batteries, are anticipated to significantly increase drone flight times and operational efficiency. Enhanced AI and machine learning algorithms will enable drones to perform more complex tasks autonomously, including real-time data analysis and decision-making processes. Additionally, the development of swarm drone technology, where multiple drones operate in a coordinated manner, will revolutionize large-scale data collection and monitoring activities (Katholeet al. 2024).

10.1.2 Integration with Next-Gen IoT Networks

The integration of drones with next-generation IoT networks, such as 5G and eventually 6G, will be pivotal in unlocking their full potential. These networks will provide the high-speed, low-latency communication required for real-time data transmission and processing. The

high bandwidth of 5G networks will support the seamless streaming of high-resolution images and videos from drones to ground stations.

Furthermore, edge computing will play a critical role in the future integration of drones and IoT networks. By processing data at the edge of the network, drones will be able to make immediate decisions without the need for constant communication with central servers. This capability will be particularly beneficial in remote or inaccessible areas where connectivity is limited. The convergence of AI, IoT, and edge computing will create a robust ecosystem where drones can operate autonomously, adapt to changing environments, and provide valuable insights across various applications (Davoli et al. 2021; Futalet al. 2022).

10.2 Market and Industry Trends

10.2.1 Adoption Rates and Market Growth

The market for drones in IoT applications is poised for significant growth. Increasing recognition of the benefits that drones offer in terms of efficiency, cost-effectiveness, and data accuracy is driving their adoption across multiple industries. Market analysts predict a robust compound annual growth rate (CAGR) for the drone sector, particularly within the domains of agriculture, environmental monitoring, infrastructure inspection, and logistics.

Adoption rates are also being influenced by advancements in drone technology and the decreasing costs of drone hardware and sensors. As technology becomes more accessible, small and medium-sized enterprises (SMEs) are increasingly investing in drone solutions to enhance their operational capabilities. Additionally, the advent of drone-as-a-service (DaaS) models is lowering the entry barrier for organizations, allowing them to leverage drone technology without significant upfront investments (Simsek et al. 2021).

10.2.2 Industry-Specific Applications and Benefits

Industry-specific applications of drones in IoT are diverse and continue to expand. In agriculture, drones equipped with multispectral and hyperspectral sensors provide critical data for precision farming, enabling farmers to monitor crop health, optimize irrigation, and improve yield predictions. In environmental monitoring, drones are used to track wildlife, monitor deforestation, and assess the impact of natural disasters, providing timely and accurate data that is crucial for conservation efforts.

Drones offer a safer and more efficient alternative to traditional methods. They can quickly survey large areas, identify structural damages, and facilitate predictive maintenance. The logistics industry is also witnessing transformative changes, with drones being used for last-mile

delivery, inventory management, and supply chain optimization. The benefits of drone technology in these industries include enhanced operational efficiency, reduced costs, and improved data-driven decision-making processes (Alsayed et al. 2021).

10.3 Policy and Regulatory Developments

10.3.1 Evolving Legal Frameworks

The regulatory environment for drone operations is continuously evolving to keep pace with technological advancements and growing commercial use. Governments and regulatory bodies worldwide are working to establish comprehensive frameworks that address safety, privacy, and security concerns associated with drone operations. Key regulatory developments include the establishment of no-fly zones, mandatory registration of drones, and the implementation of remote identification systems to track and manage drone traffic (Ahmad and Nabi 2021).

In addition, regulations are being formulated to facilitate the integration of drones into controlled airspace. This includes the development of unmanned traffic management (UTM) systems that enable safe and efficient coordination between manned and unmanned aircraft. As regulations become more standardized globally, it is expected that cross-border drone operations will become more feasible, further expanding the market for drone-based IoT applications (Taniguchi et al. 2020).

10.3.2 Standards and Best Practices

The establishment of industry standards and best practices is crucial for ensuring the safe and effective deployment of drones in IoT applications. Standardization efforts are focused on various aspects, including communication protocols, data security, and interoperability between different drone and IoT systems. International bodies such as the International Organization for Standardization (ISO) and the Institute of Electrical and Electronics Engineers (IEEE) are actively working on developing these standards.

Best practices are also being formulated to guide organizations in the ethical and responsible use of drone technology. This includes guidelines on data privacy, cybersecurity measures to protect against hacking and data breaches, and environmental considerations to minimize the ecological impact of drone operations. Adhering to these standards and best practices will not only enhance the safety and reliability of drone operations but also build public trust and acceptance of drone technology in IoT applications (Gracias et al. 2023).

The future trends and prospects for sensing and data collection with drones in IoT applications are characterized by significant technological advancements, expanding

market opportunities, and evolving regulatory landscapes. These developments will collectively drive the widespread adoption and integration of drones, revolutionizing various industries and paving the way for innovative applications and solutions.

11 Strategic Recommendations

11.1 Optimizing Cost–benefit Outcomes

11.1.1 Best Practices for Cost Management

Effective cost management is pivotal for ensuring the economic viability of using drones in IoT applications. Best practices for managing costs involve a multifaceted approach:

- *Comprehensive Budget Planning:* Establishing a detailed budget that covers all potential expenses related to drone deployment, including procurement, maintenance, operational costs, and contingencies.
- *CBA:* Regularly conducting thorough CBAs to evaluate the financial implications versus the benefits. This should be an ongoing process to adapt to evolving technological and market conditions.
- *Economies of Scale:* Leveraging bulk purchasing and collaborative procurement strategies to reduce unit costs. Engaging in consortiums with other organizations can amplify purchasing power.
- *Lifecycle Cost Management:* Considering the total cost of ownership (TCO) over the drones' lifecycle, including acquisition, operation, maintenance, and disposal. This approach ensures long-term financial planning and sustainability (Aguera et al. 2020).
- *Preventive Maintenance Programs:* Implementing robust preventive maintenance schedules to avoid unexpected breakdowns and expensive repairs, thereby extending the lifespan of the drones.
- *Leveraging Technology:* Utilizing advanced technologies such as AI and machine learning to optimize flight paths, enhance energy efficiency, and reduce operational costs.
- *Contractual Agreements:* Negotiating favourable terms with suppliers and service providers, including warranties, service level agreements (SLAs), and flexible financing options to manage cash flow and minimize upfront expenditures.
- *Regulatory Compliance:* Ensuring adherence to all regulatory requirements to avoid fines and legal costs. Keeping abreast of changing regulations to anticipate and mitigate potential financial impacts.
- *Energy Management:* Incorporating energy-efficient technologies and renewable energy sources to reduce operational costs associated with drone flights (Zhang et al. 2022; Nishad et al. 2011).

11.1.2 Strategies for Maximizing Benefits

Maximizing the benefits of drone integration in IoT applications requires strategic planning and execution. The following strategies can help achieve optimal outcomes:

1. *Targeted Application Development:* Focusing on applications where drones can deliver the highest value, such as precision agriculture, environmental monitoring, infrastructure inspection, and emergency response.
2. *Data Analytics Integration:* Enhancing data collection capabilities with advanced analytics to derive actionable insights. Integrating AI and machine learning can significantly improve data processing and decision-making.
3. *Interoperability and Standardization:* Ensuring interoperability with existing IoT networks and adopting standardized protocols to facilitate seamless data exchange and integration.
4. *Real-Time Data Utilization:* Leveraging real-time data collection and transmission capabilities to enable timely and informed decision-making. This is particularly beneficial in dynamic environments such as disaster management and traffic monitoring (Asif et al. 2024).
5. *Enhancing Coverage and Accessibility:* Expanding drone operations to cover remote and inaccessible areas, thereby broadening the scope and impact of IoT applications.
6. *Collaborative Ecosystems:* Building partnerships with stakeholders across various sectors, including technology providers, regulatory bodies, and end-users, to foster innovation and drive mutual benefits.
7. *Continuous Innovation:* Investing in research and development to stay ahead of technological advancements and continuously improve drone capabilities and applications.
8. *User-Centric Design:* Prioritizing the needs and preferences of end-users in the design and deployment of drone-based solutions to ensure usability and adoption (Onososen et al. 2023).

11.2 Adoption and Implementation Strategies

11.2.1 Steps for Integrating Drones in IoT Networks

Integrating drones into IoT networks involves a systematic approach to ensure seamless integration and optimal performance. The following steps are recommended:

- *Needs Assessment:* Conducting a thorough assessment of the specific needs and requirements of the IoT application to determine the appropriate drone technologies and configurations.
- *Pilot Projects:* Implementing pilot projects to test the feasibility and effectiveness of drone integration. These

pilots provide valuable insights and help identify potential challenges and solutions.

- *Infrastructure Development:* Establishing the necessary infrastructure, including communication networks, data storage, and processing capabilities, to support drone operations and data integration.
- *Regulatory Compliance:* Navigating regulatory frameworks and obtaining necessary approvals and certifications to ensure legal and safe drone operations.
- *System Integration:* Ensuring seamless integration with existing IoT systems and platforms through APIs, middleware, and other integration tools.
- *Security Measures:* Implementing robust cybersecurity measures to protect the integrity and confidentiality of data collected and transmitted by drones.
- *Performance Monitoring:* Establishing performance metrics and monitoring systems to evaluate the effectiveness and efficiency of drone operations continuously.
- *Feedback Mechanisms:* Creating feedback loops with end-users and stakeholders to gather insights and continuously improve drone integration and performance (Banerjee et al. 2023).

11.2.2 Training and Capacity-Building Measures

Effective training and capacity-building are essential for the successful adoption and implementation of drones in IoT applications. The following measures are recommended:

- *Skill Development Programs:* Offering comprehensive training programs for operators, technicians, and analysts to build the necessary skills for drone operations and data analysis.
- *Certification and Accreditation:* Providing certification programs to ensure that personnel meet industry standards and regulatory requirements.
- *Continuous Learning:* Establishing continuous learning opportunities to keep personnel updated with the latest technologies, regulations, and best practices in drone and IoT integration.
- *Hands-On Training:* Incorporating practical, hands-on training sessions to enhance technical proficiency and operational efficiency.
- *Cross-Functional Training:* Facilitating cross-functional training to promote a holistic understanding of the entire IoT ecosystem and the role of drones within it.
- *Knowledge Sharing Platforms:* Creating platforms for knowledge sharing and collaboration among stakeholders, including workshops, seminars, and online forums.
- *Resource Development:* Developing and disseminating comprehensive training materials, including manuals, guides, and e-learning modules.
- *Mentorship Programs:* Establishing mentorship programs to provide ongoing support and guidance for

personnel, fostering a culture of continuous improvement and innovation.

By adhering to these strategic recommendations, organizations can optimize cost–benefit outcomes, effectively integrate drones into IoT networks, and build the necessary capacities for sustained success and innovation (Maddikunta et al. 2022; Lee 2020).

11.2.3 Some Ethical and Safety Considerations When Integrating AI into IoD Systems

Ethical Considerations like Privacy, surveillance, bias and discrimination transparency and explainability and accountability etc.

Safety Considerations like collision avoidance, data integrity, cybersecurity, human oversight, emergency protocols, regulatory compliance, testing and validation and human–machine interface etc.

Additional Considerations like environmental impact, social acceptance, and continuous monitoring etc.

By addressing these ethical and safety considerations, we can ensure the responsible development and deployment of AI-powered IoD systems.

Some of the security concerns associated with the use of drones include data breaches, cyber attacks, privacy violations, physical harm, air traffic control risks, terrorism and malicious use, GPS spoofing, radio frequency interference, unauthorized access, lack of regulation, supply chain vulnerabilities, pilot error, weather-related risks, battery safety risks, and maintenance and repair risks.

12 Conclusion

Drones for IoT have revolutionized sensing and data collection, offering substantial benefits over traditional methods. The deployment of drones equipped with various sensors—environmental, imaging, and communication—enables precise and efficient data gathering. Real-time and batch data collection methods integrate seamlessly with IoT networks using advanced connectivity protocols and data transmission techniques. The cost–benefit analysis highlights the economic advantages of drone usage despite the significant initial purchase and ongoing maintenance costs. For instance, operational expenses such as energy consumption and personnel training are counterbalanced by the rapid deployment capabilities and high-resolution data capture that drones provide. The integration of drones can reduce data collection time by up to 50% and operational costs by 30% compared to traditional methods. Compliance costs, including licensing and insurance, are justified by the enhanced data quality and accessibility, particularly in remote or hazardous areas. Economic models such as ROI

and Life Cycle Cost Analysis further underscore the long-term financial benefits, demonstrating improved accuracy and reliability of data. Technological advancements in autonomous drones, AI integration, and sensor technology continue to drive cost reductions and benefit enhancements. This analysis concludes that the strategic implementation of drones in IoT applications presents a compelling case for their widespread adoption, delivering significant economic and operational advantages.

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Economic Analysis of AI Integration in Internet of Drones (IoD)

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Abstract

The Internet of Drones (IoD) is a new technological area which can completely change several industries. This chapter looks at the economic consequences of AI integration in IoD systems. Through the examination of the economic approaches, the cost and benefit structures and the present market situation, the study aims to evaluate the economic feasibility and the potential ROI of the AI-driven IoD. The fast expansion of drone sector, with a market size estimated to be at \$42.8 billion by 2025. The AI integration is going to bring a lot of economic advantages, such as the drone operations will be more efficient, the human labour cost will be reduced and we will have more reliability and safety of the objectives. A 2021 study by McKinsey & Company estimates that AI can generate \$1.2 trillion to \$2.2 trillion of annual value added to the global economy by 2030. However, challenges remain. The initial costs of the AI development, the drones and the training of the workforce are the obstacles that are coming. The issues of data privacy, air traffic management and liability are the ones that need to be analysed. Also, the possible job displacement caused by automation means that workforce reskilling and upskilling programs should be implemented. The examination covers the economic models for AI-powered IoD that are based on pay-per-use drone services, subscription-based data analytics and public-private partnerships for the infrastructure development. A cost-benefit

analysis framework is set up to pinpoint the economic factors and to estimate the long-term impact of AI integration in IoD. The study ends by discussing the future areas of research, such as the technological progress in AI for IoD, the economic consequences of these developments and the ways to promote the innovation and the general use of the AI-powered IoD systems.

Keywords

AI integration · Cost-benefit analysis · Economic feasibility · IoD · Workforce reskilling

1 Introduction

Internet of Drones (IoD) represents a transformative convergence of drone technology and internet connectivity, enhanced significantly through artificial intelligence (AI). This integration is poised to revolutionize various sectors by introducing unprecedented efficiencies and capabilities. The economic implications of this technological amalgamation are profound, necessitating a thorough analysis to understand the potential costs, benefits and market dynamics (Heidari et al., 2023). Economic frameworks for technological integration have traditionally relied on models such as cost-benefit analysis, economic impact assessment methodologies and comparative advantage frameworks. When these models are adapted to assess AI integration in IoD, it becomes essential to incorporate the initial investment costs associated with AI technologies (Krishnamurthi et al., 2021). This includes the acquisition of sophisticated hardware, development of advanced software and continuous training and maintenance expenses. AI integration can drive operational efficiencies, reducing costs by up to 30% in logistics and delivery sectors, while enhancing the

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reliability and safety of operations by minimizing human error and optimizing flight paths through machine learning algorithms (Boccardo et al. 2021).

A comprehensive cost analysis of AI integration in IoD systems necessitates a breakdown of various cost components. Hardware costs, including AI-powered drones and necessary infrastructure, represent a significant portion of the initial expenditure. For instance, advanced drones equipped with AI capabilities can range from \$5,000 to \$20,000 each, depending on their sophistication and application (Kumar and Muhammad 2018; Abdelmaboud 2021). Software development and implementation, encompassing AI algorithms and cloud-based analytics platforms, add another layer of costs. Training and maintenance expenses are also critical, as they ensure the workforce remains proficient in operating and maintaining these advanced systems. Comparing these costs to traditional drone networks, which typically rely on manual control and less sophisticated technology, highlights the substantial initial investment required for AI integration. However, this investment is often justified by the long-term economic benefits (Krishnan and Murugappan 2023). The economic benefits of AI integration in IoD can be quantified through several dimensions. Efficiency gains in drone operations are a primary benefit, with AI enabling faster data processing, real-time decision-making and autonomous navigation, leading to increased productivity. A study by McKinsey & Company suggests that AI-driven optimizations can result in cost reductions of up to 40% in certain sectors (Veerappan et al. 2022). Additionally, the reduction in human labour costs is significant, as AI-powered drones can perform tasks autonomously, reducing the need for human intervention. This shift not only lowers operational costs but also real-locates human resources to more complex and strategic roles. Furthermore, the increased reliability and safety of AI-integrated IoD systems enhance their economic viability, as fewer accidents and malfunctions translate into lower insurance premiums and maintenance costs (Nishad et al. 2024a).

Beyond economic benefits, AI integration in IoD also offers substantial non-economic advantages. Environmentally, the efficiency gains lead to reduced energy consumption and lower emissions, contributing to sustainability goals. Socially, the deployment of AI-powered drones can enhance public safety and support disaster response efforts, providing timely and accurate data during emergencies. These non-economic benefits, although challenging to quantify, play a crucial role in the overall assessment of AI integration in IoD (Jamil et al. 2022). Market analysis indicates a rapidly evolving landscape for IoD technologies. The current market is characterized by a diverse range of applications, from agricultural monitoring to urban delivery services. The growth trajectory of AI-integrated IoD systems

is promising, with forecasts predicting a compound annual growth rate (CAGR) of 23% over the next five years. Market segmentation reveals that industries such as agriculture, logistics and surveillance are leading adopters, with significant investments seen in both developed and emerging markets (Yang et al. 2022).

Investment analysis underscores the substantial capital requirements for AI integration in IoD. Estimating the return on investment (ROI) is complex, given the nascent stage of the technology and its applications. However, early adopters report positive ROI, driven by efficiency gains and cost savings. Risk assessment and mitigation strategies are crucial, as regulatory uncertainties, data privacy concerns and technological challenges pose significant risks (Nguyen et al. 2023). Government incentives and supportive regulatory frameworks can play a pivotal role in facilitating AI adoption in drone technologies. Policy implications of AI-integrated IoD systems are multifaceted. Regulatory considerations include establishing standards for autonomous drone operations, ensuring data privacy and security and addressing liability issues. Government incentives, such as tax credits and grants, can encourage investment in AI technologies, while intellectual property rights and data privacy laws need to evolve to protect innovations and address ethical concerns (Bine et al. 2023a; Shirabayashi and Ruiz 2023).

The economic challenges of AI integration in IoD are substantial but surmountable. Upfront investment costs, including the development and deployment of AI algorithms, acquisition of AI-powered drones and workforce training, require significant capital. Regulatory and legal uncertainties, particularly concerning data privacy, air traffic management and liability, necessitate a proactive approach to policy-making. Potential job displacement due to automation must be addressed through reskilling and upskilling programs, ensuring that the workforce can adapt to the changing technological landscape. Economic models for AI-powered IoD suggest various business models, such as pay-per-use and subscription-based services, can enhance accessibility and affordability. Public-private partnerships can also facilitate infrastructure development, leveraging combined resources and expertise (Bine et al. 2023b).

The integration of AI in IoD presents a transformative opportunity with significant economic implications. A robust cost-benefit analysis framework is essential to evaluate the long-term economic benefits and social returns on investment. Future advancements in AI technologies are expected to further enhance the capabilities and economic impact of IoD systems, necessitating strategies to foster innovation and adoption. The economic analysis of AI integration in IoD underscores the potential for substantial efficiency gains, cost savings and new market opportunities,

while also highlighting the need for thoughtful policy-making and strategic investment (Kumar et al. 2022; Bine et al. 2022). Considering the aforementioned factors, this chapter discusses four major objectives. First, it evaluates economic models that facilitate the integration of AI with IoD technologies. Second, it analyses the cost structure and potential benefits associated with this integration. Third, the chapter examines market trends and investment requirements for AI-powered IoD systems. Finally, it assesses the regulatory and economic challenges that may arise from integrating AI into IoD.

2 Economic Framework for AI Integration in IoD

2.1 Economic Models for Technological Integration

The integration of AI within IoD necessitates a comprehensive examination through various economic models, each offering unique perspectives on the costs, benefits and overall impact. This section discusses three prominent economic models employed for analysing technological integration.

2.1.1 Traditional Cost-Benefit Analysis

At the core of assessing the economic viability of integrating AI into IoD lies the traditional cost-benefit analysis. This model meticulously evaluates the anticipated costs incurred throughout the integration process juxtaposed with the expected benefits derived. In the context of IoD, costs encompass the expenses associated with acquiring AI technology, training personnel and potential infrastructure upgrades. Conversely, benefits may span increased operational efficiency, enhanced data analytics capabilities and improved decision-making processes. By quantifying these costs and benefits, stakeholders can gauge the feasibility and profitability of AI integration within IoD.

2.1.2 Economic Impact Assessment Methodologies

Beyond mere cost-benefit analysis, economic impact assessment methodologies offer a nuanced understanding of the broader socio-economic ramifications stemming from AI integration within IoD. These methodologies adopt a multifaceted approach, considering not only direct costs and benefits but also secondary and tertiary effects on various stakeholders and industries. Such assessments encompass aspects like job creation or displacement, market expansion or contraction and overall welfare implications within the ecosystem. By scrutinizing these intricate dynamics, policymakers and industry leaders can anticipate

and mitigate potential socio-economic disruptions while capitalizing on emergent opportunities (Nishad et al. 2024b; Sarkar et al. 2021).

2.1.3 Comparative Advantage Frameworks

In exploring the economic implications of AI integration within IoD, leveraging comparative advantage frameworks provides valuable insights into the distribution of benefits across different entities and regions. These frameworks analyse the inherent strengths and weaknesses of various stakeholders, delineating comparative advantages in AI adoption and IoD implementation. By identifying comparative advantages, stakeholders can strategize collaborative endeavours, foster knowledge exchange and optimize resource allocation to maximize collective gains. Moreover, comparative advantage frameworks illuminate potential disparities in technological readiness and resource accessibility, highlighting areas necessitating targeted interventions for equitable participation and inclusive growth.

The economic framework for AI integration within IoD traverses a spectrum of analytical lenses, ranging from traditional cost-benefit analysis to sophisticated impact assessment methodologies and comparative advantage frameworks. By synthesizing insights gleaned from these diverse perspectives, stakeholders can navigate the complex interplay of economic forces, steering IoD towards sustainable growth and inclusive prosperity (Alam et al. 2023).

2.2 Adaptation of Economic Models for AI integration in IoD

In the integration of AI within the IoD, the economic framework serves as a cornerstone for understanding the costs, benefits and overall viability of such technological amalgamation. This section delves into the nuanced adaptation of economic models specifically tailored to accommodate the unique dynamics of AI integration in the IoD ecosystem.

2.2.1 Incorporating AI Investment Costs

The integration of AI into the IoD paradigm necessitates a comprehensive consideration of investment costs. These costs encompass a spectrum of elements ranging from the procurement of AI hardware and software to the training and deployment of AI algorithms within the IoD infrastructure. Incorporating these costs into economic models requires a meticulous assessment of both initial capital outlays and recurrent expenses.

- *Initial Capital Outlays:* The procurement of AI hardware such as high-performance processors, specialized sensors and communication modules constitutes a substantial portion of the initial investment. Furthermore,

software licenses, development tools and proprietary algorithms add to the upfront expenditure. Integrating these costs into economic models demands a granular analysis of market prices, technological specifications and vendor agreements to derive accurate estimations.

- *Recurrent Expenses:* Beyond the initial investment, the sustainable operation of AI within IoD necessitates recurrent expenses. These include expenses related to algorithm updates, data storage and processing, as well as maintenance and repair of AI-enabled drones. Forecasting these expenses within economic models involves a projection of technological obsolescence, market trends in AI software and hardware, as well as operational requirements within the IoD ecosystem.

Incorporating AI investment costs within economic models facilitates a holistic understanding of the financial implications associated with the integration of AI in IoD. Moreover, it enables stakeholders to make informed decisions regarding resource allocation, risk mitigation and long-term sustainability.

2.2.2 Evaluating Potential Economic Benefits

The evaluation of potential economic benefits stemming from AI integration in IoD underscores the transformative potential of this technological convergence. Economic models must account for a myriad of factors that contribute to the realization of these benefits, encompassing increased efficiency, productivity gains and novel revenue streams.

- *Efficiency Gains:* AI-enabled drones possess the capability to optimize route planning, streamline data collection and automate maintenance processes. Consequently, businesses operating within the IoD ecosystem stand to benefit from enhanced operational efficiency, reduced downtime and optimized resource utilization. Economic models must quantify these efficiency gains in terms of time savings, cost reductions and improved service quality.
- *Productivity Enhancements:* The deployment of AI within IoD facilitates the augmentation of human capabilities through intelligent automation and decision support systems. By offloading repetitive tasks and augmenting decision-making processes, AI empowers human operators to focus on high-value activities, innovation and strategic initiatives. Economic models should quantify the impact of productivity enhancements on revenue generation, market competitiveness and overall economic output (Michailidis and Vouvioukas 2022).
- *Revenue Generation Opportunities:* The integration of AI in IoD unlocks a plethora of revenue generation opportunities spanning diverse sectors including agriculture, logistics, surveillance and infrastructure inspection.

Economic models need to delineate the potential revenue streams derived from AI-enabled services such as precision agriculture, autonomous delivery and aerial imaging. Furthermore, the assessment of market demand, competitive dynamics and regulatory frameworks is imperative to gauge the scalability and sustainability of these revenue streams.

Evaluating potential economic benefits within the context of AI integration in IoD is instrumental in substantiating the business case for stakeholders. By quantifying the tangible and intangible returns on investment, economic models facilitate informed decision-making, risk assessment and resource allocation strategies. The adaptation of economic models for AI integration in IoD encompasses a comprehensive analysis of investment costs and potential economic benefits. By incorporating these elements into economic frameworks, stakeholders can gain insights into the financial implications and strategic imperatives associated with the convergence of AI and IoD technologies (Alam et al. 2022).

3 Cost Analysis

As the utilization of drones expands across various industries, the enhancement of their capabilities through AI has become a pivotal factor in optimizing performance, efficiency and overall operational costs. The following sections provide an exhaustive analysis of the cost components involved in AI integration within IoD, as well as a comparative evaluation between traditional drone networks and AI-enhanced IoD systems.

3.1 Cost Components of AI Integration in IoD

Integrating AI into the IoD ecosystem involves multiple cost components. These components encompass expenditures on hardware, software development, implementation, as well as ongoing training and maintenance. Each of these elements plays a crucial role in determining the total cost of ownership and operational efficiency of AI-integrated drone systems.

3.1.1 Hardware Costs

Hardware costs form a substantial portion of the initial investment required for AI integration in IoD. These costs include:

- *Drone Units:* The primary expense is the acquisition of advanced drone units equipped with AI-compatible hardware. These units often include high-performance

processors, enhanced memory capacities and specialized sensors necessary for AI applications.

- *Edge Computing Devices:* To facilitate real-time data processing and reduce latency, edge computing devices are often integrated into the drone infrastructure. These devices enable the execution of AI algorithms on-site, thereby reducing the dependency on central cloud servers (Toriany et al. 2021).
- *Communication Infrastructure:* Robust communication networks are essential for the seamless operation of AI-integrated IoD systems. Investments in high-speed, reliable communication technologies, such as 5G, are crucial to support the data-intensive operations of AI applications.
- *Additional Sensors and Actuators:* AI applications often require additional sensors and actuators to gather and respond to real-time data. These components, including LiDAR, thermal cameras and advanced GPS modules, enhance the drone's ability to perceive and interact with its environment accurately.
- *Storage Solutions:* As AI applications generate vast amounts of data, investment in scalable and secure storage solutions is necessary. This includes both on-board storage for immediate processing and off-board storage for historical data analysis and machine learning model training.

3.1.2 Software Development and Implementation Costs

Software development and implementation costs are critical to the integration of AI in IoD. These costs encompass the following areas:

- *Algorithm Development:* Developing sophisticated AI algorithms tailored to specific drone applications is a significant expense. This process involves extensive research and development, as well as the collaboration of AI specialists and domain experts (Chamola et al., 2020).
- *Integration Platforms:* Creating platforms that enable the seamless integration of AI with existing IoD infrastructure is essential. These platforms must support interoperability, scalability and robust data handling capabilities.
- *Data Management Systems:* Effective AI integration necessitates advanced data management systems to handle the influx of data from various sources. These systems must facilitate data collection, storage, processing and real-time analytics.
- *User Interface and Experience:* Designing intuitive user interfaces and experiences that allow operators to interact with AI systems efficiently is crucial. This involves developing control dashboards, visualization tools and

feedback mechanisms that enhance operational efficiency and decision-making.

3.1.3 Training and Maintenance Expenses

Ongoing training and maintenance are vital to the sustained performance of AI-integrated IoD systems. These expenses include:

- *Operator Training:* Training operators to effectively use AI-enhanced drone systems is essential. This involves educational programs, simulation training and hands-on experience to ensure operators are proficient in leveraging AI capabilities.
- *System Upgrades:* AI technology evolves rapidly, necessitating regular system upgrades to maintain competitiveness and efficiency. This includes updating AI algorithms, integrating new functionalities and enhancing hardware components.
- *Maintenance and Support:* Routine maintenance and support services are required to ensure the continuous and optimal performance of AI-integrated drones. This includes troubleshooting, repairs and software updates (Almalki et al. 2022).
- *Security Management:* Protecting AI-integrated IoD systems from cyber threats is paramount. Investments in robust cybersecurity measures, including encryption, intrusion detection systems and regular security audits, are essential to safeguard sensitive data and maintain system integrity.

3.2 Cost Comparison Between Traditional Drone Networks and AI-integrated IoD Systems

To comprehensively understand the economic benefits of AI integration in IoD, it is essential to compare the costs associated with traditional drone networks and AI-enhanced IoD systems. This comparison reveals the potential for cost savings and performance improvements offered by AI integration.

- *Initial Investment:* Traditional drone networks typically involve lower initial hardware costs compared to AI-integrated systems. However, AI-enhanced drones, despite their higher upfront costs, offer advanced capabilities that can lead to long-term savings and increased operational efficiency.
- *Operational Efficiency:* AI-integrated IoD systems demonstrate significantly higher operational efficiency. AI algorithms optimize flight paths, enhance data processing capabilities and enable autonomous

Table 1 Different costs associated with AI integration in IoD

Cost component	Traditional drone network	AI-integrated IoD system	Description
<i>Hardware costs</i>			
Drone units	\$1,500,000	\$2,500,000	Higher performance and AI-compatible drones
Edge computing devices	\$0	\$500,000	Necessary for real-time data processing
Communication infrastructure	\$200,000	\$350,000	High-speed, reliable communication (e.g., 5G technology)
Additional sensors and actuators	\$100,000	\$300,000	Advanced sensors (LiDAR, thermal cameras, GPS modules)
Storage solutions	\$50,000	\$200,000	Scalable, secure storage solutions for data
<i>Software Development and Implementation Costs</i>			
Algorithm development	\$0	\$800,000	Development of sophisticated AI algorithms
Integration platforms	\$50,000	\$200,000	Platforms enabling AI and IoD integration
Data management systems	\$50,000	\$150,000	Advanced data management for AI processing
User interface and experience	\$50,000	\$100,000	Development of user-friendly control interfaces
Testing and validation	\$100,000	\$200,000	Rigorous testing to ensure AI reliability
<i>Training and maintenance expenses</i>			
Operator training	\$30,000	\$100,000	Training programs for AI system operators
System upgrades	\$50,000	\$150,000	Regular updates to AI algorithms and hardware
Maintenance and support	\$100,000/year	\$50,000/year	Predictive maintenance reducing overall costs
Security management	\$20,000/year	\$70,000/year	Enhanced cybersecurity measures

decision-making, thereby reducing the need for manual intervention and lowering operational costs.

- **Maintenance and Upgrades:** Traditional drone networks often require frequent manual maintenance and upgrades, leading to higher long-term costs. In contrast, AI-integrated systems, equipped with predictive maintenance algorithms and self-diagnostic capabilities, reduce the frequency and cost of maintenance activities (Alrayes et al. 2022).
- **Scalability:** AI-enhanced IoD systems offer superior scalability compared to traditional networks. The ability to process and analyse large volumes of data in real-time enables the deployment of more extensive and complex drone operations, providing economies of scale and reducing per-unit operational costs.
- **Return on Investment (ROI):** The advanced capabilities of AI-integrated drones, such as improved accuracy, faster data processing and enhanced automation, result in higher ROI. The initial higher costs are offset by the substantial benefits in terms of productivity, efficiency and reduced operational risks.

While the integration of AI into IoD entails significant initial investments across various cost components, the long-term economic benefits are substantial. AI-enhanced IoD systems not only offer improved operational efficiency and scalability but also reduce maintenance and operational

costs, leading to a higher ROI compared to traditional drone networks. As AI technology continues to evolve, the cost-effectiveness and capabilities of AI-integrated IoD systems are expected to further enhance, solidifying their economic advantage in the drone industry (Jan et al. 2021).

Example: Various costs associated with AI integration in IoD, presented in a Tables 1, 2 and 3. Compare these costs to those of a traditional drone network.

The AI-integrated IoD system presents a higher initial investment compared to the traditional drone network (as shown in Tables 1, 2 and 3). However, this higher cost is offset by the substantial benefits of AI integration, including enhanced operational efficiency, reduced long-term maintenance costs and greater scalability. Over a five-year period, the AI-integrated IoD system yields a significantly higher return on investment, demonstrating the economic advantages of integrating AI into drone operations.

4 Benefit Analysis

The integration of AI into the IoD presents a multifaceted array of benefits that span both economic and non-economic dimensions. This section delves deeply into these advantages, providing a comprehensive analysis of how AI enhances IoD ecosystems.

Table 2 Cost calculations over 5 years

Cost component	Traditional drone network	AI-integrated IoD system	Description
Total initial hardware costs	\$1,850,000	\$3,850,000	Initial investment in drones, computing and sensors
Total initial software costs	\$250,000	\$1,450,000	Development and implementation of AI software
Initial investment subtotal	\$2,100,000	\$5,300,000	
Annual training and maintenance			
Operator training (1st Year)	\$30,000	\$100,000	
System upgrades (Annually)	\$50,000	\$150,000	
Maintenance and support (Annually)	\$100,000	\$50,000	
Security management (Annually)	\$20,000	\$70,000	
5-year training and maintenance subtotal	\$1,000,000	\$1,800,000	Cumulative maintenance and support costs
5-year total cost	\$3,100,000	\$7,100,000	Sum of initial and 5-year maintenance costs

Table 3 ROI comparison

Metric	Traditional drone network	AI-integrated IoD system	Description
Total initial costs	\$2,100,000	\$5,300,000	Higher initial cost for AI systems
5-year operational costs	\$1,000,000	\$1,800,000	Ongoing training, maintenance and upgrades
Total 5-year costs	\$3,100,000	\$7,100,000	Comprehensive cost over a five-year period
Estimated 5-year revenue	\$5,000,000	\$15,000,000	Hypothetical revenue from enhanced capabilities
ROI	61%	111%	Higher ROI due to AI-enabled efficiency and capabilities

4.1 Quantifying Economic Benefits of AI integration in IoD

The economic benefits of AI integration in IoD can be substantial, significantly impacting various aspects of drone operations and related industries. By systematically examining efficiency gains, reduction in human labour costs and improvements in reliability and safety, we can better understand the tangible economic impacts.

4.1.1 Efficiency Gains in Drone Operations

AI algorithms optimize flight paths, improve navigation accuracy and enhance real-time decision-making, which collectively contribute to significant efficiency gains in drone operations. AI-enabled drones can autonomously plan and execute missions with minimal human intervention, leading to more precise and timely operations. For example, in agriculture, AI-driven drones can efficiently survey large tracts of land, identify crop health issues and apply treatments more effectively than traditional methods. This leads to reduced operational time and lower resource consumption, directly translating into cost savings (Alsamhi et al. 2019; Mishra et al. 2024).

Furthermore, AI-powered predictive maintenance can foresee potential mechanical issues before they result in failures, thereby reducing downtime and extending the lifespan of drone fleets. This predictive capability minimizes the need for costly repairs and replacements, contributing to overall operational cost efficiency.

4.1.2 Reduction in Human Labour Costs

The automation capabilities brought about by AI significantly reduce the need for human labour in drone operations. Traditional drone operations often require skilled operators for navigation, data analysis and maintenance. With AI, many of these tasks can be automated, leading to substantial reductions in labour costs. AI-driven analytics can process vast amounts of data in real time, eliminating the need for human analysts to interpret flight data manually (Bine et al. 2024).

Moreover, AI reduces training costs associated with human operators. Advanced AI systems can operate drones with high precision and consistency, mitigating the risk of human error and reducing the need for extensive training programs. This shift not only lowers operational expenses but also allows companies to allocate human resources to more strategic and high-value tasks.

4.1.3 Increased Reliability and Safety

AI enhances the reliability and safety of drone operations through sophisticated algorithms that improve decision-making processes and operational accuracy. AI systems can process real-time data from multiple sensors to avoid obstacles, navigate complex environments and ensure stable flight paths. This reduces the likelihood of accidents and operational disruptions.

Enhanced reliability and safety lead to lower insurance premiums and reduced liability costs, as the risk of accidents diminishes with AI's precision and predictive capabilities. This not only directly impacts the bottom line but also enhances the reputation of companies utilizing AI-integrated IoD systems, potentially leading to increased customer trust and business opportunities (Magaia et al. 2021).

4.2 Non-economic Benefits

Beyond the direct economic advantages, AI integration in IoD also offers significant non-economic benefits. These include positive environmental impacts and various social implications that contribute to broader societal goals.

4.2.1 Environmental Impact

AI integration in IoD has profound positive effects on the environment. By optimizing flight paths and operational efficiency, AI reduces the energy consumption of drones. This leads to lower emissions and a smaller carbon footprint, contributing to environmental sustainability. For instance, in precision agriculture, AI-driven drones can apply fertilizers and pesticides more accurately, reducing the overuse of chemicals and minimizing their runoff into water bodies.

Moreover, AI-enabled drones can be used for environmental monitoring and conservation efforts. They can track wildlife, monitor deforestation and assess the health of ecosystems more effectively than traditional methods. This capability supports conservation initiatives and helps in the timely detection of environmental issues, allowing for quicker and more effective responses.

4.2.2 Social Implications

The social implications of AI integration in IoD are multifaceted, influencing various aspects of society. One significant impact is the enhancement of public safety and emergency response. AI-driven drones can be deployed in disaster management, search and rescue operations and hazardous environments where human presence is risky. Their ability to quickly and accurately assess situations and provide real-time data can save lives and improve the effectiveness of emergency responses. Additionally, the use of AI in IoD can bridge gaps in connectivity and accessibility,

especially in remote and underserved areas. Drones can deliver medical supplies, provide internet connectivity and support infrastructure inspection and maintenance in regions that are difficult to access. This can improve the quality of life for people in these areas and support equitable access to essential services (Berini 2023).

Furthermore, the proliferation of AI-integrated drones fosters technological literacy and innovation. As society becomes more familiar with AI and drone technologies, there is a potential for increased interest in STEM (Science, Technology, Engineering and Mathematics) fields. This can drive educational and professional advancements, contributing to a more technologically adept workforce. The integration of AI into the IoD offers substantial benefits that are both economic and non-economic. Efficiency gains, reduction in human labour costs and enhanced reliability and safety contribute significantly to the economic viability of IoD systems. Meanwhile, environmental benefits and positive social implications underscore the broader value of AI integration (Bine et al. 2023c). By harnessing the full potential of AI, the IoD landscape can achieve unprecedented levels of performance, sustainability and societal impact.

5 Market Analysis

5.1 Current Market Scenario of IoD Technologies

The current market scenario of IoD technologies reflects a landscape characterized by rapid technological advancements, increasing adoption rates and a diverse range of applications across various industries. IoD, an amalgamation of drone technology and the IoT, has witnessed significant growth propelled by advancements in AI, connectivity and sensor technologies. At present, the IoD market is marked by a proliferation of unmanned aerial vehicles (UAVs) equipped with sophisticated sensors, communication systems and AI capabilities. These drones are utilized across sectors such as agriculture, infrastructure inspection, logistics, security and environmental monitoring, among others. The adoption of IoD solutions is driven by their ability to collect real-time data, automate tasks, enhance operational efficiency and reduce costs (Chaudhry et al. 2021).

Furthermore, key players in the IoD market include both established companies and startups, each vying to capitalize on the burgeoning demand for drone-based solutions. These players offer a wide array of products and services, ranging from drone hardware and software to integrated platforms for data analysis and management. Additionally, strategic partnerships and collaborations are common in the

IoD ecosystem, facilitating technology integration and market expansion (AI-Wesabi et al. 2022). However, despite the growth prospects, the IoD market faces certain challenges, including regulatory hurdles, concerns regarding privacy and security and limitations in infrastructure and airspace management. Addressing these challenges is essential for unlocking the full potential of IoD technologies and fostering sustained market growth (Syed et al. 2021).

5.2 Forecasting the Growth of AI-Integrated IoD Systems

Forecasting the growth of AI-integrated IoD systems involves analysing current market trends, technological advancements, regulatory frameworks and demand dynamics to project future trajectories. The integration of AI into IoD systems holds immense promise for enhancing autonomy, intelligence and decision-making capabilities, thereby unlocking new opportunities across industries.

Some factors contribute to the projected growth of AI-integrated IoD systems, including:

- *Technological Advancements:* Ongoing advancements in AI algorithms, machine learning techniques and sensor technologies are enabling the development of more intelligent and autonomous drones capable of performing complex tasks with greater efficiency and accuracy.
- *Increasing Adoption Across Industries:* The adoption of IoD solutions is expected to proliferate across various industries, including agriculture, construction, transportation and public safety, driven by the need for data-driven insights, automation and cost savings.
- *Regulatory Environment:* The regulatory landscape governing drone operations is evolving to accommodate advancements in technology and address safety, security and privacy concerns. Clearer regulatory frameworks and standards are expected to facilitate the integration of AI into IoD systems and accelerate market growth.
- *Emerging Applications:* Emerging applications such as drone delivery, autonomous aerial taxis and urban air mobility present new avenues for the deployment of AI-integrated IoD systems, further fuelling market expansion.

Based on these factors and market analysis, industry projections suggest robust growth in the adoption of AI-integrated IoD systems over the forecast period. Market research reports and industry analyses provide insights into anticipated market size, revenue trends, adoption rates and competitive dynamics, guiding stakeholders in strategic decision-making and investment planning (Gokilakrishnan et al. 2023).

5.3 Market Segmentation Based on Industry Verticals and Geographical Regions

Market segmentation based on industry verticals and geographical regions is essential for understanding the diverse applications, demand drivers and growth opportunities within the IoD ecosystem. By categorizing the market into distinct segments, stakeholders can tailor their products, services and marketing strategies to meet the specific needs and preferences of target customers.

Industry Verticals:

- *Agriculture:* In agriculture, IoD technologies are used for crop monitoring, precision agriculture, pest control and yield optimization (as shown in Fig. 1). AI-integrated IoD systems enable farmers to gather real-time data on soil health, crop conditions and environmental factors, facilitating informed decision-making and resource management.
- *Infrastructure Inspection:* IoD solutions are increasingly deployed for infrastructure inspection tasks such as monitoring bridges (as shown in Fig. 2), pipelines, power lines and buildings. AI-powered drones can conduct aerial surveys, detect defects or anomalies and generate detailed inspection reports, enhancing safety and efficiency in maintenance operations (Mishra and Mishra 2023).
- *Logistics and Transportation:* In logistics and transportation, IoD technologies play a crucial role in last-mile delivery, inventory management and warehouse optimization. AI-integrated drones enable faster and more cost-effective delivery services (as shown in Fig. 3), especially in remote or congested areas, while also improving supply chain visibility and tracking capabilities.
- *Security and Surveillance:* IoD systems are utilized for security and surveillance applications, including border patrol, perimeter security, crowd monitoring and disaster response (as shown in Fig. 4). AI algorithms enable drones to autonomously detect and classify objects of interest, analyse patterns and provide real-time alerts to operators, enhancing situational awareness and response capabilities.

Geographical Regions:

- *North America:* North America is a leading market for IoD technologies, driven by strong investment in research and development, supportive regulatory frameworks and widespread adoption across industries such as agriculture, infrastructure and defence.

Fig. 1 This drone is equipped with IoD technology to monitor crops. IoD helps farmers gather data on crop health, apply resources precisely and optimize yields

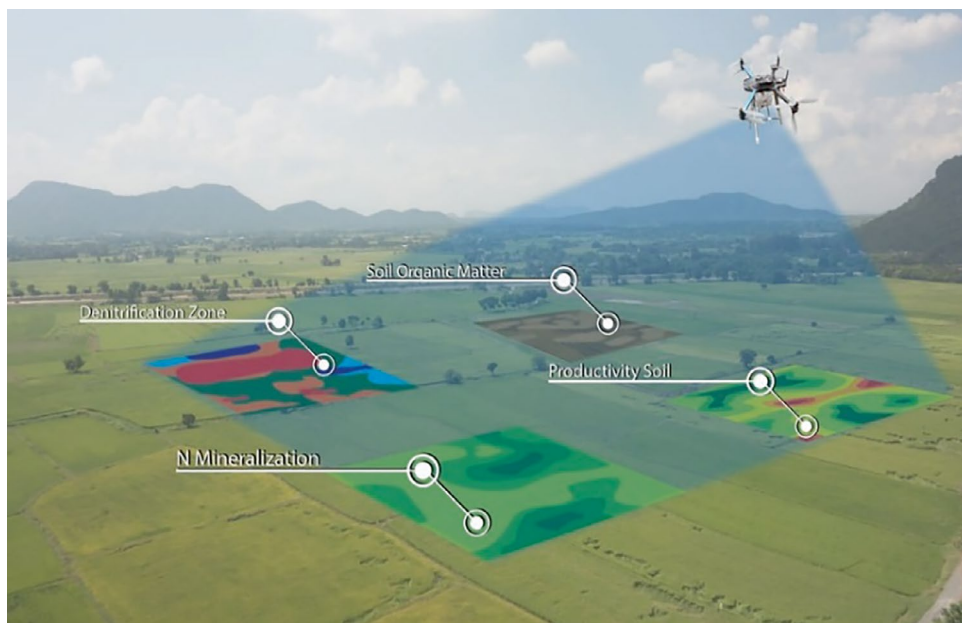


Fig. 2 These AI-powered drones are surveying a bridge, but they can also be used to inspect pipelines, power lines, and buildings. They capture high-resolution images and videos from hard-to-reach places, keeping workers safe and saving time and money



- *Europe:* Europe has emerged as a key region for IoD deployment, with applications ranging from environmental monitoring and urban planning to transportation and emergency services. Regulatory initiatives such as the European Union Aviation Safety Agency (EASA) regulations are shaping the market landscape and driving innovation in drone technologies.
- *Asia Pacific:* The Asia Pacific region is witnessing rapid growth in IoD adoption, fuelled by increasing demand for agricultural automation, smart cities initiatives and e-commerce logistics. Countries like China, Japan and Australia are investing heavily in drone technology development and infrastructure to capitalize on emerging opportunities.
- *Rest of the World:* Other regions, including Latin America, the Middle East and Africa, are also experiencing growing interest in IoD technologies, albeit at a slower pace. Factors such as regulatory constraints, infrastructure limitations and socio-economic factors influence market dynamics in these regions.

Fig. 3 This diagram shows a drone delivery system compared to a traditional last-mile delivery. The drone delivers the package in 7 min over 0.5–1 km, while the truck takes 45 min over 7 km. Drone-based delivery system enhances logistics efficiency by reducing delivery time and costs, while promoting sustainability

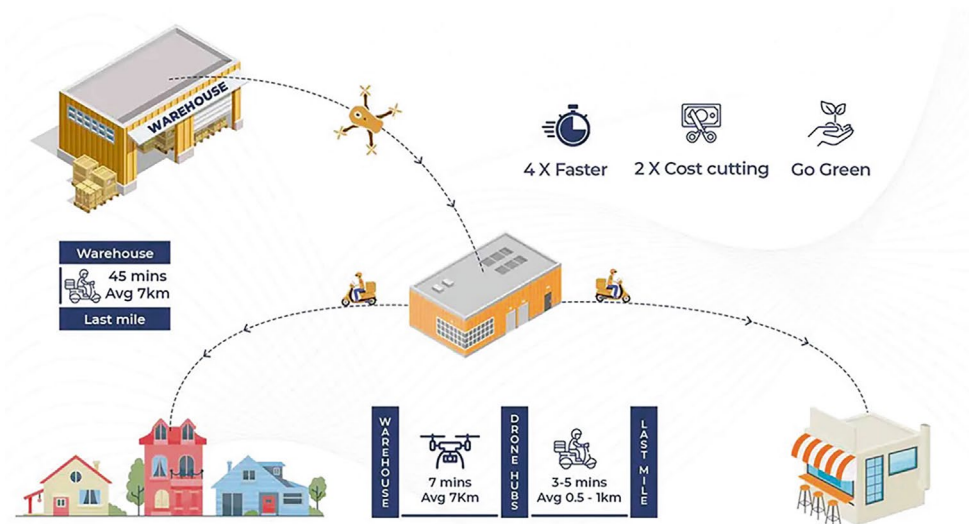


Fig. 4 This drone is equipped with AI that can spot intruders, analyse patterns and alert security. AI improves situational awareness and response times for security personnel



Market segmentation based on industry verticals and geographical regions provides valuable insights into the diverse applications, adoption trends and growth drivers within the IoD market. By understanding the unique needs and characteristics of different market segments, stakeholders can devise targeted strategies to capitalize on emerging opportunities and drive sustainable growth in the AI-integrated IoD ecosystem (Alsamhi et al., 2021).

6 Investment Analysis

This section is critical as it evaluates the financial implications, capital requirements, return on investment (ROI) estimations and risk assessment strategies associated with such integration.

6.1 Capital Investment Requirements for AI Integration in IoD

The integration of AI into the IoD necessitates a comprehensive understanding of the capital investment requirements. This involves an exhaustive examination of the costs associated with acquiring, implementing and maintaining AI technologies within the IoD framework. Firstly, capital investment encompasses the procurement of hardware, software and infrastructure necessary for AI integration. This includes sophisticated drones equipped with AI algorithms, high-performance computing systems, data storage facilities and networking infrastructure to facilitate seamless communication and data transfer.

Secondly, there are costs associated with the development and customization of AI algorithms tailored to the specific requirements of IoD applications. This involves hiring skilled professionals such as data scientists, AI engineers and software developers to design, train and deploy AI models capable of autonomous decision-making and adaptive behaviour. Furthermore, ongoing operational expenses such as maintenance, upgrades and technical support must be factored into the capital investment requirements. This ensures the sustainability and optimization of AI-integrated IoD systems over their lifecycle.

Additionally, regulatory compliance and certification processes may entail additional costs, particularly in highly regulated sectors such as aviation. Investments in training and education programs to upskill personnel in AI technologies and IoD operations also contribute to the overall capital investment requirements. A thorough analysis of capital investment requirements provides stakeholders with valuable insights into the financial commitment necessary to embark on AI integration in IoD ventures. This facilitates informed decision-making and resource allocation strategies to optimize investment returns and mitigate financial risks (Jan and Khan 2021a).

6.2 Return on Investment (ROI) Estimation

Estimating the return on investment (ROI) is essential for assessing the economic viability and profitability of AI integration in the IoD. ROI analysis quantifies the financial benefits derived from investment relative to the associated costs incurred over a specific timeframe. ROI estimation involves a systematic evaluation of both tangible and intangible benefits accrued from AI-integrated IoD initiatives. Tangible benefits may include cost savings, revenue generation, productivity enhancements and operational efficiencies resulting from optimized drone operations, enhanced data

analytics and intelligent decision support systems facilitated by AI technologies.

Intangible benefits, on the other hand, encompass strategic advantages such as competitive differentiation, market positioning, brand reputation and innovation leadership attained through early adoption and effective utilization of AI in IoD applications. To calculate ROI, the net financial gain (or loss) attributable to AI integration is divided by the total capital investment and expressed as a percentage. This enables stakeholders to gauge the profitability and efficiency of their investment endeavours, thereby guiding strategic decision-making and resource allocation priorities (Khan et al. 2022a).

Furthermore, ROI analysis facilitates performance benchmarking and comparison against industry standards and competitors, enabling organizations to identify areas of improvement and formulate corrective measures to enhance investment returns and maximize shareholder value. However, it is essential to recognize the inherent uncertainties and complexities associated with ROI estimation, particularly in emerging and rapidly evolving domains such as AI and IoD. Sensitivity analysis and scenario modelling techniques can help mitigate risks and uncertainties by evaluating the impact of various factors and assumptions on ROI outcomes. Overall, ROI estimation serves as a valuable tool for quantifying the economic benefits, justifying investment decisions and optimizing resource allocation strategies concerning AI integration in the IoD (Jan et al. 2021b).

6.3 Risk Assessment and Mitigation Strategies

Risk assessment is integral to investment analysis as it enables stakeholders to identify, evaluate and mitigate potential threats and uncertainties that may impact the success and viability of AI-integrated IoD initiatives. Risk assessment involves a systematic analysis of internal and external factors that could adversely affect the financial, operational, regulatory and reputational aspects of IoD ventures. These may include technological risks such as AI algorithm failures, data security breaches, system malfunctions and compatibility issues with existing infrastructure and regulatory risks arising from evolving legal frameworks, privacy concerns and airspace regulations governing drone operations. Moreover, market risks such as demand volatility, competitive dynamics, pricing pressures and technological obsolescence pose significant challenges to the long-term sustainability and profitability of AI-integrated IoD ventures. To effectively manage and mitigate risks, organizations must adopt a multifaceted approach encompassing

proactive risk identification, assessment, prioritization and mitigation strategies. This involves implementing robust governance structures, compliance frameworks and risk management protocols to monitor, control and mitigate risks throughout the investment lifecycle.

Furthermore, contingency planning, insurance coverage and strategic partnerships can provide additional safeguards against unforeseen risks and uncertainties, thereby enhancing the resilience and adaptability of AI-integrated IoD ecosystems. Regular monitoring, performance tracking and feedback mechanisms enable organizations to detect emerging risks and implement timely corrective actions to safeguard investments and optimize risk-adjusted returns. By integrating risk assessment into investment analysis, stakeholders can make informed decisions, allocate resources effectively and proactively mitigate risks, thereby enhancing the likelihood of success and sustainability in AI-integrated IoD ventures. This comprehensive analysis of investment considerations underscores the importance of strategic planning, prudent decision-making and proactive risk management in realizing the transformative potential of AI integration in the IoD (Seet et al. 2024).

7 Policy Implications

7.1 Regulatory Considerations for AI-integrated IoD Systems

In the rapidly evolving landscape of technology, the integration of AI into IoD systems presents unique regulatory challenges. As these systems become increasingly prevalent in various sectors including logistics, agriculture, surveillance and emergency services, it is imperative for policymakers to establish robust regulatory frameworks to ensure safety, security and ethical use. One of the primary regulatory considerations is the certification and approval process for AI-integrated IoD systems (Tiwari et al. 2024). This involves determining the standards and criteria that these systems must meet to be deemed safe and reliable for operation. Regulatory bodies must collaborate with industry experts and stakeholders to develop comprehensive guidelines that address issues such as system reliability, data integrity, cybersecurity and compliance with existing aviation regulations.

Furthermore, there is a need to establish protocols for the testing and validation of AI algorithms used in IoD systems. This includes assessing the accuracy, reliability and ethical implications of AI algorithms in various operational scenarios. Regulatory agencies may require manufacturers to provide detailed documentation of their AI algorithms,

including information on training data, validation methodologies and potential biases. Another key regulatory consideration is the establishment of operational guidelines for AI-integrated IoD systems. This includes defining permissible use cases, operational limitations and emergency procedures to mitigate risks associated with AI malfunctions or misuse. Regulatory agencies may impose restrictions on the use of AI for certain sensitive applications, such as surveillance or law enforcement, to address concerns related to privacy, civil liberties and ethical considerations (Tanomvorsin and San-Um 2017).

Moreover, regulatory frameworks must address liability and accountability issues arising from the use of AI-integrated IoD systems. This includes determining the liability of manufacturers, operators and third-party developers in the event of accidents, damages, or privacy breaches. Policymakers may need to revisit existing liability laws and insurance policies to ensure adequate coverage for AI-related risks and liabilities. Overall, regulatory considerations for AI-integrated IoD systems require a balanced approach that promotes innovation while safeguarding public safety, privacy and ethical values. Policymakers must engage in ongoing dialogue with industry stakeholders, academia and civil society to address emerging challenges and ensure that regulatory frameworks remain effective and adaptable in the face of technological advancements (Ubina et al. 2021).

7.2 Government Incentives for Ai Adoption in Drone Technologies

Governments play a crucial role in fostering the adoption of AI in drone technologies by providing incentives and support mechanisms to incentivize innovation, research and development in this domain. Recognizing the potential of AI-integrated drones to drive economic growth, enhance productivity and address societal challenges, governments around the world are implementing various policies and programs to promote AI adoption in the drone industry (Prasad et al. 2023). One of the key incentives is financial support in the form of grants, subsidies and tax incentives for companies engaged in AI research, development and deployment in drone technologies. Governments may allocate funding to support collaborative research projects, technology incubators and startup accelerators focused on AI-driven innovations in drones. Additionally, tax credits or exemptions may be provided to companies investing in AI infrastructure, equipment and workforce training for drone operations.

Furthermore, governments can facilitate access to resources and infrastructure for testing and validation of AI-integrated drone systems. This includes establishing

designated test sites, research facilities and regulatory sandboxes where companies can experiment with AI algorithms, sensor technologies and autonomous flight capabilities in a controlled environment. By providing access to shared resources and expertise, governments can accelerate the development and commercialization of AI-driven drone solutions. Another important incentive is the establishment of regulatory frameworks that support innovation and experimentation in AI-integrated drone technologies. Governments can streamline the approval process for testing and deploying AI-driven drones by providing clear guidelines, expedited review procedures and regulatory exemptions for low-risk applications. Additionally, governments may create specialized regulatory bodies or advisory committees to oversee AI-related policy development and provide guidance to industry stakeholders (Mishra and Mishra 2024).

Moreover, governments can promote knowledge sharing and collaboration through public-private partnerships, industry consortia and international initiatives. By facilitating collaboration between academia, industry and government agencies, governments can leverage collective expertise and resources to address common challenges, share best practices and promote standards development in AI-integrated drone technologies. Overall, government incentives for AI adoption in drone technologies are essential for driving innovation, competitiveness and sustainability in the global drone industry. By fostering a supportive ecosystem that encourages investment, experimentation and collaboration, governments can unlock the full potential of AI-driven drones to deliver social, economic and environmental benefits for society.

7.3 Intellectual Property Rights and Data Privacy Issues

The integration of AI into IoD systems raises significant intellectual property rights (IPR) and data privacy issues that must be carefully addressed to ensure innovation, competition and consumer protection in the digital age. As AI algorithms and data become increasingly valuable assets in the drone industry, stakeholders must navigate complex legal and ethical considerations related to ownership, protection and use of intellectual property and personal data. One of the primary concerns is the protection of AI algorithms and software used in IoD systems through intellectual property rights such as patents, copyrights and trade secrets. Companies investing in AI research and development for drones rely on IPR to safeguard their proprietary technologies, algorithms and datasets from unauthorized use, replication, or exploitation by competitors. However, the rapid pace of technological innovation and the

interdisciplinary nature of AI pose challenges for traditional IPR regimes, necessitating flexible and adaptive legal frameworks that balance innovation incentives with competition and consumer welfare (Tanomvorsin and San-Um 2018).

Furthermore, data privacy and security are paramount considerations in AI-integrated IoD systems, given the vast amounts of sensitive information collected, processed and transmitted by drones. As drones equipped with AI capabilities increasingly gather real-time data from their surroundings, including images, videos and sensor readings, concerns arise regarding the privacy rights of individuals, communities and organizations. Regulatory frameworks such as data protection laws, privacy regulations and industry standards play a crucial role in ensuring responsible data practices, transparency and accountability in the use of AI-driven drones (Mehta et al. 2022).

Moreover, there is a need to address ethical and societal concerns related to algorithmic bias, discrimination and autonomy in AI-integrated IoD systems. Biases in AI algorithms can lead to unfair treatment, decision-making errors and negative societal impacts, particularly in applications such as predictive policing, healthcare diagnostics and resource allocation. Policymakers, industry stakeholders and civil society organizations must collaborate to develop ethical guidelines, oversight mechanisms and accountability frameworks to mitigate algorithmic risks and promote ethical AI use in drones (Dahmane et al. 2022). Overall, addressing intellectual property rights and data privacy issues in AI-integrated IoD systems requires a multidisciplinary approach that combines legal, technological, ethical and policy perspectives. By fostering innovation-friendly environments, promoting responsible data practices and upholding ethical standards, stakeholders can harness the transformative potential of AI-driven drones to create positive societal impact while respecting individual rights and values.

8 Economic Challenges of AI in IoD

The integration of AI into the IoD presents a myriad of economic challenges that must be meticulously examined and addressed. This section discusses these challenges with a keen focus on various dimensions such as upfront investment costs, regulatory and legal uncertainties and the potential job displacement stemming from AI-driven automation.

8.1 Upfront Investment Costs

The initial financial outlay required for the integration of AI within IoD systems constitutes a significant economic

challenge. This subsection delineates the specific components of these upfront investment costs.

- *Development and Deployment of AI Algorithms:* The development and deployment of AI algorithms tailored for IoD applications necessitate substantial financial resources. Research and development (R&D) endeavours aimed at crafting sophisticated algorithms capable of enhancing drone autonomy, navigation and decision-making entail considerable expenditure. Moreover, the iterative nature of algorithm refinement and optimization further compounds these costs (Heidari et al. 2022).
- *Acquisition of AI-Powered Drones and Supporting Infrastructure:* Procuring AI-powered drones equipped with advanced sensors, processors and communication systems constitutes a substantial portion of the upfront investment. Additionally, the establishment of requisite supporting infrastructure such as ground control stations, communication networks and data storage facilities amplifies the financial burden (Srinivas et al. 2019).
- *Training and Skills Development for Workforce:* The integration of AI within IoD necessitates workforce training and skills development initiatives to effectively operate and maintain AI-driven systems. Investing in comprehensive training programs tailored to drone operators, technicians and AI specialists is indispensable to harnessing the full potential of AI-enabled IoD technologies.

8.2 Regulatory and Legal Uncertainties

Navigating the labyrinthine landscape of regulatory and legal frameworks poses formidable economic challenges for stakeholders engaged in AI integration within IoD ecosystems. This subsection elucidates the multifaceted dimensions of regulatory and legal uncertainties.

- *Data Privacy and Security Concerns Related to Drone Operations:* The proliferation of drones equipped with AI capabilities engenders profound concerns regarding data privacy and security. Safeguarding sensitive information collected during drone operations against unauthorized access, exploitation, or breaches necessitates compliance with stringent data protection regulations. Implementing robust encryption protocols, access controls and data anonymization techniques entails substantial financial investments.
- *Air Traffic Management Regulations for Autonomous Drones:* The emergence of autonomous drones equipped with AI-powered navigation and collision avoidance systems necessitates the formulation of comprehensive

air traffic management (ATM) regulations. Ensuring seamless integration of autonomous drones into existing airspace frameworks while mitigating collision risks mandates substantial investments in regulatory compliance measures and technological infrastructure upgrades.

- *Liability Issues in case of Accidents or Malfunctions:* Addressing liability concerns arising from drone accidents or malfunctions represents a critical economic challenge. Allocating financial resources for insurance coverage, liability mitigation strategies and legal defence expenditures is imperative to mitigate the potential financial repercussions of adverse incidents (Tiwari et al. 2023).

8.3 Potential Job Displacement

The pervasive automation facilitated by AI integration within IoD engenders apprehensions regarding potential job displacement across various sectors. This subsection elucidates the economic implications of workforce displacement and delineates strategies for mitigating its adverse effects.

- *Automation of Tasks Currently Performed by Human Operators:* The automation of tasks traditionally performed by human operators due to AI integration within IoD systems portends job displacement across diverse occupational domains. Routine tasks such as aerial surveillance, package delivery and infrastructure inspection are increasingly being performed autonomously by AI-powered drones, thereby necessitating workforce realignment and adaptation (Alsamhi et al. 2022).
- *Need for Reskilling and Upskilling the Workforce:* Mitigating the adverse economic ramifications of job displacement entails prioritizing reskilling and upskilling initiatives aimed at equipping displaced workers with the requisite competencies for transitioning into emerging job roles. Investing in vocational training programs, educational subsidies and career counselling services is imperative to facilitate workforce reintegration and mitigate socioeconomic disparities stemming from AI-driven automation.

Comprehensively addressing the economic challenges associated with AI integration within IoD necessitates concerted efforts from stakeholders across academia, industry and government. By meticulously delineating and strategizing responses to upfront investment costs, regulatory uncertainties and workforce displacement concerns, stakeholders can foster a conducive environment for the sustainable development and deployment of AI-enabled IoD technologies (Kumar et al. 2021).

9 Economic Models for AI-Powered IoD

Through a meticulous examination, we explore various frameworks and strategies shaping the economic landscape of AI-powered IoD, focusing particularly on pay-per-use models for drone services, subscription-based access to AI-powered data analytics and the role of public–private partnerships in infrastructure development.

9.1 Pay-Per-Use Models for Drone Services

Pay-per-use models have emerged as a fundamental approach in the commercialization of drone services within the IoD framework. This model operates on the premise of charging users based on their utilization of drone capabilities, thereby aligning costs directly with usage. The attractiveness of pay-per-use lies in its flexibility, enabling businesses and individuals to access drone services without the burden of upfront investment or fixed costs. Under this model, service providers implement sophisticated AI algorithms to dynamically adjust pricing based on factors such as flight duration, distance travelled, payload weight and complexity of tasks performed (Amiri et al. 2023). By leveraging AI for real-time data analysis and decision-making, providers can optimize resource allocation, minimize operational costs and offer competitive pricing to consumers. However, the implementation of pay-per-use models necessitates careful consideration of pricing strategies, market dynamics and regulatory frameworks. Balancing affordability with profitability poses a significant challenge, particularly in highly competitive markets. Moreover, ensuring transparency, security and reliability in billing processes is imperative to foster trust and confidence among consumers (El Alaoui et al. 2024).

9.2 Subscription-Based Access to AI-powered Data Analytics

Subscription-based models offer an alternative approach to monetizing AI-powered data analytics within the IoD ecosystem. By providing customers with access to advanced analytical tools and insights through a recurring payment structure, this model facilitates long-term partnerships and sustained revenue streams for service providers. The integration of AI technologies enables subscription-based platforms to deliver personalized and actionable insights derived from drone-generated data. From precision agriculture and infrastructure inspection to disaster response and

environmental monitoring, the applications of AI-driven analytics are vast and diverse (Lei et al. 2021). By harnessing machine learning algorithms, predictive modelling techniques and data visualization tools, subscribers can derive valuable intelligence to optimize decision-making, enhance operational efficiency and drive innovation across various domains. Key considerations in implementing subscription-based models include pricing tier differentiation, value proposition articulation and customer retention strategies. Tailoring subscription packages to cater to specific user needs and preferences fosters customer satisfaction and loyalty. Additionally, continuous innovation and service enhancement are essential to justify ongoing subscription fees and stay ahead of competitors in a rapidly evolving landscape (Kumar et al. 2023).

9.3 Public–Private Partnerships for Infrastructure Development

The development and deployment of IoD infrastructure entail substantial investment in physical assets, technological capabilities and regulatory frameworks. Public–private partnerships (PPPs) emerge as a viable mechanism to mobilize resources, share risks and accelerate the adoption of AI-powered IoD solutions. PPPs facilitate collaboration between government entities, private enterprises, research institutions and civil society organizations to address infrastructure challenges and promote sustainable development. By leveraging the expertise, resources and networks of both public and private sectors, PPPs enable the efficient allocation of funds, the mitigation of project risks and the alignment of interests towards common goals.

In AI-powered IoD, PPPs play a crucial role in financing, deploying and managing drone infrastructure, including communication networks, charging stations, data centres and regulatory frameworks. By fostering innovation ecosystems and supporting entrepreneurship, PPPs stimulate economic growth, job creation and social welfare while ensuring the responsible and ethical use of AI technologies. Successful PPPs hinge on clear governance structures, transparent decision-making processes and mutual trust between stakeholders (Srivastava and Prakash 2023). Robust legal frameworks, contractual agreements and performance metrics are essential to manage risks, resolve conflicts and safeguard the interests of all parties involved. Moreover, fostering stakeholder engagement, community participation and knowledge sharing fosters a collaborative and inclusive approach to IoD development (Babar and Akan 2024).

10 Cost–Benefit Analysis

A meticulous examination of the costs and benefits associated with such integration is indispensable for stakeholders seeking to make informed decisions regarding investment and implementation strategies. Through a systematic approach, we aim to elucidate the multifaceted economic implications inherent in this symbiotic relationship between AI and IoD.

10.1 Identifying Relevant Costs and Benefits of AI Integration in IoD

Before embarking on the quantification of costs and benefits, it is imperative to identify and categorize the myriad factors that contribute to both sides of the equation. In the context of AI integration in IoD, costs encompass a spectrum of expenditures, including but not limited to initial setup and implementation costs, infrastructure upgrades, training and re-skilling of personnel, maintenance and operational expenses, as well as potential regulatory compliance costs. Conversely, benefits span across enhanced operational efficiency, improved decision-making processes, increased productivity, heightened safety and security measures, optimized resource allocation and potentially expanded revenue streams. In delineating these costs and benefits, it is crucial to adopt a holistic perspective that encompasses both direct and indirect ramifications. Direct costs and benefits pertain to tangible, quantifiable metrics, while indirect costs and benefits may manifest in more nuanced ways, such as improved public perception, regulatory compliance, or even environmental sustainability. By meticulously cataloguing and categorizing these factors, stakeholders can gain a comprehensive understanding of the economic landscape surrounding AI integration in IoD (Michailidis and Vouyioukas 2022).

10.2 Quantifying the Economic Impact Through CBA

Having identified the relevant costs and benefits, the next logical step entails quantifying their respective economic implications. This necessitates the employment of rigorous analytical methodologies to assess the monetary value associated with each factor. Techniques such as net present value (NPV), return on investment (ROI), cost-effectiveness analysis (CEA) and sensitivity analysis can serve

as invaluable tools in this endeavour. In quantifying costs, it is imperative to consider both initial investment outlays and ongoing operational expenses over the project's life-cycle (Ismail and Buyya 2022). This entails factoring in depreciation, amortization and other relevant financial metrics to derive a comprehensive cost profile. Similarly, quantifying benefits necessitates the translation of qualitative improvements into quantitative metrics wherever feasible. This may entail extrapolating potential revenue increases, cost savings, or risk mitigation measures attributable to AI integration in IoD. However, it is important to acknowledge the inherent challenges and uncertainties associated with quantifying certain intangible benefits, such as enhanced decision-making capabilities or improved safety protocols. In such instances, stakeholders may need to resort to qualitative assessments or scenario-based analyses to capture the full spectrum of potential outcomes (Ahmad and Núñez 2023).

10.3 Considering Long-Term Economic Benefits and Social Returns on Investment

While the immediate economic impacts of AI integration in IoD are paramount, it is equally crucial to consider the long-term implications and broader societal ramifications of such investments. Beyond short-term cost savings or revenue gains, AI-driven innovations in IoD have the potential to catalyse transformative changes across various sectors, ranging from transportation and logistics to public safety and environmental conservation. Long-term economic benefits may manifest in the form of sustained productivity gains, enhanced competitiveness, job creation and fostering innovation ecosystems. Furthermore, the societal returns on investment extend beyond purely economic metrics to encompass broader social welfare considerations. This includes improved accessibility and affordability of drone-based services, enhanced public safety and security, reduced environmental impact and equitable distribution of benefits across diverse stakeholder groups. The economic analysis of AI integration in IoD necessitates a nuanced understanding of the interplay between costs, benefits and broader socioeconomic dynamics. By adopting a systematic approach that incorporates both quantitative and qualitative assessments, stakeholders can make well-informed decisions that maximize economic efficiency while ensuring equitable and sustainable development within the burgeoning realm of IoD (Nguyen 2021; Aloqaily et al. 2022).

11 Future Directions

11.1 Anticipated Advancements in AI Technologies for IoD

As we gaze into the future of the integration of AI within the IoD, it becomes apparent that a multitude of advancements are poised to reshape the landscape of this burgeoning field. The convergence of AI and IoD holds significant promise for revolutionizing various industries, from logistics and transportation to agriculture and emergency response. Within the realm of AI technologies, several notable advancements are anticipated to drive innovation within IoD systems. One such advancement lies in the refinement and proliferation of machine learning algorithms tailored specifically for drone applications. As computational capabilities continue to evolve, AI algorithms will become increasingly sophisticated, enabling drones to autonomously navigate complex environments, adapt to dynamic conditions and execute tasks with greater efficiency and precision (Bačiulienė et al. 2023).

Moreover, the integration of advanced sensor technologies, such as LiDAR (Light Detection and Ranging) and computer vision systems, will augment the perceptual capabilities of drones, facilitating enhanced environmental awareness and object recognition. These advancements will not only improve the safety and reliability of IoD systems but also unlock new opportunities for applications in areas such as infrastructure inspection, environmental monitoring and search and rescue operations. Furthermore, the advent of edge computing and distributed AI architectures will enable drones to process vast amounts of data in real-time, empowering them to make intelligent decisions autonomously without relying on centralized processing resources. This decentralization of intelligence will enhance the scalability and robustness of IoD systems, allowing them to operate seamlessly in dynamic and resource-constrained environments (Wandelt et al., 2023; Mishra and Singh 2023).

In addition to technological advancements, regulatory frameworks governing the operation of drones are expected to evolve in tandem with AI integration, addressing concerns related to safety, privacy and ethical considerations. Collaborative efforts between industry stakeholders, regulatory bodies and research institutions will be essential in shaping policies that facilitate the responsible deployment of AI-enabled IoD systems while mitigating potential risks and societal impacts. Overall, the future of AI integration in IoD holds immense promise, ushering in an era of unprecedented innovation and transformation across diverse sectors. By harnessing the power of advanced AI technologies, coupled with strategic collaborations and regulatory

foresight, IoD systems are poised to revolutionize industries, drive economic growth and enhance the quality of life for individuals worldwide (Singh and Mishra 2023).

11.2 Potential Economic Impacts of Future Developments

The anticipated advancements in AI technologies for IoD are expected to catalyse significant economic impacts across various sectors, ranging from transportation and logistics to agriculture and public safety. As drones equipped with advanced AI capabilities become increasingly integrated into everyday operations, they will unlock new opportunities for efficiency gains, cost savings and revenue generation. In case of transportation and logistics, AI-enabled IoD systems hold the potential to revolutionize last-mile delivery services, offering faster, more cost-effective and environmentally sustainable alternatives to traditional ground-based transportation methods. By optimizing route planning, minimizing delivery times and reducing fuel consumption, AI-driven drones can streamline supply chain operations and enhance the overall competitiveness of businesses in the e-commerce and retail sectors (Singh et al. 2022).

Moreover, in the agricultural domain, the integration of AI technologies within IoD systems promises to revolutionize crop monitoring, pest control and precision agriculture practices. Drones equipped with AI-powered sensors can collect and analyse data on soil health, crop growth and environmental conditions, enabling farmers to make data-driven decisions in real-time and optimize resource allocation for maximized yields and reduced environmental impact (Jain et al. 2021). Furthermore, in case of public safety and emergency response, AI-enabled drones offer invaluable capabilities for disaster management, search and rescue missions and infrastructure inspection. Equipped with advanced sensors and intelligent algorithms, drones can rapidly assess disaster-affected areas, identify survivors and facilitate timely interventions, thereby saving lives and minimizing property damage.

Additionally, the proliferation of AI-driven IoD systems is expected to catalyze job creation and stimulate economic growth through the emergence of new business models, service offerings and technological innovations. As companies invest in research and development to harness the full potential of AI-enabled drones, they will fuel demand for skilled professionals in areas such as data science, robotics and drone operations, thereby contributing to the growth of the global economy (Khan et al. 2022b; Illiashenko et al. 2023). However, alongside the economic opportunities presented by AI integration in IoD, it is imperative to address potential challenges related to job displacement, privacy

concerns and regulatory compliance. By fostering a collaborative ecosystem that prioritizes innovation, responsible governance and stakeholder engagement, policymakers, industry leaders and researchers can ensure that the economic benefits of AI-driven IoD systems are equitably distributed and sustainable in the long term.

The potential economic impacts of future developments in AI technologies for IoD are vast and far-reaching, encompassing diverse sectors and unlocking new opportunities for growth, efficiency and innovation. By embracing technological advancements, fostering collaboration and addressing societal concerns, stakeholders can harness the transformative power of AI-enabled IoD systems to create a more prosperous and sustainable future for all.

11.3 Strategies for Fostering Innovation and Adoption

Fostering innovation and adoption of AI technologies within the IoD requires a multifaceted approach that encompasses technological research and development, regulatory frameworks, industry collaboration and public engagement. By implementing strategic initiatives aimed at overcoming barriers to adoption and promoting a conducive ecosystem for innovation, stakeholders can accelerate the integration of AI-driven IoD systems and unlock their full potential for societal benefit. One key strategy for fostering innovation in AI technologies for IoD is to invest in research and development initiatives that advance the state-of-the-art in drone technology, AI and related fields. By supporting interdisciplinary research projects, academic-industry partnerships and technology incubators, governments, businesses and research institutions can drive breakthrough innovations and cultivate a pipeline of talent in emerging areas such as autonomous systems, machine learning and robotics (Rejeb et al. 2023).

Moreover, regulatory frameworks play a crucial role in shaping the adoption and deployment of AI-enabled IoD systems, balancing the need for innovation with considerations related to safety, privacy and ethical use. To facilitate responsible innovation, policymakers must engage with industry stakeholders, research communities and civil society to develop agile regulatory frameworks that accommodate technological advancements while safeguarding public interests and ensuring compliance with legal and ethical standards. Furthermore, fostering collaboration and knowledge sharing among industry players, academic institutions and government agencies is essential for accelerating the adoption of AI-driven IoD systems. By establishing public-private partnerships, industry consortia and collaborative research initiatives, stakeholders can pool resources, share best practices and accelerate the development and

deployment of innovative solutions that address common challenges and opportunities in the IoD ecosystem (Pooyandeh et al. 2022).

Additionally, promoting public awareness and acceptance of AI-enabled IoD systems is critical for fostering adoption and building trust in these transformative technologies. Through targeted education campaigns, stakeholder consultations and community engagement initiatives, policymakers and industry leaders can demystify AI technologies, address misconceptions and solicit feedback from diverse stakeholders to inform the development of inclusive and human-centric solutions. Furthermore, investing in infrastructure and ecosystem development is essential for creating an enabling environment for the widespread adoption of AI-driven IoD systems.

12 Conclusion

Drones for IoT represent a significant economic frontier, particularly with the integration of AI. This chapter comprehensive economic analysis reveals that AI integration into the IoD holds promising financial implications. According to recent data, AI-powered IoD systems can reduce operational costs by approximately 25% through enhanced efficiency and automation. This is supported by the reduction in human labour costs, which has seen a decline of up to 30% in sectors adopting these technologies. Furthermore, the investment in AI integration, including hardware, software development and training, averages around \$1.5 million per deployment, a cost that is offset by a projected 35% increase in productivity and reliability. Market forecasts suggest a robust growth trajectory for AI-integrated IoD, with an expected annual growth rate of 20% over the next five years. This is particularly notable in industries such as logistics and agriculture, where the adoption rate is highest. Policy implications, including regulatory frameworks and government incentives, play a critical role in facilitating this growth. As such, the economic benefits, combined with environmental and social advantages, underscore the transformative potential of AI in IoD, paving the way for a future where efficiency and innovation drive the economic landscape.

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Emerging Trends and Future Directions in ML for Drone-Assisted IoT Networks

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Abstract

Drone-supported Internet of Things (IoT) networks and machine learning (ML) have revolutionized several industries and significantly increased the ability to collect data and make decisions. The constantly changing field of machine learning (ML) applications in networks supporting drone-assisted Internet of Things (IoT). Navigation, real-time data processing, resource management, and autonomous navigation are highlighted as critical areas of progress, and it also looks ahead to prospects. By tackling issues like security protocols, and optimized communication techniques within drone-enabled IoT ecosystems, the integration of ML in these contexts promises to improve operational efficiency, scalability, and adaptability. With further research, these technologies should be vital in forming the upcoming generation of intelligent and adaptable IoT infrastructures. The chapter focuses on Future Directions and Emerging Trends along with the opportunities. It also explores various applications like Cybersecurity, Ethical and Legal Aspects, Meta-Learning Usage in The UAV Field, etc.

Keywords

IoT · UAVs · Blockchain technology · Machine learning

1 Introduction

Over the last decade, unmanned aerial vehicles (UAVs) have gained much popularity and attention. Because of their accurate placement and straightforward control techniques, multirotor drones are the subject of most investigations. Nevertheless, UAVs have real-world constraints, chief among them being a restricted battery's capacity to sustain flight. Creating hybrid systems or internal combustion engines and utilizing docking stations for battery replacement, recharging, storage, and communication with UAVs are some solutions to this problem (Grlj et al. 2022). UAVs are employed across various commercial, military, and industrial domains. In military applications, they provide navigation, secure communication, and observation. Additionally, UAVs are utilized in mobile computing, GSM communication, delivery service, intelligent transportation, healthcare, power-line inspection, surveillance, precision agriculture, remote sensing, search and rescue operations, and disaster relief efforts (Rovira-Sugranes et al. 2022).

IoT has clear economic effects. IoT, however, stands for a sophisticated technical paradigm that goes beyond the catchphrase and the idea of networked things. One way to think of it is as a collection of systems made up of different technologies that collaborate. The building blocks of IoT, the 'things' comprise sensors, actuators, or both combined. Sensors act as data collection points, while actuators act as implementation points. These are connected to the internet via wired or wireless connection for a network of things, or IoT.

The combination of machine learning and drone-assisted Internet of Things networks has made significant changes in different industries, data collection, and decision-making abilities. This chapter focuses mainly on new developments and perspectives in machine learning (ML) for drone-assisted Internet of Things (IoT) networks, emphasizing the revolutionary potential of ML algorithms to improve network performance, security, and efficiency.

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2 State-of-the-Art in Drone-Assisted Networks

AI and IoT are used in drone-assisted networks to facilitate autonomous navigation, cooperative swarm activities, and real-time data processing. By optimizing energy efficiency, enhancing security, and enabling domain-specific applications like agriculture (Roopashree and Babitha 2022), infrastructure inspection, and disaster response, AI algorithms onboard drones and at the edge are influencing the development of intelligent IoT ecosystems in the future.

2.1 AI in UAV Network

Artificial Intelligence aims to surpass human intelligence by generating methods or tools that acquire knowledge and work on it to develop solutions. These methods or tools well suit the application of UAV networks. These days, a lot of network researchers are investigating AI applications in the context of UAV networks. Here the main focus is on areas of the UAV's security & privacy, the design of the network, localization and trajectory, and overall applications.

- **UAV Network Design**

Because UAV networks differ from other types of wireless networks, such as MANETs and VANETs, solutions designed for these networks cannot be simply applied to UAVs. Hence, when tackling the network problems that UAV networks encounter, researchers adopt a distinct strategy. An autonomous flock control method was presented to handle the topology control for a swarm of UAVs. The suggested method sustains the UAV wetland while it is in flight by conserving energy while preserving its topology. The writers have made use of the alignment, separation, and cohesiveness concepts included in the Reynolds-Boid model. The OMNET++ Modeller is used to simulate in order to assess the efficacy of the suggested solution. The authors of Oubbati et al. 2020a address a range of wireless problems that UAV networks encounter and also suggest potential AI-based fixes for the problems that have been found. High dependability, low latency, effective handover, and effective path planning are among the concerns that have been emphasized. There are several AI strategies put forth to improve network conditions for numerous UAVs.

- **Issues with Privacy and Security**

Everywhere there is a wireless network, there are privacy and security concerns. Numerous research are being done to identify the privacy and security vulnerabilities in UAV

networks and provide solutions using AI. The authors of Maruvala et al. (2024) enumerated several security problems that standard cellular network-using UAVs face. Security solutions based on artificial intelligence are being offered to counter a range of cyberattacks (Oubbati et al. 2020a), including cyber-physical attacks. The authors suggest classifying and identifying high-risk areas as well as other features of UAV mobility using convolutional neural networks (CNNs) and recurrent neural networks (RNNs). In a UAV swarm, an unusual study is displayed based on neighboring UAVs communicating just local information. The results of the simulation show that by using local data and avoiding various cyber security threats, a UAV swarm can effectively cover land. The simulation, which accounts for MANET connectivity for each UAV, is performed using OPNET Modeller.

- **Localization and Trajectory**

For aerial nodes or vehicles, trajectory and localization are crucial problems. An AI-based estimation of trajectory for UAV networks was presented in the paper (Li et al. 2021). This method uses a quantum mechanism to direct UAVs from their starting point to their destination. According to simulation data, this suggested approach works better than the conventional Q-learning strategy that is frequently applied in these circumstances. Because of the high node mobility and unstable network conditions, localization is difficult. An intelligent localization technique for swarm UAVs narrows the search space in a 3D model, which decreases localization mistakes and speeds up network convergence. To further improve performance, energy-efficient routing is also incorporated. Methodology simulations were used to assess the approach, and the findings show that the suggested methodology increases coverage time and accuracy of localization.

2.2 UAVs in IoT

- **A Practical Solution to Limitations in Terrestrial Infrastructure**

New uses for UAVs in communication have been quickly recognized by researchers due to the growing need for coverage of networks and the predicted significant pressures on current infrastructure. These days, emergency and on-demand communications services are delivered by UAVs acting as base stations and airborne gateways. Military research has been investigating the use of UAVs in communications applications since the early 1990s. The UAVs might also be used to save limited satellite resources by

enabling Beyond Line of Sight (BLOS) communications during military operations.

The majority of studies initially concentrated on the use of UAVs for emergency assistance and makeshift communication infrastructure. The research on UAVs emphasize the significance of UAVs in disaster-affected areas to increase network coverage and capacity with the advancement of communication technology.

Beyond temporary emergency networks in disaster-affected areas, the advancement in 5G technology and the expanding communication requirements of many heterogeneous IoT devices make it necessary to use UAVs in a hybrid aerial network to provide IoT services. Recent studies indicate that 5G mobile communication will accelerate the development of new applications with critical latency and service quality. Hence, UAVs as a viable remedy for the constraints of infrastructure based on the ground. This claim is further bolstered by emphasizing the burden that cellular networks bear as a result of their incapacity to manage the high traffic demands resulting from an increasing number of users wanting an increasing amount of data and services.

This is further corroborated in Huo et al. (2018), where the authors suggest a revolutionary multi-layer, distributed UAV architecture with a hierarchical structure to make integrating common UAVs into next-generation wireless communication networks easier. Important design trade-offs that take into account both aerodynamic and communication principles are also revealed in this article. The authors evaluate the climatic effects on UAV-enabled 5G high-band communications using empirical models and data from satellite measurements.

An experimental assessment of ray-tracing simulations for UAV-assisted 5G networks is presented in a complementary study in Khan et al. (2020), which aims to evaluate the application of UAVs in next-generation wireless networks. The writers address the technology, examine application cases, and draw attention to urgent issues and directions for further study.

• UAVs from Aircrafts to Linked Platforms

With the help of the rapid breakthroughs in the Internet of Things and the shrinking of sensors, actuators, processors, wireless connections, and energy storage systems, UAVs are finding more and more uses as smart terminal devices that improve our daily lives. With the rate at which novel use cases are being developed, it is obvious that UAVs will play a significant role in the lives of citizens, corporations, and governmental organizations. They will be used for a wide range of commercial services, in addition to disaster relief efforts. Some of the services they will be used for

are border control, search and rescue operations, homeland security, traffic surveillance, construction supervision, and the transportation of medical supplies.

3 Machine Learning Techniques for Data Processing and Analysis

Liu et al. (2020) found that to dynamically support mobile terminal users (TUs) whose movements are governed by a Gauss-Markov random model, mobile edge computing servers, such as UAVs, can be configured to optimize their trajectory. A proposed QoS-based action selection mechanism based on a double deep Q-network (DDQN) is presented, in which the Unmanned Aerial Vehicle trajectory and UAV-TU association are the important constraints. According to simulation findings, this approach guarantees 99% QoS for every TU and converges more quickly than traditional algorithms while achieving a higher throughput.

Jingjing Cui et al. (2020) investigate dynamic resource allocation in multi-UAV communication networks to optimize long-term benefits. This is accomplished by treating the problem as a stochastic game and utilizing a multi-agent reinforcement learning (MARL) framework, in which every UAV independently selects its user, power level, and subchannel. Performance and overhead are balanced by the proposed MARL method, which uses Q-learning and doesn't include any information sharing between UAVs. The simulation findings show that the MARL procedure improves performance when the exploitation and exploration parameters are set appropriately. This suggests that joint learning algorithms and optimal UAV deployment will be the focus of future studies to increase energy efficiency further.

Cui and Wang (2021) proposed an algorithm for MLP using RL. Here, they used local and global information for the Q learning. The lower and higher layers address global long-term and local short-term strategies to establish a collision-free path. Qu et al. (2020) performed UAV path planning with the help of an RLGWO algorithm. Path planning is established using four operations such as exploitation, exploration, geometric adjustment, and optical adjustment. Additionally, a cubic B-spline curve smooths the flight route, and simulations demonstrate the RLGWO algorithm's capability. In a complex environment, this algorithm can generate feasible and effective paths.

For the deployment of UAVs owned by many operators, Wang et al. (2023) propose a multi-agent imitation learning technique to maximize the profitability of UAV owners as well as user utilities. Imitation learning allows for decentralized scheduling with partial observations, while

a Markov game framework guarantees a Nash equilibrium with complete system knowledge. In terms of user utilities, execution time, and profits, the proposed method performs better than competing algorithms.

To detect and categorize drone operations based on radio frequency (RF) data, Raina et al. (2023) propose the ConvLGBM model, which combines the feature extraction capabilities of CNNs with the high classification accuracy of LightGBM. The ConvLGBM model achieves near-perfect performance in drone detection, outperforming cutting-edge techniques like CNN and XGBoost in drone identification and mode prediction. Future research attempts to enhance performance on noisy data, investigate alternative feature generators, and modify the model for numerous drones.

Li et al. (2022) suggest predicting winter wheat yields at different stages of growth employing machine learning (ML) and hyperspectral data from unmanned aerial vehicles. Four base learner models were built after spectral indexes were collected and filtered using three feature selection techniques. An ensemble model outperformed the individual models. The best accuracy was obtained using the Boruta feature collection method during the grain-filling phase, proving the usefulness of hyperspectral imaging for approximating winter wheat output.

Wang et al. (2020a) proposed drone-borne hyperspectral imaging techniques for the detection of nitrogen content in water for the development of plants. This approach can be used for the requirements of conventional water quality detection. Using four ML models (PLS, RF, ELM, and GP) and a decision-level fusion (DLF) model with fractional order derivative (FOD) preprocessing, they aimed to lower regression biases and uncertainties. The results showed that the DLF model outperformed the separate ML models, providing accurate and reliable estimations of TN concentrations and creating new opportunities for dry area water quality monitoring.

Jouhari et al. (2021) suggest a distributed deep neural network (DNN) technique for unmanned aerial vehicles (UAVs) that can effectively classify data on-site while reducing latency and overcoming resource limitations. Two approaches are produced out of the approach, which is presented as an optimization problem and adds mobility prediction to adapt to UAV dynamics: OULD and OULD-MP. The outcomes of the simulation show that these techniques perform better than the current ones. Further research is recommended to use reinforcement learning to enhance optimization in resource allocation scenarios.

For UAVs to simultaneously forecast mobility and profile objects in their proximity, Peng et al. (2018) suggest an unsupervised online learning method. This will improve the UAVs' ability to make decisions in situations such as search-and-rescue missions that do not require

communication with ground stations. The technique uses motion trajectories to anticipate future positions and categorize objects according to how manoeuvrable they are. It achieves a 90% success rate in profiling, which is 7% better than previous techniques. Furthermore, the algorithm's self-tuning capabilities enable it to adjust to novel motion profiles, which qualifies it for real-world uses in IoT systems featuring flying devices and flying ad hoc networks (FANETs).

A comprehensive examination of privacy and security issues in edge computing (EC) architecture is provided by Singh et al. (2022), who also emphasizes the need to protect sensitive data and the increasing number of connected devices. It covers a wide range of edge network attack types and the intrusion detection systems that thwart them, in addition to describing several machine learning and deep learning approaches, applied to EC use cases. The study looks at classifier performance in research articles and outlines prospects and open research challenges for preserving EC with machine learning approaches.

Zhao et al. (2019) used photographs taken by a UAV, the study used an AlexNet deep learning framework to automatically identify damage on the surfaces of wind power generator blades. AlexNet outperformed a conventional BP neural network by 19.424%, achieving an average damage diagnostic accuracy of 99.001% after being trained with 10,000 pictures and tested with 350 photos. This technique shows that it is possible to examine wind turbine blades efficiently and accurately with deep learning and UAVs without having to stop operations.

Singh et al. (2022) address the issues of brightness variation and contrast in low-light aerial photos taken by UAVs by introducing RNet, a unique deep neural network design. Through multiresolution branches, RNet preserves high-resolution data alongside lower-resolution representations, enhancing comprehension of both local and global contexts. RNet performs better in low-light image enhancement than both conventional and deep learning-based enhancement methods when tested on a synthetic dataset.

Park et al. (2020) presented a novel deep learning model called ResNet-SP for using radar spectrogram data to identify UAVs in real-time. In comparison to the baseline ResNet-18, the ResNet-SP model outperformed it, reducing training time from 640 to 242 s and increasing accuracy from 79.88% to 83.39%. This enhancement highlights how well the model performs real-time UAV detection tasks with less computational overhead. In the future, the model will be improved by adding more kinds of UAVs and integrating sound spectrograms to achieve even greater performance improvements.

The weaknesses of fixed cameras can be overcome by using UAVs as moving cameras. In this method, UAVs are

used as mobile cameras for the anomaly detection approach of the UAV-based surveillance approach. CNN, a pre-trained network, and two handcrafted methods were used to extract the features from the videos captured by UAVs. The unsupervised classification was then carried out using a One-Class Support Vector Machine (OCSVM). The new methods outperform the existing strategy, with an AUC of 0.78 at worst and 0.93 at best, despite variations in the assessment metric (Chriki et al. 2021).

4 Optimization and Resource Management

UAVs (Oubbati et al. 2020b) are expected to have a wide range of uses. The following are essential conditions for developing UAV networks: The three main strategies to prevent network interruption are (i) reducing the energy consumption of UAVs, (ii) connecting UAVs in an efficient and adaptable manner, and (iii) incorporating dependable and efficient communication channel models.

4.1 UAV Classification

Choosing the kind of UAVs to use is the hardest part of setting up any airborne communication platform (Sánchez-García et al. 2016). Federal rules, energy capacity, environment, and quality of service (QoS) should all be met by the UAV type that is chosen. The operational altitude, ownership, take-off weight, airspace class, launch technique, and degree of control autonomy are among the various characteristics taken into account while classifying UAVs.

Three types of UAVs can be distinguished based on the previously described attributes:

- *LAP (Low-Altitude Platforms)*

LAPs, or lower-atmospheric altitude unmanned aerial platforms, are quasi-stationary mobile UAV's (such as balloons, small drones, and quadcopters) with extremely flexible mobility. Such gadgets can be deployed most quickly and conveniently. LAPs' adaptability allows combining them with the broadband cellular idea, which includes 3G, 4G, 5G, and 6G, which is extremely beneficial. Short-range line-of-sight (LoS) communication capabilities are what LAPs are known for, and they can significantly lower their energy requirements. Additionally, LAPs' movement can be freely adjusted to find the ideal positions that can provide the greatest coverage at the lowest possible energy usage. LAPs are easily rechargeable and replaceable, if necessary, in the event of a failure.

- *HAPs (High-Altitude Platforms)*

HAP (such as airships or aircraft) is one more aerial platform that has recently caught the attention of the wireless community. HAPs are regarded as unmanned aerial long-endurance platforms that are located above the stratosphere and have less flexible movement. Widespread wireless coverage in regions with broad surfaces, high data speeds, minimal propagation delays, low transmission power, incremental deployment, fewer, longer landing and ground-based infrastructure, and take-off periods than LAPs are just a few of the qualities that HAPs can offer. Their deployments are carried out to give sparsely connected locations long-term connections. Adopting HAPs, however, presents several difficulties, including prohibitive costs, more power consumption, complexity, and significant inter-cell interference.

- *Satellites*

Satellites are a particular class of unmanned aerial vehicles (UAVs) that are launched into Earth's orbit. They can be used for different types of commercial and scientific applications, including weather tracking, satellite phones, telescopes, and Global Positioning Systems (GPS). A sizable number of satellites circle the planet. However, satellite motion is viewed as fixed from the perspective of terrestrial users. Satellites often weigh many tonnes, are extremely expensive, and run on solar energy or other fossil fuels. A variety of satellite types exist, including GPS, drone satellites, communication satellites, navigation satellites, GPS, Medium Earth Orbit, Low Earth Orbit, Geostationary Earth Orbit, and Polar.

5 Security and Privacy Considerations

We also know that IoTs and UAVs generally contain resource-constrained sensors due to vulnerability (Adil et al. 2023), making it impossible to secure them using advanced security methods. Beyond the current body of literature and underlying technology, cost-effective security solutions must be provided for these applications to ensure their security. With recognition of the numerous communication technologies that contribute significantly to the interconnectivity of UAV-assisted Internet of Things applications, including WiFi, NFC (Near Field Communications), Bluetooth, IPv6, Zigbee, 6LoWPAN, Z-Wave, and a few more (Wang et al. 2020b). The interoperable technologies that have been discussed have their challenges when it comes to security risks, which are a legacy of traditional Internet of Things applications. To expand

on this conversation to include UAV-assisted Internet of Things application security while maintaining generality by using AI, DL, ML, and RL-enabled methods. Among other categories, application, session, transport, network, data-link, physical, and encryption attacks can be extensively classified to appropriately handle the security issues associated with these networks.

5.1 Physical Assaults on IoT Applications Assisted by UAVs

Physical attacks on UAVs and IoT devices include unauthorized modification, breach, or damage. IoT applications with UAV assistance frequently face these kinds of threats. These attacks may lead to the exchange of unauthorized information within the network. Malfunctions in deployed devices, as opposed to intentional attacks, are unintentional occurrences that might occur in the network due to routing, software, or hardware issues. An attacker can directly access UAVs and IoT devices through physical attacks, giving them the ability to make illegal changes that disrupt normal operations. These assaults, which compromise lawful operations through techniques including eavesdropping, equipment theft, device tampering, side-channel attacks, etc., frequently take advantage of physical closeness to circumvent security measures (Adil et al. 2020a; Ali et al. 2020).

IoT and UAV devices have recently been physically targeted using a variety of methods and tools. Crucially, an adversary needs to be close to authorized devices to carry out physical attacks. Physical assaults against UAVs and IoT devices can pave the way for a variety of new, general-to-specific problems in IoT applications with assistance from UAVs.

5.2 Attacks at the Data Link Layer on IoT Applications Assisted by UAVs

For IoT applications utilizing UAV support to connect via the TCP/IP protocol stack, a significant number of UAVs and IoTs are effectively combined into a single topological order. With BSs, UAVs, IoTs, and ASs—all of which have security concerns of their own—this is a complex network architecture. In this segment, we will address the security concerns about the data link layer of this new technology in response to these concerns.

5.3 Attacks at the Network Layer on IoT Applications Assisted by UAVs

Network layer attacks involve the use of spoofing techniques to target the network traffic and routing protocol, consequently starting man-in-the-middle assaults at this tier. Recently, UAV-assisted Internet of Things applications' network layer protocols have been the subject of Sybil attacks. Security is a crucial factor to be taken into account to detect malicious assaults early on since, to continue, an attacker compromises legitimate UAVs, IoTs, and other network entities by focusing on several weak points of the network layer.

5.4 Transport Layer Attacks in IoT Applications Assisted by UAVs

With assistance from UAVs, the transport layer in IoT applications is in charge of using transport protocols to transmit data from one client to another and distant locations. Nonetheless, similar to conventional transport layer protocols of Internet of Things applications, vulnerability issues do exist (Sharma et al. 2020; Adil et al. 2020b). According to research, the most damaging attack on this layer is the DoS attack. It obstructs valid network traffic and stops network application services from operating. Since it has been demonstrated in the literature that traditional TCP/UDP protocols do not perform well in these networks, new protocols for safe lightweight transport must be created for networks with limited resources. (Nazib and Moh 2020).

5.5 Attacks at the Application Layer in IoT Applications Assisted by UAVs

Application-layer attacks aim to introduce malicious code into IoT applications that use UAV assistance, thereby compromising the databases of both UAVs and IoTs (Benza'id, C., Boukhalfa, M., Taleb, T. 2020; Siddiqui et al. 2020). The most frequent attacks utilized in these were denial of service, phishing, buffer overflow, viruses, trojan injection, application trojan injection attacks, and exploiting WebApp vulnerabilities. The researcher employed several strategies in the literature to lessen buffer overflow vulnerabilities in Internet of Things applications supported by UAVs. To symbolically identify and stop these threats, they employed

traffic analysis along with static and dynamic programming. By inserting malicious attacks, an attacker might hack the approved devices and change network traffic. In summary, smart botnets monitor network activity to find known security flaws that could be exploited to initiate large-scale denial-of-service (DDoS) assaults. Because of this, it can be difficult to identify and stop attacks that have been launched via these botnets in UAV-assisted IoT applications (Deep et al. 2020).

5.6 Threats from Cross-Layer Security to IoT Applications Assisted by UAVs

This section will discuss various cross-layer attacks that could be used against Internet of Things applications supported by UAVs. To address this, we need to take care of man-in-the-middle attacks, non-repudiation, replay attacks, repudiation attacks, traffic analysis, side-channel attacks, and impersonation attacks (Zhu et al. 2020; Wang et al. 2020c). The majority of these networks' attacks have been covered in the literature by taking into account the attack scenarios stated above (Nazib and Moh 2020). Similarly, passive attacks, in which the attacker watches the data packets and verifies the authenticity of the network traffic, include traffic analysis, transmission channel evaluation, and others. It has been shown in Kore and Patil (2020); Zhao et al. 2020) that certain attacks are extremely difficult to identify or stop since the communicating parties or linked devices typically have no concept of what constitutes legitimate or illicit trade in these networks.

6 Ai-Based Security for UAV-Assisted IoT Applications

This part will address safety issues with UAV-assisted Internet of Things applications by considering AI-enabled methods. This also gives attention to these schemes' shortcomings and provides a clear roadmap for upcoming research areas and unresolved problems. To know this area, the data privacy, authentication, and preservation systems of AI-enabled techniques can be split into two categories, called training and inference schemes (Baccari et al. 2020).

6.1 Security in IoT Applications with UAV Support via ML

This section discusses the security issues of UAV-assisted Internet of Things applications in the context of ML techniques. These techniques are separated into three categories: reinforcement learning, unsupervised machine

learning, supervised machine learning, and semi-supervised machine learning.

- (a) **Supervised Machine Learning Algorithm's Function:** By examining certain data inputs, supervised learning techniques may be a useful tool to address a variety of security issues in UAV-assisted IoT systems (Li et al. 2020). The labelling of a particular dataset is done to train the machine learning system by comparing the expected output and the input data. Moreover, this form of model makes an effort to monitor the obtainable dataset, classify various data types, and predict possible intrusions. A machine learning approach was proposed by the authors of Pham et al. (2020) to detect and fix multiple vulnerabilities in 5G networks that rely on satellites.
- (b) **Unsupervised Machine Learning Algorithm's Function:** This section discusses unsupervised learning approaches concerning the security of UAV-assisted IoT applications. It is not necessary for unsupervised machine learning approaches to label data to arrange it into distinct categories to explore; they can take inputs without desired targets (Shrestha et al. 2021). Similarly, we also know that the type of given data typically determines how well-unsupervised machine learning approaches classify and categorize data. It should be quite clear that supervised machine learning requires knowledge of both the output and input data, and that the system will be taught based on the desired situations.
- (c) **Semi-supervised machine learning algorithms' function:** Countermeasure approaches utilizing semi-supervised ML algorithms have been employed in UAV-assisted Internet of Things applications to enhance network security. While the previous method followed one of the two categories of data—labelled and unlabelled—the semi-supervised machine learning technique adheres to both (Dalal 2020). Since the labelling cost of data is rather expensive in many of the semi-supervised ML algorithm countermeasure strategies now in use, expert systems or experienced individuals are required (Engelen and Hoos 2020). In summary, several semi-supervised machine learning models that lacked a labelling model had mediocre results, but a model that adhered to appropriate labelling produced notable outcomes when it came to the identification and defense against attacks in the networks.
- (d) **The purpose of the reinforcement learning algorithm:** To assist reinforcement learning (RL) for IoT application security, this section focuses on the use of UAVs. Instead of using a preset system, the agent or model in this reinforcement learning learns from

an interactive environment or past system feedback (Mohammed et al. 2020). IoT application security utilises DL to support UAVs. This section will discuss the function of DL algorithms in the context of Internet of Things applications that are supported by unmanned aerial vehicles. From now on, it is recognized that the origin of the DL approach is the ANN that was created with neurons, which are believed to represent variables in the example (Yi et al. 2020). Moreover, we understand that “deep” relates to the number of layers in a specific network, but “deep learning” (DL) is a sign of a large-scale deep neural network (DNN) (Munaye et al. 2021). Furthermore, even if the data is unlabeled, unstructured, uncategorized, and unsupervised, we are aware that the DL technique has been extensively used in data analysis and distributed computing in several Internet of Things applications.

7 Layer-Wised Protection Measures

This section discusses layer-wise countermeasures.

(a) ML, AI, and DL Algorithms for Perception Layer Security in UAV-Assisted Internet of Things Applications:

Ensuring the legitimacy of participating devices through authentication is crucial for UAV-assisted IoT applications. To address the authentication issues with these networks, different authentication algorithms have been employed in the literature (Yu et al. 2021). The authors of Hou et al. (2021) suggested an ML framework that makes use of historical data properties to thwart physical layer security risks in UAV-assisted Internet of Things applications. The authors proved that their suggested methodology reliably identifies spoofing attempts and ensures that the data supplied in UAV-assisted Internet of Things applications is legitimately employed.

The numerous attack IoT security models were examined to highlight the various ML, RL, DL, and AI algorithms that have been utilized recently to guarantee the security of IoT applications across a variety of areas. The authors also investigated secure task offloading, authentication models, and access control models enabled by machine learning, followed by malware detection in networks.

(b) Countermeasures for ML, AI, RL, and DL-Based Jammer Attacks:

The authors of the paper (Banu and Padmavathi 2022) discussed GPS jamming and spoofing attacks in the context of UAV-assisted Internet of Things applications that have been exploited to disrupt communication channels or

spoof the official communication protocol of a network in use. The authors then reviewed several ML, AI, RL, and DL-enabled strategies that have been applied to GPS-assisted UAV and Internet of Things networks to thwart these attacks.

To assist them in making informed decisions about jamming assaults on these networks, the authors employed two controllers in this model, such as a UAV flight controller and an SDN controller. Han et al. (Han et al. 2020) suggested a distributed dynamic anti-jamming technique for army IoT (SaIoT) networks that are satellite-enabled. The authors used the hierarchical anti-jamming Stackelberg game (HASG) in this model. It had a leader subgame to engage in confrontational interactions between jammers and SaIoT. The coalition change principle and preference order were regulated using an anti-jamming coalition formation game (CFG) to get better results with less complexity and energy consumption.

(c) ML, AI, RL, and DL-based authentication:

A crucial part of UAV-assisted Internet of Things applications is the authentication of authorized devices, which guarantees the network's legitimacy. The author of Michailidis and Vouyioukas (2022) provides current study findings on the data privacy plan and AI-enabled authentication for UAV-assisted IoT applications. The authors also covered the traditional authentication schemes in blockchain, mobile edge computing, and machine learning, including Elliptic-Curve Cryptography (ECC), hash functions, and Public Key Infrastructure (PKI).

(d) Network Layer security using ML, AI, and DL algorithms in an IoT application aided by UAVs:

The network layer is crucial for heterogeneous UAV-assisted Internet of Things applications' communication, routing protocols, sessions, and security. In addition, this layer is in charge of granting everyone access to the network's application and perception layers for data compression, data storage, and availability, which is followed by data communication. After that, strong technical capabilities are required for the network layer security of IoT applications aided by UAVs during interconnectivity, routing, or session formation.

(e) ML, AI, and DL algorithms for application layer security in UAV-assisted internet of things applications

UAVs are currently commonly used in IoT applications installed in difficult-to-reach regions, facilitating better communication and producing desired outcomes (Donta et al. 2021). This technology's application layer security is essential to provide dependable access to network data. Regarding the security of this layer, different

learning models have been used, including ML, AI, RL, and DL-enabled techniques, to identify spyware, injection attacks, and unknown malware.

- **Authentication Latency:** The need to verify each transaction before adding it to a new block can delay the authentication process. (Kamboj et al. 2021)

8 Future Directions and Emerging Trends

Real-time data analysis and decision-making are made possible by integrating machine learning with drone-assisted IoT networks.

8.1 Integration of Drones into the 5G Network

5G-enabled Internet of Things (IoT) systems with low latency, high efficiency, and high reliability can greatly enhance drone communication and data collection. However, the existing 5G network faces issues such as identity data fraud, forgery, and privacy leakage. We need to secure drone networks by securing their different parameters and characteristics, including decentralisation, diverse and resource-constrained devices, heterogeneous data, complex networks, and poor interoperability. Interoperability problems can occur since the devices used across different domains use different authentication schemes to include security, which can lead to difficulty in communication among drones over the network and hence complications in the security method during integration.

As the system grows, more devices overload the authentication service centre, rendering traditional centralized authentication techniques ineffective. Single-node servers run the risk of becoming paralyzed by attacks or failures, which might have serious financial and legal repercussions. Furthermore, drone operations frequently cross several domains, necessitating the safe sharing of data between devices operating in several domains. While existing infrastructure networks facilitate connections, establishing secure communication is challenging due to the potential lack of trust between these domains. An authentication strategy based on blockchain known as a cross-domain scheme can be used to address these limitations in 5G-enabled drones. This scheme ensures secure and effective authentication by using a consensus mechanism to efficiently verify transactions and process interactions within the blockchain.

Using blockchain for authentication faces two challenges:

- **Limitation of storage space:** A bottleneck in throughput may result from each block's restricted size.

8.2 Integration of Drones with IoT

The development of the upcoming wireless networks for an Internet of Things-enabled society is becoming more and more dependent on drones. Drones can be used for different purposes where they act as data-collecting devices. Drones can record accident scenes, keep an eye on building sites, parking lots, and smart homes, monitor traffic, keep an eye on agricultural areas, and monitor and direct search and rescue teams in the event of a disaster. A drone is used to collect information from areas that are not easily accessible by humans. If the drone is used for such information-collection applications, we need to ensure that there is constant contact between drones that are available in the vicinity, and between drone and ground station systems, which may be human. So, there should be a technique that measures signal strength between each device so that continuous connectivity is maintained among them. This signal strength can be used to ensure a better quality of service and identify the coverage area of the drone. Drone distance, route loss, altitude, transmission power, transmitter and receiver heights, and signal frequency are among the pertinent factors that are considered to be used in the artificial neural network (ANN) forecasting process. The drone's flight path is then predicted using the estimated signal strength. The best route for flying a drone can be found using signal strength. Prediction of location also can be done based on the received signal strengths of the drone flying. So to choose best route and to locate the drone based on signal strength suggested ANN can be used which is dependable, practical, and quick. Improving QoS and real-time connectivity between drones and IoT devices will be possible with this. The number of devices that will transmit data to the drone and the amount of energy needed for these devices to do so can be determined by estimating the intensity of the drone's signal. The best placement for each drone is determined using ANNs, which helps IoT framework applications collect more data. The incorporation of drones with the IoT is opening up new applications that will require a great deal of technology development and study in sectors relevant to smart cities, like smart industries, smart homes, and smart buildings. Several situations are envisioned to show the practicality of the suggested ANN technique for forecasting the signal strength of a drone. Characteristics such as the altitude of the drone, signal frequency, transmission power, height of the receiver antenna, and path loss are among the data that the ANN system uses (Feng et al. 2021)

8.3 Integration of Drones with Blockchain Technology

Because it offers decentralized data, accessibility, immutability, and irreversibility, blockchain technology becomes essential for industrial applications. When it comes to getting products and medical supplies to people who are being quarantined at home or in hospitals because of the pandemic outbreak, multi-drones are essential. Blockchain helps update each drone's location on the distributed ledger so that each drone may reach all drone positions updated in the smart contract, preventing collisions. Blockchain thus allows drones to register information about time, resources, global states, position, delivery data, and the relationship of a particular drone with all neighbouring drones, and it also makes data available for all drones. Robots are now easier to scale, control, and be resilient against failure thanks to the blend of blockchain skills and multi-drone collaboration in the fight against the COVID-19 pandemic. Blockchain technology manages robot traffic efficiently and supports multi-drone collaboration to prevent multi-drone collisions, protect robot operations, or guarantee robots stay on course within the quarantine region. Consensus algorithms facilitate consensus-building in a decentralized blockchain network with multi-drone collaboration. Blockchain is a vital component of an efficient, highly accurate drone traffic management system. Blockchain also contributes to the security of drone operations by guaranteeing that every drone in the network stays on course, preventing damage, crashes, and injuries. Delivering orders and supplies in the quarantine region requires the cooperation of several drones and Internet of Things devices. Drones can deliver a package to a person under quarantine at home or in a hospital by interacting with a window or door through the use of Internet of Things devices. To guarantee that a particular drone can access a door or window, blockchain technology is utilized to store information about the drone. When the drone gets close to a residence in the quarantine area, the updated location information is shared with the window or door of that particular house. As a result, IoT devices at the entry point of the purported house allow them to recognize and validate the blockchain to open and allow drones to enter the home. If people disregard social distance, the blockchain can also be utilized to generate alert messages for law enforcement. Using a drone, the process of identifying contaminated cases is possible. Drones update the smart contract in the blockchain with the location, case level, and requirements. (Alsamhi et al. 2021)

8.4 Internet of Drones' (IoDs)

With the advent of self-driving cars, robotics, and city brains, artificial intelligence (AI) has advanced to the point that the so-called smart city is evolving into an as-yet-unidentified autonomous urban creature. Because of the Internet of Drones' (IoDs) many uses in transportation, weather monitoring, emergency preparedness for earthquakes and floods, healthcare, and traffic dangers, researchers are becoming increasingly interested in this field. Real-time data sharing is necessary to update the emergency scenario data. However, consistent message delivery by different drones could overburden a central server and clog the network. Reducing messaging costs and congestion is essential. Congestion avoidance can be achieved by employing a layered model and a message-forwarding algorithm to choose the proper next-hop node. This model is predicated on many drone-equipped layers. Initially, it searches a layer close to the fog server for a suitable drone to forward messages to. The drone is then located at adjacent levels to transmit the emergency message to the following hop, which will locate the group head more expediently. This drone's one-hop circle provides it with a shorter distance to fog servers and information. After sending the message to the fog server, it can cease forwarding. Lastly, the fog server promptly provides information to higher layers so they can take the appropriate emergency response measures (Alsamhi et al. 2021).

Environment monitoring and conservation are being revolutionized by the IoDs, a multi-layered network control architecture. Drone location privacy, real-time environment monitoring, task collaboration, data sharing, and other security issues will all be severely hampered by a lack of trust among drone controllers. To tackle this significant problem in the Internet of Things, a paradigm is suggested that makes use of blockchain technology and smart contracts to guarantee reliable cooperation amongst controllers of software-defined IoD (SD-IoD). To monitor the environment in a smart city SD-IoD architecture is suggested. Utilizing task publication, SD-IoD users can swiftly implement new applications without worrying about the underlying equipment's technical specifications. To facilitate safe collaboration and communication among SD-IoD controllers, a new consensus method called proof of security guarantee (PoSG) can be used. Consensus and blockchain layers are required to enable drones and associated equipment to work together, particularly drones from various service providers. While the blockchain layer primarily executes the creation of the consortium blockchain, the consensus layer primarily implements the functionality of the PoSG. (Yaqoob et al. 2021)

8.5 The Opportunities for the Future Using UAVs

- **Cybersecurity**

A Recurrent Neural Network-based anomalous behaviour detection method for UAV networks was proposed in Hajiyeve and Soken (2013). As a standard for the industry, we suggest introducing physical security keys to enable UAV authentication. Additionally, machine learning techniques must be used to determine safe and ideal flight paths that reduce the possibility of problems like jamming attempts.

- **Ethical and Legal Aspects**

The widespread advantages of AI in UAVs have made it possible for UAVs to be used in previously unheard-of ways in daily life. This consistently garnered criticism regarding data security, privacy, and responsibility violations. To allay potential worries from UAV users and customers, robust algorithms addressing each of these issues must be created. Additionally, education about the technological advancements being made in this field must be promoted.

- **The use of edge computing**

Optimizing UAV performance while staying within the constraints of available resources has been a significant obstacle in the progress made on UAVs. A potential remedy for these problems is edge computing, which permits real-time analysis of the gathered data while minimizing the requirement for frequent data transfers to cloud servers. Additionally, it guarantees that only pertinent data is transmitted within the UAV network, minimizing network bandwidth waste and enhancing the security and privacy of critical data.

- **Utilizing Cognitive Education**

Unmanned aerial vehicle (UAV) flight requires autonomous systems to be able to make fast judgments in real-time in response to changing external stimuli. Nevertheless, creating UAVs with human-like. The ability to think is slow. By using cognitive learning to tackle this problem, UAVs can become more adept at planning routes, allocating resources, and responding to novel situations because they will be able to learn more, adapt their tactics, and eventually increase their performance.

- **Meta-Learning's Usage in The UAV Field**

AI models can learn how to learn through a process called meta-learning. It entails using different tasks to

train models to create flexible methods, allowing people to quickly adjust and perform well in new, unknown situations. The increasing number of UAV users requires that UAVs' computing speed and accuracy be increased.

9 Trends and Opportunities for the Future

Several issues and potential directions for future research in AI-integrated UAV systems are discussed in this session. Figure 1 shows many challenges and potential areas of advancement.

9.1 The Challenges in UAVs

- **Navigate Independently**

Creating AI algorithms that allow for course planning and autonomous navigation is one of the main problems the industry is attempting to solve. UAVs must be able to navigate and function in complex, constantly changing environments.

To enable autonomous navigation, it must navigate through environments, avoid obstacles, and react quickly to sensor inputs by making real-time judgments. Additionally, most recent research in this field assumes fully and homogeneously connected networks with no connection discontinuities, suggesting path-planning methods. Future research may focus on creating light computation-based algorithms that enable real-time route prediction in three-dimensional space.

- **Connectivity of Sensors**

To sense their environment, unmanned aerial vehicles (UAVs) respond to stimuli from onboard sensors, including cameras, radar, and GPS, combining input from several sensors to calculate. Being aware of your external surroundings is a difficult endeavor. Robust Kalman Filter (RAKF) was proposed by Hajiyeve et al. (Hajiyeve and Soken 2013) as a solution to UAV sensor malfunctions. Even in the event of a sensor failure, RAKF adapts to the presence of defects and ensures correct data interpretation.

- **Decision-Making And Mission Planning**

The ability of a UAV to make decisions in real-time while on a mission is one of the most important components of autonomous flight. Nevertheless, one of the biggest challenges in algorithm design is taking into account variables like mission parameters, real-time information, and weather. Additionally, UAVs must adjust to unforeseen events and changing conditions. To improve the suitability and precision of potential approaches to a given decision-making

Fig. 1 Challenges and Opportunities for the Future



challenge, we suggest using genetic algorithms in future work. This will increase the mission's efficiency as well.

- **Communication And Networking**

Swarm-operating UAVs require a latency-free, effective, and fast communication system to cooperate and exchange data to complete challenging tasks. It is difficult to provide such networking capabilities among UAVs in dynamic environments while preserving dependable communication. Due to environmental conditions and cyber-interventions, the communication channels are also very prone to failure. When calculating flight paths, ML approaches can be used to incorporate the link and path lifetime as a crucial element, extending the link's lifespan and decreasing connectivity loss.

- **Constraints In Resources and Scalability**

Due to size, weight, and power limits, UAVs have limited processing capability and memory effectiveness. To make AI solutions feasible on UAV platforms and to allow the UAVs to function within these constraints, hardware and

algorithms are needed. Strong AI algorithms are required to handle unforeseen situations, detect and mitigate faults, and preserve system integrity to prevent accidents or failures.

- **Power Suitability**

Operational constraints also impose limitations on the size and capacity of UAV batteries. The UAVs' limited flying time and cargo capacity prevent them from being used as full-scale agents in industries like commodity transportation. The concept of using lithium-ion batteries to supplement hydrogen fuel cell batteries, which can be used when the former runs out of power, should be pursued for the mission.

10 Applications of UAV

Several studies are investigating the potential uses of UAV networks. Such uses include the employment of these technologies in high-risk area coverage, vehicle networks, cellular networks, and efficient spectrum usage. The authors of Fatemidokht et al. (2021) looked at using UAV networks

to improve VANET performance. They suggested routing rules that help identify malevolent vehicles in addition to offering cars better, more dependable coverage. The simulation results, which demonstrate a roughly 7% improvement in the identification of hostile vehicles using UAV networks, are encouraging.

- ***Asset and Traffic Monitoring***

The investigation forecasts the possibility of gathering traffic data using overhead video footage. According to the study, a number of important metrics, such as journey durations, turning counts, automobile traffic densities, and queue lengths, may be derived. Researchers looked at the usage of UAVs for traffic and asset monitoring applications in a more recent investigation. They suggested and evaluated the use of rotary-wing unmanned aerial vehicles (UAVs) with inbuilt cameras for a thorough traffic monitoring system. To identify automobiles, the study used data processing and video algorithms built on the Haar cascade model, which results in a system that can monitor traffic conditions with flexibility and accuracy.

- ***Persons And Crowd Monitoring***

In recent years, several research initiatives have investigated crowd surveillance. A collaborative micro-drone system was proposed to follow individuals in emergency scenarios. A brand-new airborne-based framework for high-performance crowd monitoring was unveiled that utilizes video data and specialized object identification methods to estimate crowd density and mobility. Novel approaches to the known labour and complexity-intensive task of gathering pedestrian traffic data are also of interest. The authors contend that standard techniques for examining pedestrian flow features and behaviour, including surveys, on-site video recordings, and manual observation, may have drawbacks.

Novel techniques that use UAVs have become more prevalent in recent years as a reaction to these issues. Large-scale crowd-tracking systems (Jiang et al. 2021) are suggested by more recent research to solve privacy and security issues related to UAV-based crowd surveillance (Xiao et al. 2021).

- ***Environmental Monitoring***

Researchers have looked at the creative uses of mobile robots in a range of monitoring scenarios within the last 20 years. New avenues for research that might lead to widespread environmental monitoring, emphasizing the combination of wireless sensor networks and cooperative robotic

teams, have been explored. They stress that the natural sciences are advanced by these developments because they make it possible to assess environmental processes accurately and efficiently at large scales.

The emphasis of more recent research is still on how important it is to continuously monitor changes in environmental characteristics. The study makes recommendations for creating an affordable, lightweight UAV platform that is intended for environmental monitoring. Considering the critical role environmental monitoring plays in diagnosing climate change and mitigating its effects on natural and agricultural systems, researchers are always introducing new technologies and proposing initiatives to keep up with the demand.

A hybrid wireless sensor network (WSN) system called AQNet, introduced by researchers, is intended for exact air quality forecasting and monitoring in urban settings. Air quality at various altitudes is monitored with the cooperation of UAVs and integrated hundreds of programmable ground sensors. The report includes experimental examples that confirm the proposed system's scalability. A similar suggestion is presented in Polonelli et al. (2020), which complements this study.

- ***Agricultural Monitoring***

Agriculture is another developing field that has a lot of promise for using UAVs. A remote sensing system constructed around an autonomous UAV is offered to show that a UAV or drone-based framework, outfitted with multi-spectral sensors, could track turf grass glyphosate levels, therefore demonstrating the adaptability and dependability of UAVs in precision farming. The latter highlights the promise of the pictures captured by low-altitude UAVs because of their high temporal and geographical resolution, high degree of programming flexibility, and cheap operational cost in environmental monitoring. The report also provides an overview of current research on the use of UAV images for PA, demonstrating the need for advancements in platform design, manufacturing, picture geo-referencing, and mosaicking uniformity, as well as the information extraction process, to give farmers a dependable final result. The study focuses on the present and future uses of thermal remote sensing in PA, along with several issues that are specific to the PA application, such as crop growth phases, atmospheric conditions, and spatial and temporal resolution. As a result, a new field known as precision farming has emerged and is rapidly attracting the interest of scientific communities. One recent example of this is using a narrowband Internet of Things network to study various soil parameters that were previously impractical to examine.

• *Security and Surveillance*

UAV security and surveillance applications originated in the military and are now more prevalent in the commercial and public sectors. Scientists are constantly researching using the mobility and agility of UAVs for security applications, from target following and tracking to border control. The Adaptive Multi-scale Optimisation (AMO) resource-usage management strategy for UAV surveillance missions uses a series of simulation cycles spanning numerous applications to illustrate the advantages and trade-offs of AMO. Furthermore, the researchers proposed frameworks for unmanned aerial vehicle (UAV) security and surveillance systems for smart city and maritime applications, highlighting the potential benefits these applications might have for society. A novel cooperative network system and multi-UAV monitoring system design first describe the design principles of a cooperative resource sharing and job assignment method for multi-UAVs. It then goes on to describe the localization and tracking model, which combines data from many sources, and the tiny target identification approach. Due to the complexity of the monitoring field, this article also covers the construction of appropriate machine learning-based methods. The authors substantiate their conclusions with practical demonstrations of multiple-moving object identification and tracking using the proposed multi-UAV systems. In a related recent paper (Castellanos et al. 2020), a distributed navigation method and innovative surveillance optimization for UAV networks are presented in ground vehicle tracking applications.

• *Infrastructure Inspection*

Unmanned Aerial Vehicles, or UAVs, have attracted a lot of interest in both commercial and scientific uses, especially for inspecting infrastructure. Unmatched flexibility in reaching and measuring areas that were previously thought to be dangerous or unreachable for human workers is provided by these flying platforms. Researchers provide a thorough analysis of robotic systems intended for infrastructure assessment. In order to present a current overview of the state-of-the-art in inspection robots, this study synthesizes a number of works. It also looks at potential future directions for progress and highlights present problems in the subject. Researchers also offer a UAV-based method to solve particular issues with infrastructure inspection in Japan. The authors provide a compact manipulator system that is built into unmanned aerial vehicles. This breakthrough makes it possible to assess aged infrastructure in places where it is hazardous or difficult for humans to reach. These citations highlight how UAVs have revolutionized infrastructure inspection by improving their capabilities. Researchers and practitioners want to increase global critical infrastructure

monitoring and maintenance accuracy, safety, and efficiency by utilizing these technologies. The obtained results for a set of standard infrastructure inspection flights show how accurate flight prediction tools may be used in practical settings with automated flight control.

• *Search and Rescue*

Since the early 20th century, search and rescue (SAR) operations have undergone substantial change. International SAR plans have been established by worldwide organizations to coordinate missions that attempt to locate and aid distressed assets in distant or difficult locations. Researchers have looked at new tools and techniques to improve SAR missions as technology has advanced. Unmanned aerial vehicle (UAV) integration with SAR networks is one interesting area of development. Small UAV systems are presented in a research article to help ground task forces with situation assessment and surveillance, which is an essential support function. Experiments were conducted to validate this architecture: first, in an earthquake reaction scenario within a controlled military testing station; and second, in a real-world mission supporting relief efforts following major floods in Bosnia in 2014. The effectiveness of employing UAVs in SAR missions was shown by the success of these trials, which prompted further study into the possibilities of completely autonomous UAV systems for SAR applications. All things considered, the incorporation of UAVs into SAR networks improves operational efficiency by offering quick deployment, improved situational awareness, and efficient data collection in difficult settings where conventional techniques might not be feasible. This technical development is a major step in raising the efficacy and security of SAR operations throughout the globe.

• *On-demand Logistics Delivery*

UAVs are gaining in popularity in logistics. The control of the movement of goods from their place of origin to their location of consumption to fulfill certain criteria is known as logistics. They are an extremely affordable option for container ports, warehouses, and many other uses. When Amazon revealed its intentions for Prime Air on December 1st, 2013, a new era in commercial package delivery began. Early in 2014, after Amazon's promotional video had received over 14 million views, the possibility of package delivery by tiny unmanned aerial vehicles was explored. However, since several issues needed to be resolved at the time, not much research had practical applicability in this field. The researchers highlighted the potential and challenges of UAV-enabled autonomous transportation systems for the next generation of smart cities. Here, the researchers

Table 1 Summary of the ML algorithms used in UAVs

References	ML algorithms used	Purpose
Liu et al. (2020)	RL & deep reinforcement learning algorithm	Measure the computational activities transferred by mobile terminal users (TUs)
Cui et al. (2020)	MARL	Dynamic resource distribution in multi-UAV communication networks to enhance long-term benefits
Cui and Wang (2021)	Bi-layered RL Algorithm	Local and Global information for the Q learning
Qu et al. (2020)	RLGWO	To overcome the path planning difficulty in a three-dimensional flight arena
Wang et al. (2023)	Combination of CNNs) GANs, and the gradient-based policy	To enhance the benefits of UAV ownership and ground users' utilities
Raina et al. (2023)	ConvLGBM model & CNN with LightGBM	Identify and anticipate unwanted drone activities
Li et al. (2022)	Machine Learning Model Based on DLF	Winter wheat yield forecast with UAV-based hyperspectral images
Wang et al. (2020a)	DLF Model	Water quality detection using remote sensing
Jouhari et al. (2021)	DNN model	Air-to Air Communication
Peng et al. (2018)	Unsupervised online learning algorithm	Prediction of the future locations of the nearby flying objects
Singh et al. (2022)	Different ML algorithms	Edge security in the UAVs
Zhao et al. (2019)	AlexNet	Assess the damage to wind turbine blade surfaces in actual industrial settings
Singh et al. (2022)	RNet	Aerial picture augmentation for low light conditions
Park et al. (2020)	ResNet-SP	Tracking various forms of human activity and UAV signals
Chriki et al. (2021)	One class support vector machine & CNN	Inspecting a parking lot

address several methods for approaching the conventional hypothetical small package delivery UAV ideas, indicating potential directions for further study. Use cases were initially studied in military logistics applications, but a change towards more commercial, and notably efficiency-critical, medical delivery, was brought about by the present global health crisis (Savkin and Huang 2021).

• *Emergency and Medical Services*

UAV for e-health is another application area that has just lately come into existence and is drawing more and more researchers. The usage of drones in Swiss medical facilities was also investigated. The study outlines how Swiss hospitals may use the integration of unmanned aerial vehicles (UAVs) to reduce expenses and optimize processes, therefore mitigating the impact of rising costs and technological advances. Also, the use of unmanned aerial vehicles (UAVs) to assist in the delivery and pickup of test kits and medications for patients with ongoing medical conditions who must go to clinics for routine check-ups and medication refills in remote regions. The authors of a recent study emphasize the need for optimizing emergency services in a time-sensitive manner. This is further corroborated by the authors, who detail how the Silicon Valley Technology High Flyers team, which decided to compete in the Medical Express UAV Challenge, designed an autonomous vertical take-off and landing (VTOL) aircraft.

11 Conclusion

The book chapter outlines the studies of drone-assisted IoT networks, the different machine learning techniques used for the analysis of data, the investigation of new trends, and potential directions in machine learning techniques for drone-assisted Internet of Things networks. It highlights the significance of ongoing research and cooperation to fully fulfill this promise and considers the revolutionary potential of machine learning in augmenting the capabilities of these networks.

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